Jaemin Lee

Artificial Intelligence and International Law

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About the Author

Jaemin Lee is currently Professor of Law at School of Law, Seoul National University in Seoul, South Korea. He obtained LL.B., LL.M. and Ph.D. from Seoul National University; LL.M. from Georgetown University Law Center; and J.D. from Boston College Law School. His major areas of teaching and research are public international law, international economic law and international dispute settlement. Upon graduation from the College of Law, Seoul National University in 1992, he joined the Korean Ministry of Foreign Affairs as a foreign service officer. His post in the ministry included deputy directorship of the Treaties Division and the North American Trade Division. Between 2000 and 2004, he practiced law with Willkie Farr & Gallagher (Washington, D.C.) as an associate attorney of the firm's international trade group. From 2004 to 2013 he taught international law at School of Law, Hanyang University in Seoul. He has published articles and books (including book chapters) on various topics of international law and international dispute settlement. He is currently serving as a member of the Executive Committee of the Society of International Economic Law, and as a member of the Steering Committee of the ISDS Academic Forum. Previously, he served as president of the Korean Society of International Economic Law, and vice president of the Korean Society of International Law.

Chapter 1
Introduction



Recent advancement in Artificial Intelligence (hereinafter AI) is a huge leap forward for our daily lives, affecting a broad range of sectors in the international community. General purpose technologies (hereinafter GPTs), or technologies that change both household life and ways of conducting business, are especially important for their wide societal impacts. According to a Brookings report, following the steam engine, electric power, information technology (hereinafter IT), AI is the latest GPT development.[1]

The first question to ask ourselves is how to define "AI". This term is used in a broad manner in diverse contexts. The Oxford Dictionary defines AI as "the theory and development of computer systems able to perform tasks normally requiring human intelligence".[2] Indeed, "computer systems" that are capable of performing "tasks normally requiring human intelligence" is so broad that it could cover almost everything around us today.[3] The two key words ((i) computer systems and (ii) human intelligence) may explain the whole surroundings of modern life. More specifically, the term can be defined as follows: "AI refers to systems designed by humans that, given a complex goal, act in the physical or digital world by perceiving their environment, interpreting the collected structured or unstructured data, reasoning on the knowledge derived from this data and deciding the best action(s) to take (according to pre-defined parameters) to achieve the given goal. AI systems can also be designed to learn to adapt their behaviour by analysing how the environment is affected by their

[1] *The author would like to thank the Hong Jin-ki Legal Research Foundation for its generous support for the research and publication of this book. All internet links cited in this book were last visited on December 31, 2021.

See Indermit Gill, *Whoever Leads in Artificial Intelligence in 2030 will Rule the World Until 2100,* Brookings (Jan. 17, 2020), available at https://www.brookings.edu/blog/future-development/2020/01/17/whoever-leads-in-artificial-intelligence-in-2030-will-rule-the-world-until-2100/.

[2] Oxford Dictionary of English 89(2nd ed.2003), *cited by* Jack M. Beard, *Autonomous Weapons and Human Responsibilities*, Georgetown Journal of International Law, Vol. 45 (2014), p.624.

[3] Ibid.

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previous actions."[4] The three key words here in turn would be: (i) systems, (ii) data, and (iii) choosing the action to take. The second item (data) and third item (action choice) would arguably together mean human intelligence. Thus, the definitions are largely the same. Other definitions of AI are not that different. This book employs the same definition for AI.

AIs are welcomed as a new catapult for a future society. Not only can production efficiency be significantly enhanced through AI, but also AI can enable people to achieve tasks that has not been possible before: drones can be deployed where humans cannot enter so as to collect data, or satellite images can be analyzed real-time to predict abnormal natural phenomena with a surprising level of accuracy. At the same time, AI is also feared as a tool or scheme for new and additional regulation in a way that has not been contemplated or envisaged before. AI allows real-time data collection of people's movements, speeches, behavior and various activities. The data so collected can be processed promptly for many purposes. As a consequence, governments (or any other authority in possession of the data) can track us. Consider what is happening globally during the COVID-19 pandemic.

In the meantime, AI systems, such as drones, autonomous entities, and killer robots, are now generating a variety of legal complexities in the absence of meaningful norms and regulation. This phenomenon is being observed in all legal systems, be it domestic or international. In this context, in the age of AI international law also confronts a myriad of new challenges to address, owing to recent technological developments—which include new AI programs, algorithms, and machines—in global and domestic societies. In this regard, identifying and grasping legal implications for international law ensuing from the fast advent and application of AI, and analyzing existing norms of international law from the perspective of this new phenomenon are of utmost concern.

With this in mind, the aim of this book is to explore how AI is rippling through our societies, and how it affects and implicates the existing principles and norms of international law, though the entire spectrum and scope of the advent and application of AI is not clear yet. In this context, new forms of inter-state conflicts and new subjects and objects of international law are discussed in this book, along with select noteworthy up-to-date developments worldwide in this field. More specifically, various outstanding issues of international law in the new era of AI, categorized under headings of human rights, state sovereignty, state responsibility, dispute settlements, law of war, and north–south divide, are addressed respectively in the book. This book does not claim that it covers all the issues of international law arising from AI; instead, it focuses on the most conspicuous topics ushered in by the emergence of AI. The major contribution of this work is claimed to be its effort to provide the first overview of the overall impact of AI on international law and to show how international law should react—in order to update itself and fill the legal vacuum

[4] High-Level Expert Group on Artificial Intelligence (AI HLEG), *A Definition of AI: Main Capabilities and Scientific Disciplines* (Dec. 18, 2018), available at https://ec.europa.eu/newsroom/dae/document.cfm?doc_id=56341.

in the increasingly hostile (anti-globalization, regional conflicts, pandemic, etc.) environment of the international community of today.

This book aims to examine one of the most controversial and important issues of the moment—AI. AI is already rapidly developing, spreading and permeating in our daily lives. It is being utilized expansively in many different sectors of our societies. Meanwhile, the rise of AI raises complex questions and poses new challenges in the legal area: new products and services involving AI require new regulations and standards to minimize potential negative side-effects and maximize possible benefits of the new technology. Not only in terms of domestic law, but also in international law, these legal questions keep emerging. How to address these legal issues at this point is expected to have a long-lasting impact on law and thus the global community. Mindful of this timeliness and importance, this book focuses on the impact of AI on international law and seeks ways to develop or fine-tune international law to adequately accommodate the new challenges of the AI era.

Chapter 2
The Advent of AI and Its Present and Future Application



The twenty-first century has seen (and is seeing) an unprecedented and groundbreaking advancement in technology, namely, the advent of AI. In many sectors of our daily lives, AI has already reached a level beyond being mere tools for human activities, as some forms of AI can even carry out automated decision-making processes, without human interventions. It is expected that the advent and permeation of AI will change our lives to a significant extent, and its use is becoming ever more common in all respects of people's daily activities. In fact, its use is now almost ubiquitous, though with varying degrees, ranging from medical treatment to dispute settlement, and from private to public sectors. The United States, China, the European Union and many other countries have already introduced such novel technology to fulfill the government tasks and enhance government functions. This government function component now enabled by AI is the key concern of this book, as it is such component that implicates international law.

The fast advent and application of AI has also raised concerns over its side effects. The flip side of enhanced efficiency and effectiveness of government functions by AI is how to protect the data so accumulated, most notably possible infringement on privacy. Furthermore, there are other negative collateral consequences flowing from the wide application of AI, such as unemployment and other societal disruption due to technological divide between different groups of the society. The utilization of AI in the decision making process of governmental agencies could amount to virtual delegation of the governmental authority, which would then trigger administrative law issues on the domestic law front and state responsibility on the international law front. These side effects, except for the issue of privacy infringement, have hardly been discussed. As much as AI brings societal benefit in many sectors, it also invites new challenges. These challenges should merit careful attention in order to assess the full impact of the emergence of AI and introduce a framework for the regulation of AI.

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1 Definition of AI

John McCarthy and colleagues first coined the term "artificial intelligence" in 1956. They described AI as follows: "An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. … For the present purpose the artificial intelligence problem is taken to be that of making a machine behave in ways that would be called intelligent if a human were so behaving."[1]

This definition of 1956 still catches the basic elements of AI even in 2022. There are, however, diverse definitions of the term for different purposes. Despite the wide usage of the term AI, there is no clear or official definition of it yet. Brown (2021) proposes that there needs to be a legal definition for AI to clarify its 'legal personhood'.[2] A legal definition is indeed what is needed at the moment given AI's increasing legal implications in various fields. As a matter of fact, AI, which cannot be simply regarded as a mere tool or instrument, has social presence one way or another whatever we may call.[3] This broad coverage and omnipresence arguably makes any definition ever more difficult.

According to a report from the Brookings Institution, AI can be defined either narrowly or broadly:

> More specifically, that there is a key difference between narrow AI such as translation services, chatbots, and autonomous vehicles and general AI—"self-learning systems that can learn from experience with humanlike breadth and surpass human performance on all tasks." General AI raises broader existential concerns, such as how to align the goals of such a system with our own to prevent catastrophic outcomes, but general AI remains a technology still to be developed in the distant future. … Narrow AI also includes specific tools such as out-of-sample validation to validate models, stochastic gradient descent for training models on streams of data, and graphical processing units (GPUs)—originally developed for video games but which have proven well-suited to support the types of massive parallel computations needed to train [Deep Neural Networks] (DNNs).[4]

The above definition shows the wide variance of the definition of AI itself. Such being the case, any discussion of AI would be subject to an inherent challenge of an amorphous scope. At any rate, it would be fair to say that we are now using AI without a clear definition. Although many credit McCarthy and his colleagues for initially coining the term, the definition of AI has changed, evolved, and shifted over time.[5] Dictionary definitions also present variances. Some focus on simulation or imitation by computers while others focus on machine's carrying out human tasks. For instance, while Merriam-Webster defines AI as "1. A branch of computer science dealing with the simulation of intelligent behavior in computers; 2. The capability of a machine to imitate intelligent human behavior", the Encyclopedia Britannica

[1] Press [1].

[2] Brown [2].

[3] Gunkel and Wales [3].

[4] Meltzer [4].

[5] See Marr [5].

defines it as "the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings."[6]

The leading companies in the field also apply different definitions. For example, Google, while avoiding a clear definition of what they mean by AI, differentiates AI, machine learning, and deep learning: "While artificial intelligence (AI) is a convenient and commonplace term, it has no widely agreed-upon technical definition. One helpful way to think about AI is as the science of making things smart. Much of the recent progress we've seen in AI is based on machine learning (ML), a subfield of AI where computers learn and recognize patterns from examples, rather than being programmed with specific rules. There are many different ML techniques, but deep learning is a particularly popular one right now. Deep learning is based on neural network technology, an algorithm whose architecture is inspired by the human brain and can learn to recognize pretty complex patterns, such as what 'hugs' are or what a 'party' looks like."[7] In short, different levels of AI utilization can each lead to different defInition of AI. Amazon, another global Information and Communication Technology (hereinafter ICT) giant, offers a more concrete definition: "Artificial Intelligence (AI) is the field of computer science dedicated to solving cognitive problems commonly associated with human intelligence, such as learning, problem solving, and pattern recognition. Artificial Intelligence, often abbreviated as 'AI', may connote robotics or futuristic scenes, AI goes well beyond the automatons of science fiction, into the non-fiction of modern day advanced computer science."[8]

Thus, it is sometimes mentioned that policy makers need to define AI more clearly. Indeed, any proper regulation of AI begins with the clear scope of the term itself.[9] As long as the term itself does not have a consensus-based definition, subsequent discussion of how to regulate it may continue to confront confusion. As a matter of fact, AI can take a variety of forms ranging from automated chess players to algorithms producing media contents, to medical devices issuing a medical diagnosis, and to platforms making decisions for government agencies.[10] All these activities using computers and machines are collectively called AI or AI-enabled equipment. The fluctuating faces of AI imply that there are different levels of risks involved for different applications of AI. Any discussion of AI, therefore, should start with clarification as to what levels of AI the proposed topic aims to address. Otherwise, ensuing dialogues may simply sound similar to "I am using computers for my work," an abstract description that is not amenable to reliable legal guideline. The focus of the discussion in this book is the ability of the AI that makes decisions itself for or on behalf of the government, which triggers the application of international law (by

[6] Ibid.

[7] *See* the website of Google—*Exploring 6 AI Myths,* available at https://ai.google/static/documents/exploring-6-myths.pdf.

[8] *See* the website of Amazon Web Services—*What is Artificial Intelligence?*, available at https://aws.amazon.com/machine-learning/what-is-ai/.

[9] Schuett [6].

[10] Ibid.

invoking state responsibility). In that sense, the level of AI in this sense is said to be situated at the higher (if not highest) level.

2 Current Status of the Development of AI

AI is now ubiquitous: it is emerging throughout our society. Recent advancements in this area have been made in the form of deep learning, where systems automatically learn from experience, going beyond the role of a mere assistant that helps human lives. Deep learning is also enabling AI to judge and decide autonomously in various areas of our society. Many ICT giants are making increasing investments in technologies regarding AI, which is expected to spur a further breakthrough. At the current rate and pace, it is expected that AI will take over a large portion of our daily lives in the near future, both in the private sector and public sector.

In this era of the so-called Fourth Industrial Revolution, new goods and services are developed and sold one after another with the application of AI—through the internet of things (hereinafter IoT), smart sensors, and other data collection and processing devices. Not only new goods and services, but also existing goods and services also find new frontiers through the adoption of AI and IoT. New AI products such as drones,[11] autonomous vehicles,[12] and multi-functioning home appliances[13] are

[11] At present, drones are recognized as a key instrument and major item in the 4th Industrial Revolution, showing a rapid growth in various areas. Particularly regarding drones, it is said that a new drone ecosystem' is being established in connection with the so-called 'area of ICBM (IoT, Cloud, Big Data, Mobile/Machine Learning)' going beyond the area or mere drone development, production, and sales. As one recent example, Qualcomm in the US seeks to combine its own smartphone technology with autonomous flight of drones and considers adopting deep-learning devices. *See* the website of Qualcomm—*Deep Learning on Embedded Devices: The Rise of Intelligence at the Edge*, available at https://www.qualcomm.com/invention/research/projects/deep-learning).

[12] Interests in the market of autonomous vehicles continue to increase. This market is expanding rapidly in diversified areas by applying such concepts as smart car, connected car, autonomous car, and so forth. 'Smart cars' adopt IT, software, and electronic devices as part of the vehicle. 'Connected cars' connect the internet and mobile devices to the vehicle. 'Autonomous cars' drive the vehicle autonomously with no intervention of a driver.

[13] For instance, Samsung Electronics Co., Ltd. introduced IoT in October 2017, making known its project called "Connected Thinking" through which home appliances can be manipulated more easily. This system makes it possible to manipulate various home appliances simultaneously by means of a smart-phone. Samsung Electronics is developing various compatible items such as smart TV, smart refrigerator, and so forth. *See* Maribel Lopez, *Samsung Gets Serious About Smart Home Battle, Announces Enhanced Voice, IoT Cloud And New Hardware*, Forbes (Oct. 27, 2017), available at https://www.forbes.com/sites/maribellopez/2017/10/27/samsung-gets-serious-about-smart-home-battle-announces-enhanced-voice-iot-cloud-and-new-hardware/#693489e03f4f.

It is also reported that based on a wealth of information and data, customized service will be made available for such smart devices as smart thermostat, smart window blind, smart lighting, smart coffee machine, and so forth. The Forbes Technology Council, *IoT and You: Making Your House Smarter with Technology*, Forbes (Jun. 28, 2018), available at https://www.forbes.com/sites/forbestechcouncil/2018/06/28/iot-and-you-making-your-house-smarter-with-technology/#417734412cbf.

emerging as major items in the market, and it is expected that they will gradually become major items in the global trade.[14] The application is not just confined to business sectors; these new products are also presenting themselves in the battle field as well. In fact, this phenomenon is being observed in almost every area where AI or IoT can be combined with existing goods and services.[15]

Meanwhile, the advent of these new products and services is changing the forms and modes of our daily lives significantly. These goods and services create new kinds of roles for the users and consumers in combination with AI and through IoT. For instance, a new form of parcel delivery service by means of drones based on AI-enabled data processing programs has arrived in the market.[16] Autonomous vehicles, being wired and connected to diverse databases and processors, are no longer a simple transportation means but operate as a moving office or cozy resting space for those in the vehicles. IoT-based home appliances, rather than performing certain designated conventional functions only, now provide various types of information to users and cater to new demands of users in connection with or independently from such conventional functions. Products are intertwined with services, and services are embedded in products, all of which are then provided through digital means. Even in the clothing industry which has been regarded as handling the most basic products, the application of IoT is bringing about innovative changes. For example, new smart clothing products function in response to human activities, bio-health data and conditions of surroundings. Information relating to customers who consume clothings is then accumulated and delivered directly to manufacturers and used in developing new products and services. This information is valuable in a plethora of sectors: bio-health

[14] The US Congress as well supports the idea that areas related to the 4th Industrial Revolution will be a major issue in the future global trade market, and thus that the domestic legal system of the US Needs to be modified accordingly. *See* Atkinson D. Robert, *2016 Testimony before the Committee of Ways and Means Trade Subcommittee*, Hearing on Expanding US Digital Trade and Eliminating Barriers to Digital Exports (Jul. 13, 2017), available at https://waysandmeans.house.gov/event/hearing-expanding-u-s-digital-trade-eliminating-barriers-u-s-digital-exports/; The International Trade Commission (USITC) also prepares various reports on global digital trade in 2018. *See* the USITC website, available at https://www.usitc.gov/press_room/spotlight/global_digital_trade_2_business_to_business_market.htm (Jul. 30, 2018).

[15] McKinsey, a US Consulting company, predicts that by 2025, IoT will form a new business ecosystem in connection with various existing industry sectors. This new ecosystem will embrace the general industry including B2B service (legal/accounting/business management consultation), B2B machinery and equipment, B2C transport, corporate financing, clothing, restaurant business, hotel business, communication business, health, culture business, and so forth. This suggests that the era of the 4th Industrial Revolution involves significant change in the service industry. *See* McKinsey & Company, *The Internet of Things: How to Capture the Value of IoT* (2018), p. 26, available at https://www.mckinsey.com/~/media/mckinsey/business%20functions/mckinsey%20digital/our%20insights/the%20internet%20of%20things%20how%20to%20capture%20the%20value%20of%20iot/how-to-capture-the-value-of-iot.ashx.

[16] According to the US Washington Post, Amazon.com has obtained a patent on the delivery service by means of drones that respond to human behaviors. Drones, as the medium, make possible the communication between the delivery service supplier and consumers. *See* Hamza Shaban. *Amazon is Issued Patent for Delivery Drones that Can React to Screaming Voices, Flailing Arms,* The Washington Post (Mar. 22, 2018), available at https://www.washingtonpost.com/news/the-switch/wp/2018/03/22/amazon-issued-patent-for-delivery-drones-that-can-react-to-screaming-flailing-arms/.

data would be crucial for the health equipment industry. These new roles and functions that IoT-based products and services carry out are features that have never been seen or contemplated in the past. If anything, such roles and functions have had to be purchased and consumed separately independent of these products and services.[17] Generally speaking, in the conventional trade agreements, the products around us are goods and thus regulated by goods trade norms, and the roles or functions that we use are services and are thus regulated by different services trade norms.[18] This conventional binary structure of trade agreements are now history. In short, the emergence of new products and the provision of services through the products, and the creation of new services by dint of new products are one of the major characteristics of digital trade and digital economy that has just begun. The co-mingling, collapsing and blending of goods and services using digital equipment and device will continue to expand as we go along. The digital component that creates these new products and services and that bridges between products and services is in turn anchored in intellectual property rights (IPR), another core pillar of trade agreements. As such, the trilogy of modern day trade agreements—goods, services, and IPR—are shaping a new landscape for the global economy in the new age.

As such, digital economy or trade involves a new form of production, supply and consumption that is distinguished from those in the previous days. Most notably, goods and services are provided and consumed simultaneously. They are installed and embedded in one another. Thus, digital goods are consumed along with related services on the basis of the capabilities to collect, analyze, and transmit relevant data sets.[19] Services become more available to the general public because of ubiquitous digital gadgets. Consider the broad range of new services that come to our aid because of smartphones. As IoT advances further and its mutual linkage through a web of tight networks is strengthened further, this blending of goods and services will continue to intensify. When the two are combined, experiments and creation

[17] As service is embedded in goods and goods are embedded in service, new issues also have been raised. For instance, the combination of goods and services has raised new issues regarding value estimation of goods and confirmation on the traditional way of estimating the amounts of tariffs. *See* Joshua P. Meltzer, *Maximizing the Opportunities of the Internet for International Trade*, E15 Expert Group on Digital Economy, Policy Options Paper, International Centre for Trade and Sustainable Development (ICTSD) & World Economic Forum (Jan. 2016), p. 12.

[18] At present, 'service' is not clearly defined except that it is understood as based on the 'physical existence of a service supplier and the 'intangibility' of service itself. Basically, the scope of service is quite comprehensive. The WTO GATS is applicable to every service area except certain types that are provided as part of governmental exercise of authority. For more details on service areas to which the GATS is applicable, *see* the WTO website: *Services: Services Sectors*, available at https://www.wto.org/english/tratop_e/serv_e/serv_sectors_e.htm. For more details on the WTO agreement on the concept of 'service,' *see* the WTO website: *Services: Rules for Growth and Investment*, available at https://www.wto.org/english/thewto_e/whatis_e/tif_e/agrm6_e.htm.

[19] In the context of digital trade, personal information is valuable as highly as a new monetary unit. Some scholars view that cross-border exchange or transaction of data should be regarded as an independent form of trade. *See* Margaret B. Sedgewick, *Transborder Data Privacy as Trade*, California Law Review, Vol. 105 (2017), p. 1513; Nivedita Sen, *Understanding the Role of the WTO in International Data Flows: Taking the Liberalization or the Regulatory Autonomy Path?*, Journal of International Economic Law, Vol. 21 (2018), p. 324.

of new types of goods and services will also be facilitated.[20] This may be called the phenomenon of 'convergence' or 'integration' of goods and services.[21] Even the databases and computing facilities of big data, which constitute a core mechanism that makes such convergence and integration possible, carry aspects of both goods and services.[22] The clouding computers and storage facilities are hardware, thus goods, but what are contained in them are elements of provision of services. Netflix's clouding facilities in Singapore are goods (i.e., computers, memory chips and non-memory chips) but the movies distributed from the facilities to the Asian watchers are services (i.e., audio-visual services) Such being the case, it is also a natural consequence that the terminal equipment on the part of the consumers plays the roles of both goods and service suppliers. The people watching Netflix movies after the subscription payment are consuming both goods (i.e., set-top boxes) and services (i.e., contents of movies). At the same time, all these goods and services packaged together are protected via an IPR protection regime. In sum, the digital economy that is unfolding now is seeing the situation where goods, services and IPRs emerge together, travel through borders together, and cater to the needs of consumers together. And most importantly, the thing that has made the goods, services and IPRs all the more valuable is what is contained in the combined triangular framework—personal information. This has been made possible by the advent of IoT, and now it rises to the next level with the fast development and permeation of AI.

In this era of the Fourth Industrial Revolution that is transpiring at present, preferences and demands of consumers will be formulated and expressed in various ways, and technologies will be developed in order to keep up with them. Therefore, it is expected that new concepts and types of goods and services that do not currently exist will likely to emerge as we go forward. New goods and services will then lead to yet

[20] Devices with, in front, the adjective 'smart' that is being widely distributed recently such as smart phone, smart car, and smart home are items of 'goods trade' under the current WTO agreement system. However, as the term 'smart' is used in a sense that such devices satisfy needs of humanity, they also involve characteristics of 'service based on the collection and transmission of data on the other hand. *See* Nivedita Sen, *supra* note 12, p. 32.

[21] For this reason, the new term, 'servitization of goods', has appeared. As a major example, medical devices provide medical service as a package. *See* South Centre & United Nations Economic Commission for Africa (UNECA), *The WTO's Discussions on Electronic Commerce*, Analytical Note (SC/AN/TDP/2017/2) (Jan. 2017), pp. 5–6.

[22] Whether a database is a type of goods or service is also ambiguous. Some classify a database as a type of 'goods' with the focus on hardware characteristics while some view it as service with the focus on networking among users. *See* Wolfgang Alschner, Joost Pauwelyn, and Sergio Puig, *The Data-Driven Future of International Economic Law*, Journal of International Economic Law, Vol. 20 (2017), p. 227.

another set of newer goods and services.[23] For instance, establishing a new information and communication network that exceeds current scopes and capabilities will help overcome technical limitations in terms of speed, volume and capacity. This improvement may come with the development of a new generation of semiconductors. At the new level reached by the new network, the utilization of IoT may become far more intensified and upgraded than what exists now. This will then lead to the development of new goods. The emergence of these new IoT-based goods will in turn make new services available, and such new services will then facilitate the experiment and development of new goods. As such, goods and services develop separately, consecutively and collectively, establishing what is called a 'digital virtuous cycle.' At the center of this new cycle lies AI.

Moreover, as the conventional distinction between goods and services gets more and more vague, there is a high probability that people and companies start to view these new digital packagings (an item where goods, services and IPRs are combined together) from a new angle. They will purport to bring new digital packagings to the market. Governments also may examine these packagings from a new perspective. Over time, these changing views and perspectives may demand new regulatory regimes and monitoring schemes. Even this new regulatory regime/scheme is also being empowered and operated by AI. Here also, the AI works two-ways: (i) AI has inaugurated a new regulatory regime, and (ii) the new regime itself is also relying on AI. Both of these two segments merit careful legal attention. The former is important indeed since it deals with exploration of new regime commensurate with the arrival of AI. The latter is equally important as it discusses whether and how AI is properly utilized in discharging governmental function. If anything, the critical question of who to be held accountable when something goes wrong appears in both segments. In any event, so far it appears the second segment has received less attention than the first one, both in domestic and global planes. In facilitating international law discourse, both segments should also receive an equitable amount of attention. Arguably, the second segment has received relatively less attention in international law dialogues involving AI.

As a consequence, new norms are also being contemplated, debated and adopted in the area of international law. Existing norms do not dovetail with the new digital landscape of the global community. One area where such drive for change is being vividly felt is the trade and investment front. This is not surprising because digital changes are

[23] It is noteworthy that in the era of AI, data in data flows itself is a core element of new goods or service. Around the globe and with no limitations by national borders, a tremendous amount of data is collected through Facebook, Google, etc. Such data is converted into core elements of other goods and services and becomes available, going through the process of searching, payment, transaction, and advertisement. Data itself is a key factor to international trade, and its flows and transactions are performed not merely in each country's markets but in global unified markets (for instance, cyber space or cyber market). The formation of data in this manner with every subject in the society such as governmental agency, corporation, household, and individual participating in transactions is a significant difference from the existing trade system where only some of them participate. *See* Dan Ciuriak, and Maria Ptashkina, *The Digital Transformation and the Transformation of International Trade*, International Centre for Trade and Sustainable Development (ICTSD) & Inter-American Development Bank (IDB), Issue Paper (Jan. 2018), pp. 3–4, 6.

currently affecting the economic sectors as the term digital economy itself connotes. On this front. traditional trade forms and methods are going through significant transformation. The current trading regime featuring a large-scale vessel shipment of mass-produced goods from the manufacturing facilities in an exporting country to an importing country does not apply to most of the digital trade. The preferred modes of trade in the digital area are electronic transmission, aircraft transport, and international mail. The contents that we purchase from Amazon, Google and Netflix are arriving at our homes through internet connection, and they collectively cost a lot. Although traditional transport means and trading modes will also be utilized continuously since goods transportation is still inevitable (digital goods are also goods to be shipped), their importance and share are gradually being outshadowed by the the new transports and modes.[24] Particularly, services trade will be further conducted through a network-based electronic way. Almost all services are now available on-line, which means that foreign service providers may also access to the service market of another country with ease and convenience. The COVID-19 pandemic has further accelerated the transition since January 2020. Accordingly, the existing trading system that is mainly based on vessel transport/customs procedures of large-size freight is gradually giving way to internet/delivery procedures of digital items. As the Agreement on Trade Facilitation (hereinafter TFA) of the World Trade Organization (hereinafter WTO) that came into effect in 2017 points out, computerization and the rapidity of customs formalities are of the greatest importance for modern day businessmen.[25] Note that the TFA is geared toward to facilitating customs processing of goods trade in a traditional sense. The bulk of digital trade do not involve international shipping in the first place, hence no traditional customs procedure at the border. Thus, facilitation is pushed to the limit in the digital trade. The rationale of adopting TFA speaks volume about the importance of the digital trade. Governments may not necessarily like the disappearing customs borderline as it means less revenue of tariffs at the border. But the strong consumer demand for the abolished or streamlined customs procedure for digital trade is virtually forcing the governments to accommodate the new trend.[26] Needless to say, modernization

[24] The new technology of 3D printing that is being widely spread will make the boundary between trade means and trade objects vaguer. If trading is practiced through the 3D printer technology, the final product will be goods but it will be data (more like service) that is actually traded across the border. *See* Nivedita Sen, *supra* note 12, p. 330.

[25] The Agreement on Trade Facilitation has been introduced with the aim to simplify and computerize import/export customs formalities and to harmonize procedures among countries. Section I (Articles 1 to 12) specifies Articles V, VIII, and X of the GATT in more detail to accelerate customs procedures. Section II (Articles 13 to 22) specify developing country preferential provisions, and Section III (Articles 23 to 24) includes provisions regarding establishment of a permanent WTO board of directors for trade facilitation as well as final regulations. For more details, visit the website of the WTO: *The Agreement on Trade Facilitation: An Overview*, available at https://www.wto.org/english/tratop_e/tradfa_e/tradfatheagreement_e.htm.

[26] In July 2018, Crimson Logic in Singapore introduced the block-chain trade platform for the first time in the world. This platform links ASEAN countries with China, pursuing trade safety and transparency. This principle reflects the joint response policy of governments in that region to market demands. *See* Samburaj Das, *'World's First' Trade Blockchain Launches in Singapore to*

of customs-related equipments and systems, and establishment of ample databases are pivotal in moving goods smoothly across the globe. In that respect, the TFA is indeed an important achievement. That said, the agreement does not respond sufficiently to the demand from the people and market in terms of digital trade. What we are facing now is a new type of trade, which requires a new norm. Merely peppering the existing norms with digital condiments are better than doing nothing, but it is far from addressing the real demand from the market and the global community.

Goods trade is closely connected to traditional notion of national borders. The same also applies to services trade. The WTO's General Agreement on Trade in Services (hereinafter GATS) defines "trade in services" as "supply of a service" through four different modes.[27] The GATS also defines "supply of a service" as activities "including production, distribution, marketing, sales, and delivery of service."[28] In other words, a series of diverse activities to deliver services from providers to consumers are all part of the "supply of a service." Four modes of services supply are: Mode 1—cross-border supply; Mode 2—consumption abroad; Mode 3—commercial presence; and Mode 4—movement of natural persons.[29] To which of these four modes a specific service supply corresponds depends on how and in what country's territory the service supplier and consumers interact at the time of "service supply."[30] As such, the existing services trade system is also being operated on the basis of national borders. Services trade is also hinged on the traditional notion of national borders. Among the four modes, "Mode 1: cross-border supply" is different from the others in that the service supplier and consumer in this context do not move physically across the border. However, Mode 1 assumes that there is a one-time or temporary contact from the buyer to the seller for the supply of service via telephone or the internet. In this regard, Mode 1 is different from the digital trade, in which service is available at all times within the country where the service is consumed (with the equipment of service supply as part of the goods package). The seller and

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Link China and ASEAN Nations, CCN (Jul. 19, 2018), available at https://www.ccn.com/worlds-first-trade-blockchain-launches-in-singapore-to-link-china-and-asean-nations/.

[27] Paragraph 2, Article I of the GATS specifies 4 modes of service supply applicable to service trade as below.

2. For the purposes of this Agreement, trade in services is defined as the supply of a service.

(a) from the territory of one Member into the territory of any other Member.

(b) in the territory of one Member to the service consumer of any other Member.

(c) by a service supplier of one Member, through commercial presence in the territory of any other Member.

(d) by a service supplier of one Member, through presence of natural persons of a Member in the territory of any other Member.

Among these, it is Mode 2 in Paragraph (b) that a service consumer goes personally to the country of service supply, purchases, and consumes the service.

[28] GATS, Article XXVIII (b), ("supply of a service" includes the production, distribution, marketing, sale and delivery of a service").

[29] Ibid.

[30] *Scheduling of Initial Commitments in Trade in Services,* Explanatory Note, MTN, GNS/W/164 ("Modes are essentially defined on the basis of the origin of the service supplier and consumer, and the degree and type of territorial presence which they have at the moment the service is delivered").

buyer are virtually residing together in the same cyberspace. Thus, new digital trade items are distinct from even the supply of services through Mode 1. Again, the hard reality begs for the introduction of new norms for the digital trade. Similar consideration apples to other branches of international law with equal force. Existing norms in traditional treaties and customary international law would have to be re-examined with the digital glasses, and necessary changes need to be made before it is too late.

3 Current Status of the Utilization of AI

Digital economy that has been already reshaping our businesses and activities is advancing to the next chapter with the utilization of AI. Digitized sectors and value chains are incorporating and adapting to AI increasingly, and the utilization of AI is transforming the business and decision making dynamics in these sectors and value chains. AI is not just confined to certain digitized sectors and industries alone. Fast advances in AI have the potential to be applied across numerous fields that have not been considered digital-prone previously, such as health care, education, financial transaction, telecommunications, personal transportation, dispute settlement and more. For instance, the utilization of AI in medical settings (i.e. Robo-Doctor) and in arbitration/mediation for settling disputes between parties (i.e. Robo-Arbitrator) is under robust discussion at the momentalthough it is still in the initial stage.[31] This is the case even in the area of handling government/state affairs. As digital technology advances to our homes and offices, certain governmental or state tasks such as taxation and traffic regulations are already being assisted or even handled by means employing AI. The scope of government/state affairs adopting AI is expanding.

The recent outbreak of a global pandemic has offered yet another rationale that has accelerated and spread the utilization of AI. In an attempt to slow down the dissemination of the COVID-19 virus, individuals in many countries have chosen to or been forced to stay inside their homes and offices maintaining social distancing guidelines. Many started to and continue to work from home, and school classes and business meetings are moved online. Zoom classes and meetings have become a new normal of the pandemic-stricken society. As the international community scrambles to cope with the pandemic, it faces numerous challenges, some old and some new. One of most evident challenges at this time is how to decrease human interactions through a face-to-face format. To limit the direct human contact, and thereby prevent the spread of the disease, stores and restaurants have introduced online ordering. Online education is no longer a subject of public debate—it is reality in all levels of schools. Demand for health counseling services and treatment using telecommunications is also soaring. Governments are now taking a more positive view toward medical treatment through online diagnosis—any side effect is regarded as marginal on account of the unprecedented magnitude of medical treatment demand from the

[31] See Snider et al. [7].

COVID-19 pandemic. In all these faucets of pandemic countermeasures, digitization is key, and AI is the key to digitization. With its current pace, it is no doubt that AI will play a bigger role in the post-COVID-19 era in all sectors of national and global community. Government agencies themselves are looking to digitization and AI for carrying out their respective tasks. In a sense, the pandemic has imposed a forced trial run for AI for many governments for many national businesses.

Indeed, according to a report by the Organisation for Economic Co-operation and Development (hereinafter OECD), AI can help us combat the global pandemic in diverse stages.[32] If data collection, processing and analysis while maintaining social distancing are the core ingredients of COVID-19 countermeasures of many countries, perhaps AI is best suited to handle the task. As seen from the experience of China, South Korea and Taiwan, ICT infrastructure plays an essential role in detecting the virus' transmission and monitoring its spread almost real-time. ICT infrastructure also enables efficient allocation of national resources. The contents that flow through the infrastructure (from individuals/hospitals to government agencies, and from government agencies to individuals/hospitals) are none other than data created, stored and interpreted by AI. All this can be done by minimizing person-to-person contact. Granted, possible privacy infringement raises a critical legal issue, but the new technological horizon has been certainly tested and proven during the 2020-2022 pandemic. Moreover, AI-powered early warning systems or diagnoses can help increase the predictability of new outbreaks in the future. New pandemics may be inevitable but their negative impact could be contained as much as feasible through AI-supported preparation and research. Likewise, in the recovery phase from the current pandemic, AI can contribute to filtering and fighting misinformation so as to elicit an accurate lesson from the event for future guidance. It can also guide us to mend national healthcare systems including virtual diagnosis and treatment. A post-COVID-19 research based on AI-processed data may also tell the governments about how to identify and protect those vulnerable, either health-wise, economically or socially in a pandemic. If a white paper is ever prepared from the current pandemic, its contents could only become meaningful when AI-processed data and analyses are provided.

Unfortunately, the global community lacks a coordinated effort in combating the global pandemic such as COVID-19. Each state is left alone in the dark to find the way of survival on its own. Not so speak of global cooperation, even regional cooperation or state-to-state cooperation has been hardly seen. International organizations have also proven not ready for a global health disaster of this scale. Neither the United Nations (hereinafter UN), nor the World Health Organization (hereinafter WHO), nor the OECD has shown a strong leadership for concerted global action. Situations are not that different for other international organizations. Recommendation issuance and information collection have been the lion's share of international organizations' work for the pandemic. Part of the reason is the embedded limitation of international

[32] *OECD Policy Responses to Coronavirus (COVID-19)—Using artificial intelligence to help combat COVID-19, OECD* (Apr. 23, 2020), available at http://www.oecd.org/coronavirus/policy-responses/using-artificial-intelligence-to-help-combat-covid-19-ae4c5c21/.

law: namely, 'fragmentation' of international law where each branch of international law operates in its respective legal silo. The splintered responses of states and international organizations have brought back the importance of national borders to the minds of general public, slowing down or even reversing the wave of globalization. As the coronavirus was spreading, governments were forced to take action without paying much attention to the norms of international law. For example, restrictive trade measures immediately emerged both on the export and import fronts, sometimes skipping or postponing analyses of possible implications on trade agreements or investment agreements. Faced with the fast spread of the coronavirus in early 2020, many countries, both developing and developed, restricted exports of medical items and daily hygiene essentials (e.g. face masks)—items, articles and equipment collectively called Personal Protective Equipment (hereinafter PPE)—to meet the domestic demand first.[33] Some countries imposed import restrictions on other goods (such as clothes) as well out of the fear of virus infection. Certain Least Developed Countries (hereinafter LDCs) also took similar measures. Notably, only part of these measures were properly notified to the WTO as required by the WTO Agreements. Rampant trade restrictions were not merely limited to goods trade. Services trade was also significantly affected by the suspension or prohibition of international travels. These restrictions of goods and services trade revealed the fragility and weaknesses of Global Supply Chains (hereinafter GSC).[34]

> With COVID-19, another shoe has dropped. The pandemic has exposed a lack of resilience within the global economy's non-financial "real" sector: the world of production and distribution of goods and services. […] Moreover, the aggressive drive to reduce costs at the expense of resilience has not been limited to transnational supply chains. As the physician and author Siddhartha Mukherjee recently noted, an analogous phenomenon with even more dire consequences has afflicted health care, where hospital administrators have increasingly embraced the same "market-driven, efficiency-obsessed culture" held by many corporations. As a result, he writes, "Questions about 'best practices' in management have become questions about best practices in public health. The numbers in the bean counter's ledger are now body counts in a morgue."[35]

As the above statement indicates, COVID-19-afflicted disruption spans over both goods and services. Movement of goods has been curtailed, suspended or prohibited. Even in the area where free flow is still permitted, the cost has increased. In an attempt to slow down the virus spread, several states issued travel bans or closed borders against high-risk countries. Other states imposed immigration bans claiming them as virus-deterrent measures.[36] Travel and immigration regulation directly affect services trade, as the trade presupposes cross-border movement of natural persons (i.e., service providers). Each country created its own domestic bubble insulated from an outside world. GSCs were disrupted or re-configured among reliable partner countries. Not only goods and services, but also IPRs also felt direct impact. Shortages of vaccines and medicine sparked a heated global controversy over the need

[33] See Hogan [8].

[34] See Janeway [9].

[35] Ibid.

[36] See Somin [10].

and scope of protection of IPRs of these items held by global pharmaceutical companies. In essence, the three pillars of the current trade agreements—goods, services and IPRs—face unprecedented challenges stemming from the pandemic. The sometimes arbitrary and sporadic appearance of diverse restrictive measures that were put in place by different countries demonstrates the lack of coordination on the part of the international community, be it a state-to-state level or an international organization-to-organization level.[37] The WTO's voice has been feebly heard during the pandemic process. The two superpowers, the United States of America (hereinafter US) and the People's Republic of China (hereinafter China) are engaged in a fierce competition on multiple fronts, most notably in the trade area. COVID-19 has further strained the bilateral relationship, and they have not been in a position to show leadership as G-2 countries. Circumstances in other international organizations and the G-2 confrontation in other sectors of global business are not that different. The WHO and the Sino-American conflict in the pandemic policy offers yet another example. Partly due to the two countries' stand-offwithin the organization in the aftermath of the coronavirus outbreak, the WHO has arguably failed to play a 'central' role as the global health policy coordinator. The Geneva-based international organization has mainly presented health policy guidelines and recommendations for member states and announced statistics of the pandemic victims, but policy coordination among states and among international organizations, and fact-findings coupled with indication of future course of action are largely missing. Each state has been left (and to some extent forced) to make its own decisions, and as of March 2022 the pandemic still persists in the midst of such compartmentalized and uncoordinated responses from the international community.[38]

In all these pandemic-related global surroundings, AI has become all the more relevant. Lockdowns, social distancing and health regulations have bloomed online meetings, virtual gatherings, and internet-based business activities, and AI is the key component of the new global landscape. The new landscape, in turn, is fast-forwarding the development and application of AI in diverse sectors. The utilization of AI is also becoming more evident in the public sector. Governmental agencies of various countries have realized the benefit of using AI in taking care of their respective public tasks, and joined the global trend.[39] In line with their national strategies for the leadership in the global digital economy, AI is finding its inroad into the governmental business. Utilization of AI can be envisaged in a plethora of public sectors,[40] and governments indeed declare they will incorporate AI into

[37] See Lee [11].

[38] See Lee [12].

[39] See New [13].

[40] See Zusammen. *Für unser Österreich. Regierungsprogramm 2017–2022* [Together. For Our Austria. Government Program 2017–2022], available at https://www.bundeskanzleramt.gv.at/documents/131008/569203/Regierungsprogramm_2017–2022.pdf/b2fe3f65-5a04-47b6-913d-2fe512ff4ce6, pp. 75, 81; European Commission for the Efficiency of Justice (CEPEJ), *European Ethical Chapter on the Use of Artificial Intelligence in Judicial Systems and their Environment* 34 (CEPEJ(2018)14, (Dec. 3, 2018), available at https://rm.coe.int/ethical-charter-en-for-publication-4-december-2018/16808f699c; Bundesregierung [Federal Government], *Eckpunkte der*

their respective administrative systems to fulfill public mandates and to provide public services. Just like in the private sectors, personal information, databases and algorithms are being increasingly employed in public administration and decision making process,[41] and their arrival in the public domain brings about significant impact, both positive and negative. Recent development in key states in this respect is examined below. As can be seen, the AI-driven public administration indeed opens up a new opportunity and invites unprecedented homeworks and challenges.[42]

3.1 United States

Over the years, the US government has been actively extending applications of AI to carry out government functions. Having seen the private sector benefit a lot from the application of AI to increase the efficiency in the workplace, the public sector, which is also under constant pressure for cost reduction and efficiency, has been following suit.[43] In fact, Deloitte, one of the major global consulting firms, predicts that AI will decrease the amount of public labor by reducing 25% of current working hours in the US federal government.[44] The White House recognized that "[i]n recent years, the private sector has demonstrated that AI can be used to improve the quality of services, assist in the creation of new products, and increase efficiency. Likewise, AI provides opportunities to help the Federal Government be more responsive, effective, and efficient for the American people."[45]

Mindful of this contribution, the previous Trump Administration encouraged federal agencies to actively develop and apply AI technology to facilitate the work of government officials. For example, the US federal government is on its way to digitize its government records, which facilitates data sharing and enables data analysis in

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Bundesregierung für eine Strategie Künstliche Intelligenz [Key Points for a Federal Government Strategy on Artificial Intelligence] (Jul. 18, 2018), at 1, no. 1a, available at https://www.bmbf.de/files/180718Eckpunkte_KI-StrategiefinalLayout.pdf; Karla Pequenino, *Governo Financia Inteligência Artificial na Saúde, Mobilidade e Educação*, PÚBLICO (Oct. 24, 2018), available at https://www.publico.pt/2018/10/24/tecnologia/noticia/governo-inteligencia-artificial-dados-administracao-publica-1848718.

[41] *See* UK House of Commons Science and Technology Committee, *Algorithms in Decision-Making* (HC 351, May 2018), available at https://publications.parliament.uk/pa/cm201719/cmselect/cmsctech/351/351.pdf.

[42] *See* UK House of Commons Science and Technology Committee, *Algorithms in Decision-Making: Government Response to the Committee's Fourth Report 2* (HC 1544), Sep. 2018), available at https://publications.parliament.uk/pa/cm201719/cmselect/cmsctech/1544/1544.pdf.

[43] See Eggers et al. [14].

[44] See Eggers et al. [15].

[45] The White House -Office of Science and Technology Policy, *American Artificial Intelligence Initiative: Year One Annual Report* (Feb. 2020), p. 23, available at https://www.whitehouse.gov/wp-content/uploads/2020/02/American-AI-Initiative-One-Year-Annual-Report.pdf.

an attempt to cope with complicated social problems of mondern days.[46] The Social Security Administration has developed tools to handle backlogs of pending cases, and to minimize inter-judge disparities in decision-making processes, an improvement which would be difficult, if not impossible, in the absence of AI.[47] The Department of Energy (hereinafter DOE) established an independent office—the Artificial Intelligence and Technology Office (hereinafter AITO)—to "serve as a nexus for coordinating AI activities and accelerating intradepartmental and interagency collaborations".[48]

The sudden outbreak of COVID-19 in January 2020 has since accelerated the digitization of governmental functions in the United States. For example, due to the social distancing requirements to slow down the spread of the coronavirus, various courts in the US have introduced virtual hearings for the court proceedings in which parties, legal counsel and the judges meet virtually.[49] Online hearings and video proceedings had already been adopted by some state courts of the United States even prior to the COVID-19 spread, but the pandemic turned these formats into a new standard.[50] Techniques and schemes for virtual hearings have made significant stride during the pandemic. Even if the pandemic is brought under control, the virtual option, whose benefit has already been observed and tested, will stay with us now. As such, the virtual alternative is being offered and adopted in more and more segments of diverse judicial proceedings.[51]

These efforts for AI incorporation are also visible at the law enforcement of the state or city level government. The city of Pittsburgh has introduced an AI-enabled traffic lighting system that helps increase the efficiency of traffic control.[52] In an attempt to decrease the crime rate, the city of Chicago has constituted a predictive analytics unit running spatial algorithms on violent crimes and robberies, the data from which can be utilized to predict and deter crimes by mobilizing limited police resources to identified targets, a situation described in the movie "Minority Report".[53] The Los Angeles Police Department's Real-Time Analysis Critical Response (RACR) Division is also heavily data-dependent in its mission to prevent and fight crimes in the city.[54] RACR is known to work in collaboration with Palantir, a private digital firm, for the data collection and processing. In the same vein, in 2015, the New York Police Department introduced ShotSpotter, a pilot program

[46] According to the *Strategic Plan 2018–2022* of the National Archives, it plans to digitize over 500 million pages by fiscal year 2024. *See* The National Archives & Records Administration, *Strategic plan 2018–2022* (Feb. 2018), available at https://www.archives.gov/about/plans-reports/strategic-plan/strategic-plan-2018-2022.

[47] See Curle [16].

[48] *Supra* note 49.

[49] See Quaintance [17].

[50] Ibid.

[51] Ibid.

[52] See Eggers and Beyer [18].

[53] Ibid.

[54] See Ferguson [19].

using electronic sensors to detect the sound of gun fire, which accumulates and examines data sets of numerous circumstances of human interaction in New York.[55] While AI-based law enforcement tools and schemes certainly lend a significant assistance to government agencies, they also trigger valid concern over possible infringement on human rights and discrimination of certain racial and/or ethnic groups through data-reliant targeting.[56]

Another area where US government's active mobilization of AI stands out is border control and immigration regulation. Economic slowdown since the 2008 global financial crisis and ensuing protectionism have led the US government to tighten border control and immigration regulation. Information processing based on a web of databases of multiple agencies together with US missions abroad is only made possible through application of AI. Outcomes of such processing informs agencies in charge who to admit, who to refuse and who to re-examine. While an official decision is made by an immigration official or US consul in missions abroad, it is based on the AI-powered data processing. Reportedly, Unmanned Aerial Vehicles (hereinafter UAVs) are being introduced to border control to facilitate the surveillance of remote areas in the southern part of the United States.[57] As a matter of fact, unmanned surveillance aircrafts flying over the US-Mexico were commissioned for the border patrol service as early as in 2006.[58] The federal agency in charge—US Customs and Border Protection (hereinafter CBP)—is operating UAVs over the aerial zones covering a sizable portion of the US-Mexican border, and these flying machines are taking more active roles. A continuing question mark is how to protect the privacy of those people, many of whom US nationals, tracked by the drones. Legal restrictions are certainly in place, but they are not detailed or stringent enough to be responsive to the voice of concerns raised by some watchers.

Not surprisingly, despite worries relating to human rights implications, national security and defense areas are the ones in which the US government is extending the use of AI for some time. The Department of Defense (hereinafter DOD) established the Joint Artificial Intelligence Center (hereinafter JAIC) in 2018, which played a pivotal role in the DOD's release of AI Strategy in the following year. JAIC's mission is "to both deliver new AI-enabled capabilities to DOD end users, as well as to incrementally develop a common foundation of shared data, reusable tools, frameworks, libraries, and standards that are essential for scaling the impact of AI across DOD."[59] Put differently, AI in the DOD's view does not simply mean the adoption of new technologies for particular military hardware; it also covers the software and the entire system. According to one senior DOD official, the advent of AI will basically

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[55] *See* Tatiana Schlossberg, *New York Police Begin Using ShotSpotter System to Detect Gunshots*, New York Times (Mar. 16, 2015), available at https://www.nytimes.com/2015/03/17/nyregion/shotspotter-detection-system-pinpoints-gunshot-locations-and-sends-data-to-the-police.html.

[56] See Rademacher [20].

[57] *See Border Patrol Testing Drones to be Its 'Eyes in the Sky'*, Fox News (Jan. 19, 2018), available at https://www.foxnews.com/tech/border-patrol-testing-drones-to-be-its-eyes-in-the-sky.

[58] See Ghaffary [21].

[59] The White House website, *AI for American Industry*, available at https://www.whitehouse.gov/ai/ai-american-industry/.

change the way the United States prepare for and carry out future warfare: "[t]he impact of AI will extend across the entire Department, spanning from operations and training to recruiting and healthcare. The speed and agility with which we will deliver AI capabilities to the warfighter has the potential to change the character of warfare. We must accelerate the adoption of AI-enabled capabilities to strengthen our military, improve effectiveness and efficiency, and enhance the security of our nation."[60]

Amidst worsening relations with China, the former Trump Administration clearly expressed its interest in and commitment to invest for AI for national security and defense purposes. The *FY 2020 Administration Research and Development Budget Priorities* memo set American leadership in AI as one of the priority areas for government research and development (hereinafter R&D),[61] which calls for massive investment in AI, quantum information science, and strategic computing as critical components of US national security. It is widely believed that the US drive for AI was triggered by its strong belief that its continuing status as a global hegemon will depend on its capabilities in AI. As a result, the intelligence community (hereinafter IC) and intelligence agencies of the United States are also actively calibrating and adopting AI for their activities. *The Augmenting Intelligence using Machines* (hereinafter AIM) Initiative, released in January 2019, outlines how the IC aims to incorporate AIM capabilities across the board.[62]

In sum, various levels of governments and agencies of the United States have been incorporating AI robustly. On the one hand, AI is viewed as a facilitator of governmental businesses. On the other hand, it is perceived as a strategic tool to maintain US leadership in the world. In terms of the function AI is performing, the categories of utilization have emerged. One category is to utilize AI to simply assist government officials—just like computers and electronic devices that we use daily. The other category is to delegate governmental authority to AI-enabled machine or system, or at least to place a decisive weight on the data (or outcome) processed from such machines or systems, in order to carry out the public tasks and mandates accorded a specific governmental agency under a statute. As further explained below, these two categories of the AI functions pose different legal questions. The former hardly raises a complex legal issue other than instances of malfunction or technical failure. Product liability or insurance under domestic law may come into play. They are also certainly important issues for user-customers inside agencies in individual cases, but may not cause structural contemplation from a new angle. Conversely, the latter category would quite possibly pose complex legal issues such as legal liabilities for decisions thus made and remedies for victims of such decisions, as it is difficult to identify, both technically and legally, who should be held accountable for the mishap and to what extent. The focus of the concern of international law regarding AI is

[60] Ibid.

[61] The White House, *The FY 2020 Administration Research and Development Budget Priorities,* (Jul. 31, 2018), available at https://www.whitehouse.gov/wp-content/uploads/2018/07/M-18-22.pdf.

[62] Full text available at https://www.dni.gov/files/ODNI/documents/AIM-Strategy.pdf.

more geared toward the second category - whether and to what extent a sovereign state is held legally liable under international law (e.g., violation of a treaty) because of the application and utilization of AI in its administration of national business. Two types of AI functions are mixed up and widely spreading in the United States.

3.2 European Union

The European Union (hereinafter EU) has also seen the rise of AI in its administration of official mandate in the region. The European Commission, the EU's executive arm, published a White Paper in 2020, in which it called for promotion of the utilization of AI in the public sector.[63] In particular, the European Commission underscored the AI application in the healthcare and transport sectors. It stated in the pertinent part of the White Paper:

> F. Promoting the Adoption of AI by the Public Sector
>
> It is essential that public administrations, hospitals, utility and transport services, financial supervisors, and other areas of public interest rapidly begin to deploy products and services that rely on AI in their activities. A specific focus will be in the areas of healthcare and transport where technology is mature for large-scale deployment.
>
> • Action 6: The Commission will initiate open and transparent sector dialogues giving priority to healthcare, rural administrations and public service operators in order to present an action plan to facilitate development, experimentation and adoption. The sector dialogues will be used to prepare a specific 'Adopt AI programme' that will support public procurement of AI systems, and help to transform public procurement processes themselves.[64]

As the above statement clarifies, the EU's AI drive is not simply confined to specific sectors such as healthcare or transportation. They are presented as merely key examples for the future endeavor. If anything, a wide range of public sectors are expected and slated to import AI for the fulfillment of their public mandates. The AI-embracing phenomenon is also appearing in respective member states, with varying degrees. Some member states are more active than others in this regard. Some examples of using AI for public tasks of governmental agencies are: the British government has introduced AI-chatbots to assist patients in non-critical circumstances so as to better allocate resources in emergency care;[65] Norway's postal service agency, Posten-Norgen, in partnership with an automation company, Buddy Mobility, decided to deploy delivery bots for mail delivery;[66] Germany's national postal service also uses robotic assistance tools.[67]

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[63] *See* European Commission, *White Paper On Artificial Intelligence—A European Approach to Excellence and Trust,* COM(2020) 65 Final, (2020), p. 8.

[64] Ibid.

[65] *See supra* note *34.*

[66] *See supra* note 33.

[67] Ibid.

In particular, the European Commission has also promised to pursue a fair and responsible development and use of AI to contribute to its Sustainable Development Goals (SDGs), a policy objective championed by the EU in various policy areas.[68] The EU is also committed to fair and equitable use of AI that contribute to the preservation of human rights, another key objective of the EU diplomacy. These are all laudable courses of direction, but the question mark is what the term "fair and equitable use" of AI actually means. In any event, what is clear is that the EU's public mandates and policy objectives are strongly hinged on 'proper' application and utilization of AI. Arguably, AI's permeation into the public sector is, at least, clear.

Meanwhile, the European Commission is also aware of a legislative gap or legal vacuum to control the risks of AI properly. Therefore, it has continuously called for an update of rules and regulations in specific respects.[69] Among a series of outstanding topics, what stands out in the EU's debates is the attribution issue: the realization that it is sometimes difficult to attribute actions of AI to concrete legal entities or individuals at this point. In other words, liability laws need to be updated to be responsive to the novel situations ushered in by AI. The current EU and member states' municipal laws may be able to handle safety risks, product liability, and insurance issues through some adjustment of the existing legal regime.[70] The bigger challenge is to trace down those responsible when AI is applied for a decision making process of somebody else. Who should be held liable for an accident or negative consequence—the AI operator, the AI-using machine operator, or the ultimate decision maker? This question again goes back to governmental agencies that make decisions using AI or AI-supported machines. Is the agency to be held liable for problematic decisions in domestic administrative law proceedings? Once the scope is widened to the international plane, is the government held liable under international law for the measure it adopts through the application and utilization of AI? This fundamental question also hovers around the horizon of the EU regulatory landscape.

Furthermore, there is a lot of uncertainty involved on the allocation of legal responsibilities when AI is involved in the currently controversial GSCs. In the above-mentioned White Paper, the European Commission noted:

> Uncertainty as regards the allocation of responsibilities between different economic operators in the supply chain: in general, EU legislation on product safety allocates the responsibility to the producer of the product placed on the market, including all components (e.g. AI systems). But the rules can for example become unclear if AI is added after the product is placed on the market by a party that is not the producer. In addition, EU product liability legislation provides for liability of producers and leaves national liability rules to govern liability of others in the supply chains.[71]

The above statement of complexity in the White Paper relates to a situation where a product is manufactured using parts from different sources in different countries.

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[68] *See supra* note 67, pp. 8–9.

[69] Ibid., pp. 14–15.

[70] Ibid., pp. 12–13.

[71] Ibid., pp. 14–15.

Assuming that each part used employs AI components respectively, it would obviously become more complex to determine who is to blame when the final product does not function as it ought to. The current complexity of pinpointing the responsible person as a result of outsourced parts to different sources will be multiplied by the insertion of the AI layer in each part production; now even the part manufacture might try dodge legal responsibility by citing AI designer or developer. The level of complexity moves to yet another level when the final product is used in a public agency's administration of public affairs. The faulty product is or is not used in the decision making process of a public entity, and the decision thus made entails victims among the general public. The pursuit of placing legal burden in this context is anything but simple. The domestic administrative law proceedings and international law proceedings would encounter significant complexities. In the new era of AI, even an otherwise simple term could turn out to be controversial. Consider the term 'risk' or 'safety'. These terms are now undergoing changes because of the increasing threats of cyber breaches and attack and intensifying sensitivities of personal information. Thus, even in the conventional area of legal issues associated with a final product, where attendant legal issues are supposed to be relatively simple, the future legal debates may not be so simple since such basic terms as 'risks' and 'threats' are now understood rather expansively or at least differently. Mindful of these changes, the European Commission has proposed referring to the experience and resources of the EU Cybersecurity Agency (ENISA) to assess the new AI-related threat landscape.[72]

3.3 China

While on a different note, China has also been actively introducing AI in its governmental activities. Its central government is closely coordinating and cooperating with local governments as well as China's digital giants such as Baidu, Alibaba and Tencent.[73] The close coordination and cooperation between the public and private sectors, a unique feature of China's economic governance lying at the core of the US-China confrontation, are also presenting themselves in the digital arena. The tightly knit web has enabled agencies of the Chinese governments to collect and process data in large quantities in an efficient manner. A relatively low level of sensitivity for privacy and personal information and the strong regulatory grip of the central government have permitted China to get a leg up in the digital economy and digital society. China's tight COVID-19 control is the most recent evidence for this.

Broad data collection in China benefits both private and public sectors. By way of example, AI-based Social Credit Score system displays robust application in private

[72] Ibid.

[73] *See* Fabian Westerheide, and Cognitive World, *China—The First Artificial Intelligence Superpower,* Forbes (Jan. 14, 2020), available at https://www.forbes.com/sites/cognitiveworld/2020/01/14/china-artificial-intelligence-superpower/#696b7fea2f05.

business activities as well as government regulatory activities.[74] Its private sector application includes visa application assessment, interest rate determination, and online personal profile checking.[75] The vast amount of personal information thus collected underpins the regulatory regime in a wide range of sectors of China. While credit scoring systems are in fact common in many countries, China's Social Credit Score system is unique because of its centralized data collection spanning over almost entire segments of the society. Viewed from the purely profit maximization perspective of private entities and from the purely efficiency optimization perspective of public entities, none would be more valuable than the 'big data' collected from reliable sources nationwide.

The specific AI technology that has attracted the most global attention would be China's facial recognition technology widely used in law enforcement agencies of the country. It is reported that a vast national database has been created through the combination of advanced technologies and regulatory frameworks mandating collection and submission of facial data.[76] Consequently, government agencies have reached the point where it is capable of identifying each one of over a billion citizens in several seconds, remarkably improving the efficiency and capacity of law enforcement agencies. It is not just national law enforcement agencies stationed in Beijing, but police stations on the field are also relying on the new technology for their garden variety police works.[77] Its unprecedented efficiency notwithstanding, to the extent that the facial recognition technology can be used for a possible racial, ethnic or religious profiling, it raises a concern on the part of human rights activistsis.[78] It is AI that provides the digital infrastructure for this whole new scheme. AI itself is value neutral but it can be a double-edged sword depending on how to utilize it. That is all the more reason for the global community to formulate international norms on the application and utilization of AI. It is not just something that happens in the far future, but is already taking place around us all. Unless reliable legal guidelines are introduced sometime soon, the kind of quibbles between China and western countries are likely to persist in various digital sectors. China's active absorption of AI in the public sector also presents the same legal test bed: when, who and how to blame once AI-triggered decision making process goes awry, either under domestic administrative law and, more importantly, under international law.

It seems that the discussion on the liability issue has not gained much traction in China yet. Part of the reason may be China's relatively underdeveloped jurisprudence on government legal liability as a state with strong, central government leadership. Yet another reason may be the fact that predominant focus on China's AI utilization has been placed on privacy and human rights concerns so that other structural

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[74] See Campbell [22].

[75] *See supra* note 63.

[76] See Mozur [23].

[77] *See* Library of Congress, *Regulation of Artificial Intelligence in Selected Jurisdictions* (Jan. 2019), pp. 39–40, available at https://www.loc.gov/law/help/artificial-intelligence/regulation-artificial-intelligence.pdf.

[78] See Mozur [24].

legal issues have gone largely unnoticed yet. Meanwhile, China seems to be particularly careful about data security and protection. It approaches the issue from the perspective of national security. For instance, China's Cybersecurity Law mandates that network operators and businesses in 'critical sectors', which not only include public services and government affairs, but also entities operating in public communications, energy, transport, and the like, must be scrutinized and monitored for any possible interference, damage or unauthorized access to the national network, in fact covering both the public sector (i.e., the government and its agencies) and public-related sector (i.e., State-Owned Enterprises (SOEs)).[79]

3.4 Japan

Japan has also introduced a number of initiatives for deploying AI to implement a variety of societal agenda.[80] Japan's AI projects include educational reforms, collection and consolidation of data, renewal of regulatory review process, and introduction of data-related privacy standards. Japan's application and utilization of AI made a significant stride in its preparation for the Tokyo 2020 Olympics, held in July 2021. By way of example, the Japanese government deployed robots for patrols and facial recognition systems to detect and prevent terrorism. In line of this, Tokyo Metropolitan Police and Panasonic, one of the major Japanese ICT companies, cooperated to develop a new 'crowd forecasting system' which monitors suspicious activities in public areas.[81] The Kanagawa Prefecture Police in turn adopted its own predictive policing based on AI in its preparation for the 2020 Olympics. The AI-based law enforcement systems tap into information from big data encompassing statistics on weather, geographical conditions, certain aspects of crimes and accidents, as well as information from social media, a situation which would not have been considered possible previously.[82] Some local governments have been introducing English-speaking chatbots to provide better service for foreign visitors. Others, such as the

[79] See Zhang et al. [25].

[80] *See* The Government of Japan, *How Japan Uses AI and Robotics to Solve Social Issues and Achieve Economic Growth*, Harvard Business Review (Feb. 4, 2020), available at https://hbr.org/sponsored/2020/02/how-japan-uses-ai-and-robotics-to-solve-social-issues-and-achieve-economic-growth.

[81] *See* Synced, *One Year Countdown: Readying AI Security for the Tokyo 2020 Olympics* (Jul. 20, 2019), available at https://syncedreview.com/2019/07/20/one-year-countdown-readying-ai-security-for-the-tokyo-2020-olympics/.

[82] *See Kanagawa Police to Launch AI-Based Predictive Policing System Before Olympics*, The Japan Times (Jan. 29, 2018), available at https://www.japantimes.co.jp/news/2018/01/29/national/crime-legal/kanagawa-police-launch-ai-based-predictive-policing-system-olympics/#.XrSAPqhKiUk.

city of Takamatsu applied AI for daily administrative function such as evaluation of applications for the city's daycare facilities.[83]

Faced with the increasing geopolitical volatility in northeast Asia, Japan has also plans to increase the utilization of AI to enhance its defense capabilities. Japan is said to be drawing up plans to establish an 'unmanned aircraft department' inside a defense-related agency, which will be in charge of using drones to monitor and patrol remote island regions.[84] Although it is still unclear how the current effort to increase the application and utilization of AI in its defense area would affect Japan's traditional self-defense-only defense policy, what is clear at the moment is that AI is being pushed to take up more roles in the defense area. In particular, the new scheme could lend significant help address and mitigate the chronic social problem of the rapidly aging population. Prompted by this new development, Japan has been actively participating in international discussions on regulating automated weapons, a hot topic in the area of modern warfare as evidenced in the recent Russia-Ukraine military conflict in March 2022. While Japan's firm position, or any other country's firm position for that matter, has not been officially announced in the global fora yet, it appears that Japan takes a rather cautious approach so as not to legally bind itself to banning lethal autonomous weapons systems (hereinafter LAWS) at this stage.[85]

Japan is also one of the early adopters of facial recognition technology in its immigration processes. The recognition scheme found its place in immigration as early as the fall of 2017.[86] This scheme resorts to AI to compare the machine-read images of faces to the bio data encrypted in passports at airports and other entry points. The technology is also applied to monitor departing foreign visitors, whose bio data have been collected through photographs taken at the time of initial entry. If anything, the reliance on AI-related technology for immigration policy has deepened during the COVID-19 pandemic. Japan's examples also indicate government agencies increasing interest in adopting AI technology for official activities.

3.5 Russia

Along with China, Russia is generally regarded as a country that takes a 'utilization-but-caution' approach to AI, mindful of its possible implication for national security.[87] Under the same rubric of national security, however, Russia's AI approach arguably has not reached the level of China where various types of

[83] *See Editorial: AI Systems Hold Great Promise for Local gov'ts, but Efficiency isn't Everything*, The Mainichi, (Aug. 13, 2019), available at https://mainichi.jp/english/articles/20190813/p2a/00m/0na/012000c.

[84] See Hong [26].

[85] See Akimoto [27].

[86] See Osumi [28].

[87] See Polyakova and Meserole [29].

pan-society surveillance technologies have come to existence.[88] And yet, as with any other country Russia has realized the enormous benefit of AI technologies for surveillance purposes. Russia has been investing to advance a project called "Safe City" since 2015, the key component of which is nation-wide video surveillance systems equipped with an automatic information transfer feature.[89] The project installed approximately 170,000 cameras in the city of Moscow, the majority of which has facial recognition capability.[90] Growing confrontation with the West would only accelerate the spread of surveillance-oriented AI application. In fact, Russian President Vladimir Putin is calling for more investment to develop and adopt AI-powered surveillance systems.[91]

Interestingly, the surveillance cameras with facial recognition technology have proven its effectiveness in light of the global pandemic. The cameras transmit real-time data to police authorities, and the data so submitted constitute key digital assets for strong enforcement of quarantine orders. It was reported that through the data Moscow Police were able to identify several hundred violations of the quarantine orders.[92] While the effectiveness of the technology as a means to fight against the deadly coronavirus, the obvious risks of misuse or abuse of the surveillance technology for other regulatory purposes have invited legitimate voice of concern.[93]

As a military strong power, Russia is also actively seeking AI application and utilization in its military.[94] In particular, the Russian military is considered to be highly interested in semi- and fully autonomous vehicles for military use: Russia is on its way to developing an unmanned ground vehicle called Nerehta, which will operate as an AI-based platform for combat, intelligence gathering, and logistics management.[95] Reports also indicate the Russian military's interest in applying and utilizing AI for its swarming capabilities (situations where conventional fighter jets and UAVs are mobilized together for a joint operation), electronic warfare and information dissemination.[96]

Of course, security and military are not the only areas of AI application and utilization. Other governmental service areas are being occupied by AI technologies fast. The Moscow city government has been pursuing a 12-year project, termed as "Smart City 2030", to provide citizens with personalized access to digital services

[88] Ibid., p. 7.
[89] Ibid. p. 8.
[90] Ibid., p. 8.
[91] Ibid., p. 9.
[92] See Reevell [30].
[93] Ibid.
[94] *See* Congressional Research Service (CRS), *Artificial Intelligence and National Security*, (Nov. 21, 2019), p. 24, available at https://fas.org/sgp/crs/natsec/R45178.pdf.
[95] Ibid., pp. 24–25.
[96] Ibid., p. 25.

related to health, education and environment, arguably all core elements of governmental function.[97] A new data platform capable of these new services to the public is also being developed.[98] A public-private partnership comes into play as well. Moscow city government formed a partnership with Russia's largest bank, Sberbank, to collect, anonymize and analyze a wide range of personal data such as bank payments, loans parking fees, health-related spendings, fines, and federal and regional tax payments.[99] Needless to say, information collected by such a centralized digital platform will give the authorities an accurate picture of the social and economic circumstances of the city. This information may easily turn into a pool of digital assets that can be analyzed and mobilized for political purposes.[100] The memory is still vivid among cyber security experts as regards the alleged Russian entities' intervention in the 2016 US presidential election.[101] The proven cyber operations capabilities of the Russian entities might find the stored information to be a tempting digital asset for governmental purposes as well as business opportunities. By any account, information warfare has already taken its root in the Russian foreign policy, and with the help of AI, the repercussions could be way wider and deeper.[102]

Another area in which the Russian government is actively applying AI is scientific exploration. The Russia's state-owned space corporation, Roscosmos, has displayed robust plans to apply and utilize AI to its "Milky Way" space situational awareness program.[103] It is claimed that AI can help reduce errors in calculations to predict collisions between space debris and operational satellites. The space exploration programs and activities will be further facilitated by a wide range of AI-related technologies and machines. All in all, Russia is also relying on AI for a variety of governmental and public functions in various sectors.

3.6 South Korea

One of the signature characteristics of the Korean society is its 'fast' culture. Almost everything is quick and speedy. There is just one exception to this national trait: its slow changing web of regulatory frameworks. By and large, laws and regulation fall way behind technological development in the market, and the demand from

[97] *See* The City of Moscow, *Russian Federation: The SDGs to Promote Balanced Urban Development,* from the *2nd OECD Roundtable on Cities and Regions for the SDGs,* available at https://www.oecd.org/cfe/cities/Moscow-Issue-Note.pdf.

[98] See Gaaze [31].

[99] Ibid.

[100] Ibid.

[101] Ibid.

[102] See Polyakova [32].

[103] *See Russia to Develop Space Surveillance Satellite to Monitor Space Debris as Part of Milky Way SSA Network,* Space Watch Global (Jun. 2020), available at https://spacewatch.global/2020/06/russia-to-develop-space-surveillance-satellite-to-monitor-space-debris-as-part-of-milky-way-ssa-network/.

the businessmen. Start-up companies' first encounter with the laws and regulations is regulatory red tapes of various forms. By definition, start-up companies' business models are something novel or fresh, which means that specific provisions in laws and regulations are non-existent or at least blurry. Absence of provisions or unclear wording in the existing provisions are quite natural because after all they were written when the new business models did not exist or were not even contemplated. The gaps between norms and reality are almost inevitable. Such gaps, however, often lead to agencies' refusal of authorization or permission for new business activities. This is deemed rather ironic for a country making a living from ICT.

Uber's failure in the Korean market and its subsequent withdrawal from it in 2015 is often invoked for the slow-moving red tapes in Korea. The withdrawal was mainly prompted by the strong opposition of taxi companies that invoked relevant laws and regulations for for-hire ride service generally. Those laws and regulations were drawn up long time ago with the taxi companies in mind, so they do not contemplate new ride-sharing services provided by Uber and the likes. Individual car owners providing ride service to other individuals for a fee, the crux of Uber-style business model, is simply prohibited by the law. Uber's departure did not end the confrontation. In 2018, just three years after Uber's departure, taxi companies' protests flared up again. This time the protest was directed at a Korean company launching a web-based, car-pooling service. Many reasons are cited, one of which is again the potential risks of such new services, namely, the possibility of illegal activities or even crimes involving unregistered or unregulated drivers; more cars on the road during rush hour would mean traffic jams and other problems. The opposing rationale got further traction due to an alleged violation of relevant laws and regulations. These events showed the difficulty start-up companies, many of whom ICT ones, should face in the Korean market. It is an environment that affects the application and utilization of AI negatively.

However, the tide is changing. The Korean government has become ever more conscious of the importance of data in the future digital economy and global trade. Views have been expressed comparing data in the digital economy to oil in conventional industry, and various agencies have rolled out a new data-driven economic policies for diverse sectors. The government has announced plans to make massive investment for the establishment of the data industry, including the construction of a "data expressway."[104] It is considered to be investment for the future general infrastructure for the nation. While the regulatory red tapes still remain a headache for many in the business sector, the realization is taking root in the Korean society that the elimination of the red takes are prerequisite for the digital economy, and that digital economy is where future economic development and trade can bloom. Thus, Korea is undergoing rather rapid changes in this area.

Due to Korea's early success in mitigating the spread of the novel coronavirus, its use of ICT in handling the pandemic received keen attention. Since the outbreak in January 2020, the Korean government has been actively applying and utilizing AI

[104] See President Vows Increased Support for Big Data Industry, The Korea Herald (Aug. 31, 2018), available at http://www.koreaherald.com/view.php?ud=20180831000695.

technology to provide public services to the citizen. Immediately after the spread of COVID-19, the Seoul Metropolitan Government introduced an AI-based monitoring call system to keep track of individuals in self-quarantine.[105] In fact, AI technologies had already arrived even before the pandemic. In 2019, the Seoul Metropolitan Government launched AI-based taxi services to better allocate vehicles according to demand patterns.[106] At one point, Seoul won the title of "the world's best e-government city" for several years and this effort for digitized governmental business offered a momentum for continued effort for a smart city.[107] The smart city project includes using big data collected through 50,000 IoT sensors spread city-wide to analyze citizens' demands such as construction and re-routing of public transportation, operation of city parking system, air quality prediction and control, and crime prevention.[108]

At the national level, the South Korean government had been introducing AI into government functions well before the novel coronavirus. In the 2016 plan, the South Korean government announced plans to proactively apply AI in the public sector.[109] As with other countries, AI is also advancing to the Korean military too. In the military sector, South Korea hopes to replace labor intensive tasks with AI in preparation for the streamlining of military organizations and reductions of military personnel according to the future plan announced in 2014.[110] Faced with ever-present military confrontation with North Korea, South Korea aims to enhance its military capability with the adoption of new technologies of AI by 2030, applying it in diverse defense areas ranging from surveillance and monitoring of the Demilitarized Zone (hereinafter DMZ) to lending assistance to command-level decision-making process.[111] LAWS are being deployed along the DMZ and coastal areas for the purpose of carrying out sentry duties. It is expected that the military will remain to be the most robust users of AI in Korea. Other areas in which the Korean government is planning to apply and utilize AI for public purposes include, but are not limited to, policing (e.g. crime prevention and crime response), social welfare (e.g.

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[105] *See* Seoul Metropolitan Government, *Seoul Introduces the COVID-19 "AI Monitoring Call System"* (Apr. 8, 2020), available at http://english.seoul.go.kr/seoul-introduces-the-covid-19-%E3%80%8Cai-monitoring-call-system%E3%80%8D/.

[106] *See* Seoul Metropolitan Government, *Seoul Introduces AI-based Taxis for the First Time in Korea* (Mar. 14, 2019), available at http://english.seoul.go.kr/seoul-introduces-ai-based-taxis-for-the-first-time-in-korea/.

[107] *See* Seoul Metropolitan Government, *Seoul, the World's Best e-Government to a Smart City; 50,000 IoT Sensors to be Installed throughout the City* (Mar. 13, 2019), available at http://english.seoul.go.kr/seoul-the-worlds-best-e-government-to-a-smart-city/.

[108] Ibid.

[109] Government of the Republic of Korea Interdepartmental Exercise *Mid- to Long- Term Master Plan in Preparation for the Intelligent Information Society* (2016), pp. 40–42, available at https://english.msit.go.kr/cms/english/pl/policies2/__icsFiles/afieldfile/2017/07/20/Master%20Plan%20for%20the%20intelligent%20information%20society.pdf.

[110] In 2014, the Korean military announced plans to reduce its number of troops more than 17% by 2022. Ibid., p. 40.

[111] Ibid., p. 40.

legal service, tax service, public opinion monitoring, demographic analysis), transportation (e.g. real-time traffic data collection, traffic minimizing system), and postal services.[112]

Moreover, in the *National Presidential Initiative for AI*, the government expressed its strong desire to actively apply and utilize AI to provide better public services:

> Fourth, the Government will become AI-oriented. Right after the inauguration, my Administration launched the Presidential Committee on the Fourth Industrial Revolution. It selected 'Data, Network, and AI (DNA)' as three new innovative industries and has supported them. Last year, 'AI R&D Strategy' and 'Data Industry Vitalization Strategy' were put in place at the pan-government level and have steadily been implemented. The Government itself will actively use and support AI at every opportunity. We will transform ourselves into an AI-based digital government, going beyond the world's best e-government. We will provide high quality services, starting with those areas that can directly impact the people's lives such as the environment, disasters, safety and national defense, so that the public can sense the changes. The Government's public services will also be transitioned to internet- and smartphone-based ones.[113]

The above statement is a blueprint for the government for AI-induced public service in the future. Indeed, the application and utilization of AI is increasing and spreading in many public sectors of Korea. The trend is not simply confined to governmental agencies. It is also being observed in government-affiliated entities and public entities. For instance, the Bank of Korea announced plans to use AI and big data to devise and implement monetary policies.[114]

The increasing presence of AI in the public sector also entails new problems. The common mixture of mechanical errors, technological glitches and human mishaps makes it more difficult to pinpoint the wrong-doer or those responsible. It is also not entirely impossible for an agency to hide behind AI-produced data or AI-enabled machines. By way of example, as drones and automated weapons systems have become popular in the front lines in step with the dwindling national population, vigilance failures and field accidents spark debates as to who or which to blame.[115]

3.7 Summary

To summarize the recent development in select countries, an emerging situation is that governments are starting to apply and utilize AI in carrying out their government functions. National security and defense areas are arguably the most prominent adopters of AI technologies in many countries. China and Russia are particularly

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[112] Ibid., pp. 40–42.

[113] Ministry of Science and ICT, *Presidential Initiative for Artificial Intelligence (AI)* in *National Strategy for Artificial Intelligence* (2019), pp. 1–3. Full text available at https://english.msit.go.kr/cms/english/pl/policies2/__icsFiles/afieldfile/2020/03/23/National%20Strategy%20for%20Artificial%20Intelligence_200323.pdf.

[114] See Min-hyung [33].

[115] See Kyu-Seok [34], Yoon and Martin [35].

active in this type of AI application. The domestic regulatory regime of the two countries permits the collection and storage of a vast amount of personal information and the information so collected in turn constitute the databases for the reference and usage by government agencies. National security and defense sectors stand to benefit from the data more than anything else. Mounting uncertainties swirling around the global community recently have re-kindled the policy orientation toward national security and defense. With that, the value of AI technologies continues to rise in the two countries. To the extent that the information and data are utilized for further surveillance and control of ordinary citizens, a concern over human rights and privacy have become a topic of constant debates. It should be noted that the elevated national security alert and processing of personal information for security purposes are not merely confined to these two countries or any other country for that matter. Other countries with arguably more democratic political system and legal guardrails, too, are also looking to AI for its vast potential as a vehicle, enabler or facilitator for law enforcement activities within their territories. Terrorism prevention and immigration control occupy the top place of major countries' national agenda. The United States, European Union, Japan and South Korea are importing AI technologies and machines for key governmental functions. No wonder AI is now regarded as a critical issue of national security. Likewise, it is not surprising either that semiconductors, the mechanical component that makes AI operate, have taken the center stage lately. This phenomenon explains why semiconductors sit at the center of US-China confrontation, and why Washington and Beijing see them as items of national security rather than just computer parts. In a word, AI has come to signify national security and national defense.

Meanwhile, AI also is contributing to the improvement of general welfare. Europe is at the forefront of applying AI in a more human-centric approach. AI can assist agencies to provide better public services to people, if applied appropriately. The White Paper published by the European Commission offers a good starting point in guiding governments of EU as well as non-EU member countries to find a proper course of action for the future in dealing with these new technologies and formulate best practices in this regard. AI in the public sector is not just a super computer that can increase the efficiency of government work and reduce bureaucratic hurdles. It also means, more critically, the way government agencies draw and implement plans for public mandates is going to be changed. Regulatory landscapes are going through stages of transformation. As a matter of fact, a number of countries, including Japan, Russia, and South Korea, aim to create "smart" cities with the new AI technology. While it is still unclear what a smart city will look like, the concept arguably embraces both (i) automation of daily living and governmental activities to increase the convenience and efficiency, and (ii) data-driven regulatory process. Both require deep integration of AI. What raises a more structural issue from a legal perspective is the latter. It is in this context that international law stands to be affected by the new development.

In sum, AI can upgrade and facilitate significantly the function agencies carry out in the public sector, and we are observing fast and deep penetration of AI technologies and equipment into various segments of government service. The legal

discourse as to how to understand the implication of this new development and how to devise a new framework to address such implication is still in a budding stage waiting to bloom. It is time international law starts to digest these issues and initiate relevant discussions.

4 Evolving Debates on AI

As with most technologies, AI is a double-edged sword. It offers tremendous benefits, but the benefits also come with acute risks and side-effects. One of the most notable advantages of AI lies in its high analytical power. AI can process a large amount of data at a remarkably high speed with an unprecedent level of accuracy. As a result, AI is prepared and indeed expected to make better predictions and decisions, leading to increased efficiency and productivity for the areas in which it is employed. Nevertheless, because AI, as a man-made machine or a system, is not fault-proof, one needs to be cautious about possible risks and side-effects. Some risks and side-effects stem from technical errors and failures. They could be fixed or at least gradually ameliorated with the improvement of relevant technologies. On the other hand, there are risks and side-effects that are structural in nature, embedded in the AI-driven human society itself. Consider labor market disruptions, human labor displacement, authorities' tightened monitoring, privacy infringement, and the global community's South-North divide exacerbation. Insertion of an AI-layer in the decision making process complicates the person or entity which is indeed legally liable under either domestic law or international law. These risks and side-effects should receive adequate attention at this juncture as the global community proceed with AI-driven society. Marveling at the technological development and looking to risks and side-effects of the former category (relating to technological aspects) would not provide an accurate picture for a future global landscape filled with AI-technologies, machines and systems.

As AI takes a bigger role in the society, the second category of structural problems would loom large. If an automated screening process is uniformly used in selection processes (e.g. scholarship evaluation or visa approvals) with accurate data sets and high technological precision enabled by AI, one could expect a mechanical application and stable outcome. There may not, however, room for acceptable outliers or reasonable flexibility. Alternatively, AI can be trained through deep learning for a wider margin of discretion and broader spectrum of judgment. It may overcome shortcomings of mechanical application and accommodate outliers. The other side of the coin, however, is infusion of perception and bias embedded in the data sets themselves. If we use data that is contaminated with structural discrimination in the society in order to train an AI machine, the resulting outcome will also be tainted with discriminatory effect. In a sense, AI in this context is a mirror of underlying human society or human judgment. The question mark, therefore, is not so much about how to attain further technological improvement as about how to program AI machines and systems to achieve the pre-determined objective. In other words, a

human element still comes in and remains as a critical component in an AI decision making process. It could be called a combination of human and AI judgment processes: (i) existing human judgment still seeps into final decision making process of AI machines and systems, (ii) individuals and corporations designing AI programs also play a critical role in the final outcome, and (iii) AI machines and systems still make their own decisions. Assuming that the final AI decision again constitute the logical groundwork for ensuing decisions of people and entities, it cycles back to the human segment. The decisions made by people and entities in turn constitute a new data set, ready to be picked up by future AI machines and systems. The cycle thus goes on and on. Any decision using AI machines and systems operated on data sets drawn from human societies can be said to be an AI-human 'hybrid' or 'joint' decision or otherwise anchored on 'hybrid' or 'joint' decision making process. The mixed decision making process signals an inherent challenge when we face the question of who is to blame once something goes wrong. As noted above, there may be other structural problems even if the AI functions seamlessly as it is originally programmed free from any human society-induced taints. Since AI and the Fourth Industrial Revolution are expected to bring unprecedented changes to our society, clashes with existing social structures and frameworks seem inevitable. In an uncharted territory, bumps along the way will be unavoidable. An example would be the employment issue. When automation replaces human labor in more sectors and more tasks, the unemployment rate stands to shoot up. In the AI society, some people's prompt gain could mean other people's immediate loss. The widening schism may cause social and political unrest. How to distribute the overall gains from AI among the society in general is a pending homework at the moment.

On the other hand, the global community should also be wary of over-regulation. While many schemes of the domestic regulation are based on legitimate concerns and policy objectives, often times they tend to go above and beyond the proper parameters of regulation, interpreting existing laws and regulation conservatively, tightening regulatory grips and discouraging innovation when things are murky. It is particularly harmful to the AI area where creativity and innovation are the driving force. Few, if any, existing laws and regulations account for the new circumstances of the AI era, so the gap between the norms and reality is inevitable here and in fact wider than any other area. The gaps have been interpreted and applied in an unfavorable manner toward start-up businessmen and ICT companies eager to explore new application and utilization of AI. More often than not, government agencies refer to legitimate policy objectives as a rationale for regulation, but the question is not whether policy objectives are benign but whether the regulatory wherewithal thus chosen are malignant. In this process, what is emphatically underscored is the concern over possible risks and side-effects of AI, both known and unknown. When in doubt, a default answer is usually no, instead of yes, to be on the safe side, which is understandable for government agencies and their officials. Arguably, the benefit is invisible and materialized in a longer term, but the problem is visible and prompt. Naturally, this is an uphill battle for new ideas and endeavors. Various sectors of a modern society encounter this obstacle, but AI simply stands out because of its very nature of new attempt and absence of precedents. Concerns over expected risks

and side-effects are the obligations of any responsible government. But, if pushed to the extreme, concern can become overconcern. That is the point where regulation turns into 'over-regulation.' Over-regulation may stifle innovation and creativity, a critical threat to AI development. A risk or side-effect should be clarified in advance, and any regulation should be tailored to address the risk and side-effect precisely.

When ideas and business models are tried for the first time, with new technologies and formats, they are mostly in the twilight zone of existing regulation. In the twilight zone, depending on the policy orientation of the government agency at issue, the answer to a question may well go in two-different directions of yes and no. Different answers originating from different policy orientation provide a good recipe for intense disputes between states. As we proceed in the new AI landscape, similar questions will keep on coming up. ICT companies will have to walk a regulatory tight rope each time their products and services reach the market. Laws and regulations should not be overly restrictive and bureaucratic. Absent concrete evidence, there is no reason for an overblown concern. Of course, what should not be forgotten in this process is how to ensure that existing industries are not marginalized in a way that threatens the welfare and livelihood of workers in those industries. That does not necessarily mean that interest of existing industries and their workers should always come first. As with any other governmental policy, a balance should be struck. This is something that requires extensive debates in the current AI dialogue. At least, one thing seems to be clear: it would not be prudent to allow too much concern or overconcern predicting a worst case scenario to operate as a blanket justification to maintain existing regulation or block new one.

Thus, AI is not free from existing social justice problems, nor is it above regulation. At the same time, tight regulation is a great hindrance to the innovation of AI technology. Regulation should be kept in a reasonable boundary. It is why we need to weigh the benefits and risks of AI keeping a totality of circumstances in mind. In the following section, AI's benefits as well as concerns over the risks and side-effects are examined in more detail to grasp the ongoing debate surrounding the pros and cons of AI.

4.1 Benefits from AI

AI's benefits are broad and unmistakable. It helps replace humans in painstaking tasks, reduce labor hours, and enhance work efficiency. As for the players in the business sector ever pushing themselves to the extreme for profit maximization, none would match AI in terms of devising future business strategies. Pressured by budget constraints and dwindling workforce, the public sector is also looking to AI as a source of last resort. Multifarious application and utilization of AI among governmental agencies of many countries are based on the expectation that AI will contribute to acceleration, cost-saving, and efficiency of administrative services.[116]

[116] Eggers et al. [36], Mehr [37], Reis et al. [38], Engstrom [39].

After all, with the advancement of AI technology and automation, such expectations have mostly come to fruition.

As energy becomes a hot topic of the national economy, and as climate change policies become a priority in national agenda, many governments are paying attention to AI technology to enhance energy efficiency in both the supply and demand side of the relevant industries.[117] As rare earths becomes a coveted item as materials essential to produce digital gadgets and devices, ICT industries and their governments canvass the pockets of natural resources of their counterpart countries. AI technology helps map the location, size and value of underground resources. AI is also applied and utilized to the design and operation of wind and solar farms to increase the production of renewable energy. Furthermore, AI also allows for a more efficient use of energy. These energy projects could contribute to lowering carbon emissions, the hallmark of the current environmental policy as a result of the 2015 Paris Agreement. AI has made it possible to explore new schemes in this regard such as green energy purchase credits, energy purchase decision adjustments, stable electricity supply and consumer needs accommodation.[118]

If climate change demands an emergency response, and the response in turn demands an efficient use of energy, AI seems to be the only solution except for the creation of new sources of future energy. With accurate data sets and fair burden sharing proposals, AI could also allay the complaints of domestic interest groups that stand to receive negative impact from climate change policies. On balance, when it comes to a carbon emission reduction policy, AI could help ensure better adjustment and adaptation strategies for national governments.[119] Some governments are already mobilizing AI technologies to facilitate their climate change drive.[120] AI technology is not just confined to the decarbonization area. Other environmental issues are being assisted by AI as well. Accordingly, a network of sensors is being developed and deployed to monitor the changes of the ecosystem and biodiversity of the target region, all made possible with the introduction of AI technology. Furthermore, digital platforms are being created to collect and analyze wildlife species distribution, forestation conditions, and weather changes. As much as environmental protection, irrespective of specific sub-topics, heavily requires scientific knowledge and analyses, AI plays a critical role and will continue to do so in the future.

Undeniably, some even view the judgment and decisions of AI as more accurate than those of humans.[121] Of course, the accuracy of the AI decisions depend on the prior programming of the AI technologies and systems by a human developer. Assuming that the programming part and software component are carried out flawlessly, and technological or mechanical errors do not occur, judgment and decisions so made by AI might be potentially more accurate than those made by human operators. This trend implies that in the near future AI may expand its

[117] See Victor [40], Wolfe [41], Mehmood et al. [42].

[118] Ibid.

[119] Ibid.

[120] See Reardon [43].

[121] See Burri [44].

presence to more and more decision making positions so as to review and judge a wide range of governmental affairs. It could be called a "robo-regulator", "robo-judge", "robo-arbitrator", "robo-mediator" or "robo-evaluator" depending on the role AI plays. Properly programmed and operated, AI decision makers can reduce human errors, enhance work efficiency, and ensure analysis accuracy. These are certainly direct benefits of AI application and utilization.

The recent pandemic of COVID-19 also shows the immense power and broad utility of big data. AI technology, machines and systems are playing a crucial role in counter-COVID-19 measures globally. Countries that seem to be containing the deadly virus better than others, namely Taiwan, South Korea, Australia, China, Singapore and the like, have collected and analyzed a vast amount of data to track the movement of the virus-borne individuals. While China has arguably adopted the most stringent version of data collection and analysis, other countries, though in more benign forms, have also followed suit. By way of example, the South Korean government required quarantined citizens to install a "Self-Quarantine Safety Protection" smartphone application programs to monitor people's locations movements.[122] On its part, the government of Taiwan required its citizens to download a phone-location tracking applicaton, and the Central Epidemic Command Center, a Taiwanese government agency, has been given the authority to gather people's location and movement data.[123]

4.2 Concerns Over Risks and Side-Effects

As with any innovative technology, AI is not fault-free. Nor is it problem-free. Technological errors of AI machines are indeed a problem, but they can be readily identified and fixed. Given the fast pace of technological development, one way or another a solution will come our way. Product liability and insurance policies will cover the situation. Human errors are another set of problem. As the old adage goes, "to err is human". It is inevitable and natural for human beings to make mistakes even with perfectly designed machines and systems. AI is not an exception. Proper training on the job and personnel management policies (incentives and penalties) will address the situation. Negligence jurisprudence of tort law, and manslaughter and negligence injury of criminal law will handle the relevant cases. These two types of mishaps, though common, are arguably amenable to an easy solution according to convetional frameworks, be it legal or logistical. A completely different set of problems are those that are structural and systemic in nature. This means that even if AI is applied and utilized exactly as intended, without any error, a wave of challenges will hit the core of our existing societal and national structure. Machines replacing human beings, data-driven economic activities, tightly-knit surveillance networks and human rights infringement offer the best examples in this regard. They change how businesses

[122] See Timberg and Harwell [45].

[123] Ibid.

carry on their activities, how people interact with each other, how individuals pursue their personal objectives, and most notably how government exercise their regulatory authority. They also change how states interact with other states. Accordingly, structural consequences in both domestic and international levels are on their way.

The important task for the global community at this juncture of the AI-era is to identify these structural risks and side-effects (i.e. without abuse or misuse of AI), and to explore ways to avoid or reduce them going forward.[124] The abrupt outbreak and spread of the pandemic have expedited the recognition of these problems. COVID-19 emergency situations forced governments to turn to AI technologies for virus control purposes and societal disruption management purposes. In the aftermath of the coronavirus pandemic, the UK government designed an algorithm to 'predict' student performances based on their previous test scores and school data, because final exams of many primary and secondary schools had to be canceled due to the pandemic.[125] While the algorithm may have succeeded in predicting the outcome and producing reasonably reliable proxy scores, criticism for the decision persisted because it does not square with the very circumstances of an exam—even for the same individual exam-taker, the scores are likely to vary for better or worse. Perhaps the exam system began on the notion that a selection or evaluation process should proceed independent of prior academic records. In fact, this kind of past record-based prediction contradicts the principle that decisions "affecting individuals must be primarily based on individualized factors".[126]

It is crucial to remember that AI, just like any other statistics models or computer programs, is not fault-proof. As mentioned above, technological or human error may always take place, which may then skew the outcome. If such an error was involved, the predicted test score and following grading may not be accurate, and the whole purpose of the 'alternative' exam system would fail. Thus, the whole discussion and criticism presuppose that such error did not take place in the first place. Even so, structural problems do arise (i.e., using AI to analyze prior records and predict future scores) and invite our careful contemplation. Such record-based prediction schemes could cause significant impact on various evaluation processes that the private and public sectors implement. Consider similar schemes in college admission process, recruitment process, banking financing process, immigration process, and building permission approval process. If their application is extended to more sensitive areas such as medical treatment or criminal proceedings, an acute social controversy will follow. AI-assisted automated decision-making stands to cause impact on the lives of individuals to a significant level, a situation that requires a careful balancing analysis between the risks and benefits in a holistic manner.[127]

In particular, it is important to take into consideration the effects of outliers in these prediction schemes. Every data set includes some outliers for various reasons. These outliers may have a notable effect on predictions, which could then skew

[124] Gesser et al. [46].

[125] Ibid.

[126] Ibid.

[127] Ibid.

the final results. Individual performances may be downgraded or upgraded by the variance spectrum of these outliers. For example, the algorithm designed by the UK government significantly lowered the scores of high achievers from historically underperforming schools. Most probably, the scores of these high performing students may have been discounted. Thus, techniques such as stress testing and black swan modeling so as to limit the impact of outliers are important in AI-based assessments.[128]

Possible errors and inherent limitations of AI technologies and systems to the extent that they rely on statistics may be overcome as further technological and statistical improvements are achieved. Mounting stress on the existing societal framework and economic system may tell a different story. Labor market disruptions are already taking place in many different forms. AI-based technologies, machines and systems are expected to replace not only low-skilled workers (consider ordering kiosks at fast food restaurants or taxi drivers) but also high-skilled workers or professionals (consider doctors, accountants or lawyers). A significant portion of the labor market has already been taken over by AI-based or AI-related work force or machines. Sometimes these labor market shifts come with other social policy changes. For example, when the minimum wage was rapidly increased in South Korea since 2017, a lot of businesses decided to turn to automation to replace their workers. While the government's intention in increasing the minimum wage was obviously to better protect the workers, the policy led to an increase in unemployment, partly triggered by the fast advent of AI.

Another growing concern over AI is the strong power of the data possessor and the constant possibility of their abuse of the power. Whoever holds data is in the driver's seat in the digital economy. No wonder private companies make massive investments to collect and process consumer and industry data. Governments are no different. They have also realized the utility and importance of the data. As one of the most valuable data sets for private and public entities is personal information, privacy infringement has become a hot potato recently. Domestic laws and regulations to protect privacy (e.g., the EU's General Data Protection Regulation (hereinafter GDPR)) are being introduced by several jurisdictions. Treaties are being formulated to address privacy issues as well (e.g., Free Trade Agreements' E-Commerce Chapters or stand-alone Digital Economy Partnership Agreements that include notable exceptions (including for privacy protection) to the free flow of cross-border data and to prohibition on server localization requirement). Here again, the recent COVID-19 pandemic has brought the issue of privacy to the fore.[129] Tracking, monitoring, and publishing location information of individuals have been widely adopted. Granted, the unprecedented nature of the recent pandemic may justify extraordinary measures to collect and process personal information, the experience has vividly proven the potency and danger of handling personal information. In other words, the capability of collecting data and collecting better data faster than competitors give a decisive advantage in the fierce competition in the market.

[128] Ibid.

[129] See Zastrow [47], Perrigo [48].

Thus, privacy protection and data integrity guarantee are the hallmark of the digital economy, and expose the vulnerability of the AI-saturated business and regulatory activities.

How, then, should we address the issue of privacy? The question boils down to how to find a balance between utilization of data and protection of personal information. The former is essential part of the digital economy and the latter stems from basic human rights. A balancing point should be found somewhere in the scale between the two competing objectives. As always, 'balancing' is a tricky subject. Different countries have different approaches and perspectives. Consequently, their respective balancing points slide along the scale in one direction or the other by a couple of notches. By way of example, the US traditionally favors more liberal data utilization while the EU takes a more cautious stance. Their positions are well reflected in their respective domestic legislations. Japan, Canada and Australia each take positions compromising the US and EU approaches. China and Russia, on the other hand, take a more protective approach, basically banning in principle the transfer of data out of their respective jurisdictions. Many countries are paying close attention to this regulatory development mindful of different approaches, and trying to come up with their own formula. In South Korea, the general public has been increasing its voice to demand a stricter legal framework due to the increasing sensitivity about privacy and personal information.[130] The nation's legislative body, the National Assembly, organized a special task force committee (*Special Committee on the 4th Industrial Revolution*) to study the issue and set the future direction of policies. The committee's final report contains legislative recommendations for areas where personal information is collected and utilized including proper data protection measures. A stringent penalty in cases of violation is also suggested for the teeth of prospective legislations. New laws and regulations were thus introduced, and the recommendations worked as a platform in the process.

AI causes disruptions at the international level as well. Some argue that the cost of AI-based weapons deployment is low enough to empower non-state actors to seek the path, changing the conventional way of thinking concerning the asymmetry of power balance between state and non-state actors in warfare or conflict.[131] In a sense, warfare may become cheaper and easier with the introduction of AI weapons systems, a worrying situation indeed. Cyber attacks have also become a form of new conflict even among sovereign states. Smaller states are finding cyber warfare more appealing in their confrontation with bigger states.[132] Bigger states are also turning to cyber warfare because of its low-intensity and invisible nature. It is possible that in the future, virtual battlefields may become the flash points of the military conflict, ranging

[130] See Recommendations to the Government for the Fourth Industrial Revolution, Presidential Committee on the Fourth Industrial Revolution, available at https://www.4thir.go.kr/cdn/recommendation(eng).pdf.

[131] See Wagner and Furst [49], Denise Garcia, *Lethal Artificial Intelligence and Change: The Future of International Peace and Security.* International Studies Review, Vol. 20, Issue 2 (Jun. 2018), pp. 334–341; James Johnson, *Artificial Intelligence & Future Warfare: Implications for International Security,* Defense & Security Analysis, Vol. 35 (2019), pp. 147–169.

[132] Ibid.

from dissemination of fake information to paralyzing military control systems with malicious codes.[133] Legal guidelines are being explored and introduced, but they still contain loopholes in details.

Another significant risk and side-effect from the fast introduction of AI is the possibility of deepening North-South divide. Fast innovation of technologies, machines and systems defines the future global landscape of business and regulatory activities. With the already established ICT infrastructure and viable ICT corporations, developed states are likely to enjoy competitive advantage in leading the new landscape of the global community. They possess the manpower, technology and financial resources to participate in the new digital regime with more vigor and investment. The problems identified in this subsection may also be more efficiently addressed by the developed states with their stable legal infrastructure and expertise. They will stand to set the norms for the new digital economy in the AI era. As a consequence, developing states and LDCs may be impacted more negatively than their counterparts. Just take a few examples in this regard. Since the current global value chains (hereinafter GVCs) and GSCs rely on the abundance of low-skilled workers in developing states, the displacement or reduction of human labor in these low-income countries as a result of global automation will undermine the basic concept of the conventional GVCs and GSCs. If AI fosters the reduction of labor costs, the developed states would seriously ponder whether they should bring manufacturing facilities back to their home countries. The remaining reliance on developing states and LDCs will further shrink in this process. Moreover, quite possibly the more we rely on new technologies and machines, the higher the demand for energy becomes. Since the developing nations have unstable energy supply, their ability to utilize AI and construct new digital infrastructure might be already limited unless there is meaningful outside support. As a consequence, it is likely that global inequality will be worsened by the proliferation of AI. If this trend persists, the already wide gap between global haves and have-nots will further widen. Future discourse on AI at the global level should not lose sight of this chronic problem, further exacerbated by the introduction of AI. All in all, AI promises new opportunities to business entities and government agencies. At the same time, there will be various structural problems and tensions that require careful analyses and thorough preparation for the global community to make a smooth transition to the digital economy with AI-technology.

5 Utilization of AI in the Future

AI is making inroads into almost all segments of our society. Though with varying degrees, the application and utilization of AI are one of the most symbolic development that defines the global community as of March 2020. This trend will stay on its course for the future. The pandemic has only expedited the scheduled transition. It should be noted that the penetration of AI is not merely dependent on technological

[133] Ibid.

development: sometimes other societal changes or policy shifts may jump-start the AI transition, and sometimes seemingly unrelated events may accelerate the spread of AI. This is the reflection of the new reality that AI is now ubiquitous and available: companies and businessmen can easily turn to AI for a solution or an alternative.

As mentioned above, South Korea's recent experience with the minimum wage rule demonstrates this point well. Recently, the minimum wage in Korea has increased substantially following the recommendation from the Minimum Wage Commission.[134] The increase is part of a pan-governmental effort to narrow the gap between the privileged and underprivileged in the Korean society as the widening gap threatens the social cohesiveness. The minimum wage increase is one way to alleviate the problem by providing real benefits to low-wage workers and their families. Overall, it is indeed a step in the right direction from the welfare policy perspective.

Nevertheless, the minimum wage hike has ushered in unexpected automation in small and medium-sized enterprises (hereinafter SMEs). Fearing a fast increase of labor costs, owners of SMEs have been trying to make both ends meet. One way of reducing the cost is to slash the workforce—the layoffs. Reducing working hours or business hours is another way of reducing the labor cost. Yet another way is to pursue faster and deeper automation.

.Not only SMEs but also small street shops and restaurants have begun an automation experiment. Compared to its peers, Korea has been relatively slow when it comes to automation in shops and restaurants. Interestingly, self-check-in kiosks are only recently installed at Korean airports whereas major airports in other countries have long placed the machine. Likewise, self-ordering kiosks at restaurants are still a relative rarity in Korea (contrast with neighboring Japan). This is rather a peculiar phenomenon for a country known as the most tightly connected in the world with the fastest internet. Given Korea's ICT infrastructure and potential, the newly begun automation drive, once set in motion, will penetrate the business sectors rapidly.

Similar transitions are also taking place in other countries, In the global workplace, kiosks and automated machines are increasingly replacing human labor. In November 2016, upon the increase of the minimum wages in certain jurisdictions, the first response from fast food businesses such as McDonald's and Wendy's was to expand the installation of ordering kiosks. Touch-screen service machines and ordering kiosks are nothing compared to the prospect of driverless cars or AI-powered AlphaGo, but it is still a symbolic and meaningful development for countries standing at doorsteps of AI-era.

Should automation mean reduced hiring of workers, it may entail domestic political consequences. Even a simple, self-service assistance machine threatens routinized low-wage jobs. So, if shop owners turn to automated, a direct consequence at least in the short term is job losses for workrers in certain segments of

[134] The commission is raising the minimum wage from 6,470 won per hour this year to 7,530 won next year, a 16.4% increase and the largest increase since 2001. The rise will continue until the minimum payment reaches 10,000 won. As of now, Korea's per hour minimum wage is roughly in the upper middle among the OECD countries, when per capita GNP and GNI are factored in.

the economy. In other words, the very group of people that were to be protected by the social welfare policy will in fact take the brunt of the change, at least in the initial stage. It would be a recipe for a social tension and class confrontation. These examples tell us of the multi-dimensional effect on our society stemming from AI. Policy adoption and implementation even in the conventional sectors and areas will now have further complication and more twists because of the AI component.

Thus, as the application and utilization of AI increases, the call for responsible AI will get stronger, which will pressure not only the private sector such as tech companies, but also governments.[135] As the AI-era progresses cybersecurity will become more important, and the need for transparency in AI application and utilization will be further emphasized.[136] Consequently, one could expect legislators in many jurisdictions to face greater pressure to revise existing regulatory framework and create new ones in order to address new issues associated with AI. States will compete in terms of not only technological superiority, but also new standard setting.

It follows then that international law is bound to undertake relevant updates and revisions. Sooner rather than later, select international agreements, ranging from technical cooperation agreements to free trade agreements (hereinafter FTAs) and to covered agreements of the WTO, need to be re-examined for possible amendment accordingly. Amendment of these existing treaties should aim to update and revise the current provisions so that they keep up with the AI-era. As necessary, new agreements should also be concluded from scratch. Through this process, new norms can and should be introduced one after another in order to support goods and services trade in harmony with the advent of the AI technologies and also to facilitate digital trade independent of conventional goods and services trade. In this process of creating and formulating new norms, each country needs to be apprised of the trend of norm/standard changes and be ready for active participation and prompt responses. Those who fail to participate in the process will lag behind their counterpart countries as 'rule followers'. On the other hand, those actively preparing and participating will position themselves as 'rule setters', and be able to stipulate provisions in a way that reflects their core interest in the new digital age.

If new norms applicable to the new digital era are not introduced in a timely manner, there will be ongoing conflicts between and among countries due to the widening gap between reality and norm. Since AI technologies and machines are so innovative a phenomenon arriving with such a short notice, existing treaties were not able to and did not reflect them properly. Consequently, any dispute arising from AI-related issues may not find sufficient legal guidance from existing treaties. In this situation, disputes in the international community are bound to increase. For example, as the introduction of AI can complicate the issue of confirming state responsibility, international disputes concerning state responsibility related to the activities involving AI technologies and machines could be further intensified or otherwise take a long time to be resolved. Trade disputes are also going to intensify. In the digital era, goods and services are converging, clouding the existing borderlines between

[135] See Fleming [50], Calo [51].

[136] Ibid.

them. Sometimes new items being traded in the digital economy fall under neither of the two categories. The current trading regime based on the Uruguay Round trading regime seems to be inadequate, if not totally obsolete, to address various novel issues of the digital trade. Already, a wide range of trade disputes involving digital items which are raised through the WTO and FTAs have shown a steady increase in both quality and quantity. Perhaps, this is just the beginning and more disputes are on their way. Only upgraded trade norms will provide sufficient guidelines and respond to new disputes. Thus, it is essential that the international community update and adjust the existing international regulatory framework in order to be responsive to the increased application and utilization of AI . This task should be implemented as soon as possible. Needless to say, similar work should also be undertaken in domestic laws and regulations.

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Chapter 3
AI's Impact on the Society and Human Behavior



In general, AI is defined as systems and/or mechanical devices that are capable of calculating, learning, and making decisions autonomously.[1] This definition is differentiated from that of automation. Basically, automation stays in the area of assisting human activity and judgment, not making decisions autonomously. Since automation also involves technical innovation and development, and AI is also enabled by technical innovation and development, there are overlapping areas between the two. The critical distinction between the two, however, lies in the self-judging capability. One would thus call automation as a way station to AI. Automation is a necessary condition for AI but not a sufficient one. Such being the case, while the AI discourse at the moment places significant weight on automation, the more critical part (and the one that deserves more attention) is the self-judgment and self-decision portion. Core legal issues arising from the introduction of AI, both under domestic law and international law, are centering around the self-judgment and self-decision element. Deep down, who actually makes the decision directly leads to the question of who bears legal liability. In an armed attack by means of a drone, for instance, if the operator is a human soldier and the drone is merely manipulated by him or her, it is this operating solider that takes any legal responsibility, if any. In contrast, if the drones are programmed already in a way that collects enemy information, chooses a target and initiates an attack, then the question of responsibility becomes more complex. Granted, human elements are still involved here - the manufacturer and programmer of the drones - but the final decision of an attack was made by the machines. Should the human liability be mitigated one way or another, or should the machine even be held liable ultimately? This question hovers around the modern battle field and will more so in the future. There are many types of automated and autonomous decision making machines and systems (so-called "killer robots") that are designed to collect information, process the information, make judgment, and initiate attacks, all autonomously. A mistaken attack would trigger a controversy regarding who

[1] See Hill and Marsan [1].

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J. Lee, *Artificial Intelligence and International Law*,
https://doi.org/10.1007/978-981-19-1496-6_3

should take legal responsibilities for the mistake and on what basis.[2] It is uncertain whether the state that deployed the killer robots in the first place should take legal responsibility; whether the soldier who supervised the killer robots (and/or his superior) should take the responsibility; or whether neither the state nor the soldier takes any responsibility as long as the state has completed the required process of inspection and confirmation prior to deployment and the solider has exercised his or her due diligence in its operation.[3] The manufacturer of the killer robots may also be implicated in this process for its share of responsibilities if the design or manufacturing process is somehow found to be faulty.[4]

Indeed, perhaps all of the players mentioned above are implicated in the final outcome, so any legal liability may have to be borne collectively. In other words, every step of the way from the manufacturing of killer robots to their operation in the battle field involves a varied level of human participation, triggering the question of a respective extent of legal liability. The complexity increases exponentially if the same logic applies to the parts and components that have their own AI-powered sub-function themselves. The existence of many intermediary players could possibly dilute the amount and sense of legal liability. As the old maxim says, everyone's responsibility is nobody's responsibility. Furthermore, the current usage of the term AI is rather broad and vague. This seems inevitable given AI is just a new phenomenon. As a consequence, new technologies and machines involved in any segment of the long cycle which possesses self-judging traits are simply called AI or AI technologies and systems. Different technologies and machines here accord different levels of self-judging capability. The different levels of self-judging capability in turn affect the scale weighing legal liability. Therefore, a more pertinent question is not whether AI is utilized, but AI with what level of decision making powert is utilized. In short, the AI spectrum is quite broad. AI technologies, machines and systems that we encounter fall along the spectrum between the two extremes: (i) the stage of simple automation using AI and (ii) the stage of completely independent judgment, decision, and action by AI. To simplify, one could think of AI as a tool for human beings (the former) and AI as a decision maker free of human intervention (the latter). The former hardly raises a novel legal questions except for

[2] See Alston [2], Hammond [3].

[3] It is expected that the advent of AI will cause significant changes in war and military cooperation in the future. Autonomous weapons based on AI involve a high risk of violation of the International Humanitarian Law and the International Law of Human Rights as well as issues of state responsibility. 'Autonomous weapon' means a weapon that is capable of imitating cognitive thinking of humans and deciding its own behaviors in order to achieve a preset goal by means of AI. Development activity in this area has continued in consideration of the advantage that the risk of soldier casualties is minimized with the workload reduced. It is expected that sooner or later, complete automation can be achieved technically. See McAllister [4].

[4] Such legal responsibilities due to a defect in the manufacturing process are not discussed in this paper. Basically, such responsibilities are taken by humans since there is no significant difficulty in applying the domestic laws of a state with the jurisdiction. In this case, however, this fact itself does not make conclusive the fact that it is a state responsibility. See Marchant et al. [5], Krishnan [6], Kastan [7], Schmitt and Thurnher [8].

technology-related legal issues, but the latter does raise a rather fundamental legal question. The two categories of AI application and utilization are discussed below.

1 AI as a Tool for Human Activities

The fist category concerns the situation where AI-technologies, machines and systems are intended just to help human beings conduct activities and carry out business without employing the AI's own analysis or judgment. Most importantly, the final decision-maker still remains the human being. In short, AI is only a tool—though innovative it is—for human activities. When AI is mentioned by regulators and businessmen, it often refers to AI of this type. It is simply sophisticated and advanced automation. It undertakes repeated work with precision, and thus helps people reduce their workload. Consider industrial robots at factories and kiosks at airports and shops. They in fact carry out the work otherwise conducted by conventional workers. While automation has long been at works and factories, the prior automated machines merely repeated their mechanical action as programmed. They do not engage in data collection, processing and analyses. On the other hand, current automation at workplace includes these functions. Based on data thus collected and analyzed the automated machines adjust their mechanical action. Automated machines can provide different options to consumers based on the data the machines collected on consumers' individual preferences for activities or items. Our google or Youtube search results are memorized and next time we visit the websites the items that we may find attractive appear at the top of the screen or new search results. While a data portion is inserted anew, the machines still do not make a decision by themselves; they only carry out the pre-destined function in accordance with the programmed instruction. In fact, this is the most commonly mentioned AI application and utilization in business and our daily lives. While this data-driven automation is far more advanced than conventional mechanical automation, the essence still remains the same: it just operates as an assistant as opposed to a decision maker. Quite a few examples of automation that we encounter in our daily lives at home and on the streets are the machines that provides convenience to us, but the final decision, if one is needed, is still reserved to a human operator or a business owner. Although human intervention is not needed at the time of operation of the machine, and the machine simply carries out a mechanial task according to the program (such as an airport kiosk dispensing a boarding pass after checking the passport and the ticket), any substantive judgment is only to be made by a human if circumstances require a decision making one way or another (consider when the passport is being rejected by kiosk because of a suspicious description on the biographic page). Monitoring programs applying machine learning also stands within this category of AI as long as a final decision belongs to a human. Due to the legal liability issues with AI technologies and machines, and other uncertainties regarding AI technologies and machines, regulations tend to require that a final decision be taken by a human being. For example, in the city of Takamatsu, Japan, AI initially ranks the applications for

the service of the city's day-care facility according to the pre-established criteria, but final decisions on the applications are still made by human evaluators.[5] Drones and automated robots deployed at high-risk regions are also another examples of AI as a tool, as the operating soldier decides on an attack or a withdrawal. In these instances, legal liability attaches to the people operating the AI technologies and machines. The manufacturer or the programmer of the AI machine may assume his or her own legal liability if there is a problem in the process of manufacturing and programming of the machine, but with respect to the final decision stemming from the machine the legal liability goes entirely to the operator. A man cannot blame his computer when his decision is found to be wrong; it is just his responsibility to have a computer that is error-free and bias-free. The moment he adopts the results of the computer and makes a decision based on the results, all responsibilities are his own.

Likewise, when a state agency utilizes AI as an instrument or assistant in discharging its public duties, only the decision of the agency remains, and whatever outcome from the AI that underpins the decision are simply absorbed by the agency.[6] Regardless of technical advancement or equipment innovation, a final decision is still made by the agency, more specifically a public official in charge. Thus, the subject that carries out the decision-making is an agency or an official of a government of a sovereign state, the legal liability of which is attributed to the agency or official under respective domestic law, and to the state at issue under international law. The agency or officer exercises action or inaction in reference to information or suggestions provided by AI. As long as the final action or inaction is viewed as that of an agency or a public official, its attribution to the state would not entail a perculiar problem as regards legal liability.

As such, no matter how many new devices or machines are applied and utilized in the long process of the decision making, there is no fundamental change in the context of legal liability (state responsibility under international law) as long as the subject of the final decision remains an agency or a governmental official. Therefore, in case AI is applied and utilized only as an 'instrument' of the governmental agency or official, it would pose a relatively easy question in terms of legal liability in the sense who should bear the responsibility for the final decision affecting a person or an entity. Any result, product or outcome of AI is not legally independent, and it is simply incorporated into and subsumed within a final decision of the agency or the official.

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[5] *See Editorial: AI Systems Hold Great Promise for Local gov'ts, but Efficiency isn't Everything*, The Mainichi, (Aug. 13, 2019), available at https://mainichi.jp/english/articles/20190813/p2a/00m/0na/012000c.

[6] *See* International Law Commission, *Draft Articles on Responsibility of States for Internationally Wrongful Acts*, November 2001, Supplement No. 10 (A/56/10), (hereinafter, "2001 ILC Draft Articles"), Article 4 stipulates.

Article 4 Conduct of Organs of a State.

1. The conduct of any State organ shall be considered an act of that State under international law, whether the organ exercises legislative, executive, judicial or any other functions, whatever position it holds in the organization of the State, and whatever its character as an organ of the central Government or of a territorial unit of the State.

2. An organ includes any person or entity which has that status in accordance with the internal law of the State.

Whether it is the agency or the official that makes the final decision does not matter at least under international law, because both of them constitute 'state organs' whose action or inaction are directly attributable to the state. As long as other conditions are met, there does not seem to be a peculiar difficulty in assessing and finding state responsibility in this case of AI as a mere instrument.[7]

2 AI Without Human Intervention

Conversely, there are also other examples of AI that is free from human intervention even in the decision making process. Or human intervention is so minimal or subsidiary that the essence of the decision is made by AI and man is merely an announcer of the decision. This phenomenon is observed in the most recent AI (and most certainly upcoming AI) with far more advanced technologies and machines than those that existed previously. In fact, some AI technologies, AI machines, and AI systems have advanced remarkably and reached the level of full automation with self-judgement and self-decision making function. The fundamental difference between the above-mentioned AI as a tool and this self-thinking and self-judging AI is that the final decision-maker is not a human being but a non-human object called AI.

In the discussion of autonomous vehicles, for example, the level of automation increases gradually up to complete autonomous driving (called the 'fifth' level of automation), in which every decision-making on driving is performed by the vehicle itself without human intervention. The key concept here is AI's 'autonomy'. It should be noted that the level of autonomy involves an assessment of not only the machine's mechanical capabilities, but also the degree of interaction between a human operator and the machine.[8] In the military context, autonomy usually refers to "the capacity of a machine to operate in the real-world environment without any form of external (human) control, once it is activated."[9] Some up-to-date LAWS retain full autonomy. More LAWS with this level of full autonomy will come our way. Accountability problems then arise due to the fact that LAWS of this sort can operate without any direct human intervention.[10] Human Rights Watch (hereinafter HRW), an international human rights non-governmental organization (hereinafter NGO), even argues that any LAWS with 'full autonomy' should be considered illegal under international humanitarian law.[11] In other words, the absence of human decision in and of itself should be considered problematic, according to the rationale. Even if a categorical prohibition approach is not accepted, what is at least clear is that as to the AI at this level, a new legal framework is required at this juncture. The same also applies to the medical field as well. As AI technology advances, the medical treatment using AI

[7] Ibid.

[8] Anderson and Waxman [9].

[9] Lin et al. [10].

[10] Hammond [11].

[11] *Supra* note 146, p. 1104.

has been making gradual improvement; it began by performing the role of assisting doctors' medical treatment, and it now has reached the point where it can diagnose a disease and even prescribe drugs, arguably a final decision at medical treatment. At this point, there does not seem to be a fundamental difference between such performance of AI and typical doctors' medical services. The need for a new legal system to regulate this type of AI is again presented.

Thus, the term "AI" as it is currently being used is somewhat vague and broad, and thus covers various new automation devices and their utilizations comprehensively in every different level of automation. In the final level, in which AI-enabled machines and devices make judgment and decisions by themselves and even perform certain acts based on such judgment and decisions, the issue of who (or what) should take legal responsibility is raised. Highly advanced AI which is emerging in various fields, and which is going beyond the level of simply assisting humans in business settings is capable of making autonomous judgments and decisions. When this trend continues so as to bring more and more self-judging machines, it might be ultimately necessary to recognize that thse AI entities, whatever we may call, should take the legal responsibility (of course on the assumption that other requirements are met). In this context, manufacturers and programmers would be left off the hook since the machine operated as it was originally planned and intended. Of course, should the self-judging process malfunction and hurt someone, the manufacturers or programmers may still be held accountable. Setting aside the malfunctioning situation, however, they would not be held legally liable. Neither would be an installer or user who are not involved in the decision making process, unless he or she has agreed to assume all legal responsibilities, or installing or using the self-judging machine itself is somehow considered problematic (where the person has or should have the knowledge that the machine malfunctions). On a different note, if the fully autonomous, self-judging AI machine is (or should be) regarded as an agent of a governmental agency, its action becomes attributable to the state.

As the above discussion reveals, the issue of state responsibility raises an acute legal problem in the new AI era, and the higher level of autonomy AI possesses, the more complicated the issue becomes. As explained earlier, for AI with lower levels of autonomy, the existing basic principles and jurisprudence can be applied with relative ease. AI with higher levels of autonomy, however, raises novel challenges. As it is reasonable to expect an increase in applications and utilization of AI with higher level of autonomy, equipped with original and stand-alone thinking power independent from the manufacturers and operators, sooner or later we will have to contemplate the systemic implications that this development entails.

The higher level AI function can proceed to the public sector. For instance, AI technologies, machines and systems may conduct immigration procedures including document evaluation and interviews, making final decisions whether to issue visas. They can conduct law enforcement activity by screening background checks of suspicious individuals and make decisions whether to bring them into custody. These decisions sometimes may have an outer appearance of human decisions, but such an appearance would not change the essence of the decision if the human factor is simply to read and tell the decisions of AI. Indeed, there are other similar cases where

AI takes over state affairs and public mandates. AI makes comprehensive judgment and independent decisions by tapping into the accumulated big data as well as newly acquired information.

Issuing a final decision by an AI-empowered machine or system can be broken down to two categories: (i) a situation where judgment and decisions are notified from the AI machine or system directly to the person or entity concerned; or (ii) a situation where the AI's judgment and decisions are first conveyed to the governmente officer officially, which is then conveyed to the person or entity concerned. It should be noted that the two situations share commonalities, if not entirely identical. In the latter scenario, the official in charge, in delivering an official disposition, does not conduct a substantive review on his or her part, but instead accepts the AI's judgment without meaningful review or examination. Under these circumstances, doubts will emerge concerning whether there is an independent judgment or decision making action by the governmental official in the genuine sense. When it comes to action or inaction of a state agency which is attributable to the state as set forth in Article 4 of the Draft Articles on Responsibility of States for Internationally Wrongful Acts, with Commentaries, adopted by International Law Commission at its fifty-third session in 2001 (hereinafter "2001 ILC Draft Articles"), the focus is placed on the 'outward' characteristics and the status of the actor without any additional consideration of the internal decision-making process or internal work division of that organization.[12] Nonetheless, the outward characteristics benchmark cannot mean that it even covers an empty shell situation. The extent and the degree of separate intervention or review on the of the official should be calibrated. At a minimum, the official must contemplate on the contents of AI's judgment and decision, and then undertake his or her own decision. If this is missing, an argument can be made that a human decision making does not exist in this context. The absence of the government official's judgment or decision making process in a true sense would make the latter category virtually undistinguishable from the former category. At least the critical key is whether the official in charge conducts an independent review even if minimal, organize his or her own thoughts, and then issue a final decision.[13] Thus, the second situation can be said to be subsumed into the first situation. As noted above, the firs situation involves a complex legal problem since it results in AI's self-judgment and self-decision which is then conveyed to the person or entity concerned through direct contact. The discussion below thus focuses on this first scenario.

First of all, how could the status of AI performing such 'direct contacts' be understood? One way would be to view an AI machine or system with such function as a subject (be it human or not) that is 'employed' by the government for its official affairs. It is not a natural person and does not receive wages either, but the AI machine or system is recognized as a subject that is eligible to exercise 'authority' on behalf of the government. In this instance, the AI machine or system is entrusted 'exclusively' with governmental affairs (as an independent decision maker), and the government also recognizes the AI's authority to make decisions on certain matters

[12] Ibid.

[13] *See* 2001 ILC Draft Articles, Article 11.

or aspects of certain matters. Stated differently, this is as if the AI machine or system were recognized as a subject (i.e., issuer) of administrative measures according to the Administrative Law although it is not a worker specified in labor law, nor a government official stipulated in government organizational law. In this case, the state (the government) is accountable for the action and inaction (omission) of AI to the extent it affects a person or an entity through an AI-initiated and AI-decided administrative measure.[14] Here, AI may be understood as a type of "automated public official (so-called Robo-regulator or Robo-government employee)." In accordance with the underlying intention of Article 4 of the 2001 ILC Draft Articles, how AI is evaluated under domestic laws of a country concerning its governmental organization or governmental structure will be a determinative factor, which also shows a wide range of country-specific variance.[15] The results of such evaluation may well be varied depending on the relevant provisions in the respective domestic laws of each country. First of all, of course, it would be necessary to define the concept of handling public affairs in an 'employed status' or carrying out the task 'exclusively' for a state agency. Other key terms would have to be defined as well. In existing documentations of many states, being employed by a state exclusively for its public affairs is applied basically to natural persons only. If an AI machine is not regarded as someone or something working for the government (as it is not a natural person), an immediate question then is who should bear responsibility for the final decision that it has issued. Quite possibly, the agency is off the hook as the AI is not regarded as a government official, so its decision is not automatically attributed to the agency. It is possible that the agency is held accountable if it has made an error in installing and operating the machine in the first place, but it may not be necessarily the case. Officials in the chain of command within the agency would also be off the hook because none of them have made interventions or otherwise endorsed the AI's decision—again, it is AI's own and independent decision. Obviously, the AI machine is not held accountable as it is not a natural person. A victim of the governmental measure may find it difficult to pursue his or her legal remedy. One way to address this problem is to pronounce that the agency is liable for any outcome of AI decisions regardless of the level of automation, a sort of 'no-fault' liability rule. In any event, this obvious legal vacuum needs to be filled. The situation applies to both the domestic administrative law context and the international law context.

Then what about the cases where AI is understood as a subject that is neither in 'direct contact' nor employed by a state or governmental agency? This is the situation where a final decision is superficially or ostensibly made by a human being but the essence of the decision is made by AI and such decision is merely accommodated by the human being. Note that if a human operator engages his or her own analysis and decision making process, it is the situation of using AI as a tool, a different scenario covered above. In this section, on the other hand, the human element is just a facade or an outer cover. To the extent that decisions and treatments through

[14] *See* 2001 ILC Draft Articles, Article 4, s*upra* note 129.

[15] Ibid. Article 4 also states the status under domestic laws ("whatever position it holds in the organization of the State").

a human are exercised in a way that is virtually identical or tantamount to direct contacts, the above discussion on 'direct contact' will stand applicable here too. AI in this context may be regarded as a subject that exercises the 'entrusted' governmental authority even if there is no official relationship of employment. The specific distribution of legal liability among different players is still complex in this case. But the fact that the human facade remains will render the legal question relatively easy; as long as the official announces the decision, his or her action is directly attributable to the state, and the state is accountable for such acts.[16] A complex question will continue to arise if the official invokes AI's own decision and his or her mechanical approval as a defense. The more the stand-alone feature of AI's decision is underscored, the more unclear who should bear the legal liability and how, unless a 'no-fault' liability system is adopted. The legal vacuum concern still remains valid in this situation of no 'direct contact'. If anything, one way to add clarity to the murky situation is to make certain the legal basis for the delegation of authority to AI.[17] When AI is given the authority to make judgments and decisions on certain specific issues or aspects of those issues according to laws, enforcement ordinances, rules, etc., it would be an easy case. Quite possibly, any delegation of authority, even if it exists, may be stipulated in a less-conspicuous document such as guidelines, policy bulletins, internal memo, or even in the form of unwritten agency practice. Nonetheless, the existence of the documented evidence for delegation (and in the case of unwritten practice, with the support of other corroborating evidence) could help address the legal question. Even if the AI machine or system somehow goes beyond the scope of the legitimately entrusted authority in making judgments and decisions, and in exercising performance or nonperformance, the agency and the state may nonetheless be held accountable through the jurisprudence of 'vicarious liability' within the domestic legal system or 'acts of arrogation' under international law.[18]

Even if an AI machine or system is neither 'employed' exclusively for governmental affairs nor officially entrusted by law with certain decision making process and formulation of governmental measures, there still is a possibility that legal liability is found alternatively for the state. In other words, the government's 'effective control' over the AI machine or system is recognized, AI's decision can be attributed to the government for legal liability purposes. In this case, acts based on such AI's judgments and decisions, which are seemingly irrelevant to the agency or the government, are determined to be attributable to the state.[19] Under international law jurisprudence specifically, therefore, as mentioned earlier, even if the AI machine or system neither officially belongs to the government nor is it 'employed' by the government (situation not falling under Article 4 of the 2001 ILC Draft Articles is applicable),

[16] *See* 2001 ILC Draft Articles, Article 5; See Hammond [12], Coeckelbergh [13].

[17] *See* 2001 ILC Draft Articles, Article 5.

[18] *See* 2001 ILC Draft Articles, Article 7.

[19] *See* 2001 ILC Draft Articles, Article 8; *Military and Paramilitary Activities in and Against Nicaragua (Nicaragua v. United States)* (hereinafter *'Nicaragua* case'), Merits, Judgment, I.C.J. Reports 1986, p. 41.

but instead operates in a private sector with no legal basis for an authority delegation (situation not falling under Article 5 of the 2001 ILC Draft Articles), its acts can still be attributable to the state as long as it is under an 'effective control' of the government (situation falling under Article 8 of the 2001 ILC Draft Articles). This aspect is especially noteworthy: under the current national system in which a governmental agency manages, regulates and supervises the business entities within the jurisdiction that collect and process the information and big data of individuals, it is probably the entire AI ecosystem that can be claimed to be under governmental control. Given the increasing national attention to the sensitivity of the personal information, any regulation and supervision by the agency may well satisfy the threshold of an 'effective control'. In particular, when it comes to personal information and big data, many countries are currently introducing new regulations on protection of privacy and restriction of commercial use of personal information. Although the regulatory focus is placed on 'protection' of information, it may be viewed as a measure of tightening control in the sense that governmental regulation is becoming ever stricter. Some countries even present specific instructions on how to collect and manage personal information together with harsh penalty for violators. On top of this, the personal information and big data are increasingly being perceived from the perspective of national security, making the situation further more delicate and complex.

Thus, the current digital landscape and the recent trend where countries are attempting to tighten their grips on the use of personal information and big data suggest that AI's judgment, decisions, and acts based on such information and data can be attributable to the state almost automatically. Whether there is an effective control is a deciding factor, but it can be relatively easily satisfied because of the unique feature of the digital environment. Furthermore, it should be noted that even judgments, decisions and acts of AI totally irrelevant to the government or separate from the government with no medium of control involved could nonetheless be possibly attributable to the state in the event that a state agency permits the judgments, decisions and acts to stand as they are even with the specific knowledge on them, or otherwise endorse them *post-facto*. In such cases, it may be viewed that the government and the state has accepted and confirmed the judgments, decisions and acts of AI as if they were its own.[20]

The 'medium of control' and 'effectiveness of control' over AI by a state or government mentioned above has been examined from various different angles and perspectives. If one of the various media of control with the threshold requirement of effectiveness specified in the current 2001 ILC Draft Articles is found to be involved, a judgment, decision or act of AI will be found to be attributable to the state. Consequently, if the said jugdment, decision or act is determined to be in violation of any relevant international laws, be it treaty or customary international law, it will lead, ultimately, to state responsibility. Accordingly, a follow-up measure to remedy the

[20] *United States Diplomatic and Consular Staff in Tehran*, Judgment, I.C.J. Reports 1980, p. 35, para 74.

situation will be then necessary in order to release the state from the state responsibility so found. With different terminology and conditions though, a similar jurisprudence also exists in the domestic administrative law. Thus, above discussions will apply to the domestic discussion *mutatis mutandis.*

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Chapter 4
New Issues Raised by AI



Any big change in society raises novel issues and important questions, and the emergence of AI is no exception. The difference this time would be that the current change caused by AI is coming and spreading faster than any other change in previous times. The changes cover the entire spectrum of our society. Notably, these changes cover philosophical, ethical, technological and legal aspects. First, in any discussion of AI there arises an ultimate philosophical question of what a human being is or what it means to be a human. The fast technological development is now introducing machines and systems that undertake functions that humans used to carry out. The machines and systems are not just confined to facilitating human action or assisting human decision; they are gradually substituting humans in making judgment, issuing decisions and disposing a matter. This trend will continue on its course and we will encounter AI-enabled machines and systems that further resemble human beings in memorizing, analyzing and deciding, the types of unique function that we have so long believed only humans possess. That borderline is now being blurred. The increasingly close resemblance of AI to humans (not in outer appearance but in thinking ability) makes the fundamental question about the very nature of human being more crucial than ever. Obviously, this is an important philosophical question. Since our definition of human or human being provides the underlying basis for many of our social norms and institutions, the answer to this philosophical question will have significant and lastinggreat repercussions in all discussions and debates of AI now and in the future. Likewise, the emergence of AI provokes important ethical questions as well. If we admit the close resemblance of AI to humans, we might need to ask ourselves whether it is ethical to attempt to develop and create another 'being' that thinks and decides like human beings. A similar question has already been raised in the experiment of cloning human cells. The same dilemma also arises if we are able to produce a non-human unit which then acts like a human. By coping what is stored in the memory section of one's brain, advancing technology in the future may even create an entity that copies an individual in terms of memory, feeling and emotion. The recent emergence of 'metaverse' raises this very question; there is someone in the digital environment, who re-produces the same person in the real

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J. Lee, *Artificial Intelligence and International Law*,
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world, and the digital people so created then establish a new digital society or club. The co-existence of humans and entities similar to humans, and of real us and digital us forces us to face an ethical question. Similarly, the applicability of existing ethics to AI is another aspect that needs to be examined in the context of ethical questions. With such philosophical and ethical concerns in mind, in applying and utilizing AI we also need to ask ourselves whether technical development, innovation and advancement are being headed in the right direction. Is it appropriate or should it be permitted for us to strive to achieve any technological development? Is there an outer limit in terms of technological development? What is it that governments and corporations should abide by in developing technologies in the AI sector?

All these three questions are interrelated and intertwined. Philosophical questions lead to ethical questions which in turn lead to technological questions. Conversely, technological questions lead to ethical questions, which ultimately reaches the fundamental philosophical questions. When all these intertwined questions apply to our daily business and lives, they boil down to legal questions. What is legal or illegal when it comes to application and utilization of AI? What new law needs to be introduced to keep up with the changing reality? How should the legal framework be re-structured in light of the newly emerging philosophical, ethical and technical questions? Thus, here also the legal questions are closely tied to the philosophical, ethical and technological questions. While the legal questions are also highly related to philosophical, ethical and technological questions, they just have more practical consequences and tangible real-life implications. In particular, the legal personhood of AI is an important concept that needs to be addressed in the first place as the use of AI increases. When we think of the conventional legal personhood concept as devisible and creatable, the question can be essentially broken down to yet another following questions: (i) should we at some point recognize the legal personhood of certain AI?; and (ii) if so, to what extent should we recognize the legal personhood of AI (only in a specifically limited area or in broader areas)? A delay or a failure to adjust the legal system appropriately will lead to the creation and widening of a gap in the legal liability scheme, where AI machines or systems can be applied and utilized to escape from otherwise existing legal liabilities. It may be called an "AI-shield" phenomenon. This is just one example and there will be numerous legal questions stemming from the AI application and utilization. As it is the legal norms that provide guidance to members of a society, underlying philosophical, ethical and technological questions should find their places in the legal plane and be digested and reflected in the legal system if they were to be applied in practical sense in our daily lives. The legal questions apply to both domestic law and international law with equal force.

1 Philosophical Questions

The advent of AI, in particular the above-mentioned highest level of AI, or AI with the capacity for independent thinking and decision-making similar to human thinking

and decision-making, raises important philosophical issues. Simply put, the AI of this sort raises the very question of what a human being is or what it means to be a human being,[1] because the definition of autonomy, the key word that determines legal liability, has been developed and closely tied to the definition of the nature of a human being. In other words, defining autonomy should fundamentally address the question of self-governance, self-analysis, and self-decision making—in short, the freedom to make one's own decisions.[2] AI raises none other than this very question. It is no coincidence that the Oxford Dictionary defines AI as "the theory and development of computer systems able to perform tasks normally requiring human intelligence."[3]

The starting point in this regard is that humans have been regarded as having a special existential status mainly because their ability to think autonomously and independently unlike other living or non-living things on earth. In addition, humans pursue rationality through such thinking. If other entities than humans that could think autonomously appeare at our doorstep and pursue similar rationality through similar thinking, it is questionable whether a special status accorded to humans is still exclusive to natural persons or whether at least some part of it should be shared with new thinking entities. After all, this issue brings us back to the fundamental question of what humanity is and on what grounds humans are treated specially. These fundamental issues related to AI are being debated seriously and profoundly in the field of philosophy. The answers to these questions will help shape the future of AI in terms of how we understand it.

Subsequent questions flowing from the core philosophical question in this context are as follows: what is the nature of 'legal responsibilities'?; would legal responsibilities only have to be assumed by humans?; if there is an entity that thinks and acts like a human, does it also have to bear such legal responsibilities?; if an entity thinking and acting like a human is recognized as bearing legal responsibilities, would such primates as chimpanzees and orangutans also have to be recognized as with legal responsibilities to a limited extent but in proportion to their thinking abilities? Should the same partial or proportional responsibility be applied to AI? These various questions stand to be raised in relation to the application and utilization of AI.

Responsibilities have been understood to be only reserved to human beings. This assumption of 'person' reveals its existence in state responsibility as well. As a matter of fact, many provisions in the present 2001 ILC Draft Articles are based on the underlying assumption of existence of 'person' or 'persons.'[4] Although this international law instrument does not explicitly stipulate this matter, provisions therein are based on the concept of a 'person,' either natural or juridical. When a series of stipulated conditions regarding acts of such natural or juridical persons are met, state

[1] See Livingston and Risse [1].

[2] Beard [2].

[3] *Supra* note 2.

[4] International Law Commission, *Draft Articles on Responsibility of States for Internationally Wrongful Acts, with Commentaries* (hereinafter 'Commentaries'), Yearbook of the International Law Commission, Vol. II, Part Two (2001), Article 4.

responsibility is established. A natural person (e.g. public official) or juridical person (e.g. a private company or governmental agency) needs to be involved in the process of formulating and exercising a governmental measure one way or another. For instance, if AI utilized is somehow viewed merely as an instrument for a natural or juridical person who makes an ultimate decision, state responsibility is attributable to that person. However, if there is no natural or juridical person involved, and decisions are made by AI, complex questions ensue regarding state responsibility. In this instance, state responsibility can be found in accordance with conventional jurisprudence or it may face complexity because of a possible vacuum of guidelines, the essence of which is based on the said philosophical question of the nature of a human being.

Some argue that as the present social system is centered on humanity, personality or personhood cannot be endowed to AI in order to secure consistency.[5] This argument corresponds to the statement in the Law of War Manual published by the US DOD that robot weapons such as LAWS should not be accorded personality or personhood on their own accord.[6] According to this logic, responsibilities for decisions rendered by AI are still assumed by humans (who possess personality and personhood) who are involved in the decision making process or who have decided to make use of AI in the first place, no matter how widely and deeply AI is spread in our society. Ultimately, whether to hold a state for state responsibility under international law will be handled the same way, the argument goes. For example, AI may be or can be categorically determined in advance as being always as an instrument or wherewithal that functions to assist humans regardless of its form, the extent of its advancement, or the depth of its utilization.

On the other hand, some also argue that new, separate schemes need to be explored at this juncture, mindful of the autonomy that AI possessess.[7] In other words, they believe that it is necessary to accord at least a sort of limited personality or personhood to AI once it reaches a certain threshold level, that is the level of autonomous thinking and decision making.[8] Of course, these philosophical questions are not amenable to easy answers, and continued discussion and through review are in order regarding specific conditions of the recognition of a new status. In any event, however, the argument that AI can also be given the status of personality or personhood suggests the possibility that AI can be held for legal responsibilities, at least to a limited extent and under specified conditions. If materialized, this will constitute a significant development in the basic structure of our society. As this perception develops, it will bring about significant impact on the issue of legal liability of governmental agency under domestic administrative law as well as state responsibility under international laws.

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[5] Bryson et al. [3].

[6] Ibid., p. 285.

[7] For example, one issue that has been discussed recently in the EU is that new domestic laws reflecting characteristics of AI need to be reviewed and stipulated. European Commission, *Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee and the Committee of the Regions - Artificial Intelligence for Europe*, SWD (2018) 137 final, (Apr. 25, 2018), Section 3.3.

[8] Krishnan [4], Stahl [5].

In this regard, David Gunkel argues that while AI cannot, nor should it, be regarded as a natural person, whether AI can and should be regarded as legal persons is a more complicated question.[9] For instance, he points out the battlefield practice that explosive ordinance disposal (hereinafter EOD) robots are given names, as well as battlefield promotions. In other words, as far as explosives detection and demolition go, EOD robots have the same kind of social relationship with other human teammates on battlefield, just like other human soldiers. Evidently, it is just an anecdotal evidence, but it certainly indicates that there may be instances where highly advanced AI can assume a different status than a mere machine or system, perhaps something in-between humans and machines. In this regard, one could argue that over time it is practical to extend personality or personhood to AI of certain specification so that it can be accommodated to our social institutions properly and thus be subject to social regulation effectively.[10]

This is not just a remote possibility. According to industry watchers, it is expected that AI with adult human intelligence will appear within the next 50 years. In the interim, the global community will then see AI with intermediary levels of intelligence, which alone will be enough to trigger these philosophical questions.[11] Imposing responsibility on the owner of AI may offer a simple solution, but will be problematic when the owner does not have full knowledge of or control over the AI.[12] This scheme of absolute responsibility or vicarious responsibility will offer one solution, but it has its own consequences: (i) technological evolution will likely be hampered as attempts will be made to insert the human evaluation and review at the last stage no matter what, which is a safer choice but the one removed from the reality of and need for fast technological innovation at the moment; and (ii) the scheme might turn into a no-fault system where no person may feel guilty actually and compensation for a victim is made through various insurance programs. Quite possibly, in effect it may become nobody's responsibility and there is just a payor of compensation who has paid premiums for his or her AI insurance package. It is questionable whether such a no-fault system is sustainable or caters to the need of the society in general. Granting a personality or personhood to AI may offer another solution. As technology allows AI to be more autonomous and independent, it has become increasingly difficult to simply regard AI as a mere tool or instrument.[13] Complications are being added such that it is difficult to trace and determine who stands at one segment of the long process of AI development, deployment and operation of AI, and thus be held accountable, individually and/or collectively.[14] Such being the case, according personality or personhood to AI at some point may help address the responsibility problem. For example, compensations can be paid out from the operating profits of the AI machines or systems, together with or independent of

[9] Gunkel [6].

[10] Ibid.

[11] Brown [7].

[12] Ibid., p. 210; Chesterman [8].

[13] *Supra* note 169, p. 213.

[14] Papakonstantinou and Hert [9].

compensation from the human manufactures and operators. This scheme may also help clarify the exact extent and contours of the responsibility of the human players implicated in the long chain. Note that this is not to treat AI as a human as seen in scientific fiction movies; rather it is to accord special status on AI as it plays a unique set of roles in the society. Entities other than natural persons are sometimes accorded personality or personhood. Consider corporations that are considered legal persons for various purposes, which indicates a possibility of similar creation or recognition of personality or personhood.[15]

Brown (2021) argues that the personality or personhood of AI should be recognized with respect to property ownership:

> Nonhumans like AI must likewise undergo this socially driven transition from property to personality. Yet, that transition will likely only happen if society and the law sees AI as having the ability to possess, control, and eventually own property, rather than an object that humans control and own. This is not the same as requiring AI to have a will, or any other human traits that allows it to be deemed a strong AI. Instead, the question is limited to whether AI can own property with accompanying rights and duties, keeping in mind that property and personality are human inventions that are subject to evolving cultural and social norms.[16]

This explanation is also centered on practical consideration. It is not to elevate to AI to the level of human beings, or to regard AI something similar to humans. Instead, it is simply a suggestion that AI can become a subject of rights and obligations, as a recognized player in the society. Property ownership and information protection obligations can be assigned to AI in the future, as if companies can own property and assume obligations even if they are not human beings. In other words, as the collector, manager and hub of data, an AI can be said to own the data in some instances, and its ownership can be recognized by granting institutional personality or personhood to AI.[17]

However, some experts bring our attention to the consequential difficulties and challenges once the personality or personhood is recognized for certain AI. Chesterman (2020) argues that when we grant AI property ownership, including the accumulation of wealth, the inquiry simply does not end there, and further complications will demand our attention. Consider, for instance, how to deal with taxes and how to hold AI responsible when there is mismanagement of its property.[18] Property management can be more sensitive here than conventional situations. AI may hold a vast amount of personal information and confidential data whose breach may well lead to a possibly enormous amount of damage for many people. Furthermore, he posits that more careful deliberation should take place before granting AI personality or personhood, as the development of almost-human like AI poses many novel philosophical questions. His point hits the nail right on the head. Personality brings with it a plethora of rights and obligations, many of which are only suitable to

[15] Chesterman [10].

[16] *Supra* note 169, p. 224.

[17] Ibid., p. 234.

[18] *Supra* note 173.

human beings. This is yet another reason to explore the possibility of recognizing and according personality only to a limited extent and only for specific issues that pertain to AI function nicely. Banteka (2021) also argues that AI poses a different question from corporations (a juridical person), because it is a wholly new creation. But a counter-argument can be made that everything has its first, and that absence of a precedent does not necessarily foreclose the possibility of a new approach. She contends that granting AI a personality or personhood may not be easy given the criteria that courts have used to establish conventional personality or personhood for these entities, an argument which may also be challenges because, if anything, the gap with the conventional jurisprudence may only underscore the need to develop new jurisprudence and norms at this juncture.[19]

2 Ethical Questions

In addition to philosophical questions regarding the special status of human beings and murky status of AI in the future, AI also raises critical ethical questions. Facing AI, one may question whether it is ethically acceptable to condone the emergence of an entity that resembles a human being that thinks and acts like a human being. This whole thing is the creation and invention of us humans. Some may criticize that developing AI of this sort amounts to playing or mimicking the role of the omnipotent being. The question therefore is, can or may we create something that acts and behaves like human beings?

Another ethical question relates to an AI that makes a final decision and yet it does not possess 'human consciousness'? Is it indeed ethical for us to allow AI, or for AI to make a decision on sensitive issues involving people's fate and lives even if the AI lack consciousness, conscience or emotions that humans possess? Is it ethical to rely on data and algorithms, minus such human elements, to dispose a matter in a particular case? Once a final decision so made, it is unclear to what extent AI's actions should and could be judged from an ethical perspective.

It is undeniable and in fact inevitable that the advent of AI brings about new ethical conundrums. It is unclear at this point whether and to what extent the consequences of AI's decisions should be a subject of ethical judgement. If the actions of AI are completely exempt from ethical judgements, some people and corporations may be tempted to blame the AI and its technology to alleviate their own faults and avoid responsibilities. Likewise, some governments may be tempted to abuse or misuse AI (or let AI operate inappropriately even if they are aware of the possibility) to avoid state responsibility, leading to a situation in which an action takes place which is ethically condemnable, but legally unpunishable. If technology further advances in the future to the point where it is viewed as tantamount to or corresponding to a 'natural person' in material respects, permitting natural

[19] Banteka [11].

persons and legal persons to hide behind AI from various factual and legal consequences would certainly raise an important ethical question. Stated differently, AI can be invoked as a pretext to dodge otherwise existing legal liability because of the existing and possibly widening gap between the conventional analogue norms and the new digital landscape. Victims may go uncompensated. Wrongdoers may go unpenalized. Condoning this situation to further prosper creates an ethical problem. Under the current system, therefore, it is probable that states may view AI as a means of dodging or reducing otherwise existing obligations under respective domestic jurisdictions and international law.[20] The situation may be called "AI-shield" in that AI is invoked as an impenetrable shield to a challenge, or "AI-haven" in that companies and states attempt to hide in the AI universe as in tax haven. Escapes and avoidance of this sort are in fact similar to various contemporary cases where some countries try to avoid legal responsibility or sometimes ethical responsibility by utilizing non-governmental agencies to undertake governmental objectives, thereby denying their relevance to the states and governments. If similar situations occur in the future in the AI sector, disputes between countries and between companies will inevitably increase in the application of domestic law and international law.

As AI is rippling through every corner of our lives, ethical challenges will continue to loom over the horizon. While ethical issues are cross-cutting in nature and permeating all sectors where AI is applied and utilized, the military sector stands out as it is the first adopter of AI technology and machine and as the technology and machine engage in lethal activities. Recent incidents involving AI in the military sector offer some shadow of the associated ethical question.

AI is currently making fast inroads into the defense industry of many countries. Autonomous weapons have become a reality and newer versions are hitting the market. Taken to its technical extreme, it will not be too long before we see fully autonomous machines of war with self-analysis and self-decision capabilities storm future battlefields. The recent military conflict in Ukraine in March 2022 caused by the Russian invasion provided the most recent example of semi-autonomous weapons in combat (e.g., aerial drones conducting surveillance and attacking tanks). The pace of the adoption of new technologies and machines and its significant consequences, however, raise a voice of concern. Unlike other areas of AI application, the military's use of it belongs to a fundamentally different league because it is lethal and deadly: it is feared that these 'deploy and forget' types of machines can identify a target and initiate an attack on their own. This is a more advanced version than a "fire and forget" version where soldiers remain in control until almost the last minute - the moment of firing. In the version of 'deploy and forget', on the other hand, deployment is the final point of human intervention. The killer robot is then released on the field to operate on its own. Existing regulations for combat and the law of war may not cover the entire spectrum of these new non-human combatants. Its killing and injuring of enemy combatants prompt ethical considerations. It is not a human-to-human combat, but a machine-to-human combat. It is one thing that soldiers are killed in battlefields all

[20] *See supra* note 142, pp. 15–6.

the time, but quite another that they are killed by pre-programmed machines with no physical or emotional reaction.

One recent example in South Korea illustrates the ethical issue. Constant military tension with North Korea has made South Korea one of the active users of killer robots along the borderline. It is this sentiment of ethical questions, however, that underlies foreign academics' concern over the Korea Advanced Institute of Science and Technology (hereinafter KAIST)'s recent plan to participate in a joint research program with a private corporation to possibly develop autonomous weapons. The foreign concern targeted the development of autonomous weapons "lacking meaningful human control." To paraphrase: in operating AI machines human beings must have a final say in the decision-making process, so goes the concern. Faced with a foreign scholars' threat to boycott cooperative activities, KAIST had to officially deny any intention of conducting such research.[21]

As for another example, it has been several years since South Korea deployed sentry robots/equipment along the Demilitarized Zone (hereinafter DMZ), a military demarcation line with North Korea. In response to similar concerns from foreign observers triggered by ethical consideration, assurances have been given over the years that the machines in place remain under soldiers' control at all times. These incidents show global activist groups' continued monitoring of the development and deployment of LAWS in many countries. They also suggest that these activities are caused by ethical questions.

Consider yet another context. Some of Google's employees sent a letter in early April 2018 to its management taking issue with the company's planned cooperation with the US DOD to enhance video imagery that can facilitate the targeting process of drones.[22] The concern expressed by the employees was largely similar, prompted by an ethical reaction. Both Google and the Pentagon have stated that the research would not lead to autonomous weapons that initiate an attack without human control. Here again, the core question was absence or existence of 'human control.'

This question is also related to who should be held responsible for consequences from activities of autonomous lethal machines. By having human beings in the loop, it can be ensured that final responsibility still lies with the person in charge - a strong deterrence to any misuse or abuse of new machines. Apparently it is viewed as more ethical and more prone to provide a simple parameter for responsibility.

Needless to say, AI is taking over dangerous and difficult tasks from us. It is and will be replacing human beings in many dangerous areas and activities. What is not clear yet is who and how to supervise these autonomous machines. Recent incidents represent a lingering ethical concern: regardless of the technological development or feasibility, it is critical to put AI under the control of human beings and to ensure that a final decision is only made by human beings. This concern may also apply

[21] See Andrea Shalal. Researchers to boycott South Korean University Over AI Weapons Work. Reuters (Apr. 4, 2018), available at https://www.reuters.com/article/us-tech-korea-boycott/researchers-toboycott-south-korean-university-over-ai-weapons-work-idUSKCN1HB392.

[22] See Scott Shane and Daisuke Wakabayashi. 'The Business of War': Google Employees Protest Work for the Pentagon. The New York Times (Apr. 4, 2018), available at https://www.nytimes.com/2018/04/04/technology/google-letter-ceo-pentagon-project.html.

to other areas where AI confronts core societal values and thus weighs-up critical ethical questions. In other words, these ethical strands are of the view that there are tasks where human oversight can never be dispensed with. Handing it over to AI is in and of itself unethical, according to the logic. It is not clear at this stage how strong this ethical strand is and how long it will stay. What is clear at least, however, is that a critical ethical question hovers around AI and its independent thinking and decision making.

The rise of AI and the advent of autonomous machines throw up controversial ethical debates. AI in military applications brings these questions to the fore. As a matter of fact, a UN conference on this very topic was held in December 2021 in Geneva, Switzerland under the auspices of the 1980 *United Nations Convention on Certain Conventional Weapons*. This convention was adopted in 1980 to seek to prohibit and restrict certain conventional weapons that cause overly injurious suffering or initiate indiscriminate killing. Perhaps the discussions there will lead to an adoption of legal norms that can guide the development and deployment of these autonomous lethal weapons in the near future. The lingering ethical questions will then be able to be addressed appropriately.

In the event that current norms fail to regulate new AI situations properly and ethical questions keep up haunting the international community as a result in various areas, it will be necessary to stipulate new norms to address these ethical problems accordingly. One of the most urgent items to be handled in the process will be to define the nature of AI, which is, in fact, under serious consideration in such disciplines as legal philosophy and criminal law.

3 Technological Questions

Another line of questions that we need to address is technological ones. In the AI discourse, technological development and innovation have usually been regarded as something commendable and virtuous. Is is really so, however? Should we approve and facilitate all kinds of technological development? In short, it is worth mentioning that sometimes it is necessary to put some restraints on the development and innovation of certain types of technologies. There is an outer parameter for technology adopted by AI.

Firstly, development of a new technology and introduction of a new product using the technology sometimes may not be a sufficient condition for the application and utilization of the AI. Often times, an AI, even if fully operational on its own, may not deliver an intended objective because the operation is dependent on successful operation of other related infrastructure and sectors. Thus, an assessment of technological development should not be confined to the technological sector at issue. A broader survey of other areas and related technologies surrounding the AI technology at issue is in order. For example, although automated vehicles have been ready for commercial use for some time, they have not been fully introduced into our roads yet due to the lack of adequate infrastructure (equipment, signals and sensors on

the streets that assist driverless cars) and regulatory frameworks (new safe driving standards). Before encouraging a new technology to reach its full potential, a more cautionary approach is needed for the purpose of checking other infrastructure and systems of the society, which support the newly developed AI heading to the market. In other words, the society need to scrutinize where technological development for a particular AI machine or system is headed, whether other sectors need to be co-existent for the AI to undertake its intended function. and, if so, what sectors, before blindly allowing the technology at issue to proceed at full speed.

Another technological concern is directed at personal information. As the value of data increases in the new digital economy, personal information has become all the more important for business purposes. Consequently, protection of data is attracting keen attention from interest groups as the privacy of data source individuals is directly implicated. In order to balance the application and utilization of AI and data protection, indeed we need to reconsider the definition of what constitutes privacy and personal information.[23] What privacy is and what personal information means are obviously the starting point in exploring a new regulatory framework. What is equally important in this regard is to place an outer limit in collecting, processing and utilizing data and personal information. Setting forth such an outer limit relates to technological stipulation. To attain the goal, new technologies should not be automatically accepted simply because of its capabilities, but evaluated for their implications on data and privacy breach. Development of yet other technologies should be encouraged in order to protect data and privacy adequately. After all, at the end of the day technological experiments and discussions complement regulatory discussions and reflect the results of the regulatory discussions. What can be done or cannot be done in terms of technology is to be determined by scientists and manufacturers. What may be done or may not be done in the new environment of AI is to be determined by those who contemplate the overall architecture of the new regulatory framework. South Korea's recent experience offers an example in this respect. After several years of preparation, South Korea introduced a new legislative scheme in January 2020 to regulate data collection and protection, including, most importantly, personal information. The first legislation to protect data and personal information came as early as 2011, with the legislation of the *Personal Information Protection Act*, but it soon turned out that the law is not capable of keeping up with the fast changing landscape of the digital society. The 2020 enactment was the effort to respond to this demand. Under the new scheme, personal information can be gathered and processed, within the confinement of regulatory guidelines, for commercial purposes. As a counterweight, harsher penalties are to be imposed for violations. As for South Korea, this was a significant stride toward the digital society and the AI-era.

Critically, the new legislative scheme of South Korea purports to define the term "personal information" in a detailed manner. It now features a three-layered definition of the term: firstly, personal information; then, information organized per certain

[23] See Nohyoung Park. A Korean Approach to Data Localization. Carnegie Endowment for International Peace (Aug. 17, 2021), available at https://carnegieendowment.org/2021/08/17/korean-approach-to-data-localization-pub-85165.

categories and groups with individual identification deleted; thirdly, fully anonymous information. The third group is not a major concern as fully anonymous information is protected on its own accord. The first one continues to be subject to full protection unless express consent of the source of information is given. In terms of protecting personal information, therefore, it is covered as well. The question lies with the second group. As for this, identities are still deleted. But when information falling under dozens, if not several, categories are collected, processed and juxtaposed, identity silhouettes may gradually appear, according to critics. Full identity is not disclosed but a sketchy of an identity is obtainable. This information possesses significant commercial value. Thus, the remaining question about the new scheme is how to handle the information in the second group. For now, the government has vowed to impose harsh penalties on those who try to assemble the pieces of the puzzle from vast amount of information in this group. If the information retains such high commercial value, it seems to be a matter of time before a person or an entity purports to assemble the information falling under the second category. At a minimum, they will try to walk on a tight regulatory rope—making profits by testing the outer parameters of the legislative scheme. These attempts will be made and adjusted as necessary in accordance with technological development. What is obtainable with the current level of technological development may not be the same as what will be obtainable with the new level of technological development in the future. In other words, a legislation with seemingly clear guidelines may lose its clarity as technology advances. In order to address this problem, laws will have to be amended on a regular basis. Some laws may do this homework with ease, but others may not. In particular, important legislations in critical areas usually go through controversial legislative process, and going through similar process at a regular interval would be impractical. Thus, technological development, particularly fast one, poses a significant challenge to laws and regulations. This phenomenon is all the more vivid in AI sectors.

A similar problem also exists as regards the agency in charge of the personal information matters: Personal Information Protection Commission. Since its establishment through the 2011 enactment to oversee the protection of personal information, the commission has made significant contributions over the past decade. But its status has been marginalized due to the lack of a clear statutory authority as a guardian or watchdog of personal information. Its role has been upgraded and strengthened in 2020 legislation. Technological development also determines what an agency should or should not do, or what it does not have to do. What used to be a proper role for an agency 10 years ago may not be so at present.

It should be noted that good faith in technology or in technological development does not and cannot shield AI from criticisms and potential problems. AI developed with good faith may well turn out to be a machine or system that causes havoc throughout the society. Even in the instance where AI achieves its intended goals, statistical biases and tainted conclusion can still arise. For example, when the UK government introduced an algorithm to predict students' performance scores based on the past data, it was heavily criticized for favoring privileged students.[24]

[24] Gesser et al. [12].

Arguably, those criticisms were not based on any bad faith on the part of the technology or those who devised it. Thus, in addition to, or rather independent of, technical feasibility or mechanical attainability, each form of new technology should be assessed for its potential risks and benefits, especially if it is introduced to make automated decisions that stand to affect individuals.[25]

The guard against untamed and unregulated technological development might have us view AI's self-decision making unfavorably. Regardless of technical feasibility, the argument can go, human intervention is still required to supervise automated decision-making processes. One can also say that social acceptance of automated decisions may be contingent on the availability of appeal to a human decider, a situation where a human being can be the final decision maker.[26] This line of logic explains that issuance of a final decision for important matters is beyond a technological outer parameter. Even if it becomes doable with new technology, it should not be permitted. Discussions on the boundary of technology and how far technological development may proceed are fundamentally related to philosophical and ethical questions mentioned above. All three questions are closely intertwined and inter-connected. In this respect, when it comes to technological questions, we should also be wary of over-regulations. For example, In the early days of automobiles in late nineteenth century, a then UK law required the mandatory posting of a red-flag man in front of each vehicle for safety reasons. His job was to walk in front of the car and waive the red flag, to slow down his own car and to warn the pedestrians and horse carriages nearby. In times of new technologies, over-regulation of this sort is always a possibility. In discussing AI and related technological parameters, attention should be paid to possible attempts to introduce over-regulation out of unnecessary fear or politicized reaction.

Reasonable regulation for safety reasons should always be welcome and this is a task any responsible government must undertake. Too much concern and resulting overregulation, however, would only stifle innovation. The victims are companies and individuals with new ideas and technologies. Negative consequences will become all the more conspicuous in the AI era.

4 Legal Questions

Needless to say, AI raises important legal questions as well. The above-mentioned philosophical, ethical and technological questions all boil down to legal questions to the extent they are addressed and resolved in a practical sense. Discussions on philosophical, ethical and technological questions should lead to attendant legal discussions. There are a wide range of legal issues to be tackled in this regard. First, there is the crucial question of the legal personality of personhood of AI. Second,

[25] Ibid.

[26] Ibid.

there is a new phenomenon, caused by the digital economy, that blurs national boundaries and inter-state borders where AI-based goods and services travel without much restriction on the border. Third, there is an issue of widening gap between norm and reality, as AI requires modification of existing laws or creation of new ones, because they engender business practices and items that have not existed before.

Most importantly, one of the reasons why AI raises so many new challenges and questions at the moment is directly related to the recognition or non-recognition of its possible legal personhood. It is, needless to say, clear that AI is similar (in terms of thinking and decision making) and yet very different (in terms of feeling and emotion) from human beings. AI and human beings can be said to be similar because they both can make 'autonomous' decisions. In other words, they do not just follow given orders, which constitutes the basis for the argument that they are responsible for their actions. Granted, only certain AI, present and future, with threshold specifications for independent thinking and analytical power together with decision making capability may fall under this category. Most of technologies and machines that carry the name of AI at present, in fact, do not meet this threshold. In any event, even these certain specific AI technologies and machines, however, are fundamentally different from humans because AI is not yet considered a completely independent entity from a legal perspective. For instance, we do not think of AI as a member of our society that needs to be punished or educated for their actions. Nor does it hold rights and obligations as with other legal personalities. As it stands now, AI is something in-between, and the more technological advancement is made, the closer it gets to unique traits of human beings (namely, thinking and decision making).

As a consequence, at this juncture it may require a new concept of legal personality or personhood. Essentially, the question of legal personality or personhood involves whether AI can hold and should otherwise be subject to 'legal rights and duties'.[27] Since AI is such a novel concept and a recent phenomenon, a possible definition of AI personality or personhood within the confinement of the existing norms and regulation seems elusive. The questions and issues to be examined in this process are complex and multidimensional, but answers to these questions and solutions to these issues must be found for the global community to formulate a new legal scheme to regulate AI in the future.[28]

The question whether AI's distinct legal personality or personhood should be granted or not may elicit different answers depending on different contexts. There is the ultimate, value context, in which one should fathom the fundamental differences and similarities between AI and humans, and ensuing legal liabilities. On the other hand, there is also the low-level, practical context, in which one should examine business activities and commercial transactions.[29] For instance, one may not readily agree that AI should be given the same protection as a particular group of people needing protection (such as children), but one may be more willing to acknowledging AI's legal personality or personhood in commercial transactions.

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[27] *See* Solum [13].

[28] Ibid., pp. 1234–40.

[29] *See* Kurki [14].

The ultimate-value context is difficult to address, because it challenges our conventional concept of legal personhood based on our understanding of humanity.[30] The famous Turing test aims to see whether a computer can imitate human behavior insofar as a human being interacting with it does not recognize that it is a computer.[31] This may be an extreme end of synchronization. In the earlier process of AI development, many wondered just how closely a human mind can be imitated by AI.[32] However, for the purpose of discussing legal questions at this juncture, we probably do not have to engage in or seek such a high level of synchronization between AI and humans. Instead, we can pursue the possibility of introducing context-specific legal personalities or personhoods as a useful tool for regulation.[33]

The legal question that immediately follows the question of personality or personhood is about legal liability: how to attribute and distribute responsibility for the consequences of AI's activities. The legal debates regarding legal responsibilities for the introduction, manipulation, and operation of AI are still in the initial stage, and they have to be carried on probably for a long period of time in the future. As all or part of the decision-making is made by or entrusted a 'subject' other than humans, a question is raised regarding who (or what) will, and to what extent, take legal responsibilities for the results. Furthermore, if some decisions are made by humans while others by AI, the combination of nature of decision makers would make any analysis complex and challenging. Putting aside the philosophical question of how to define 'human' or 'humanity' as a decision-making subject, we need to focus on the legal aspect of liability and responsibility in specific circumstances. In fact, acknowledging AI's legal personality or personhood can be useful sometimes, but not always. For example, if the society ever creates a legal framework to ensure that AI machines or systems compensate for any damage it has caused, it might be useful to attribute the legal liability to AI directly.[34] Compensation can be drawn from the account holding profits accumulated from the business transactions that the AI has consummated or brokered. Human participation may be required at one point or for certain segments of the operation, the AI can still be regarded as holding a personality or personhood from the legal perspective. However, in the case of criminal liability, the issue becomes more complicated. AI cannot be incarcerated, nor can one expect any educative function from punishments.[35]

Furthermore, there is another practical problem with respect to AI's legal personality or personhood: there is a possibility of abuse of the new recognition. By way of example, people and corporations may be tempted to use AI robots as 'a legal liability shield'.[36] Or at least the AI may be invoked as a mitigating factor for the

[30] *Supra* note 151, p. 1286.

[31] *See* Bolter [15].

[32] Ibid., p. 17.

[33] Courts, including the International Court of Justice, have acknowledged divisibility of legal personhood. *See* Bryson et al. [16].

[34] *Supra note* 151, p. 1245.

[35] *Supra note* 151, pp. 1247–8.

[36] *Supra note* 156, pp. 285–287.

human responsibility. Another major difference between AI and other non-human legal entities such as corporations and international organizations is that the former may not have any real human actor to be held accountable for its actions while the latter still maintains human element because after all it is officials who operate corporations and run international organizations.[37]

As AI will likely be applied and utilized across the whole areas of our society in the future, questions about the fundamental legal personality or personhood of AI will continue to be tested, debated and calibrated. Consequently, in various academic fields, efforts have already been put forth in an effort to find answers to these legal questions, and these efforts will be accelerated and deepened with the fast advancement and spread of AI.

In addition, the advent of digital age has complicated conventional rules of national jurisdiction. National jurisdiction under domestic law and international law is basically drawn along the physical boundaries of a state, and looks at who or what is located inside the said boundaries. The 'borderless' nature of cyber activities has made this traditional jurisdictional rule outdated and detached from reality. Cyber activities virtually have no relevance to national boundaries. The beauty of digital economy is that it is freed from the shackle of physical boundaries. Thus, there is an obvious gap between the conventional rule and the reality in the market. But the rule is the rule, and governments and states have to exercise their regulatory authority within the confinement of the established jurisprudence of jurisdictional rule; they have find a territorial hook one way or another to exercise jurisdiction for cyber activities, which in fact covers almost the entire spectrum of digital economy.

One such hook is the location of a computer server. It is indeed a physical item located in one state, so it fits with the conventional rules. Thus, now the location of a server dictates who exercise the national jurisdiction, including taxation authority. ICT companies pay their corporate taxes where their servers are located but concerns are being raised that this traditional rule creates loopholes for these internet-based companies. However, suggestions for new rules are still on the drawing board, even if an important stride has been made at OECD in 2021. And at any rate, the consensus reached at OECD is just about taxation, a tiny portion of the entire iceberg of the digital economy. Until a new rule is introduced and established commensurate with the major tenets of the digital economy, conventional rules still remain the standard.

The last legal point that needs to be addressed is that the fast advent of AI requires updating and upgrading existing laws and regulations. For example, the digital economy, which is being expanded by the use of AI, engenders new challenges and problems, and as such it demands a new set of laws and regulations. One example is that while the international trade regime adopts a dichotomous approach of separating goods and services trade, and applying different sets of rules for each, digital trade items cannot be so easily categorized. They fall under both, or they fall on neither. Some of them are completely new innovation that does not fit the existing categories. New laws and regulations both in domestic and international planes should be explored and introduced as soon as possible. Even if new norms

[37] Ibid., p. 288.

are formulated and introduced, critical legal challenges still remain. First, the terms and scope of AI-related technologies and machines that are used in laws and regulations may well be unclear and even if they are at the time of introduction, they will lose their clarity sooner or later because of the fast development of technologies and societies that we live in. There will be this on-going tension between the provisions in laws and regulations and emergence of new technologies that stay outside the existing regulatory boxes of laws and regulations. Second, in this process new norms may remain incompatible with existing norms. This is inevitable because amendment of existing laws and regulations and creation of new ones are usually done in an incremental manner in specific sectors. This would mean a fragmented outcome when all sectors of the society are being evaluated horizontally at one point. How to address this fragmented approach is another critical legal question. Third, the global community will have to find a balance between data processing and data protection, a critical legal task that many governments face at the moment. Already, governments of various countries are taking diverse legislative measures at different speeds to cope with this very issue. The question is how to ensure global consensus in this area.

Legal norms related to AI are still at an early stage, and national laws and international laws have been relatively slow in developing new ones given the diverse views and technical complexities associated with these negotiations. This signifies that the legal framework that we have at present will soon have trouble keeping up with drastic changes in the digital society. AI-technologies and machines will further quicken and expedite the creation and arrival of these new problems. Given that more changes are expectedto come rapidly, the delay or mismatch could spark international disputes. In turn, these disputes could lead to a weakened and incohesive global legal regime, which could in turn slow down the process of building global consensus, initiating a vicious cycle in this area.

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Chapter 5
AI's Implication for International Law



By any measure, there are diverse ripple effects flowing from the fast advent of AI. To the extent that AI affects and changes the entire spectrum and echelon of human society and behavior, it should also bring significant changes to the international community in general and thus to the study and jurisprudence of international law as well. After all, international law is a discipline that examines and studies the international community from a normative perspective. In addition, these significant changes relate to both the present and future of international law, given the fact that this particular area is currently undergoing a drastic innovation. In other words, they affect how current international law is interpreted and applied, and how future norms of international law will be formulated and introduced.

Consider the fierce confrontation between the US and China intensifying since 20 17. The confrontation is being observed in diverse sectors of international law, one of which involves technological sectors for the now booming digital economy. Intensifying sanctions of Huawei, one of the major Chinese ICT companies, is apparently an effort by the US to contain China's technological capability in the new global community. The two countries are also competing in the semiconductor industry which manufactures memory and non-memory chips for ICT companies and other goods and services industries using IoT. Semiconductors have become an essential industrial component for the dominance of the new digital economy relying on AI. The US government has been exploring a web of new regulation to deter China's easy access to the global semiconductor technologies and manufacturing facilities. China is also responding in kind. Notably, this very effort is being exerted in accordance with or by resorting to various norms of international law, including the transboundary exercise of domestic regulatory authority, provisions in trade agreements, provisions in intellectual property protection conventions, and provisions in investment agreements, etc. These norms of international law have also mirror or sister provisions in respective domestic laws and regulations. This effort can be called an (attempted) application of present norms of international law, whether it is determined to be consistent or inconsistent with those norms at the end of the day. At least, legal discourse is being constructed and communicated robustly. In a sense, the US-China stand-off

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is about taking hegemony in a new digital society, and specifically their confrontation is revolving around diverse legal issues. Not surprisingly, this phenomenon is producing multiple disputes on many fronts. Some of them relate to present norms of international law, and others concern how to formulate new norms going forward.

The advent of the digital society and the penetration of AI have ushered in new modes of daily lives, social activities and business transactions at all corners of the society. These new modes, activities and transactions are not fully addressed or covered by existing legal norms, whether domestic or international. Unless and until new norms are introduced, mismatches and inconsistencies with existing legal norms keep springing up in the era of AI. International law is also struggling to deal with the new realities in the global community, as its basic tenet is premised upon the conventional understanding of the global community. By way of example, the exiting framework of distinguishing goods and services in international trade norms is now ill-suited. The dichotomous approach regarding goods and services has encountered fundamental limitations when it comes to digital trade: the existing binary approach to viewing goods and services as different targets or items to be traded is no longer consistent with the reality in the context of emerging digital trade or e-commerce, let alone AI-empowered new items and systems. If anything, goods and services are traded and consumed in an integrated manner as one unified item of trade. Goods are providing services, and services are sold through goods. At the center of this new transaction lies data and information whose commercial value is materialized through AI. In the new circumstances, the framework for the existing dichotomous approach is now no longer valid.[1] There is a sort of gap between norms and reality. The gap is widening with each passing day of technological innovation. Note that laws will continue to apply with legally binding force: all laws are effective and enforced until they are amended or repealed. As such, the current norms of international law will continue to apply to the respective areas of the global community even if they are detached from the current reality of the market and social activities. States and governments, entering the new national development phase of the Fourth Industrial Revolution, still have to abide by existing norms. Minor inconsistencies may be cured by elastic interpretation of the terms of the treaties at issue—through a process called 'evolutionary interpretation' which may still be found compatible with general principle of treaty interpretation under Article 31 of the *1969 Vienna Convention on the Law of Treaties* (hereinafter VCLT). Major inconsistency, however, is beyond the reach of interpretation, however elastic or flexible. Only a new treaty will cure it. A small inconsistency will grown into a major and then a bigger one as time passes by. When norms are outdated, thereby fostering legal vacuum, disputes are bound to increase. International trade is presenting this structural fissure in a colorful way.

[1] Some local trade agreements include electronic trade of traditional goods and trade of digital goods in the same chapter, viewing these as having the same attributes. However, such agreements will need adjustment in the context stated above. *See* Mark Wu, *Digital Trade-Related Provisions in Regional Trade Agreements: Existing Models and Lessons for the Multilateral Trade System*, RTA EXCHANGE, Geneva: International Centre for Trade and Sustainable Development (ICTSD) and the Inter-American Development Bank (IDB) (2017), p. 12.

Undoubtedly, efforts are being mobilized discuss and formulate new norms of international law for the future before it is too late. New norms will then be capable of regulating new technologies, machines and systems that we will encounter in the AI age. New norms for new technologies, machines or systems that did not exist before may be relatively easy to come by. All that is needed is consensus among states about how to devise new rules. If, on the other hand, new norms require major shift from existing norms, it may take a longer time and entail a more acute collision until the situation stabilizes. AI technologies and machines demand close cooperation between the private and public sector. In fact, governments of major states are now beefing up cooperation with ICT companies for the purpose of addressing novel problems of the digital society, and this cooperation sometimes includes government agencies' provision of support to these companies in the form of Public-Private Partnership (hereinafter PPP) projects. But such governmental support is in fact prohibited by the current trade norms - namely, the WTO Agreement on Subsidies and Countervailing Measures (hereinafter SCM Agreement). There is an obvious crevice between what is being demanded by national constituents and what is being required under present treaties. What is more, new norms are now being contemplated and introduced to further restrict governmental support for private industries. New norms include more sophisticated and strengthened subsidy norms[2] and SOE norms. [3] These new norms are being experimented and included in FTAs. In effect, these norms collectively demand severance of the relationship between governmental agencies and corporations. This means that governments are torn between the reality and the norm. The structural schism is most evidently being observed in the high-tech area.

Certainly, international trade is not the only area where fissures are being observed. For example, in line with the advent of new technology and automated decision-making robots, whether and how the international society regulates them has become a matter of importance. New norms are being proposed, examined and introduced through new bilateral and multilateral treaties in various secotrs. One example is the heated discussion in International Humanitarian Law (hereinafter IHL) at the moment where LAWS are being examined under the present norms and new norms to regulate LAWS are being introduced. Since AI's judgment and decision cannot be regarded as or equated with those of humans, any result from AI is merely a calculated outcome and does not rise to the level of human judgment; it follows that LAWS or any other automated weapons systems by themselves cannot be held accountable for liability, and any liability only belongs to a human operator.[4] Such a stance, however, does not necessarily dovetail with the reality that more and more AIs are in fact making judgments and decisions. Another example is the arrival of autonomous vehicles, which prompts the need to determine and adopt technical standards and technological specifications for the global community. Harmonization of standards and specifications are all the more important for safety reasons and for stable production sales. For the facilitation of data collection and processing, a

[2] *See* e.g., Horlick and Clarke [1], Howse [2], Li and Tu [3].

[3] *See* e.g., Wu [4], Qin [5].

[4] *See* Anderson and Waxman [6].

critical function of an autonomous vehicle, global harmonization is a must as data is collected and analyzed on a global scale. How to attain the needed harmonization is still an open question. More fundamentally, the nature of the cyber space and the bounds of national jurisdiction in the cyber space still remain ambiguous, and different states present different positions, which in effect means that states are bound to conflict with one another whenever cyber space becomes an issue, a situation which almost always arise as regards AI technologies and machines.[5]

1 Increase of Conflicts among States

Due to the short history of AI, the international community has yet to agree on a specific set of norms or regulations for AI related issues. More recently, there have been some notable efforts to create a set of global or regional standards for emerging technologies, but these efforts are largely fragmented and weak. As a consequence, at the moment each state has been left on its own, and the AI sector is frequently seeing conflicting laws and regulations coming from different states, blocs and regions.

Recognizing the high (and hidden) potential of the newest technologies, governments of many states have established national strategies to support and develop AI, and to apply and utilize it to guard and maximize their respective national interests. The problem is that each state has a different set of policy objectives and goals. More importantly, respective states also have different ideas and views of how to approach various issues arising from the AI era. As each country strives to take an upper hand in the fierce competition of the AI development, securing qualified human resources, allocating sufficient financial resources, establishing national digital infrastructure, and setting international norms and standards have become priority policy agenda for the government.

1.1 Lack of Applicable Norms

Compared to the rapid development of new technologies, the regulatory and legal frameworks to adequately accommodate newest inventions and innovations have been weak and feeble. The international community does not have sufficient and up-to-date norms to regulate increasing difficulties arising from moral, ethical and legal

[5] *See* Benvenisti [7]. In 2016, the General Assembly organized the fifth working group of governmental experts, initiating the discussion on information and telecommunications in the context of international security (2016–2017 UN Group of Governmental Experts on Developments in the Field of Information and Telecommunications in the Context of International Security'; hereinafter 'GGE'). The GGE is to discuss how to apply international laws to the field of information and telecommunications in each country, but participants failed to come to a basic agreement on the application of international laws to cyber space until June 2017 when it finally ended with the 4th meeting.

issues of AI. Nor is there an international organization or entity with the authority to adopt and enforce these norms. While there are efforts by individual states to explore and adopt norms and standards regarding AI, these respective and individual national efforts have not bloomed to international consensus on core issues associated with AI application and utilization.

At the moment, each state is left on its own in regulating AI and, most notably, the use of personal information and data, one of the controversial topics in the digital society. One of the important issues being raised with regard to such fragmented and splintered approaches is that individual legislative actions by respective states might be in violation of other applicable norms of international law. In particular, new laws and schemes on AI and digital society are designed to apply to foreign enterprises, mostly ICT companies, even if their territorial linkage to the legislative state in a conventional sense is either tenuous or absent. Sometimes, actions of foreign companies conducted outside the territory of the legislating state is thus brought under the regulatory authority of the state, triggering a negative reaction from the states of companies' nationalities. This may be called a digitized version of "extraterritorial application of domestic law."

Besides, to the extent that foreign business owners or operators are to be regulated by the new legislations and schemes, there is a constant risk of violation of applicable trade agreements concerning goods and services trade as well as international investment agreements (hereinafter IIAs) concerning investor protection. Furthermore, now that trade disputes and investment disputes have been on the rise recently, securing conformity to trade and investment agreements as fully as possible has become of great importance to many countries in reality. Note that these agreements are equipped with binding dispute settlement proceedings with the possibility of retaliation (trade agreements) and compensation (investment agreements).

Recognizing the lack or absence of adequate norms governing the AI-related matters, the EU is at the forefront of attempting to explore the emergence of global norms. As part of the European strategy on AI, the High-Level Expert Group on AI (hereinafter AI HLEG) recently published *The Ethics Guidelines for Trustworthy Artificial Intelligence*.[6] The guidelines call for seven key requirements that AI systems must follow: (1) human agency and oversight, (2) technical robustness and safety, (3) privacy and data governance, (4) transparency, (5) diversity, non-discrimination and fairness, (6) societal and environmental well-being, and (7) accountability. While the guidelines in and of themselves are not legally binding nor do they create specific legal obligations, they can be regarded as part of the EU's on-going attempt to engender an international consensus for "human-centric AI".[7] The guidelines provide the first step of a three-step approach to having a responsible development of AI, and once the first step is completed, the second and third steps

[6] Full guidelines available at https://ec.europa.eu/futurium/en/ai-alliance-consultation#:~:text=The%20Ethics%20Guidelines%20for%20Trustworthy,strategy%20announced%20earlier%20that%20year.

[7] *See* European Commission, *Artificial Intelligence: Commission Takes Forward Its Work on Ethics Guidelines,* Press Release (Apr. 8, 2019), available at https://ec.europa.eu/commission/presscorner/detail/en/IP_19_1893.

will in turn follow. Subsequently, the European Commission will launch a large-scale pilot project with partner institutions in which to build international consensus, through dialogue platforms such as the G-7 and G-20.

Meanwhile, the US government also purports to establish reliable standards for AI technologies. As required by Executive Order No. 13,859, the White House announced a draft *Memorandum on the Guidance for Regulation of Artificial Intelligence Applications* for all federal agencies in January 2020.[8] The guidance urges consideration and introduction of reliable and transparent regulation to promote further innovation and growth in the AI sector. It appears that the White House is advocating for a minimum level of regulation in order to keep encouraging innovation and growth of AI technologies and machines.[9] Apparently, the typical *laissez-faire* approach of the US still lives on in the AI regulation. As a matter of fact, in comparison to other members of the Organisation for Economic Co-operation and Development (hereinafter OECD) and the member states of the EU, the US has fewer laws and regulations on AI. In order to secure leadership in the new field, the Office of Science and Technology Policy (hereinafter OSTP), part of the Executive Office of the US President, has urged other nations to follow the policy stance of the US, as a way to promote innovation in AI.[10]

At the same time, the US government is vigilant against technology leaks in the AI sector. It is thus imposing certain restrictions on the international stage in order to prevent technology leaks. For instance, the US imposed an export ban on selling software that uses AI to analyze satellite imagery to foreign companies without a license from the federal government.[11] The main reason is the US concern about technology leaks to US adversaries, notably China. In January 2020, the then Trump administration banned exports of AI software, which was seen as an attempt to safeguard sensitive US technologies from falling in the hands of foreign rivals.[12] Furthermore, the US government has been working diligently for the past several years to pressure other countries to avoid the use of Chinese technology in constructing and managing their digital infrastrcture. For example, the Trump administration was keen to seek a global ban of Huawei's 5G technology.[13]

The lack of common and/or organized global norms complicates international cooperation and coordination. This problem was most vividly presented in the recent management of the global pandemic, COIVD-19, where the critical lack of global cooperation and coordination has slowed down the joint efforts to fight the virus. It has turned out that mobilizing of the international efforts to fight a pandemic

[8] Full draft of memorandum available at https://www.whitehouse.gov/wp-content/uploads/2020/01/Draft-OMB-Memo-on-Regulation-of-AI-1-7-19.pdf.

[9] *See* Vincent [8].

[10] Ibid.

[11] Ibid.

[12] *See U.S. Government Limits Exports of Artificial Intelligence Software,* REUTERS (Jan. 3, 2020), available at https://www.reuters.com/article/us-usa-artificial-intelligence/us-government-limits-exports-of-artificial-intelligence-software-idUSKBN1Z21PT.

[13] *See* Feng and Cheng [9].

of this magnitude requires immediate and comprehensive updates and adjustments in various norms of international law.[14] In the absence of the updates and adjustment, current norms of international law stand insufficient and inadequate to address the pandemic situation where a wide range of different issues flood the regulatory toolbox. Selecting one option over the other may well implicate treaties and agreements in diverse sectors. For instance, under the *1963 Vienna Convention of Consular Relations,* it is unclear whether a state could detain a foreigner and force quarantine even as a preventative measure against a global pandemic.[15] Detentions at the border, however, have become a new normal in the COVID-19 pandemic arguably without a careful assessment of legal issues. Similarly, the legality of restricting regular flights or landings by commercial airliners, another restrictive measure observed in the initial stage of the pandemic, may be questioned under the *Convention on International Civil Aviation* of 1944, which regulates international flights of commercial airlines.[16] Careful analyses of legal issues were not evident in this regard either. As a matter of fact, counter-pandemic restrictive measures may still stand as an exception (either security exception, general exception, or necessity defense) although violation of treaties is otherwise found. Even for the exceptions discussion, however, careful legal analyses are still a prerequisite. With the advent and assistant of AI, future coordination and cooperation among states and among international organizations could be facilitated through easier monitoring and sharing of information on a global scale. As much as the problem lies in the unique structural trait of international law (namely, fragmentation), an easy fix will not be within reach. It is time that the global community realized the significance of the problem of fragmentation and explored creative ways to overcome it. Having a hierarchy provision in relevant treaties in case of conflict may offer a solution. Mandating harmonized interpretation through joint consultation among treaty parties and among relevant international organizations is another option. Other alternatives could be contemplated over time as well. Again, an ultimate solution can only be achieved through a centralized treaty making process, but it is not obtainable in the reasonable future. In the interim, therefore, various mitigating options should be explored and implemented to cope with pressing global matters such as pandemics. In that respect, fast information collection and processing, accurate information analyses and projections, and comprehensive information sharing and pooling, all to be made possible through AI, may help alleviate the fragmentation problem. Needless to say, what is critical in this endeavor is how to safeguard privacy of individual subjects. Domestic legislative experiments to protect privacy while assuring necessary utilization of personal information (with an anonymization process, for instance) could provide a useful reference in this respect.

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[14] Ibid.

[15] Ibid.

[16] "The convention sets forth a "state of national emergency" exception (Article 89), so one might justify current restrictions from this emergency angle. Again, the question is transparency and reasonableness. Some restrictions are claimed to be unexpected and disproportionate." Ibid.

The COVID-19 pandemic and the global response to it for the past two and half years bring these structural problems to the fore. At present, the only international convention on the global health issue is the WHO's 2005 International Health Regulations (hereinafter IHR 2005), with 193 contracting parties.[17] It is a convention under international law, and it is the only international legal vehicle for global health challenges including a pandemic. It has turned out, however, that the convention is not comprehensive enough or detailed enough to fight a pandemic like COVID-19. While the convention sets forth basic principles regarding cooperation among states, among international organizations and between states and international organizations,[18] it is silent on how to achieve it.[19] As mentioned above, cooperation across the board in the international community by nature encounters the chronic problem of fragmentation of international law. Without addressing this issue upfront, any reference to cooperation or cooperation may well become a hollow, wishful statement without teeth. Granted, there are certain provisions in the IHR 2005 recognizing this problem. For instance, to handle the fragmentation problem, the IHR 2005 asks that its provisions and those in other relevant treaties be "interpreted so as to be compatible" (Article 57).[20] Needless to say, this is a correct statement of principle, but it does not flesh out details on how to achieve the compatibility. What is transpiring globally during the COVID-19 pandemic shows an opposite phenomenon - splintered responses from different states and different international organizations. If anything, it may be necessary for the international community to conclude a new agreement embodying specific ways of cooperation at times of a global health crisis.

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[17] Ibid.

[18] Article 56 of IHR 2005, paras.1–2 states as follows:

Article 56 Settlement of disputes.

1. In the event of a dispute between two or more States Parties concerning the interpretation or application of these Regulations, the States Parties concerned shall seek in the first instance to settle the dispute through negotiation or any other peaceful means of their own choice, including good offices, mediation or conciliation. Failure to reach agreement shall not absolve the parties to the dispute from the responsibility of continuing to seek to resolve it.

2. In the event that the dispute is not settled by the means described under paragraph 1 of this Article, the States Parties concerned may agree to refer the dispute to the Director-General, who shall make every effort to settle it.

Full text available at https://www.who.int/csr/ihr/WHA58-en.pdf.

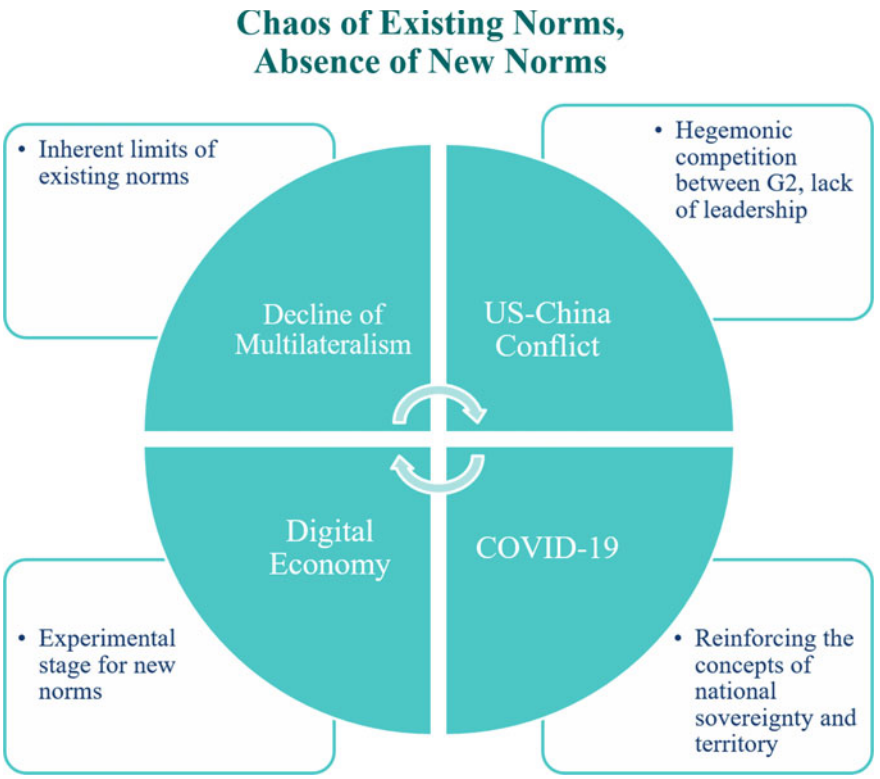
[19] *See supra* note 31.

[20] Ibid.

Article 2 of the IHR 2005 explains its "purpose and scope" to be "to prevent…the international spread of disease in ways that are…*restricted to public health risks*…" (emphasis added). The IHR 2005 thus stipulates upfront that it only addresses health issues and not any other issue. Article 56, para. 4 also states that it does not "impair the rights of States Parties under any international agreement…to resort to the dispute settlement mechanisms of other intergovernmental organizations…" Article 57 further adds that "the IHR and other relevant international agreements should be interpreted so as to be compatible" and that "…the IHR *shall not affect* the rights and obligations of any State Party deriving from other international agreements" (emphasis added). Article 3 also underscores states parties' adherence to UN Charter principles, and respect for human rights. In short, the IHR 2005 pronounces that: (i) other treaties and agreements are not affected by it, and that (ii) states should abide by the norms and rules contained therein. A typical fragmentation problem of international law underpins the IHR 2005 as well.

Notably, the advent of the new challenges stemming from the COVID-19 pandemic also indicates the possibility of utilizing AI for the prevention of a new disease and for the protection of human health on the global scale. Collection, dissemination and processing of personal information are particularly important in the context of national and global health regulation through AI technology. The sensitivity associated with personal information in the digital age will be further debated and tested when counter-pandemic measures are mobilized by heavily relying on the data containing personal information collected from various sources. If national and global health regulation has become one of the top priorities for many governments in the age of a pandemic, and if states are engaged in disputes concerning such regulation as we are seeing now, then finding a new legal scheme for the collection, dissemination and processing of personal information during a pandemic will remain one of the most important issues in exploring a new legal instrument for pandemics such as COVID-19 As much as such regulation is in turn heavily reliant on AI and personal information constituting the core components of the AI, then any prospective convention on pandemics is inevitably affected by AI and complex issues surrounding personal information. The situation also applies to other issues and sectors of international law to the extent that AI and personal information are applied and utilized, which is increasingly the trend, in any manner.

In short, it seems that we are going through a transition period where existing norms are not able to effectively regulate AI, while new norms are yet to be established fully. The multilateral system is too slow to keep up with the pace of the rapidly changing technology, and the hegemonic competition between the US and China is making any international cooperation and coordination in the multilateral system ever more elusive. What's more, while the digital economy is premised on the notion of abolishing existing national boundaries and territorial limitations, the recent pandemic has gone exactly in the opposite direction by reinforcing the concepts of and the need for national sovereignty and territory. Thus, there is a structural tension between the two most important topics at present—the digital economy on the one hand and the COVID-19 on the other hand. States, governments and business entities are torn between the two opposite policy directions. This structural tension, in turn, is intensifying the decline of the multilateralism, and the G-2 confrontation. Digital economy and COVID-19, collectively and individually, have become the hot topics of G-2 confrontation and the reason for the dwindling multilateralism. Conversely, the weakening multilateralism and increasing Sino-US stand-off, collectively and individually, have fostered a slow response to the digital economy and unorganized response to COVID-19, and failed the global community in agreeing on new norms to regulate these two fast changing areas—digital economy and pandemics. All these four issues are intertwined and interrelated. The fact that these four have come at the same time has put the global community in an unprecedentedly difficult environment. The following diagram illustrates this.



1.2 Different Views and Policy Objectives

Technology is uniform and ubiquitous in that the best and most efficient technology survives the competition in the market. Different experiments converge on a final winner over time. Technological convergence notwithstanding, states have different views and policies toward AI. Therefore, with respect to the same technology and machines that use the technology, different states have different approaches. Thus, the core issue in this regard, one might say, is not technology but how it is viewed by policy makers of respective countries. To the extent that there are differences, technological development even in one direction, does not necessarily help the global community converge on policy matters as well. The lack of legal norms on AI related sectors is the outcome of these differences, and these differences, in turn, are the outcome of the lack of legal norms. Having realized the situation, many developed countries have already begun a number of domestic programs to develop and promote the application and utilization of AI,[21] while many other governments

[21] *See* New [10], Viola [11] *cited by* Erdélyi and Goldsmith [12].

view AI from a national strategic perspective.[22] The former category of countries focus on the vast business opportunities that AI creates and introduces. A sizable commercial gain and thus national interest is at stake. The latter category of countries tend to regard AI as a crucial national asset not only in terms of military operation, but also of economic growth and societal development. Of course, each category of countries do not exclude the consideration of the policy objectives of another category. In effect, policy objectives of both categories are shared by all states, and the difference only lies at the balancing point between the two. In other words, AI is important in all areas of national development, national economy, and national security. As a consequence, major powers have been busy establising respective national strategies to promote and support the growth of AI while safeguarding national strategic interest.[23] However, as mentioned earlier, there has yet to be agreement or consensus among different states (particularly among major powers) on how to regulate AI, what to regulate in AI, and where to place the balancing point between the two competing interests, namely commercial gains flowing from liberal utilization of AI and data and security gains flowing from restrictive utilization of AI and data. All states differ in their priorities and approaches. This situation is both the reason and outcome of the lack of legal norms (domestic law as well as international law) in this area in the global community.

1.2.1 United States

From early on, the Trump administration actively sought diverse ways to facilitate and support the AI development in the US. In 2018, the Trump administration named AI the second highest R&D priority after security in a memorandum on the subject in the Administration Research and Development Budget Priorities for the fiscal year 2020.[24] The importance of American leadership in the newest technology was emphasized on the White House's website as well:

> Building on a foundation of Federal research and development (R&D) investments, America will also be the nation that leads in today's emerging technologies, from artificial intelligence (AI) and quantum computing, to biotechnology, advanced wireless communications, and space commercialization. (...) Continued leadership in AI, quantum information science (QIS), and strategic computing is critically important to our national security and economic competitiveness. Advances in these areas promise opportunities for major scientific breakthroughs and are quickly transforming American life and industry. Agencies should invest in fundamental and applied AI research, including machine learning, autonomous systems, and applications at the human-technology frontier. Agencies should prioritize QIS R&D, which will build the technical and scientific base necessary to explore the next generation of QIS theory, devices, and applications. Agencies should prioritize investment in research and

[22] *See* Walch [13].

[23] For a list of national AI strategies, *see* Dutton [14].

[24] *See AI Policy—United States*, FUTURE OF LIFE INSTITUTE, available at https://futureoflife.org/ai-policy-united-states/; *Memorandum for the Heads of Executive Departments and Agencies,* M-18–22, (Jul. 31, 2018), available at https://www.whitehouse.gov/wp-content/uploads/2018/07/M-18-22.pdf.

> infrastructure to maintain US leadership in strategic computing, from edge devices to high-performance computing, that accelerates delivery of low-power, high performance devices; supports a national high-performance computing ecosystem; and explores novel pathways to advance computing in a post-Moore's Law era.[25]

Thus, it can be said that the Trump administration recognized the wide applicability of AI and the immense opportunities that it encompasses. Indeed, President Trump remarked in the memorandum that "[w]e're on the verge of new technological revolutions that could improve virtually every aspect of our lives, create vast new wealth for American workers and families, and open up bold, new frontiers in science, medicine, and communication."[26]

In fact, President Trump was the first US president to include AI among national R&D priorities. Furthermore, AI was first mentioned in the National Defense Strategy under the Trump administration as a technology that could dramatically change the character of war.[27] On February 11, 2019, President Trump announced *the American AI Initiative*, or the national strategy on AI, through an Executive Order.[28] The Executive Order identifies "continued American leadership in AI" as of "paramount importance to maintaining the economic and national security of the US and to shaping the global evolution of AI in a manner consistent with [US] values, policies, and priorities".[29]

Interestingly, the *American AI Initiative* strategy calls for a close cooperation between the government, private sector, academia and international partners in five pillars. The five pillars are the following: (1) promote sustained AI R&D investment, (2) unleash Federal AI resources, (3) remove barriers to AI innovation, (4) empower the American worker with AI-focused education and training opportunities, and (5) promote an international environment that is supportive of American AI innovation and its responsible use.[30] The five pillar effectively covers almost the entire segments of the governmental affairs and business sectors. The wide scope of the five pillar stand to implicate a wide range of norms of international law as well. Furthermore, to improve better application of AI in government functions, the White House hosted *the Summit on Artificial Intelligence in Government* in 2019. At the summit, the federal government and agencies discussed ways to improve AI applications in the governmental activities, which stands to raise direct implication for treaties and

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[25] Ibid.

[26] The White House, *Artificial Intelligence for the American People,* available at https://www.whitehouse. gov/briefings-statements/artificial-intelligence-american-people/.

[27] *See AI Policy—United States*, FUTURE OF LIFE INSTITUTE, available at https://futureoflife.org/ai-policy-united-states/.

[28] *Executive Order on Maintaining American Leadership in Artificial Intelligence* 13,859, (Feb. 11, 2019), available at https://www.whitehouse.gov/presidential-actions/executive-order-maintaining-american-leadership-artificial-intelligence/.

[29] Ibid.

[30] The White House, *Executive Order on AI,* available at https://www.whitehouse.gov/ai/executive-order-ai/.

agreements as AI seeps into the formation of governmental measures subject to these treaties and agreements.[31]

It is clear that the US government aims to retain a world-leading competitiveness edge in AI. To this end, the US government has pledged financial contributions for R&D activities in the AI sector, which is included in its annual budget report as follows:

> - Harnessing Artificial Intelligence and High Performance Computing -
>
> The development of artificial intelligence (AI) is advancing at a rapid pace, and the 2019 Budget invests in fundamental AI research and computing infrastructure to maintain US leadership in this field. AI holds the potential to transform the lives of Americans through improved technology integration in the workplace and enhanced standards of living at home. The Budget funds basic research related to AI at the National Science Foundation and applied R&D in the Department of Transportation for the further development of autonomous and unmanned systems. In defense applications, DOD is working to deliver AI-driven algorithms to warfighting systems, which can rapidly turn volumes of data into decision-quality insights. And in the health realm, NIH is supporting the use of high performance computing to analyze large data sets to drive cancer research forward. The Budget also funds high performance computing through supporting investments in computing infrastructure, which hold the potential for AI technology use and other purposes. The Budget provides $811 million to the Department of Energy's Advanced Scientific Computing Research Program to support research and facility upgrades to supercomputing infrastructure at Argonne and Oak Ridge National Laboratories, including the development of exascale high performance computers. These supercomputers will rank among the fastest and most powerful in the world, and will leverage strong partnerships with industry and academia in their development and use.[32]

Overall, the US views AI both as an opportunity and a challenge to solidify its global competitiveness. It is striving to maintain its leading role in the development of AI, and henceforth using it to strengthen its military capabilities as well as enhance its economic interests. At the same time, Washington is concerned about China's aggressive AI development strategy. China is indeed leading in some critical categories of AI application and utilization through its governmental support schemes and availability of a massive amount of data (e.g., facial recognition technology). Generally speaking, the US has adopted a more liberal approach so as to promote entrepreneurial environment in the AI area compared to other states.[33] While this policy orientation certainly helps facilitate private entities' efforts to seek further innovation, it also causes concerns over privacy and data protection, as the exact contours of the amount of personal information and the ways it is to be used are anything but certain at this stage.

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[31] A summary of the summit available at https://www.whitehouse.gov/wp-content/uploads/2019/09/Summary-of-White-House-Summit-on-AI-in-Government-September-2019.pdf.

[32] The White House, *2019 Budget—Analytical Perspectives*, R&D CHAPTER, p. 236; Full chapter available at https://www.whitehouse.gov/wp-content/uploads/2018/02/ap_18_research-fy2019.pdf.

[33] *See* Vincent [8], Cath et al. [15].

1.2.2 European Union

The European Union's Junker Commission (2014–2019), which pushed forward a digital single market agenda in the EU, took the first step by establishing a European framework on AI. In line with the proposal for Digital Europe, the year 2018 marked the beginning of the EU's policy focus on AI.

In April 2018, 25 EU member countries signed the *Declaration of Cooperation on Artificial Intelligence*[34] to promote cooperation among themselves on European R&D and business activities concerning AI, and four others (Romania, Greece, Cyprus and Croatia) have joined since then. In the Declaration, the signatory states acknowledged the need for cooperation to create synergies across Europe. The Declaration also sets out the basic European values to be respected, such as the transparency and accountability of AI. Vice-President for the Digital Single Market, Andrus Ansip, and Commissioner for Digital Economy and Society, Mariya Gabriel, welcomed the Declaration, and emphasized cooperation among the EU states:

> In Europe, any successful strategy dealing with AI needs to be cross-border. A large number of Member States agreed to work together on the opportunities and challenges brought by AI. That is excellent news. Cooperation will focus on reinforcing European AI research centres, creating synergies in R&D&I funding schemes across Europe, and exchanging views on the impact of AI on society and the economy. Member States will engage in a continuous dialogue with the Commission, which will act as a facilitator.[35]

The Declaration was followed by a Communication Note on AI for Europe from the European Commission. In the 2018 Communication Note, the European Commission claimed that "[g]rowth in computing power, availability of data and progress in algorithms have turned AI into one of the most strategic technologies of the twenty-first century. The stakes could not be higher. The way we approach AI will define the world we live in. Amid fierce global competition, a solid European framework is needed."[36] Building on the groundwork laid down in 2018, the EU has since been actively pursuing ways to develop 'a European framework' to support and manage AI across Europe.[37]

On balance, the EU is taking a more cautionary approach compared to the US or China. For example, the European Commission pledged to set up training and retraining programs to minimalize the negative impact from AI on society and

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[34] Full declaration available at https://ec.europa.eu/digital-single-market/en/news/eu-member-states-sign-cooperate-artificial-intelligence.

[35] European Commission, *EU Member States Sign Up to Cooperate on Artificial Intelligence* (Apr. 10, 2018), available at https://ec.europa.eu/digital-single-market/en/news/eu-member-states-sign-cooperate-artificial-intelligence.

[36] *See* European Commission, *Communication from the Commission to the European Parliament, the European Council, the Council, The European Economic and Social Committee and the Committee of the Regions—Artificial Intelligence for Europe,* Brussels, COM (2018) 237 final, (Apr. 25, 2018), p.1, available at https://ec.europa.eu/transparency/regdoc/rep/1/2018/EN/COM-2018-237-F1-EN-MAIN-PART-1.PDF.

[37] Ibid.

the European job market.[38] Moreover, to retain the high standards of safety and product liability of the region, the EU aims to strengthen the regulatory framework on data protection. The well-known GDPR has effectively provided individuals with the means to object to automated decision-making. The European Commission is also trying to guarantee the free flow of non-personal data for the Digital Single Market strategy, while protecting personal security through regulations such as *ePrivacy Regulation and Cybersecurity Act.*[39]

Thus, it can be said that the EU is not taking a black-or-white approach either. It also considers the pluses and minuses of personal data collection and utilization. The EU is simply more sensitive to the protection of personal information than other jurisdictions. As such, the general level of data protection in the EU is relatively high. For instance, according to an article in the *New York Times*, a legal adviser working for the European Court of Justice (hereinafter ECJ) suggested that the data protection regulations should be abided by even when national intelligence agencies request access to such data [40]: commenting on a number of cases from the member states, the advocate general of the ECJ, Campos Sanchez-Bordona, noted that "the means and methods of combating terrorism must be compatible with the requirements of the rule of law."[41] While the legal opinion from the ECJ is not legally binding, it is expected that member states with reinforced surveillance, such as Belgium, France, and the United Kingdom (hereinafter UK), will change their domestic policies to be in line with the court's opinions.[42] Moreover, the ECJ ruled that tech companies should not be allowed to transfer data to locations outside the EU when it involves infringement of the EU privacy rules.[43] It thus stated its position as follows:

> The Advocate General considers, in the first place, that EU law applies to transfers of personal data to a third country where those transfers form part of a commercial activity, even though the transferred data might undergo processing, by the public authorities of that third country, for the purposes of national security.
>
> In the second place, the Advocate General finds that the provisions of the GDPR on transfers to third countries are aimed at ensuring the continuity of the high level of protection of personal data, whether the data are transferred on the basis of an adequacy decision or on guarantees provided by the exporter. In his view, the way in which that aim is achieved differs according to the legal basis of the transfer. On the one hand, the purpose of an adequacy decision is to find that the third country concerned ensures, as a result of the law and practices of that country, a level of protection of the fundamental rights of the persons whose data are transferred essentially equivalent to that provided by the GDPR, read in the light of the Charter. On the other hand, the appropriate safeguards afforded by the exporter, inter alia by contractual means, must themselves ensure that level of protection. In that respect, the standard contractual clauses adopted by the Commission provide a general

[38] Ibid., pp. 11–13.

[39] Ibid., p. 14.

[40] *See* Petrequin [16].

[41] Ibid.

[42] Ibid.

[43] *See EU Court Sides with Activist in Facebook Data Transfer Fight*, NEW YORK POST (Dec. 19, 2019), available at https://nypost.com/2019/12/19/eu-court-sides-with-activist-in-facebook-data-transfer-fight/.

> mechanism applicable to transfers irrespective of the third country of destination and the level of protection guaranteed there.
>
> In the third place, the Advocate General examines the validity of Decision 2010/87 in the light of the Charter. He considers that the fact that that decision and the standard contractual clauses which it sets out are not binding on the authorities of the third country of destination and therefore do not prevent them from imposing obligations that are contrary to the requirements of those clauses on the importer does not in itself render that decision invalid. The compatibility of Decision 2010/87 with the Charter depends on whether there are sufficiently sound mechanisms to ensure that transfers based on the standard contractual clauses are suspended or prohibited where those clauses are breached or impossible to honour.
>
> In his view, that is the case in so far as there is an obligation—placed on the data controllers and, where the latter fail to act, on the supervisory authorities—to suspend or prohibit a transfer when, because of a conflict between the obligations arising under the standard clauses and those imposed by the law of the third country of destination, those clauses cannot be complied with.[44]

The level of personal information protection by the GDPR is unmatched by any other counterpart legislations. It purports to fix the problem of our 'required' consent to relinquish our personal information when we access a website—most websites do not let users proceed unless they agree to provide personal information from the start. Knwoing this, the GDPR requires very specific, unambiguous and informed consent from providers. Also, consenting individuals can maintain continuous control over the information thus provided. In addition, information collecting entities are required to have Data Protection Officers in place, who are in charge of all information-related issues and matters. Noncompliance will entail harsh penalties: fines can be as high as 20 million euros or 4% of a total worldwide revenue of the company at issue. By any account, it is a meaningful sanction for violators. Given the increasing concern over the provision and utilization of personal information, the direction the EU has taken seems to be the right one in principle, assuming that fines are determined to be proportionate and reasonable under the circumstances.

As a result, EU corporations are all alerted, of course. Interestingly and importantly, this new EU regulation has also put non-EU corporations in a hurried scramble mode. In other words, those covered by the new EU regulation are not just the companies or entities operating in the EU that are collecting and processing personal data within EU territories. Those in other countries outside the EU may also be subject to the regulation if they process information collected from EU citizens or their information processing is otherwise related to EU citizens. State differently, foreign companies who have customers in the EU or business operations in the EU even remotely—which should be many by the way, given the EU market's global stature—are all subject to the GDPR and hence its strict penalty provisions.

By way of example, Korean, Japanese or Chinese companies with EU customers should also be directly implicated by this regulation. Traditionally, 'extraterritorial' application of domestic law has often prompted allergic responses from other

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[44] Court of Justice of the European Union, *Advocate General's Opinion in Case C-311/18,* Press Release No. 165/19 (Dec. 19, 2019), available at https://curia.europa.eu/jcms/upload/docs/application/pdf/2019-12/cp190165en.pdf.

countries whose companies and nationals are affected by such laws. As regards the GDPR, the conventional extraterritorial application is pushed to the next level. Given the harsh penalty provisions of the GDPR, its ultimate enforcement will probably lead to tensions and clashes between Brussels and other countries. The European Commission has begun to issue fines to companies of the EU and elsewhere. Other countries are now all eyes and ears. Once they see how the EU regulation fares in its entirety, they are then likely to take issue with the EU, and possibly initiate a legal proceeding against the EU. Furthermore, other countries may decide to follow suit with comparable enactments.

In addition, on February 19, 2020, the European Commission published a White Paper, *On Artificial Intelligence—A European Approach to Excellence and Trust,* [45] a separate report on the safety and liability implications of AI, the IoT and robotics.[46] On the one hand, the European Commission acknowledges in these instruments the opportunities and benefits created by AI. On the other hand, it is cautious about new kinds of social threats related to the connectivity and openness dramatically enhanced by the new technology. The EU thus opined:

> The emergence of new digital technologies like AI, the IoT and robotics raise new challenges in terms of product safety and liability like connectivity, autonomy, data dependency, opacity, complexity of products and systems, software updates and more complex safety management and value chains. (…) The new challenges in terms of safety create also new challenges in terms of liability. Those liability related challenges need to be addressed to ensure the same level of protection compared to victims of traditional technologies, while maintaining the balance with the needs of technological innovation. This will help create trust in these new emerging digital technologies and create investment stability. While in principle the existing Union and national liability laws are able to cope with emerging technologies, the dimension and combined effect of the challenges of AI could make it more difficult to offer victims compensation in all cases where this would be justified. Thus, the allocation of the cost when damage occurs may be unfair or inefficient under the current rules. To rectify this and address potential uncertainties in the existing framework, certain adjustments to the Product Liability Directive and national liability regimes through appropriate EU initiatives could be considered on a targeted, risk-based approach, i.e. taking into account that different AI applications pose different risks.[47]

The cautious approach of the EU and stricter regulations adopted by the EU are sometimes viewed as an obstacle for faster growth of the AI-related sectors in the EU. Although Europe hosts several world-leading AI research centers and businesses, it is still behind in private company investment compared to Asia and North America.[48]

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[45] Full paper available at https://ec.europa.eu/info/sites/info/files/commission-white-paper-artificial-intelligence-feb2020_en.pdf.

[46] Full report available at https://ec.europa.eu/info/files/commission-report-safety-and-liability-implications-ai-internet-things-and-robotics_en.

[47] Ibid., pp. 16–17.

[48] China and the US have large companies leading private investment in AI. *See* European Commission, *Communication from the Commission to the European Parliament, the European Council, the Council, The European Economic and Social Committee and the Committee of the Regions—Artificial Intelligence for Europe,* Brussels, Apr. 25, COM (2018) 237 final, (2018), p. 4 available at https://ec.europa.eu/transparency/regdoc/rep/1/2018/EN/COM-2018–237-F1-EN-MAIN-PART-1.PDF.

Therefore, the EU is trying to promote, facilitate and expand investment in the AI-related sectors.

Overall, it seems that the EU prioritizes moral and ethical guidelines to supervise the AI development in its AI projects. Compared to the US, which prefers a *laissez-faire* approach, the EU appears to be more circumspect and highlights the humancentric aspects of AI. It is largely in line with the EU's robust reputation as a forerunner and protector of human rights. Moreover, the recent attention to data security has highlighted potential problems with AI in terms of privacy. As a consequence, while the EU certainly acknowledges the benefits of AI, it is taking a cautious approach to ensure a safe and monitored development of new technologies.

1.2.3 China

China has openly expressed its ambition to seek a leading role in the AI development.[49] The Chinese government sees AI as an opportunity to enhance its prominence in international relations, especially in terms of its competition with the US. At a Politburo group study session concerning AI held in 2018, Chinese President Xi was quoted by a news agency as saying that "AI is a vital driving force for a new round of technological revolution and industrial transformation, and accelerating AI development is a strategic issue to decide whether we can grasp opportunities".[50] President Xi emphasized the importance of AI in upgrading the manufacturing capability and efficiency in China while maintaining environmental friendliness. China's realization of AI's benefit is not just confined to the manufacturing industry. China sees the vast potential of AI in the areas of healthcare, transport and communications, areas so important to countries like Chins with the world's largest population living in a vast territory.[51]

In 2017, China published a three-step program to become a global leader in the AI sector by 2030.[52] Along with the *Made in China 2025,* [53] an initiative released in 2015 for China's rise and dominance in key manufacturing sectors, the *New Generation Artificial Intelligence Development Plan* (AIDP)[54] constituted the backbone

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[49] *See AI Policy—China,* FUTURE OF LIFE INSTITUTE, available at https://futureoflife.org/ai-policy-china/; Lee [17].

[50] Xin and Chi-yuk [18].

[51] *See* O'Meara [19].

[52] The Chinese government aims to make the AI industry worth about $150 billion. It plans to expand the use of AI in numerous sectors including military and smart cities. Beijing has also invested $2.1 Billion AI-focused technology research park. *See* Walch [13].

[53] China State Council, *Made in China 2025* (Jul. 7, 2015), English translation available at http://www.cittadellascienza.it/cina/wp-content/uploads/2017/02/IoT-ONE-Made-in-China-2025.pdf. .

[54] *See Notice of the State Council Issuing the New Generation of Artificial Intelligence Development Plan* (Jul. 8, 2017), State Council Document [2017] No. 35, translated by Flora Sapio, Weiming Chen, and Adrian Lo at THE FOUNDATION FOR LAW AND INTERNATIONAL AFFAIRS, available at https://flia.org/wp-content/uploads/2017/07/A-New-Generation-of-Artificial-Intelligence-Development-Plan-1.pdf.

of China's AI strategy.[55] The opening paragraph of the AIDP clearly shows China's strong strategic interest in taking the first-mover's advantage in AI:

> The rapid development of artificial intelligence will profoundly change human social life and the world. To seize the major strategic opportunities for the development of artificial intelligence, build China's first-mover advantage in artificial intelligence development, accelerate the construction of innovative countries and the world's science and technology power, this plan is enacted in accordance with the requirements of the CPC Central Committee and State Council.[56]

To this end, the AIDP lays out the three steps as follows: First, by 2020, China expects to have achieved parity with globally advanced countries in terms of overall AI-related technology. Considering the fast application and utilization of AI recently, particularly during the pandemic period, it can be said that this objective has been achieved. Second, by 2025, China aims to achieve major breakthroughs in AI-technology in a way it becomes the main driving force for its economic growth and social progress. This stage aims to achieve further integration of AI in many sectors of Chinese economy. Apparently, various projects are underway to complete the goal. Third, China finally strives to become a world-leading center of AI in terms of both academic theories and technological development.[57] At this stage, China will cement its dominance in the new field of AI. Based on the basic plan, respective ministries and agencies have devised their own respective implementation schemes. For instance, the Chinese Ministry of Education announced a promotion plan for AI -based education and research in its report entitled *AI Innovation Action Plan for Colleges and Universities*.[58]

The Chinese government is also seeking close cooperation with the private sector and research institutions. In 2019, *the Beijing AI Principles*[59] were released by a group of AI-related stakeholders including education and research institutions, such as the Beijing Academy of Artificial Intelligence (BAAI), Peking University, and Tsinghua University, and major ICTI firms, such as Baidu, Alibaba and Tencent, where public-private joint endeavor was also underscored. The government agency-private company cooperation and coordination, which is already robust in many sectors of China, is also actively playing out in the new area of AI.

Similar to the US, China also sees AI as an opportunity to solidify its global influence and enhance economic interests. Nevertheless, compared to the US, AI development in China is largely government-centered. With its highly centralized

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[55] *See* Gregory C. Allen [20].

[56] *See Notice of the State Council Issuing the New Generation of Artificial Intelligence Development Plan* (Jul. 8, 2017), State Council Document [2017] No. 35, translated by Flora Sapio, Weiming Chen, and Adrian Lo at THE FOUNDATION FOR LAW AND INTERNATIONAL AFFAIRS, available at https://flia.org/wp-content/uploads/2017/07/A-New-Generation-of-Artificial-Intelligence-Development-Plan-1.pdf.

[57] Ibid.

[58] Liu [21].

[59] English translation (translated by Zeng Yi) available at https://www.baai.ac.cn/blog/beijing-ai-principles.

planned economy, the Chinese government has not been holding back in its investment in new technology in this area. Some already suspect that China is spending more than the US.[60] Due to the aggressive support from the government, the AI sector in China has been expanding and flourishing. For instance, Sensetime, allegedly the most valuable AI startup, which raised over 1.2 billion USD, is based in China, and more and more are coming up.[61]

The Chinese government has been known for its nation-wide collection of a huge amount of data for various regulatory purposes. Such data collection and processing have the lingering potential of triggering the concern over people's data privacy. It may well be the case that some data collection and utilization in China, if conducted in other countries, are in fact in violation of individual privacy and possibly basic human rights. As such, voice is being raised for better enforcement of private data protection.[62] Although the government certainly benefits from the application and utilization of AI because of its efficiency and speed in regulatory activities, some experts point out possible dangers of excessive data collection and processing that could lead to social instability.[63] In fact, in a 2018 survey by the China Consumers Association, 85% of respondents answered that they had experienced a type of 'personal data leak'.[64] As a result, the Chinese government responded with legislation for personal information protections, including the *2017 Cybersecurity Law* and *2018 E-Commerce Law*.[65] Combined with the *2018 Personal Information Security Specification* and *2019 Draft Data Security Management Measures,* it is argued that the Chinese regulations are "among the most comprehensive in the world against intrusive data collection".[66] Furthermore, companies such as Entropage have emerged to provide encryption of emails and messages in order to enhance data security.[67]

The width and depth of data protection is not hinged on the number of relevant legislations. Rather, they depend on how sensitive the agencies are toward personal information, what safeguards are introduced in various laws and regulations, and how individuals resort to the laws and regulations to remedy the situation when their personal information is breached illegitimately. Should these elements be found to be weak in domestic legislations, one could then argue that enforcement of the data protection is not optimal yet, even of such elements are somehow included somewhere in the enactments. This is also the direct outcome of the absence or lack of reliable legal norms in this area; states are thus left to devise and formulate their own rules respectively, opening a door for discrepancies and ultimately disputes. As with many other sectors and areas, China is seriously challenging the US in the race for world leader in the AI technology and the digital economy. If anything, AI has become

[60] *See* Gill [22].
[61] *See* Walch [13].
[62] *See* Feng [23], Campbell [24].
[63] Ibid.
[64] Geller [25], Mirchandani [26].
[65] Ibid.
[66] Ibid.
[67] *See* Feng [23].

one of the most contentious areas of the bilateral competition for global hegemony. President Xi has not been shy about China's ambition to be a world leader in this area, and Presidents Trump and Biden have not been quiet about the US' concern over China's effort either. For instance, President Trump complained to President Xi that *Made in China 2025* was 'insulting' to the US.[68] The Made in China 2025 project is geared toward manufacturing industries, but one of the core industries covered by the project is none other than the semiconductor industry. Semiconductors provide blood and neurons for the AI system, and as such they are one of the essential ingredients of the AI society. It is not surprising that the two countries are competing in the semiconductor sector. The rivalry between the two countries is likely to continue in the future. The absence and lack of adequate norms on AI foster the rivalry of the two countries in this field.

1.2.4 Japan

Japan is also striving to help develop AI technology and introduce it to broad sectors of the society. The Japanese government expects that the new emerging technologies in AI could enhance the productivity level of Japan. The former Prime Minister Shinzo Abe once said:

> The key of the second stage of Japan's growth strategy is the realization of the fourth industrial revolution. Individualized services and new business models tailored to each individual will be created as a result of IoT and robots. All factories and shops will be connected through IoT, enabling zero inventory and instantaneous order-made production. This could represent a game-changer for industry as we know it.[69]

The Japanese government claims that Japan has been a leader in the AI industry and takes pride in its world leading tech-firms, such as *HEROZ*, *Inc.* that once developed the *Ponanza* software, the shogi playing AI that defeated *Meijin* (the most prestigious shogi player).[70] Japan is also investing a lot in AI technology. For example, *Softbank* is a Japan-based investor powerhouse that has been making continued investment in the AI sector.[71] Once known as a "Robot Nation", however, Japan is now facing some serious resource strains and limitations, as it needs to update its outdated national infrastructure.[72]

As a response to the rapidly aging population and economic stagnation,[73] Japan aims to launch 'Society 5.0', a super-smart society agenda, with AI as its technological base. In January 2016, the Japanese government published its *5th Science and*

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[68] *See* Gill [22], Chine et al. [27].

[69] Garcia [28].

[70] *See* the Government of Japan. *Artificial Intelligence: A Rival for Humans, or a Partner?,* TOMODACHI SPRING (2018), available at https://www.japan.go.jp/tomodachi/2018/spring2018/artificial_intelligence.html.

[71] *See* Walch [13].

[72] *See* Minevich [29].

[73] Ibid.

Technology Basic Plan, which is a comprehensive development plan laying out the blueprint for technological development over a five-year term (2016–2020).[74]

> Around the world, initiatives that use networks and the Internet of Things (IoT), centered on manufacturing fields, are now coming out. In Japan, the use of such networking will not be limited to manufacturing. Instead, it will be extended to various other fields in order to promote economic growth, the formation of a healthy and long-living society, and social transformation. In addition, it will help the fruits of science and technology to penetrate into all kinds of fields and spheres, and thereby lead to enhanced business capability and higher quality services.
>
> We will share our vision of the future, which is characterized by the sophisticated integration of cyberspace with physical space ('the real world') and work to enhance it, while further pursuing a series of measures aimed at its realization, under the concept of "Society 5.0".[75]

In 2013, the Cabinet Office of Japan adopted the *Declaration to be the World's Most Advanced IT Nation and the Basic Plan for the Advancement of Public and Private Sector Data Utilization.*[76] Recognizing the importance of data in the development of AI, it calls for a society in which "people are enriched by data". *The Basic Act on the Advancement of Public and Private Sector Data Utilization* (Act No. 103 of 2016) provides the starting point for fostering national environment for smooth application and utilization of the data collected from both the public sector and private sector.[77]

The pooling of the public sector data and private sector data are meaningful because the scheme recognizes that any meaningful utilization of the national data is only to be achieved through collection of all relevant data in the nation, and that public-private partnership is critical in this regard. It signals that a separation of the public and private sector would be either impractical or impossible when it comes to AI and the digital economy. The Japanese government is also trying to establish some basic guidelines to minimize the risks associated with AI. For example, based on discussions from the *Conference on Principles of Human-centric AI Society,* the government of Japan published seven core AI principles in December 2018.[78] The seven principles are: (1) Do good (2) For humanity (3) Be responsible (4) Control risks (5) Be ethical (6) Be diverse and inclusive and (7) Open and share. While short, the seven principles cover all key elements of the current AI debates.

In sum, whereas the US and China are competing to be the world leader in the AI sector for strategic reasons and value AI mainly for its economic benefits, Japan's

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[74] *Outline of the Fifth Science and Technology Basic Plan,* available at https://www8.cao.go.jp/cstp/english/basic/5thbasicplan_outline.pdf; Full text of tentative English translation of the report available at https://www8.cao.go.jp/cstp/kihonkeikaku/5basicplan_en.pdf.

[75] *Outline of the Fifth Science and Technology Basic Plan,* available at https://www8.cao.go.jp/cstp/english/basic/5thbasicplan_outline.pdf.

[76] The Declaration was established in 2013, and then amended in 2016. *See* Matsuda et al. [30], Okamoto [31]. Full text of the Declaration available in English at https://japan.kantei.go.jp/policy/it/2016/20160520_full.pdf.

[77] An English draft version of the declaration available at https://japan.kantei.go.jp/policy/it/2017/20170530_full.pdf. (*See* p. 6).

[78] *See* Matsuda [30].

interest in AI is more geared toward addressing domestic concerns and homeworks. AI presents new solutions to various challenges that Japan is now facing, such as mass exodus from rural areas, an ageing population, and frequent natural disasters (earthquakes and typhoos).[79] Although it has fallen behind China recently in terms of the size of the economy, Japan is still one of the largest economies in the world and one of the most technologically developed nations. It will continue to strive to secure and exert its regional (if not global) influence through the new technology in the AI sector.

1.2.5 Russia

As is the case for other countries, AI holds many future benefits for the Russian society. For example, AI could offer solutions for new problems arising from Russia's declining population[80] Moreover, Russia has been frequently alleged to have actively engaged in information warfare as a wherewithal in its foreign policy. AI-Driven Asymmetric Warfare (ADAW), or digital information warfare, may provide Russia with a comparative advantage against the US and Europe.[81]

Apparently, President Putin is well aware of the importance of AI. At a meeting with students about science in 2017, he remarked: "Artificial intelligence is the future, not only for Russia, but for all humankind. It comes with colossal opportunities, but also threats that are difficult to predict. Whoever becomes the leader in this sphere will become the ruler of the world."[82] Thus, the Russian leadership highlights the importance of AI in global competitiveness, and in particular it notes AI's close relationship to the military sector.[83] At the St. Petersburg International Economic Forum in 2019, President Putin claimed that AI could contribute to 1.2% annual increase of the global GDP growth.[84] Putin has also emphasized the importance of public–private partnership to promote technological advances, a similar stance that China and Japan take when it comes to the R&D activities relating to AI and the cross-utilization of public and private data sets.[85]

In May 2018, the Russian government announced the Presidential Decree No.204, entitled "On the National Development Goals and Strategic Objectives of the Russian

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[79] *See* Minevich [29].

[80] *See* Polyakova [32].

[81] Ibid.

[82] *'Whoever leads in AI will rule the world': Putin to Russian children on Knowledge Day*, *RT* (Sep. 1, 2017), available at https://www.rt.com/news/401731-ai-rule-world-putin/; World Economic Forum website, available at https://reports.weforum.org/global-risks-report-2020/chapter-one-risks-landscape/.

[83] Garcia [33].

[84] *Putin Approves National Strategy for AI until 2030,* TASS, available at https://tass.com/economy/1082644; World Economic Forum website, available at https://reports.weforum.org/global-risks-report-2020/chapter-one-risks-landscape/.

[85] Ibid.

Federation in the Period until 2024".[86] Under the decree, the National Programme "Digital Economy" was launched to establish the foundations of digitization and digital-based technologies in the Russian economy.[87]

More recently, the National Strategy for the Development of Artificial Intelligence for the period until 2030 was approved in October 2019 by the Presidential Decree No. 490, entitled "On the development of AI in the Russian Federation".[88] The Russian government is also in pursuit of AI development to enhance the life quality of its population, national security, and global economic competitiveness.[89] The primary objectives of the AI national agenda have been described as the following:

> 24. The primary objectives of the development of artificial intelligence shall be: (a) supporting scientific research for the purpose of ensuring the rapid development of artificial intelligence; (b) engineering and developing software that employs artificial intelligence technologies; (c) increasing the availability and improving the quality of the data needed in order to develop artificial intelligence technologies; (d) increasing the availability of the hardware needed in order to solve problems in the field of artificial intelligence; (e) increasing the extent to which the Russian artificial intelligence technology market is provided with qualified personnel and the level of public awareness of the potential areas of use of these technologies, and; (f) creating an integrated system for regulating the social relations arising in line with the development and use of artificial intelligence technologies.[90]

According to the national strategy, the Russian government promised to observe the following principles: protection of human rights and liberties, protection of national security, enhancement of transparency, strengthening of technological sovereignty, fostering technological innovation, ensuring reasonable thrift, and facilitating support for competition.[91]

At the moment, in many respects Russia seems to be far behind in the global competition in AI development.[92] Russian investment in AI, approximated to have reached $12.5 million per year, falls far behind that of China or the US.[93] Russia aims to have $400 million worth of the AI market by 2021, and has so far raised $2 billion through the Russian Direct Investment Fund to support AI development.[94] Meanwhile, China aims to grow its AI market to the value of $150 billion by 2030.

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[86] *National Programme "Digital Economy",* OECD.AI POLICY OBSERVATORY, available at https://oecd.ai/dashboards/policy-initiatives/2019-data-policyInitiatives-24621.

[87] Ibid.

[88] *National Strategy for Artificial Intelligence Development,* OECD.AI POLICY OBSERVATORY, available at https://oecd.ai/dashboards/policy-initiatives/2019-data-policyInitiatives-24901.

[89] Office of the President of the Russian Federation, *Decree of the President of the Russian Federation on the Development of Artificial Intelligence in the Russian Federation*, (Oct. 2019). para. 23. Translation by Center for Security and Emerging Technology available at https://cset.georgetown.edu/wp-content/uploads/Decree-of-the-President-of-the-Russian-Federation-on-the-Development-of-Artificial-Intelligence-in-the-Russian-Federation-.pdf. Original available at http://www.kremlin.ru/acts/bank/44731.

[90] Ibid., para. 24.

[91] Ibid., para 19.

[92] *See* Polyakova [32].

[93] Ibid.

[94] *See* Polyakova and Meserole [34].

The difference in the amounts of investment is noteworthy. Furthermore, compared to China, the Russian government takes a less interventionist policy in terms of the level of control over its domestic digital affairs.[95]

Russia is also lacking leadership in terms of research activities as well. There are only a limited number of research institutions in Russia that rank globally in the top 500 based on the number of AI patent applications.[96] It seems that there is even a brain drain problem; it is being reported that the brightest minds in the AI sector are leaving Russia.[97]

In sum, although the Russian leadership is striving to take a part in global AI drive and competition, it is lagging behind in the competition for many reasons. While it would be wrong to simply dismiss Russia in the global AI competition based on the current performance, Russia would need political and legal reforms to effectively foster a strong AI sector for the future.[98]

1.2.6 South Korea

As the discussion of the Fourth Industrial Revolution unraveled both globally and domestically, the South Korean government published *Mid- to Long- Term Master Plan in Preparation for the Intelligent Information Society* in 2016. In the report, the South Korean government recognized the need to invest more in innovative technologies "to generate new value and ensure the competitiveness of its intelligent information society".[99] More precisely, the plan delineates the underlying aims as follows:

1. Establishment of world-class infrastructure for Intelligent IT
2. Promotion of the application of Intelligent IT to all industries
3. Proactive steps toward reforming and strengthening the social support system.[100]

Building up on the 2016 report, in 2019, the Moon Jae-in administration announced South Korea's national AI strategy, in which it declared its goal to "achieve digital competitiveness, create huge economic effect of AI, and improve quality of

[95] *See supra* note 251.

[96] *See* Dear [35].

[97] Ibid., p. 47.

[98] Ibid., p. 60.

[99] Government of the Republic of Korea Interdepartmental Exercise, *Mid- to Long-Term Master Plan in Preparation for the Intelligent Information Society* (2016), p. 3, available at https://english.msit.go.kr/cms/english/pl/policies2/__icsFiles/afieldfile/2017/07/20/Master%20Plan%20for%20the%20intelligent%20information%20society.pdf.

[100] Ibid., pp. 30–2.

life for people by 2030".[101] The national strategy consists of 100 action tasks categorized under nine strategies with respect to AI ecosystem, AI utilization, and people-centered AI. In the National Presidential Initiative for AI, President Moon once remarked:

> The era of the Fourth Industrial Revolution is indeed an age in which imagination can change the world. Korea is neither the first country to have ushered in the era of artificial intelligence nor the country with the best AI technology at present. However, the country has people capable of turning their imagination into reality and taking on challenges to pursue novelty. Even in the throes of the 1997 Asian financial crisis, the country led the Internet Revolution and now boasts world-class manufacturing competitiveness, globally unmatched ICT infrastructure and abundant data concerning e-government. If we link artificial intelligence primarily with the sectors in which we've accumulated extensive experience and competitiveness, such as manufacturing and semiconductors, we will be able to give birth to the smartest yet most humanlike artificial intelligence. The Government will join forces with developers to help them fully utilize their imaginations and turn their ideas into reality.[102]

More recently, in June 2020, the Moon administration announced the Korean New Deal, which was again composed of the Green New Deal and Digital New Deal, and promised the national investment in the amount of USD 62 billion over the next five years.[103] The initiative comes in part as a response to calls to stimulate economic growth after the global pandemic. While South Korea fared better than some countries, its highly trade-dependent economy dwindled, with soaring job losses.[104] With the Digital New Deal, the Korean government hopes to kill two birds with one stone—to revitalize its economy and to develop the newest technology for the digital society.[105] In 2020 alone, the administration suggested spending USD 2.2 billion on the Digital New Deal.[106] By the end of 2022, public investment in the Digital New Deal is expected to reach USD 11 billion.[107]

Although South Korea boasts being a global leader in the ICT infrastructure and the semiconductor manufacturing sector,[108] it does not score as highly in the AI competitiveness.[109] For example, South Korea was one of the first countries to deploy 5G, and the Moon administration plans to further develop a 5G infrastructure and promote convergence between 5G and AI.[110] However, South Korea's AI sector seem to be losing its international competitiveness, both in terms of AI research and the

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[101] Ministry of Science and ICT, *National Strategy for Artificial Intelligence* (2019), available at https://english.msit.go.kr/english/msipContents/contentsView.do?cateId=tst60&artId=2771576.

[102] *See supra* note 102.

[103] *See* Stangarone [36].

[104] *See* Kim [37].

[105] *See supra* note 260, p. 11.

[106] *See supra* note 262.

[107] Ibid.

[108] *See supra* note 260, p. 14.

[109] Ibid., p. 21.

[110] *See supra* note 262.

readiness of the society's application.[111] The South Korean government acknowledged that its "rigid, vertical regulatory system" is a weakness in this regard.[112] The truth is South Korea has rather strict regulations on AI and new technologies, such as the use of drones and introduction of new services using Big Data. While each regulation has its valid concerns, it can obviously work as a disincentive for further innovations. Unfortunately, considering this country's unique situation where so many people live in such concentrated residential areas in the Seoul Metropolitan Area together with the constant threat to national security from the North, the real question is not whether there is a risk, but whether such a risk can be managed and how. In other words, any regulation should be carefully tailored and crafted to achieve the competing goals. The same also applies to AI technologies and machines.

In sum, South Korea aims to use its existing strengths in the ICT and semiconductor sectors to develop its AI sector going forward. The South Korean government recognizes the potential benefits that AI could bring to the Korean society in terms of economic and social welfare.

1.2.7 Summary

Recognizing the benefits of AI, many countries have already established national strategies to promote development of their respective AI sectors. The US, EU, China, Russia, Japan, and South Korea all have national strategies or plans to develop and expand the AI sector. As the major countries compete to establish and secure a first-mover advantage in the AI development, international competition is expected to become fiercer as time passes by. In particular, the rivalry between the top two global powers, the US and China, will likely continue, and even intensify, in the AI sector as well.

The reason why these governments are so focused on AI is that it can bring a number of benefits in various areas. A country's international position can be significantly strengthened by a high level of AI development. Not only can AI be used to strengthen a country's military capabilities, it can also bring in tremendous economic benefits. This is why the US and China see AI as a critical asset to enhance their respective positions and power at the international level. Of course, some countries like Japan and South Korea see AI as an opportunity to boost their sagging domestic economies and improve social welfare. Different countries have different priority objectives in applying and utilizing AI.

It is noteworthy that the public sector in each country is playing different roles to varying degrees. While the US is opting for a *laissez-faire* system, leaving the market to take the reins of the AI development, the Chinese government is actively engaging in the process, mobilizing both public and private entities to get to the national goal. Meanwhile, the European Commission is playing the leading role in establishing humancentric AI and human-controlled AI. In line with its reputation

[111] Ibid.

[112] *See supra* note 260, p. 21.

for human rights, the EU is in the lead when it comes to establishing moral and ethical guidelines for the AI development.

2 Subjects of International Law

As mentioned earlier, the emergence of AI is bound to affect many aspects of international society. Not only will AI change the interaction between individuals, but it will also impact inter-state relations and change the dynamics between state and non-state actors. The problem is that existing international laws may not be adequate or sufficient to address the different concerns of all the actors in international society in these novel situations.

First, AI will affect state-to-state relationships. AI is a novel technology that needs to be regulated, and different governments are pursuing different policies at the moment, which needs to be monitored and managed to avoid possible conflicts. Some of the pre-existing bilateral and multilateral treaties may also need to be amended to accommodate new challenges raised by AI.

Second, AI also affects how natural persons are perceived and understood under the law. In addition to the fundamental question about the definition of "human", individual programmers and engineers will play a bigger role in the international society. While current international law does not directly apply to natural persons, as it is meant to regulate state behavior, the advent of AI may call for a universal norm or guidelines for AI related personnel. Moreover, starting from Europe, there has been strong calls to make the development of AI human-centric.

Third, similar to the previous point, voice is being raised to the effect that large corporations should also become a direct subject of international law, i.e., gaining a legal personality under international law. More than individual programmers and developers, tech companies and ICT firms will possess and exercise a significant amount of power and influence in the AI era. Considering the impact a large digital firm such as Google or Facebook can have, it is reasonable to call for an accountability mechanism to prevent misuse or abuse of that power. Each item is discussed below.

2.1 State/Government and AI

States' interest in AI has largely been focused on the interaction between AI and individuals or between the governments and individuals using AI or through AI. Few studies have been undertaken to look into legal matters between states or among states as a result of the advent of AI, an issue that concerns state-to-state relationship in the new era. Nevertheless, AI is bound to affect state-to-state relationships as well.

As the era of the Fourth Industrial Revolution has arrived, the government is required to play its pacesetting role more actively. Since the general system of society is changing, the necessity for a government to exert leadership more actively is

being emphasized at the moment. As a matter of fact, many countries have already taken various supportive and comprehensive measures for digital trade in line with the Fourth Industrial Revolution, and various types of R&D projects are planned and conducted accordingly.

As a result, quite a few bilateral and multilateral treaties need to be revisited and revised. The contracting parties now need to review international protocols, agreements and treaties to reflect the new trends. Already, there has been a case in which AI is causing trouble in bilateral relationships. Recently, the Court of Justice of the EU invalidated the US-EU treaty for data protection, holding that the pact fails to provide a sufficient level of protection for EU citizens. The decision indicates that even within the same country different branches of a government may have different views when choosing appropriate policies in the new digital world. While the decision was celebrated as a victory by the European data privacy advocates, it imperils the transatlantic commercial relationship and further complicates the global discussion of finding a consensus in the global community.[113]

Furthermore, the present trade agreements in general limit governmental interventions and restrict governmental support to a large degree. Various provisions of the WTO's SCM Agreementas well as FTAs regulate or prohibit governmental support for industries and agency intervention in the market.[114] Besides, National Treatment (NT) provisions in many trade and invest agreements make it difficult for a government to provide support domestic enterprises alone. If governments take the lead of digital trade and conduct various national projects for innovation of the entire societal system, such measures are likely to be contrary to such various agreements under the current regime of international law. In the Fourth Industrial Revolution, there are a number of items and areas that require the government and private sectors to cooperate, and conduct joint research projects. Thus, the possibility of conflicts and disputes between countries will increase due to different views on legitimate governmental support measures and illegitimate governmental intervention in the market. To address such issues, therefore, clearer standards for proper governmental roles need to be established as soon as possible. Legal mechanisms to meet expectations in the Fourth Industrial Revolution need to be secured in order to fulfill the original objects and purposes of trade agreements.

It is expected that the global community will have to review and examine international agreements and treaties which are pertinent to technical, legal, ethical and moral principles and which are affected by the new trends of AI. This is not only limited to trade agreements or investment agreements. Other treaties, such as military alliance treaties, extradition treaties, mutual legal assistance treaties, or environmental treaties, may need to be adjusted accordingly. Inter-governmental policy cooperation should be facilitated in order to draft a consistent regulatory framework and reduce tension brought by the drastic change in our societies due to AI.

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[113] Birnbaum [38].

[114] *See* of the Subsidy Agreement, Articles 1, 2, 3, 5, and 14.

2.2 Natural Persons and AI

The advent of AI will also affect how natural persons are contemplated, perceived and understood under law. Programmers, engineers and users should be subject to universal norms and international consensus while pursuing research and developments accrued to AI or operating AI for various purposes. However, at the moment there is no concrete set of regulatory frameworks to address possible problems such as liability or intellectual property related to the activities of AI. Thus, the epistemic community of experts in various fields should offer guidelines before the advent of technological singularity.

The EU has been at the forefront of establishing guidelines in this matter. Recently, it has released its policy objectives for AI policy, and it underscores that its current policy is a 'human-centric' approach.[115] In its press release of April 8, 2019, the European Commission makes no mention of the state-to-state issue aspects of the AI era. Instead, it focuses on various issues of importance from the perspective of individuals and companies such as "healthcare, energy consumption, cars safety, farming, climate change and financial risk management".[116] Some of them do relate to government regulation and policy but such regulation and policy are intended for issues and concerns of individual persons and companies such as "detect[ing] fraud and cybersecurity threats, and enable[ing] law enforcement authorities to fight crime more efficiently."[117] The European Commission mentions sets forth seven principles to establish trustworthy AI. They are:

(1) Human agency and oversight: AI systems should enable equitable societies by supporting human agency and fundamental rights, and not decrease, limit or misguide human autonomy.
(2) Robustness and safety: Trustworthy AI requires algorithms to be secure, reliable and robust enough to deal with errors or inconsistencies during all life cycle phases of AI systems.
(3) Privacy and data governance: Citizens should have full control over their own data, while data concerning them will not be used to harm or discriminate against them.
(4) Transparency: The traceability of AI systems should be ensured.
(5) Diversity, non-discrimination and fairness: AI systems should consider the whole range of human abilities, skills and requirements, and ensure accessibility.
(6) Societal and environmental well-being: AI systems should be used to enhance positive social change and enhance sustainability and ecological responsibility.

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[115] *See* European Commission, *Artificial Intelligence: Commission Takes Forward Its Work on Ethics Guidelines,* Press Release (Apr. 8, 2019), available at https://ec.europa.eu/commission/presscorner/detail/en/IP_19_1893.

[116] Ibid.

[117] Ibid.

(7) Accountability: Mechanisms should be put in place to ensure responsibility and accountability for AI systems and their outcomes.[118]

As one can see, these issues are arguably focused on issues of individuals and companies. The policy orientation is still focused on these issues as opposed to state-to-state or government-to-government issues, the one that raises concerns in the area of international law.

This approach is further confirmed in the most recent document of the EU in this regard. The EU released a White Paper on this issue in March 2020.[119] The White Paper aims to achieve the two competing objectives of promoting the use of AI and addressing the risks associated with such use.[120] This is a correct statement, indeed. It, however, does not provide details on how to find a balance between the two competing goals. Without specific details on the dividing line between the two equally important principles, states will be forced to be in the dark when it comes to a specific topic in a specific instance. Nor does the White Paper address a presumably larger issue implicating state-to-state relationship. The White Paper thus states:

> [A]clear European regulatory framework would build trust among consumers and businesses in AI, and therefore speed up the uptake of the technology. Such a regulatory framework should be consistent with other actions to promote Europe's innovation capacity and competitiveness in this field. In addition, it must ensure socially, environmentally and economically optimal outcomes and compliance with EU legislation, principles and values. This is particularly relevant in areas where citizens' rights may be most directly affected, for example in the case of AI applications for law enforcement and the judiciary.[121]

> The use of AI can affect the values on which the EU is founded and lead to breaches of fundamental rights, including the rights to freedom of expression, freedom of assembly, human dignity, nondiscrimination based on sex, racial or ethnic origin, religion or belief, disability, age or sexual orientation, as applicable in certain domains, protection of personal data and private life, or the right to an effective judicial remedy and a fair trial, as well as consumer protection. These risks might result from flaws in the overall design of AI systems (including as regards human oversight) or from the use of data without correcting possible bias (e.g., the system is trained using only or mainly data from men leading to suboptimal results in relation to women).[122]

> …both for high- risk AI applications and for other AI applications, effective judicial redress for parties negatively affected by AI systems should be ensured.[123]

The above reference in the White Paper underscores the nature of the current policy of the EU: addressing the issues arising from individuals. It omits the critical discussion on state-to-state aspect of the matter. While issues arising from individuals (such as privacy protection and data protection) are important, how to reconcile and

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[118] Ibid.

[119] *See,* European Commission, *White Paper On Artificial Intelligence—A European Approach to Excellence and Trust,* COM(2020) 65 Final (2020), p. 8.

[120] Ibid.

[121] Ibid., pp. 8–9.

[122] Ibid., p. 11.

[123] Ibid., p. 24.

harmonize non-matching views of countries in the global community is equally, perhaps more, important in managing the global community in terms of AI and new technologies. The current debates and discussion are somewhat blind to this aspect of the issue, as seen in the EU example.

2.3 Corporations and AI

It is predicted that AI can significantly change the power dynamics between state and non-state actors. As the significance of data increases, power asymmetries between state and non-state actors are fluctuating. In particular, large tech companies with the capacity to collect, store and analyze data are emerging as an important power in the international community. Giant tech companies can also contribute to law making through a process of standardization. Thus, with the growing importance of the private sector, corporate legal personhood, and companies' rights and obligations need to be addressed now at the international level. In other words, international law should evolve so that large tech companies can no longer hide under the corporate veil and escape the rule of law.

The main reason the international community needs to come up with ways to reign in large corporations using the 'rule of law' principle is that they will hold an immense amount of power in the era of AI, where data becomes the very source of immense power. Collecting data from scratch can cost a lot of money, but businesses are already collecting a significant amount of personal information through various means and occasions. Large digital firms such as Google, Facebook and Apple are only a few examples. Some estimate that Alibaba, Amazon, Google and Microsoft account for approximately 84% of the global public cloud market.[124] Consequently, the big tech firms can use their data as a form of leverage even against governments, home and abroad.[125]

The Cambridge Analytica scandal raised alarm around the world about how personal information gathered on Facebook can have an undeniable influence.[126] The recent allegation that Russia used fake accounts on Facebook to meddle in the US presidential election shows that the advent of AI has increased government susceptibility to digital attacks. While the incident in 2016 involved allegations that Russia, another sovereign state, was behind the scheme, in the future it is easy to imagine that any entity could act this way relatively easily, including multinational corporations or individuals attempting to use the collected data.

Because data holds a large amount of power, governments may need to cooperate closely with technology companies. For example, the recent debate between Apple and the FBI in the US shows clearly how the government and businesses may have conflicting interests. The US federal investigators requested Apple's cooperation to

[124] *See* Santeli and Gerdon [39].

[125] *See* Farrell et al. [40].

[126] *See* Matsakis [41].

gain access to encrypted iPhones, but Apple has been refusing to build in a backdoor to the encryption for user privacy reasons.[127] In fact, the incident started a wide debate on the cooperation between the private sector and the government. An article from the Washington Post notes:

> The legal battle between the Justice Department and Apple sparked a national debate about the competing interests of national security, law enforcement, personal privacy and giant tech firms. But the larger legal question of whether the government could force companies to provide access to electronic devices was never answered by the courts because, amid their fight, the FBI found a private firm that could access the gunman's phone. […] Attorney General William P. Barr and other law enforcement officials have said that widespread encryption on consumer devices makes those devices effectively off limits to investigators even with a court order, hampering investigations into terrorism, child abuse and other crimes. "We have the greatest respect for law enforcement and have always worked cooperatively to help in their investigations," Apple's statement says. "When the FBI requested information from us relating to this case a month ago we gave them all of the data in our possession and we will continue to support them with the data we have available."[128]

The increasing power of data also signifies that the public and private sectors will be in competition for talented individuals. Given that a large portion of AI technologies were developed in the private sector, the public sector has relatively fewer human resources. National governments will have to compete with private entities and businesses to recruit appropriate human expertise. The intelligence community of the US government is already well aware of the heightened competition for scarce human resources.[129] Since AI and data management skills are in high demand but in short supply,[130] the competition between the public and private sectors for competent human resources will be fiercer as we move forward. The problem is that while the AI landscape is very complex and competitive, many public organizations and governments lack incentive structures for their employees for innovation.[131] As a consequence, the public sector is at a considerable disadvantage in the competition for human resources against their private sector counterparts. This implies that the public sector has to offer a new incentive system to recruit new talents or cooperate closely with the private sector.

One way to level the playing field between the public and private sector is to clarify the legal responsibilities of corporations at the international law level. The business world, especially those sectors related to the new technology, changes at a much faster pace than governmental sector. As the tech companies accrue more power through data, national governments will have more trouble controlling them in a fast-changing environment.[132] Some employees at digital companies such as Google and

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[127] *See* Nicas and Benner [42].

[128] Barrett [43].

[129] *See* Office of the Director of National Intelligence, *The Aim Initiative—A Strategy for Augmenting Intelligence Using Machines* (2019), p.iv, available at https://www.dni.gov/files/ODNI/documents/AIM-Strategy.pdf.

[130] *See supra* note 283.

[131] Ibid.

[132] *See* Farrell [40].

Microsoft have already expressed opposition to government use of their technology for defense and surveillance purposes.[133] In this context, the international community needs to find a consensus on how to regulate corporate responsibilities related to AI application and utilization in order to guarantee responsible and equitable use of the new technology. It will be an important topic to be tackled in future treaties and agreements in this field. It will also have to be reflected in already existing treaties and agreements through their amendment when proper occasions arise.

3 Objects of International Law

Another aspect that the international law needs to take into consideration is how to regulate AI itself. It is to perceive AI as one of new objects of international law. There are many new objects emerging in the international plane, which require regulation by international law, none could be compared to AI in terms of the urgency and strong demand in the global community. As much as it affects existing legal norms such as relationship to government function and recognition of legal personhood, AI itself has become a novel item that needs to be regulated by new legal norms including international law. While it is true that AI raises a plethora of questions in a variety of issues and areas, such as international human rights law, international humanitarian law, international environmental law and international criminal law, etc., this section will focus on AI as a new item in the digital economy. This is so, because the endeavor shows well the changing nature of the existing items and targets of current treaties - that is, objects of international law - as a result of the fast arrival and integration of AI technologies and machines.

At its core, the digital economy is different from the conventional economy where goods and services are traded separately and regulated independently. First, such a binary distinction between goods and services becomes difficult, and now almost impossible, in a digital economy. A digital product equipped with AI will be both a tangible product but also a service provider. The collapse of the previous demarcation line fundamentally means that the current international trade laws, which is sharply divided between trade in goods and trade in services, needs to be updated. IIAs are also implicated as international investment law regime is also premised on the conventional mechanism of making investment, which is in turn closely related to conventional goods manufacturing and services supply. Second, the digital economy also blurs territorial boundaries. The term "digital economy" implies, by definition, the elimination, or at least weakening, of the existing national boundary lines. The advent and spread of AI underpins this new characteristic of the digital economy, because AI is based on data and information, and they are collected, processed and utilized without attachment to national boundary. As AI is not confined to territories, cross-national, cross-border (in fact, no-border) transactions will be more frequent, which will inevitably touch upon the sensitive issue of extraterritoriality. How to

[133] *See* Ghaffary [44].

understand and regulate extraterritorial application of national jurisdiction in the new context. One could even consider the establishment of an extraterritoriality zone of personal information when personal information of nationals are stored in other country's territory through the clouding facilities located there. As for that information, a state of nationality and a state with computing facilities might be permitted to exercise jurisdictions jointly or in sequence. As personal information collection, storage and utilization is by far the most sensitive topic in the current debates of digital economy and AI, diverse alternative options could be explored in prospective global discussion.

The subsections are organized as follows: First, AI is described as a new item in the international market. Then, as possible future treaties, a trade agreement on digital trade and a treaty on extraterritoriality zones of personal information are addressed.

3.1 New Items Regulated by International Law

The emergence of AI signifies that there are new items to be regulated by international law. New products that have not existed before, such as autonomous weapons systems, drones, and automated vehicles, among others, require new regulations. As the development of AI accelerates, more novel forms of robotic applications can emerge, calling for updates of international law. This means that new digital (or data-heavy, data-loaded) goods and data itself are new items that need to be regulated by international law.

The international trade regime is no exception. Global trading norms have traditionally adopted a bifurcated approach: goods trade and services trade. These two still constitute two main pillars of trade agreements. Now, a fundamental change is in the offing. The arrival of AI and IoT is blurring the conventional line between goods and services. The two are now combined, converged and integrated in one medium or in a single product. Goods provide services and services in turn put more goods up for sale. Consider driverless cars, smart home appliances, or medical staff-connected medical equipment. They are goods, outwardly speaking, but then provide services that only used to be provided by human beings. Goods and services now come in the same package to give consumers integrated benefits.

What enables this very convergence is none other than data. 'Big data', as it is called. Products (or mediums) and their handlers collect data and process it for further commercial or industrial utilization. Data so processed then forms a new platform for the sale of more goods and services. So, data is the neurons and blood of the newly integrated network of services and goods. It is no wonder that who stores and controls data has become critically important in terms of business and dominance in the market.

The need for new legal guidelines to protect personal data is becoming an increasing reality. In the digital economy, such information constitutes a critical platform for production, sales and marketing of new product and services. To stay competitive in business and in market, corporations must be keen to collect and

process personal information. On the contrary, individuals become more vulnerable to the information gathering drive and schemes that are devised and implemented ever more actively and robustly. Thus, the market naturally demands new legal guardrails.

Thus, AI sheds new light on digital goods and data as a new item in international trade. New goods, services, and trade methods that our society has never experienced or expected are appearing and will continue to do so. It will be required, therefore, to examine and adjust international trade norms to regulate them. Provisions of existing trade agreements concerning goods and services trade are applicable specifically to cross-border trading that is fragmented, and thus many of them are found to be no longer applicable to trading new types of goods and services supplied or consumed in the 'global market' with no boundaries.

In the integrative, combining, and complementary situations of digital trade unfolding at the moment, distinguishing goods from service, or vice versa, is practically meaningless and is now legally challenging.[134] The supply of goods and services and the consumption of both occur simultaneously. Not only do the two forms of trade overlap in time, but also goods and services are sold and consumed as an inseparable set. For example, goods themselves act as a sort of 'service supplier.'[135] Likewise, provision of service has become a core attribute or function of the corresponding goods.[136]

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[134] Characteristics of digital trade are based on its nature that free flows of information and data are to be secured. Data generated from digital goods and services is of economic values as it is. Such data is also the very core of goods and service. After all, goods and services are integrated with 'data' as the connecting point, and the boundary between them becomes faint gradually. *See* Ciuriak [45].

[135] Paragraph (g), Article XXVIII (Definitions) of GATS defines "Service Supplier" as below:

(g) "service supplier" means any person that supplies a service;[12.]

(*footnote* 12: Where the service is not supplied directly by a juridical person but through other forms of commercial presence such as a branch or a representative office, the service supplier (i.e. the juridical person) shall, nonetheless, through such presence be accorded the treatment provided for service suppliers under the Agreement. Such treatment shall be extended to the presence through which the service is supplied and need not be extended to any other parts of the supplier located outside the territory where the service is supplied.).

[136] GATT 1994 does not include a clear definition of 'goods' either. The recognition that characteristics of goods are relatively evident seems to be the implicit basis. One indicator that can be referred to as a direct basis for whether an item can be viewed as goods is the HS Code that is issued and included in Schedules of Concession prepared in accordance with Art. II of GATT. If a certain item obtains the HS Code and is listed in Schedules of Concession, it clearly indicates that the item is regarded as goods. In this regard, the note and memorandum of GATT below present a meaningful guideline as follows:

GATT Ad Article XVII paragraph 2.

The term "goods" is limited to products as understood in commercial practice, and is not intended to include the purchase or sale of services.

Understanding on the Interpretation of Article XXVIII.

of the General Agreement on Tariffs and Trade 1994.

4. When a tariff concession is modified or withdrawn on a new product (i.e. a product for which three years' trade statistics are not available) the Member possessing initial negotiating rights on the tariff line where the product is or was formerly classified shall be deemed to have an initial negotiating right in the concession in question. (…) For the purposes of this paragraph, "new

The problem is that the present trade regime has different rules for trade in goods and in services. For instance, as is well known, service trade is decided in accordance with the paragraph of "Market Access" in Schedules of Specific Commitments.[137] Whether or not each country that is an agreement party opens its market of certain service depends on what promise the country makes regarding each area of service. In contrast, every area in goods trade is opened with restrictions on import/exports permitted only under strict conditions.[138] As such, goods trade and service trade take a different approach from each other in terms of market access. In the context of this new trade form where goods and services are supplied and consumed simultaneously, such a different approach causes problems between the two. The market access of goods may be blocked or permitted through provisions regarding market access that are applicable to service trade.

Even the application of a non-discrimination principle is different for goods and services. The principle of non-discrimination in terms of most favored nation treatment and national treatment constitutes the core tenet of present trade agreements.[139] Accordingly, the principle of non-discrimination is specified respectively for goods trade and service trade among present trade agreements.[140] In the event that goods and services are provided simultaneously in digital trade, a discriminative measure on one side leads to a discriminative measure on the other side. A violation of most favored nation treatment for certain goods results is a violation of most favored national treatment for service that is supplied through the goods. Likewise, it is highly probable that a violation of national treatment for a certain service or service supplier leads to a violation of national treatment for goods that are the platform of service supply. It should be noted in this regard that present goods agreements and service agreements adopt different norms from each other for this principle of non-discrimination. GATS allows deviation from the principle of the most favored nation treatment when certain conditions are met.[141] Furthermore, it is made clear that basically, national treatment

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product" is understood to include a tariff item created by means of a breakout from an existing tariff line.

In other words, these provisions as well suggest that there need to be criteria of judgment based on the unique characteristics of goods distinguished from those of service as well as Schedules of Concession.

[137] *See* Article XVI of GATS.

[138] *See* Article XI of GATT.

[139] The principle of non-discrimination has been firmly established as a basic direction for free trade agreements since 1860 when the Cobden-Chevalier Treaty was concluded between France and the U.K. Which contributed significantly to the spread of free trade in the nineteenth century. The WTO system advocates the principle of non-discrimination as well as promotion of market access and transparency as the guiding principles. The principle of non-discrimination embraces both most favored nation treatment and national treatment.

[140] The GATT specifies, as the general obligation, most favored treatment in Art. I and national treatment in Art. III respectively. In contrast, the GATS specifies, as the general obligation, most favored treatment in Art. II, but national treatment in Art. XVII is a specific obligation applicable only to areas stated in Schedules of Concession, according to the conditions and restrictions in each country's Schedules of Concession.

[141] *See* Article II and Annex 2, GATS.

is in accordance with Schedules of Specific Commitments.[142] This is quite different from the case of the General Agreement on Tariffs and Trade 1994 (hereinafter GATT 1994) that sets forth basic norms regarding goods trade. With regard to goods trade, Paragraph 8, Article III of GATT specifies exceptional cases of important national treatment that are not stated in the GATS.[143] These exceptions are related to measures for governmental procurement. Such differences between goods trade and service trade in the context of the principle of non-discrimination causes confusion in a trade form where both items are provided in an integrated manner. There is always a possibility that exceptions in service trade lead to exceptions of goods trade and vice versa. Thus, it is difficult to evaluate the principle of non-discrimination accurately unless there are common norms and standards applicable to both goods and service.

Another legal issue that arises with the digital economy is that the evaluation of market conditions becomes more complex and elusive. Trade agreements emphasize and sometimes stipulate reference to relevant markets in various aspects. For instance, present agreements regarding goods trade include various indicators to evaluate the condition of market competition (e.g., evaluation of 'competition in the market' in provisions regarding national treatment; evaluation of 'material industrial damage' in anti-dumping agreements; evaluation of benchmarks in subsidy agreements; and evaluation of 'serious industrial damage' in safeguard agreements). Such market-related reference points and indicators are key elements to the operation of relevant provisions, and ultimately, they are a deciding factor in determining the conformity of a challenged measure to trade agreements. Basically, however, the 'market' stated in such trade agreements means each country's fragmented domestic market.[144] Such markets are different from this new form of global market where goods are supplied and consumed simultaneously around the world.[145] Present provisions, therefore, involve inherent limitations in reflecting the market condition properly and accurately in the new area of digital trade. The more critical problem is that the new situation is expanding and deepening in many corners of business activities and social lives, widening the already existing legal fissure.

As situations change drastically, significant change is required to the basic framework of a certain agreement itself as well. For instance, anti-dumping investigations according to the WTO anti-dumping agreement need a fundamental new review. The anti-dumping system is a concept on the assumption that the domestic market is separate from the export market. The key factor is to determine whether there is any unjust act of pricing by comparing prices in the domestic market (Normal Value, hereinafter NV) and prices in the export market (Export Price, hereinafter EP).[146]

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[142] *See* Article XVII, GATS.

[143] *See* Paragraph 8, Article III, GATT.

[144] One exceptional case is provisions stating the global market. For instance, Article 5 and Article 6 of a subsidy agreement examine whether or not a measure of subsidy grant by the government of a sued country causes negative impact on competitive relations in the global market.

[145] Unlike the existing industrial economy where competition within a region or industry were dominant, in the digital economy, temporal and physical limitations are far less or meaningless and global competition is common. *See* Ha [46].

[146] *See* Article 2 of the Anti-dumping Agreement.

In order to apply the concept of 'anti-dumping,' first of all, it is necessary to distinguish the domestic market from the export market (the domestic market of the trade country). In the event that there is no distinction between the domestic and global markets but a unified global market is operated where prices are decided individually based on many different variables, there is no room to apply an anti-dumping agreement that is applied based on the distinction between the two markets and the difference in prices between them. Basically, therefore, it is inappropriate to apply the existing anti-dumping system to the new digital trade system.

A similar issue is raised in the context of service trade as well. In the context of service trade, whether there is negative impact on competitive relations in the market is an important element in judging a violation of a trade agreement. For instance, there may be provisions requiring to take a measure into consideration concerning the domestic market conditions when a domestic regulation is to be introduced[147] or provisions requiring regulations to prevent anti-competitive practice in the domestic market.[148] The 'market' stated here as well basically means the domestic market of each country that is an agreement party. It is doubtful whether this model is still valid since the supply and consumption of service are practiced in the same form simultaneously in the global market.

Once again, this reveals the limitation of the current system that approaches trade agreements from a binary perspective based on the distinction of the domestic goods market from the services market. In reality, there is no legal provision that regulates both markets at the same time uniformly and even consistently. Even if there are market-related provisions, there is an inevitable gap between norms and reality unless they somehow reflect fast changing market conditions, industrial conditions, and up-to-date characteristics of trade and businesses. Such unrealistic market standards will eventually raise questions about the objectivity and accuracy of major indicators included in the existing agreements. As a result, most likely there will be a continual loop of domestic disputes and cross-border disputes, be it state-to-state context or state-to-business context.

When it comes to items of digital trade that is part a single global market, unlike ordinary goods and service, indicators in existing norms based on the assumption of national borders are no longer applicable or at least need to be modified to a large extent. The concept of 'market' itself that is assumed in trade agreements is changing drastically from a national market to a single market, from an analogue market to a digital market, and from a physical market to a cyber market. New indicators reflecting such changes need to be introduced to trade agreements.

In sum, the growing digital trade accelerates the introduction of an integrated global market. The present agreement system that assumes fragmented domestic markets of individual countries is inconsistent with the nature of digital trade. For example, the present trade remedy system assumes fragmented domestic markets and estimates material damage or serious damage to a certain domestic market. If a digital enterprise suffered material damage in the Japanese market but achieved

[147] *See* Article V of GATS.

[148] *See* Article IX of GATS.

rapid sales increases in the Chinese market, it is uncertain whether that digital enterprise or industry sector really suffered damage in the end. Likewise, price comparison is a challenge as well when it comes to anti-dumping investigations, and so is determining whether a subsidy grant has resulted in financial benefits. It is also difficult to determine in advance whether drastic increases of imports will be foreseeable or predictable for a safeguard measure. This is why the present trade remedy system (arguably a stable source of trade disputes) cannot adequately embrace digital trade items either.

In view of such limitations, possibly the concept of 'one global market' offers an alternative: With a reasonable estimate of prices in the global market calculated, the prices of a problematic country are compared with the estimate. For instance, items are classified into major categories, and the average prices of goods under each category traded in the global market in a certain year (e.g., 2018) are calculated to estimate a type of international official prices. Average prices of goods under each category that were sold in the year of 2018 in the problematic country (e.g., China) are then calculated and compared with international official prices under the same category. In this process, several items are reflected, if necessary, and adjustments are made after comparison. Thus, the AI era brings the concept of one global market as a new object of international law.

The long standing OECD shipbuilding agreement negotiation, which considers this comparative method based on the concept that the international shipbuilding market is one global market, presents implications in this regard. Many OECD members have long discussed the conclusion of a new shipbuilding agreement for the purpose of addressing trade disputes in the shipbuilding industry. In this process, countries that participated in this negotiation kept in mind that the shipbuilding market is one global market. They attempted to introduce the concept of substantial fair competition into a type of global market. This attempt has led to the discussion on the issue of 'price regulation' in a kind of global market. For example, the minimal price or official price of ships traded in the global market is fixed, and vessels that are traded at a lower price are subject to restrictions.[149] Since shipbuilders that save production costs in such ways as a subsidy grant, dumping bidding, and other unjust competition may compete at lower prices in the market of the importing country, and so such acts are to be regulated on a global scale by fixing the minimum price. Sales at a price lower than 'the marginal cost' or 'the average total cost' are spoken of here as 'sales at a market-distorting price.' This is a different concept from those of low-price sales in existing anti-dumping agreements or financial benefits in subsidy agreements. Although the new shipbuilding agreement was terminated in 2011 due to the dispute between the EU and China, this approach presents implications regarding digital trade items that are treated on the basis of one global market. When it comes to a trade form where both goods trade and service trade are practiced in a complex

[149] OECD, *Report on the Final Status of the Negotiations on a New Shipbuilding Agreement*, C/WP6 (2007), p. 3; OECD Special Negotiating Group, *New Shipbuilding Agreement-Consolidated Draft Text*, Working Paper 5/REV1 (Sep. 2005).

manner, it may be a new alternative to establish a comparative reference point of prices.

3.2 Treaty Conclusion

As AI raises new and important issues in international law, multilateral treaties with specified legal status, legal commitments with legally binding power, and compliance measures could be discussed to fill the legal vacuum. Of course, the level of formalization and institutionalization of such a treaty is subject to discussion. It may be that multiple separate treaties are needed to address the various issues related to AI, since regulating AI requires international cooperation in numerous fields. In this regard, this section discusses a new legal regime to address the growing digital trade, and establishment of extraterritoriality zones of personal information as two topics for prospective treaties because they present pressing needs to find global consensus.

Before diving into the specific sub-topics, it is worth mentioning concrete examples. For instance, Singapore has launched a series of negotiations to establish digital economy agreements (hereinafter DEAs). The Ministry of Trade and Industry of Singapore explains that the DEAs will contribute to alignment of rules and standards of digital systems and promote cooperation in the emerging sectors such as the AI.[150] The first DEA concluded by Singapore was the Digital Economy Partnership Agreement (hereinafter DEPA) with Chile and New Zealand.[151] In August 2020, Australia and Singapore signed Singapore-Australia Digital Economy Agreement (hereinafter SADEA),[152] which adds a new chapter to the existing Singapore-Australia Free Trade Agreement while also mindful of the Comprehensive and Progressive Agreement for the Trans-Pacific Partnership (hereinafter CPTPP) to which the two countries are parties. It also includes seven Memoranda of Understanding, one of which directly addresses cooperation on AI.[153]

The DEPA and SADEA explicitly mention AI with stand-alone articles and call for close cooperation between the contracting parties, which stipulate as follows, respectively:

Article 8.2: Artificial Intelligence of DEPA

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[150] *See* the Singapore Ministry of Trade and Industry website, available at https://www.mti.gov.sg/Improving-Trade/Digital-Economy-Agreements.

[151] Full text available at https://www.mti.gov.sg/-/media/MTI/Microsites/DEAs/Digital-Economy-Partnership-Agreement/Text-of-the-DEPA.pdf.

[152] Full text available at https://www.mti.gov.sg/-/media/MTI/Microsites/DEAs/Singapore-Australia-Digital-Economy-Agreement/Singapore-Australia-Digital-Economy-Agreement.pdf.

[153] Memorandum of Understanding between the Government of the Republic of Singapore and the Government of Australia on Cooperation on Artificial Intelligence, available at https://www.mti.gov.sg/-/media/MTI/Microsites/DEAs/Singapore-Australia-Digital-Economy-Agreement/MOUs/MOU-on-Cooperation-on-Artificial-Intelligence.pdf.

1. The Parties recognise that the use and adoption of Artificial Intelligence (AI) technologies have grown increasingly widespread in the digital economy.
2. The Parties recognise the economic and social importance of developing ethical and governance frameworks for the trusted, safe and responsible use of AI technologies. In view of the cross-border nature of the digital economy, the Parties further acknowledge the benefits of developing mutual understanding and ultimately ensuring that such frameworks are internationally aligned, in order to facilitate, as far as possible, the adoption and use of AI technologies across the Parties' respective jurisdictions.
3. To this end, the Parties shall endeavour to promote the adoption of ethical and governance frameworks that support the trusted, safe and responsible use of AI technologies (AI Governance Frameworks).
4. In adopting AI Governance Frameworks, the Parties shall endeavour to take into consideration internationally recognised principles or guidelines, including explainability, transparency, fairness and human-centred values.

Article 31 Artificial Intelligence of SADEA

1. The Parties recognise that the use and adoption of Artificial Intelligence ("AI") technologies are becoming increasingly important within a digital economy offering significant social and economic benefits to natural persons and enterprises. The Parties shall cooperate, in accordance with their respective relevant policies, through:
   (a) sharing research and industry practices related to AI technologies and their governance;
   (b) promoting and sustaining the responsible use and adoption of AI technologies by businesses and across the community; and
   (c) encouraging commercialisation opportunities and collaboration between researchers, academics and industry.
2. The Parties also recognise the importance of developing ethical governance frameworks for the trusted, safe and responsible use of AI technologies that will help realise the benefits of AI. In view of the cross-border nature of the digital economy, the Parties further acknowledge the benefits of ensuring that such frameworks are internationally aligned as far as possible.
3. To this end, the Parties shall endeavour to:
   (a) collaborate on and promote the development and adoption of frameworks that support the trusted, safe, and responsible use of AI technologies ("AI Governance Frameworks"), through relevant regional and international fora; and
   (b) take into consideration internationally-recognised principles or guidelines when developing such AI Governance Frameworks.

Other relevant articles on AI in these agreements include, but not limited to, articles on personal information protection,[154] general cooperation,[155] and cybersecurity.[156]

[154] *See* Article 4.2 of DEPA, Article 17 of SADEA on Personal Information Protection.

[155] *See* Article 33 of SADEA on Cooperation.

[156] *See* Article 5.1 of DEPA, Article 34 of SADEA on Cybersecurity.

The SADEA and CPTPP also contain provisions regarding data location and cross-border transfer of data,[157] while promoting a free flow of data for innovations[158] on the one hand, and protecting sovereign rights of a state to enforce regulations pursuant to its own policy objectives, on the other hand. And yet, the dividing line between the two competing objectives is by no means clear and is left to future practice and clarification. With this critical legal standard still being rendered unclear, disputes between states are bound to arise and intensify considering the fast increase and spread of digital trade and AI.

3.2.1 Digital Trade

As mentioned earlier, the growing digital trade calls for a new set of trade norms reflecting the integrative nature of the new technology. Of course, merely 'updating' the current trade norms can address digital trade issues to some extent. For instance, to prevent newly emerging, various digital goods from falling prey to the non-tariff barrier (hereinafter NTB) or trade remedies unjustly, which are an important issue in the global society, relevant institutions need to be adjusted in line with the changing situations of the times.[159] Particularly now that protectionism is being strengthened around the globe, the necessity for update and preempting possible NTBs could never be overemphasized. Besides, the understanding of unique problems and peculiar characteristics that result from digital trade is also being deepened and expanded.[160] It seems certain that with regard to such problems and chracteristics, discussions and negotiations between states or among states will continue to increase and intensify.[161]

Unfortunately, present trade agreements are ill-equipped to fully address digital trade issues. The current system adopts a 'dichotomous' approach, regulating goods

[157] *See* Article 4.3 of DEPA, Article 23 of SADEA on Cross-Border Transfer of Information by Electronic Means.

[158] *See* Article 9.4 of DEPA, Article 26 of SADEA on Data Innovation.

[159] New digital goods are already a major issue in existing areas of trade dispute such as tariff classification, rules of origin, technical regulation, anti-dumping inspection, countervailing duty investigation, and so forth. Manufacturing and selling of such goods are on the basis of cooperation and specialization among various entities in many countries, and it also involves advent of new goods that have not existed. Accordingly, applying existing trade norms increases variables and complicates the judgment to be made. Such situations also cause disputes and disagreements among relevant countries.

[160] In 2018, the US National Trade Estimate Report on Foreign Trade Barriers highlighted digital trade barriers. This report classifies digital trade barriers among major countries as (1) cross-border transfer and localization of data; (2) technical regulations; (3) restriction to internet-based service; and (4) digital products, encryption, etc. In general, this report focuses on restrictions of importers on US 'goods' in technical perspectives. *See* USTR, *Key Barriers to Digital Trade 2017, 2018 Factsheets*, available at https://ustr.gov/about-us/policy-offices/press-office/fact-sheets; Also, *see* González and Ferencz [47].

[161] Ibid.

and services separately.[162] The following two types of agreements have been introduced and administered under the current trade regime since the inception of the GATT in 1947: goods trade agreements that apply the same rules to all member countries under existing trade norms (e.g., GATT 1994); and services agreements that govern each member country only to the extent that it has agreed to (e.g., GATS). Accordingly, contracting parties to the current trade agreements adopt the following two separately and independently: Schedules of Concession that specify the tariff rate for each item in relation to goods trade[163]; and Schedules of Specific Commitments in the services sector that specify the level, manner and conditions of liberalization of the services market.[164] These two norms are separate from and independent of each other, and basically, there is no interaction or contact point of the two from the 'normative perspective.' [165] Goods and services are thus regulated separately. This binary structure underpins not only the WTO Agreements but also FTAs. Goods and services have been understood to be of a different, separate, and independent trade mode to each other. The fact that trade agreements apply the two sets of different norms separately and independently to goods and services tradehas been confirmed by WTO panels and the Appellate Body.[166]

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[162] Under the WTO system, goods trade is regulated according to Annex 1A and service trade under Annex 1B respectively. Annex 1A specifies multi-lateral trade agreements concerning 12 goods trade cases in total including GATT 1994, anti-dumping agreements, agreements for subsidies and offsets, agreements of technical barriers, and so forth. Annex 1B includes the GATS. This is a separate annex introduced in consideration of intangible characteristics of service that are different from those of tangible goods. Although the GATS contains no clear definition of 'service,' it specifies that there are 4 "modes" of "service supply": cross-border transaction, overseas consumption, commercial presence, and presence of natural persons. *See* para. 2, Article I of GATS.

[163] *See* Article II of GATT 1994.

[164] *See* Article XVI of GATS.

[165] In the event that the same governmental measure affects goods trade and service trade, both relevant goods agreements and service agreements may be subject to that measure. However, this does not mean that each relevant agreement regulates both goods and service. Rather, it merely means that such measures affect both trade items (just as if a trade item can be in conflict with both GATT 1994 and TBT).

[166] The WTO panel and appeal authorities make it clear that basically, the GATT and GATS are agreements aiming at separate targets but both can be applied to the same target to be regulated. In the case of *EC-Bananas III,* for example, the GATT and GATS were both applied since issues were raised regarding both bananas, which are goods, and banana marketing and distribution, which is service. This was the same with the case of *Canada-Autos*. In the case of *Canada—Periodicals*, the panel concluded that the GATT and GATS could coexist and duties under these two agreements could be applicable as long as they are counted as WTO agreements with no hierarchy applied between them (WTO Panel Report, *Canada—Certain Measures Concerning Periodicals*, WT/DS31/R (adopted 1997.03.14.), paras. 5.17–5.19). It was highlighted, however, that in such cases as *Canada-Autos*, the two agreements focused on different objects. The appeal authorities that handled this case specified that the GATT and GATS aim to regulate different objects, and that as the same measure is considered by the two agreements, the focus would be different (WT/DS139, 142/AB/R, paras.159–161).

In summary, the GATT and GATS may be applied to the same object, but in that case as well, they focus on different aspects of trade. In academic circles, there are varied opinions on the relation between the GATT and GATS. Gaffney claims that two agreements shall not be applied to certain

Under the current norm, from the perspective of each government, determining whether a trade dispute is a dispute of goods or service is crucial.[167] This is because different laws may be applied depending on the determination. Furthermore, the matter of deciding which of the two norms will be applied has become a major variable that decides the conformity of a dispute to the relevant agreement. In many WTO disputes, the countries directly concerned often express quite different opinions with regard to whether the measure is in fact in relation to goods or services.[168]

Although the fact that goods trade and services trade can affect each other is recognized in current trade agreements and jurisprudence,[169] it does not mean that their individual identities and respective independence are to be denied. The rule so far is that they are different forms of trade, so different rules apply. Sometimes, however, a particular form of trade may affect both goods trade and services trade, in which case the two different rules may apply as necessary. Critically, what is missing in this dialogue is when an item is not to be categorized to goods or services from the beginning and when the item offers a new type of trade combining the two and beyond the two. This is the area where the rule is still missing as the global community had not encountered these issues when WTO Agreements or FTAs were concluded. One example is electricity. Electricity is subject to the GATT 1994 since it is given an

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aspects of one measure. In other words, one agreement needs to be selected based on the purpose and effect of that measure. In contrast, Pauwelyn argues that a more relevant one out of the two agreements shall be applied in a special-law (*lex specialis*) approach. However, the validity of this approach is controversial because the two agreements are not in the relation of a special law and a general law. After examining these two perspectives, Vranes presented the opinion that the matter of the two agreements' overlapping needs to be determined case by case. *See* Vranes [48].

Not to mention what perspective needs to be supported, such existing discussions as well present many different views on whether there are occasions where goods agreements and service agreements are overlapped in their applications, and in such cases, how such overlapped applications can be justified. Besides, such existing discussions take into no consideration new types of trade combining the two items stated above. As stated by the panel of *Canada-Autos* in this regard, the appeal authorities of *EC-Bananas III* once judged that measures regarding goods trade are regulated by the GATT while measures regarding service trade are regulated by the GATS, but there could be the third measures to which both agreements are applicable (WT/DS139, 142/R, para. 10.233). This statement also emphasizes that the same measure may be applied to both agreements because its nature reflects their characteristics. It needs to be noted, however, that this does not mean 'the new third mode of trade' that bears characteristics of both goods and service. The new mode of trade that results from the fourth Industrial Revolution stated in this work is this 'third mode of trade.'.

[167] In the case of *China-Distribution Right*, for example, the US and China were in confrontation with each other concerning whether the sound recording distribution services of China are included in the Chinese service concessions. The WTO panel judged that the attributes of the contents included in the medium, not the physical medium itself, would be of importance. *See* WTO Panel Report, *China—Measures Affecting Trading Rights and Distribution Services for Certain Publications and Audiovisual Entertainment Products*, WT/DS363/R (adopted 2009.08.12.), paras. 7.1173–7.1181.

[168] Ibid.

[169] *Canada-Autos* (AB) recited *EC-Bananas III*, confirming that service trade and goods trade could affect each other. *See* WTO Appellate Body Report, *Canada—Certain Measures Affecting the Automotive Industry*, WT/DS139, 142/AB/R (adopted 2000.05.31.), paras. 159–162.

Harmonized System (HS) number indicating that it is a type of goods.[170] It can be measured and sold in the market with the standard measurement with a designated price. On the other hand, electric power generation, transmission, distribution, and supply are all regarded as a type of service. In fact, this is the most important service for the national economy, and many countries run the electricity service through government-related agencies or with otherwise strict governmental regulation. With the operation of various forms of service supply, electricity is provided to a consumer at the end of the day.[171] In short, electricity itself is goods, but it is conveyed from generation to final usage via a long chain of services providers. Hence, regulations on electricity to be traded involve ripple effects on both goods and services sectors. Of course, this does not mean that electricity as goods somehow becomes a service or electricity supply as a service somehow have characteristics of goods. Goods and services are two separate boxes of trade agreements. Electricity provision in the market can be divided into sub-segments—some goods segments and some services segments—and these sub-segments can fall on respective boxes accordingly. There is, however, no single box that accommodates something or some item that have both characteristics in its portfolio and that have new characteristics that go beyond the both characteristics.

Such matters are directly related to the applicable scope of services agreements. Article I of the WTO GATS indicates its wide range of applicability. According to the relevant provision of this agreement, the GATS "is applicable to every measure of member countries that 'affects' service trade."[172] Every measure that makes known its purpose by the member country to regulate service trade directly or that affects service trade, directly or indirectly even if it does not indicate such a purpose is subject to the GATS regulation. This provision shows the wide applicability of the GATS.[173] This is also suggested in the "Definition" provision of the GATS. According to this provision, there can be various forms of governmental measures that affect service

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[170] In the case of *Canada-Renewable Energy*, None of Canada, EU, and Japan protested against the idea that electricity is an item of 'goods' that can be traded. The panel that handled this case recognized unique characteristics of electricity but treated it as goods to which the GATT was applicable. *See* WTO Panel Report, *Canada—Certain Measures Affecting the Renewable Energy Generation Sector*, WT/DS412, 426/R (adopted 2012.12.19.), para. 7.11.

[171] Miranda [49].

[172] Paragraph 1, Article I of the GATS stating its scope of application specifies: "This Agreement applies to measures by Members affecting trade in services." In other words, any measure of a member country that "affects" international service trade directly or indirectly is subject to the GATS regulation. In this regard, the panel handling *EC-Bananas* stated as below:

No measures are excluded a priori from the scope of the GATS as defined by its provisions. The scope of the GATS encompasses any measure of a Member to the extent it affects the supply of a service regardless of whether such measure directly governs the supply of a service or whether it regulates other matters but nevertheless affects trade in services." Therefore, it considered that the EC banana import regime is not a priori excluded from the scope of the GATS.

WTO Panel Report, *European Communities—Regime for the Importation, Sale and Distribution of Bananas*, WT/DS27/R/ECU (adopted 1997.09.25.), paras. 7.279–7.286.

[173] WTO Panel Report, *United States-Measures Affecting Cross-Border Supply of Gambling and Betting Services,* WT/DS285/R (adopted 2004.11.10.), paras. 6.250–6.255.

trade among member countries.[174] For instance, it is indicated that governmental measures affecting service trade include "measures […] in respect of the access to and use of […] services."[175] Even if a measure that aims to restrict goods trade affects service trade as well, it does not become subject to a separate service agreement automatically.[176]

However, this merely means that an agreement on goods and an agreement on services may have different characteristics and be applied separately, and thus that they may be applicable to a certain transaction or measure simultaneously. The current trade agreement system regulates situations where goods and services affect each other on the assumption that they are provided in a separate manner. It does not consider situations where goods and services are supplied, sold, and consumed simultaneously in an integrated manner.

Thus, it is necessary to come up with a new legal paradigm that can efficiently regulate a new form of digital trade embracing both goods trade and service trade properly. It is expected that such aspects will be examined in the future in the process of negotiation and the conclusion of various trade agreements. However, it will require a long period of time and preparation. Once the basic direction is established, more efficient discussion will be possible in the practical steps ahead.

Recently concluded FTAs between countries include some regulations concerning digital trade in response to such changing environments.[177] However, these are more like one-off, temporary norms and have yet to reflect comprehensive reviews in line with the general trend of digital trade. Instead, such recent norms introduce certain special rules on items with digital characteristics among existing goods or present

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[174] Article XXVIII (c) of the GATS defines "governmental measures of a member country that affect service trade" as below:

(c) "measures by Members affecting trade in services" include measures in respect of.

(i) the purchase, payment or use of a service;

(ii) the access to and use of, in connection with the supply of a service, services which are required by those Members to be offered to the public generally;

(iii) the presence, including commercial presence, of persons of a Member for the supply of a service in the territory of another Member.

[175] Ibid. (Article XXVIII (c) (ii), GATS).

[176] In the case of *EC-Bananas*, the EC claimed that the measure was for goods trade and thus the GATS was not applicable, but the panel and appeal authorities dismissed. According to the panel's judgment, even if a measure is to regulate goods trade directly or affects goods trade indirectly, the GATS is applicable to it as long as the measure affects service trade. *See* Vranes [48].

[177] Among 17 FTAs that Korea has concluded (come into effect, and been declared as finalized) as of July 31, 2018, 11 have a separate chapter regarding electronic commerce including the Korea-Central America FTA that was most recently signed (Feb. 21, 2018), Korea-Singapore FTA (effective since Sep. 01, 2006), Korea-US FTA (effective since May 01, 2013), and Korea-Canada FTA (effective since Jan. 01, 2015). However, these briefly present special provisions concerning electronic commerce and privacy protection. Even certain provisions containing definitions of 'digital goods' or 'digital service' specifically state in a note that such provisions of definition are not a decisive judgment that such digital transactions are regarded as a kind of goods trade or service trade. In general, they reflect the perspective on digital trade still in the initial stage. For details of each FTA, visit the following website: http://www.fta.go.kr/main/.

provisions related to electronic commerce.[178] For example, there is no statement on changes in the paradigm of international trade that are directly related to digital trade. Although the CPTPP, DEPA, and DEA reflect the most advanced contents in this area, these agreements too focus on basic stipulation and regulation to 'boost' digital trade.[179] In consideration of the fact that even the most recent agreements reflecting the latest situation includes no detailed and systematic norm regarding the digital trade or AI, it would not be an exaggeration to say that this area is still in a sort of 'no norm' situation.

In understanding of the fact that this new trade operates in a paradigm fundamentally different from that of existing trade, it is necessary to introduce a new set of norms accordingly. It may be difficult to reflect this idea immediately in negotiations for a trade agreement currently in progress. Besides, it will take further time to analyze the current situation and the spin-off effect of digital trade enough to understand the contents and forms accurately. Despite such practical aspects, however, a basic review and research on structural characteristics of digital trade need to be conducted first. Since digital trade is spreading throughout society rapidly now and forming new norms requires a long period of time, it is wise to begin a review and research on this matter as early as possible. With preparation based on a review of structural characteristics, it will be possible to adjust details as the situation develops.

Although the context is somewhat different, the possibility of such integrated approaches is suggested in the WTO Agreement on Government Procurement (hereinafter GPA). The GPA specifies, in Article 1, governmental purchase of both goods and services.[180] The GPA clarifies that whether the government buys goods or service, either is subject to the same agreement. This is the same with the 2015 revision of the GPA. Of course, the GPA breaks completely away from the dichotomous structure of existing norms. The GPA also distinguishes government procurement of goods from that of services but clarifies that the same agreement is applicable to both. This agreement is also on the same dichotomous basis. It does not take into consideration new trade items that combine the two in the digital economy. Despite such limitations, however, it is of significance that both goods and services are regulated by one agreement at the same time. It indicates, although indirectly, that there is no special reason to regulate the two separately depending on trading items.

The recent pandemic has further unearthed inherent limitations of the current international trade regime and international investment regime. The reform discussion of trade regime and investment regime have been underway for some time now.

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[178] Most FTAs focus on exchange of information between parties, not introducing norms of digital trade in a substantial meaning. *See supra* note 161, pp. 8–9.

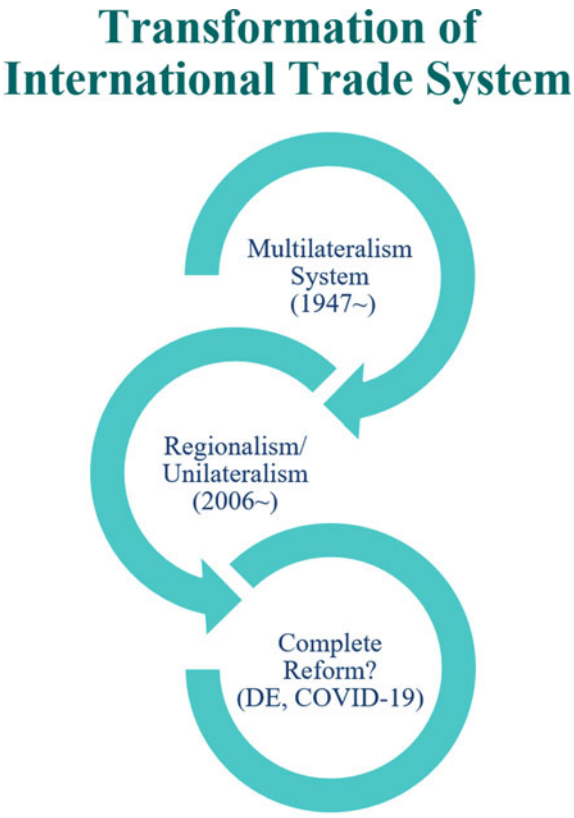
[179] For Chapter 4 (Electronic Commerce) of the TPP Agreement, *See* the following USTR website: https://ustr.gov/sites/default/files/TPP-Final-Text-Electronic-Commerce.pdf. In CPTPP as well, the chapter of electronic commerce remains as it is.

[180] Article 1 of the GPA specifies as below:

Article I Scope and Coverage.

2. This Agreement applies to procurement by any contractual means, including through such methods as purchase or as lease, rental or hire purchase, with or without an option to buy, including any combination of products and services.

These discussions were originally triggered by and have been undertaken without recognition or realization of the digital economy or AI, as these are just a new phenomenon. As the new era of the digital economy and AI call for a comprehensive review and structural shift on their own accord, arguably the time is now ripe for a systemic and architectural reform of the international trade and investment systems. Regionalism and unilateralism have been replacing the multilateralism for a while now, and the new experiments should be revive the multilateralism.



3.2.2 Subsidization Issue

The wide accommodation of automation invites a variety of legal issues under the current legal regime. One such example is the subsidization issue,[181] a hot potato topic under international trade law and investment law. In the frenzied response to social demands, governments are encouraged and even requested to provide various

[181] Acemoglu and Restrepo [50], Roberts et al. [51].

subsidies for automation in the business sector. Another automation-induced regulatory topic is a taxation issue. In fact, there are even proposals to tax those who use robots or even robots themselves so as to fund social security programs for people displaced by the use of robots in their workplaces.[182] South Korea has introduced a pilot program on this front.[183] To the extent that possible governmental support to private industries and companies is involved, these measures would inevitably implicate various provisions of the SCM Agreement under the WTO regime.

As far as artificial intelligence is concerned, one provision in the SCM Agreement merits careful consideration. It is the general infrastructure exception of the SCM Agreement. Article 1.1(a)(1)(iii) of the SCM Agreement explicitly excludes the provision of the 'general infrastructure' from the definition of the 'financial contribution by the government.' [184] The SCM Agreement is mindful not to interfere with governments' inherent authority to pursue otherwise legitimate public objectives.[185] This recognition led to the explicit exclusion of 'general infrastructure' in Article 1.1(a)(1)(iii) of the SCM Agreement.

Although it is difficult to speak in a general manner because each national infrastructure is different, an argument can be made that the construction or maintenance of a digital economy network and/or artificial intelligence operation system by a Member should be regarded as a project to establish a general infrastructure, which should thus fall under the exception of Article 1.1(a)(1)(iii) of the SCM Agreement. The ordinary meaning of the term 'general infrastructure' as interpreted by Article 31 of theVCLT [186] would essentially be "basic physical and organizational structures needed for the operation of a society or enterprise where such structures are (i) affecting or concerning all or most people or things; or (ii) not specialized or

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[182] Porter [52].

[183] *See* ibid.

[184] World Trade Organization, Panel Report, *United States—Preliminary Determination with Respect to Certain Softwood Lumber from Canada*, (WT/DS236/R), adopted April 14, 1999, at para. 7.26 ("[W]e find further confirmation of this broad meaning in the fact that the drafters of the Agreement considered it necessary to explicitly exclude 'general infrastructure' [in drafting] Article 1.1(a)(1)(iii) of the SCM Agreement.").

[185] *See* World Trade Organization, *Canada—Measures Affecting the Export of Civilian Aircraft,* (WT/DS70/AB/RW), (Jul. 12, 2000), at para. 47.

[186] The interpretation of the WTO covered agreement is governed by Articles 31 to 33 of the VCLT. Article 3(2) of the DSU provides that the DSB is to "clarify the existing provisions of those agreements in accordance with customary rules of interpretation of public international law." Isabelle Damme, "Treaty Interpretation by the WTO Appellate Body", (Oxford: Oxford University Press, 2009), at 22. *See* also World Trade Organization, Appellate Body Reports*, China—Measures Affecting Imports of Automobile Parts*, (WT/DS339/AB/R, WT/DS340/AB/R, WT/DS342/AB/R), (December 15, 20,080, at para. 171; World Trade Organization, *United States—Measures Affecting Trade in Large Civil Aircraft* ("*U.S.—Large Civil Ari craft*")(second complaint), (WT/DS353/AB/R), (March 12, 2012), at para. 586; World Trade Organization, *Canada—Certain Measures Affecting the Renewable Energy Generation Sector*, (DS426), (June 5, 2014), at para. 5.120 ("In determining the proper legal characterization of a measure under Article 1.1(a)(1) of the SCM Agreement, a panel must assess whether the measure may fall within any of the types of financial contributions set out in that provision. In doing so, a panel should scrutinize the measure both as to its design and operation and identify its principal characteristics.").

limited."[187] Depending on how the domestic collection and processing systems are established and operated, they may be claimed as constituting general infrastructure.[188] If the establishment of digital economy networks and/or an artificial intelligence operation system within a state can be characterized as government projects for general infrastructure, they could be carved out from the application of the SCM Agreement entirely.[189]

3.2.3 Extraterritoriality Zones of Personal Information

One of the most controversial topics in the discussion of digital society, digital economy and AI technology is none other than personal data. How to collect, disseminate and utilize while protecting privacy and basic human rights have become a hot topic in any discussion of digital issues or AI. In devising a new international regime in this regard, a structural obstacle that hinders meaningful progress is the fact that (i) personal information is to be flowed freely from one location to another, and thus from one country to another, so as to be commercially usable, (ii) personal information relates to privacy and is thus regarded sensitive, and (iii) consequently, the country whose nationals' personal information is being transferred elsewhere has inherent interest in getting access to the information thus transferred when necessary and in ensuring their nationals' privacy is adequately protected. The three elements illustrated above raise structural or systemic tension. The tension is recognized and is addressed in current treaties and agreement, but other than referring to these competing principles, specific guidelines or delineation points are still

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[187] *The Concise Oxford English Dictionary* defines "general" as "affecting or concerning all or most people or things; not specialised or limited". *See* Catherine Soanes and Angus Stevenson eds., "Concise Oxford English Dictionary", (11th, Oxford University Press, 2004), at p. 592. "Infrastructure", in turn, is defined as "the basic physical and organizational structures (e.g., buildings, roads, power supplies) needed for the operation of a society or enterprise". *See* ibid, at p. 730.

[188] In *EC-LCA*, when exploring the meaning of the "general infrastructure," the panel highlighted the importance of managing the program in question over time for it to be eligible for general infrastructure. For instance, it pointed out that general infrastructure at one point in time may not be general infrastructure at other times. The panel thus held that:

> [s]uch situations would have to be carefully evaluated, based on all the relevant facts in each case, in order to determine whether the provision of the infrastructure in question is general or not, and whether that determination changes over time. Thus, we consider that the proper point of reference in determining whether a provision of goods or services is 'other than general infrastructure' is the time when the act of provision that is alleged to constitute a subsidy takes place. That may be at the time the infrastructure in question is created, in the sense of being brought into existence, or a subsequent point in time, when the conditions surrounding the provision of that infrastructure are changed by the government providing it. Moreover, the determination of whether a provision of goods or services is 'other than general infrastructure' may involve a time period of limited or indefinite duration.

World Trade Organization, Panel Report, *European Communities—Measures Affecting Trade in Large Civil Aircraft*, (WT/DS316/AB/R)**,** (May 18, 2011), at para. 7.1044.

[189] *See* Article 1.1(a)(1)(iii) of SCM Agreement.

absent. Unless details are provided in this regard, disputes are bound to increase and intensify. It is imperative to find some common ground by addressing the concerns of interested parties - the states and businesses that rely on personal information on a global basis, and the states and nationals that have interest in preserving the basic integrity in using such personal information. Various creative options could be explored. One example in this regard would be to consider the concept of an agreed area of extraterritorial application of domestic jurisdiction of the state whose nationals' information is being transferred to servers and facilities located in another country. It might be called an agreed or a legitimate "zone of extraterritoriality". In discussing this topic, first of all, 'the zone of extraterritoriality' stated here needs to be defined. Although this concept has yet to be clarified, the zone of extraterritoriality being discussed in the context of personal information means that when a government provides a foreign country with personal information related to its people, the receiving government needs to abstain from compulsory execution or retrieval of individuals' personal information thus provided. For example, there may be situations where this concept is applied in the following order: a broad range of personal information of a nationality state's people is collected by a foreign company located in a foreign country. Such personal information is stored or located in the national state of the foreign company or a relevant third-party country where the foreign company has a business facility. That nationality state or a third-party state may try to to have access to and retrieve that information when necessary. In this case, such states (nationality states and third-party states) may retrieve or access to such information as long as their own domestic laws permit it according to the jurisdictional rules of the current international law. In order to prevent those foreign countries from retrieving or revealing personal information of its own nationals, the nationality state of the people whose personal information has been provided may have a legitimate concern. The originating country may want to have a legitimate safety mechanism in place as a condition for the free flow of personal information in the first place. Likewise, the originating state may also want to an assurance that it can access and retrieve its own nationals' personal information when necessary - for national security reasons and the like. Through a compromise of this sort, it may become possible to provide or share personal information with a foreign entity and perhaps foreign country in a more systematic and secure manner.

On top of this, further safety measures can be contemplates as necessary. For instance, a company may be permitted to collect and transfer personal information in a foreign country, if it agrees to process and utilize the information according to the domestic laws of the source country. At the same time, however, the national state of the company or a third-party country may be restricted to retrieve or access the personal information so provided. As seen in the Snowden case, it has been noted that in fact many countries have wherewithals and desire to access and retrieve a wide range of information collected by their companies, whether domestically or abroad. As a result, contemplation and introduction of some defensive mechanisms may facilitate finding a common ground in this new area of competing national interests. Once such a system is introduced, it is expected that the level of personal information protection will increase, and as a result individuals providing such information and

their home state could feel more secure, while business entities are allowed to collect and transfer the personal information from one country to another, and to yet another country according to their business needs.

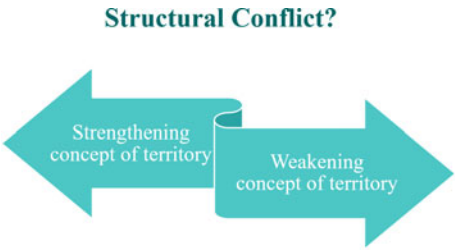
Basically, this measure is to restrict the application of general jurisdiction of a foreign or third-party country concerned. According to the principle of jurisdiction in current international law, countries are eligible to exercise such jurisdiction and thus to retrieve or access personal information located or stored within their national territories. There has been no case where the principle of jurisdiction was excluded in the international society for this and other similar reasons. From a country's perspective, such exceptional cases can be recognized as a serious infringement of its sovereignty. Besides, retrieving and accessing such personal information is directly related to the matter of national security. In particular, it would be difficult for countries to come to an international agreement that they abstain from or reduce exercising jurisdiction to that extent. The concept of extraterritoriality over personal information (or digital properties) currently under consideration can be understood as a new attempt to overcome this existing principle and relevant phenomena.

Despite the general need to introduce digital or personal information extraterritoriality, however, there is a significant challenge in that it is in conflict with the principle of jurisdiction in existing norms of international law. Especially when another country questions and objects, it will be difficult to overcome such obstacles, at least within the bounds of the current international law. Despite the realistic necessity, a strong argument can be made that this new concept is inconsistent with the principle of jurisdiction stipulated in the Customary International Law from the legalistic perspectives.

Thus, while the extraterritoriality concept seems to be necessary in the digital era, a careful approach is necessary. If a country strongly argues that this assertion contradicts the existing principle specified in international laws, the new proposal will suffer harsh criticism. It is suggested, therefore, that this idea is first shared in the abstract and then efforts are put forth to reach a consensus gradually in consideration of other countries' feedback to come up with a full treaty.

One major future challenge to be faced is to persuade various countries and obtain their consent. While overcoming this challenge, it is necessary to stipulate new provisions concerning extraterritoriality in relevant treaties or agreements. Once some bilateral or mini-lateral treaties or agreements are adopted, a new norm is established at least for the agreement parties, which will facilitate the remaining process.

Of course, it should be noted that such extraterritoriality established through a treaty is effective solely between the agreement parties. To other countries than these parties, the existing principle in the Customary International Law is still valid. As a result, the dualistic structure is established: principles applicable to treaty parties and principles applicable to the other countries. It would be difficult to realize a complete extraterritoriality system in this case. We could go as far as to say that there is a conflict between forces that strengthen the concept of national territory and that weaken it, simultaneously pulling in two different directions.



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Chapter 6
Specific Issues under International Law



Technological development has always affected state' behavior under international law. States, as with any other persons or entities, are eager to employ new technologies when needed. In August 2020, Singapore and Australia concluded a treaty through e-signature: the minsters representing the two countries did not meet at one place as is done in conventional treaty signings, but instead applied an e-signature in their respective capitals.[1] As a consequence, the treaty contains two places (Singapore and Canberra) as the concluding locations of the treaty. Likewise, states' acceptance of new technologies will change treaty negotiation and conclusion practices. Of course, the permeation of the new technologies will also take place in other areas of state behavior. Thus, states and their governments' utilization of new technologies, in and of itself, is not remarkable. It happens all the time and will continue to do so in the future. Similarly, often states also adjust their interpretation of treaty text to keep up with technological changes, which is called evolutionary treaty interpretation. All in all, this process of accommodating technological changes may be called a "technology-adaptation" process.

The same also goes for AI. The advent of AI will affect and change state' behavior in various sectors of international law, including how they formulate, interpret and implement international law. The same technology-adaptation process appears here as well. What is peculiar about AI is that the forthcoming changes are not merely confined to a conventional technology adaptation process: (i) it forces states to contemplate and introduce new norms because of the absence of norms in state activities using or relating to AI; and (ii) it also requires states to revisit the secondary norms of state responsibility because of the novel issues of attribution. The former may be called 'norm upgrades', while the latter 'attribution reformulation' mindful of its structural nature. Both are indeed important. Both are critical for international law to keep up with the fast-changing international community as a result of the advent of AI. What would be more academically meaningful, however, would be the latter, which requires the examination of the very identify of AI and which stands to apply

[1] *See* Singapore-Australia Digital Economy Agreement, Final Clause.

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to international law generally as secondary norms. Again, it should be noted that all three elements—technology adaptation, norm upgrades and attribution reformulation—are important and deserve the continuing attention of the global community to AI-compatible international law. Of the three, the third one arguably raises the most fundamental and structural questions as regards AI.

This chapter focuses on norm upgrades and attribution reformulation, as they are novel issues in the AI era and deserve careful attention of states. Because AI is used in so many issue-areas and raises new challenges, there are a number of issue areas in international law that need to be updated or revised. They range from human rights to state sovereignty, responsibility, dispute settlement mechanisms and law of war.

Human rights law is one issue area that faces many novel challenges in the AI era. As we have seen, the emergence of AI has challenged the fundamental definition of 'human'. The answer to this challenge will undoubtedly affect how human rights are regulated. Furthermore, the international human rights laws need to address new demands for a 'decent life', such as the right to privacy and data protection. If AI creates significant societal disruptions such as massive unemployment, it could severely undermine human rights standards.

On a more macro level, AI also affects state relations. As the use of AI increases, it will become easier to monitor and extract information from another state, and the expansion of the digital economy will make extra-territorial jurisdiction a contentious issue more frequently. The conventional concept of state sovereignty mainly based on its territory and people will have to be adjusted to accommodate the digital domain.

AI affects how states interact with one another in times of conflict. It is reasonable to expect that the international dispute settlement mechanisms will follow the domestic courts in the digital transformation trend. AI can cause change not only in the analytical process, but also in the preparation of evidentiary materials and adjudication. Law of war is no exception. New weapons and new battle fronts require new regulations.

Before diving into the specific areas, it is worth mentioning that there have been efforts to create new norms and change existing ones to match the new international environment because of AI. To the extent that AI is used in many different contexts in our daily lives, it certainly raises a variety of novel legal issues. Those issues span various legal sectors such as constitutional, civil, criminal, and administrative law.[2] When it raises a fundamental question of what a human being is, who (or that) is held liable for certain activities, one should turn to the area of legal philosophy. A legal discourse is not just confined to domestic law: it can and should also reach the universe of international law. As stated in the introduction, this is what this book aims to cover. Now, AI also affects international law in various ways. For instance:

[2] *See*, e.g., Council Directive 85/374/EEC of 25 July 1985 on the Approximation of the Laws, Regulations and Administrative Provisions of the Member States Concerning Liability for Defective Products, 1985 O.J. (L 210) 29, https://eur-lex.europa.eu/legalcontent/EN/TXT/PDF/?uri = CELEX:31985L0374&from = EN.

- Introduction of new norms to regulate state action using AI—for example, 11 Principles on Lethal Autonomous Weapons System (LAWS), endorsed by multiple European states (Sept. 2019).[3]
- Introduction of new norms to regulate AI-enabled and enhanced cyber activities—for example, Tallinn Manual 2.0 on the International Law Applicable to Cyber Operations (Feb. 2017).
- Introduction of new norms to regulate state action regulating AI—for example, Singapore-Australia Digital Economy Agreement (SADEA) (Aug. 2020).
- Introduction of new norms to regulate attendant issues stemming from utilization of AI—for example, EU's General Data Protection Regulation (GDPR) (May 2018).[4]
- Introduction of new norms to regulate trade in goods and services using AI—for example, Comprehensive and Progressive Trans Pacific Partnership (CPTPP) (Dec. 2018).

Indeed, all these efforts are important strides for international law. They aim to update and upgrade international law to keep up with the advent of AI. Similar efforts are taking place in diverse sectors of international law. While these norm upgrades are timely and important, the more fundamental question still resides in the 'attribution reformulation'. Arguably, this issue has received relatively little attention so far. A subsection of this chapter on state responsibility will attempt to address the issue in more detail.

1 Protection of Human Rights

AI can fundamentally challenge the human rights regime, because it can raise questions regarding the concept of what is human.[5] Moreover, it is possible to imagine a government attempting to justify autocratic rule by a super-intelligent AI if people

[3] *See* French Foreign Ministry Website, *11 Principles on Lethal Autonomous Weapons System*, available at https://www.diplomatie.gouv.fr/en/french-foreign-policy/united-nations/multilateralism-a-principle-of-action-for-france/alliance-for-multilateralism-63158/article/11-principles-on-lethal-autonomous-weapons-systems-laws; Statement Austria, Ambassador Thomas Hajnocz, *Convention on Prohibitions or Restrictions on the Use of Certain Conventional Weapons Which May Be Deemed to Be Excessively Injurious or to Have Indiscriminate Effects*, Group of Governmental Experts on Lethal Autonomous Weapon Systems, General Exchange of Views (Apr. 9–13, 2018), available at https://www.unog.ch/80256EDD006B8954/(httpAssets)/AA0367088499C566C1258278004D54CD/$file/2018_LAWSGeneralExchang_Austria.pdf.

[4] *See* Regulation 2016/679 of the European Parliament and of the Council of 27 April 2016 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data, and Repealing Directive 95/46/EC (General Data Protection Regulation) (GDPR), 2016 O.J. (L 119) 1, available at http://eur-lex. europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32016R0679&from=EN.

[5] *See* Livingston and Risse [87], pp. 151–3, Risse [62].

believe that AI is better suited in its capacity to make decisions.[6] Another, maybe a bigger, point of concern is that the development of AI raises issues that cannot be addressed effectively under the current international human rights law and humanitarian laws.[7] In particular, it becomes increasingly difficult to decide who should be held accountable for violations of these laws.

AI itself has no intrinsic value or harm to human rights. For instance, the application of AI in healthcare could improve health and longevity, because it allows for a better personalized healthcare system. However, it could also lead to serious infringement of personal freedom, as it would be easier to survey people's activities. On the flip side of the coin, AI can facilitate human rights investigations. For instance, DNA sequencing based on AI can contribute to more efficient forensic investigations to examine the circumstances of deaths.[8] In addition, machine learning automates satellite data analysis. There has already been a case at the International Criminal Court (hereinafter ICC) where a Libyan warlord was indicted based on satellite imagery and videos that were posted on social media.[9]

What matters is how and where the AI is applied. Policy leaders need to make sure that the benefits of technology are evenly distributed and contribute to the welfare of society as a whole. Although AI is expected to create some disruption in the short term, including job losses, the overall welfare, especially in Europe and the US, is also expected to increase.[10] The following are examples in which data and AI can improve people's welfare.[11]

Material living standards	Education	Health	Equal opportunities	Environmental sustainability
AI chatbots to provide better public service (immigration process, financial loan application) to a wider group of people (ex. immigrants with language barrier)	Personalized education services using AI, AI facilitators for teachers	AI-based drug trials for more efficient drug development, AI-powered diagnosis to increase accuracy	AI assisting people with special needs to reduce discrimination	AI-applied traffic optimization, AI-based infrastructure to facilitate recycling

Source Modified from Bughin et al. [44] pp. 26–27

[6] Ibid., p. 153.

[7] Garcia [28], p. 336.

[8] Ibid., p. 144.

[9] Ibid., p. 143.

[10] *See* Bughin et al. [44], p. 3, available at https://www.mckinsey.com/~/media/mckinsey/featured%20insights/future%20of%20organizations/tech%20for%20good%20using%20technology%20to%20smooth%20disruption%20and%20improve%20well%20being/tech-for-good-mgi-discussion-paper.pdf; for a general discussion on the effects of AI on the economy, *see* Furman and Seamans [50]

[11] Ibid.

While many nation states have established national strategies to develop AI, a comprehensive framework to address the risks of violation of human rights by AI is still absent.[12] Eileen Donahoe and Megan Metzger (2019) call for a human-rights-based approach for the governance of AI. They argue that a number of provisions in the Universal Declaration of Human Rights (hereinafter UDHR) speak directly to some of the most serious societal concerns about AI. According to the authors, the following provisions are related to AI[13]:

[12] *See* Donahoe and Metzger [30], pp. 115–116; McGregor et al. [60], p. 324.

[13] The original texts of the UDHR are as the follows:

Article 2

Everyone is entitled to all the rights and freedoms set forth in this Declaration, without distinction of any kind, such as race, colour, sex, language, religion, political or other opinion, national or social origin, property, birth or other status. Furthermore, no distinction shall be made on the basis of the political, jurisdictional or international status of the country or territory to which a person belongs, whether it be independent, trust, non-self-governing or under any other limitation of sovereignty.

Article 3

Everyone has the right to life, liberty and security of person.

Article 8

Everyone has the right to an effective remedy by the competent national tribunals for acts violating the fundamental rights granted him by the constitution or by law.

Article 9

No one shall be subjected to arbitrary arrest, detention or exile.

Article 10

Everyone is entitled in full equality to a fair and public hearing by an independent and impartial tribunal, in the determination of his rights and obligations and of any criminal charge against him.

Article 11

(1) Everyone charged with a penal offence has the right to be presumed innocent until proved guilty according to law in a public trial at which he has had all the guarantees necessary for his defence.

(2) No one shall be held guilty of any penal offence on account of any act or omission which did not constitute a penal offence, under national or international law, at the time when it was committed. Nor shall a heavier penalty be imposed than the one that was applicable at the time the penal offence was committed.

Article 12

No one shall be subjected to arbitrary interference with his privacy, family, home or correspondence, nor to attacks upon his honour and reputation. Everyone has the right to the protection of the law against such interference or attacks.

Article 19

Everyone has the right to freedom of opinion and expression; this right includes freedom to hold opinions without interference and to seek, receive and impart information and ideas through any media and regardless of frontiers.

Article 20

(1) Everyone has the right to freedom of peaceful assembly and association.

Article 2: The right to equal protection and nondiscrimination speaks to concerns about avoiding bias in data and ensuring fairness in machine-based decisions relied upon by governments, especially when these decisions impact citizens' rights.

Article 3: The right to life and personal security can form the foundation for addressing concerns about autonomous weapons that move beyond human control, as well as the risk of a literal arms race to develop such weapons.

Articles 8 through 11: Together, the right to effective remedy for violations and infringements of rights and the right to due process of law provide a basis for addressing concerns about transparency, fairness, and accountability in cases where AI systems impact people's rights.

Article 12: The right to privacy addresses the fundamental concern about loss of privacy in data-driven societies and the need to protect personally identifiable data.

Article 19: The right to freedom of expression, which includes the right to hold opinions and to seek, share, and access information, speaks to challenges associated with the increasing role played by algorithms in moderating content and curating what information is presented to users of digital platforms.

Articles 20 and 21: The rights to association and peaceful assembly, as well as the right to democratic participation in government, are becoming increasingly important as lenses through which to assess the impact of AI systems on human agency, the organization of civic groups, political participation, and the integrity of democratic processes around the globe.

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(2) No one may be compelled to belong to an association.

Article 21

(1) Everyone has the right to take part in the government of his country, directly or through freely chosen representatives.

(2) Everyone has the right of equal access to public service in his country.

(3) The will of the people shall be the basis of the authority of government; this will shall be expressed in periodic and genuine elections which shall be by universal and equal suffrage and shall be held by secret vote or by equivalent free voting procedures.

Article 23

(1) Everyone has the right to work, to free choice of employment, to just and favourable conditions of work and to protection against unemployment.

(2) Everyone, without any discrimination, has the right to equal pay for equal work.

(3) Everyone who works has the right to just and favourable remuneration ensuring for himself and his family an existence worthy of human dignity, and supplemented, if necessary, by other means of social protection.

(4) Everyone has the right to form and to join trade unions for the protection of his interests.

Article 25

(1) Everyone has the right to a standard of living adequate for the health and well-being of himself and of his family, including food, clothing, housing and medical care and necessary social services, and the right to security in the event of unemployment, sickness, disability, widowhood, old age or other lack of livelihood in circumstances beyond his control.

(2) Motherhood and childhood are entitled to special care and assistance. All children, whether born in or out of wedlock, shall enjoy the same social protection.

For the full text of the UDHR, *see* https://www.un.org/en/universal-declaration-human-rights/#:~:text= Article%202.,property%2C%20birth%20or%20other%20status.

Articles 23 and 25: The rights to work and to enjoy an adequate standard of living—rights whose fulfillment, unlike the fulfillment of first-generation negative rights such as freedom of expression, does not rest solely with governments—can guide governance decisions around displacement of human workers by AI.

Thus, the advent of AI raises several human rights issues. In particular, the expansive use of personal information and job losses are the most often-cited problems.

1.1 Expansive Utilization of Personal Information and Possible Invasion of Privacy

Privacy threats could surface due to pattern recognition applications and automated data acquisition programs. As a consequence, AI could further cement the authority of national states insofar as it could threaten human rights. For one, governments now have more tools to survey their own people. In particular, nation states in control of telecommunications, such as China, can easily monitor every activity of their citizens. In fact, China has been already increasing government control over its people by closely cooperating with digital firms.[14]

China is not alone in the expansive utilization of personal information. During the COVID-19 pandemic, the use of location data has contributed to the fight against its global spread, but it is easy to imagine other instances in which governments use the data for dubious purposes. Even concerning COVID-19, many raised concerns about violation of privacy.[15] For instance, it has been noted that the EU's GDPR is not compatible with strict monitoring of people's movement despite the importance of public health and disease prevention purposes.[16] Thus, European countries such as Austria, Germany, and Italy are looking into the option of using anonymized data to analyze people's movements.[17] The White House has also been in negotiation with major digital companies such as Google and Facebook, but as of March 2020, no concrete deal has been made.[18] Thus, there will be a dire need for more regulation relating to data privacy protection and confidentiality issues.

When it comes to information protection, the EU is in the lead. The EU has recently introduced the GDPR as a legal mechanism to protect and make good use of personal information at the same time. In an effort to develop the digital economy, the US allows various industrial sectors and enterprises to utilize personal information as freely as possible. In the event that a problematic or illegal act is committed, however, the US punishes such trespasses severely to set an example. Japan has recently taken

[14] *See* Westerheide and Cognitive World [32].

[15] *See* Zastrow [63].

[16] *See* Alvarez et al. [19].

[17] *See* Timberg and Harwell [15].

[18] Ibid.

decisive legal action for personal information to be utilized more freely, progressing in the direction of the model set by the US.

In contrast, China and Russia adopt policies to focus on the protection of personal information rather than its utilization. Furthermore, they have made various efforts to prevent such information from being transferred to overseas enterprises or foreign governments. It is assumed that these countries take such stances with the aim not only to prevent their own sensitive information from flowing into foreign enterprises but also to take the lead in the digital economy by ensuring that such information is utilized exclusively to protect their own industrial interests.

As such, major countries around the globe are keenly aware of the importance of personal information in the context of the fourth Industrial Revolution and the digital economy, and thus they have introduced various types of legal mechanisms to regulate it. In addition to adopting methods to help domestic industries and enterprises take advantage of such personal information, they have also made efforts to achieve the opposite goal—protecting such personal information securely. The attempts of these countries are based on their strategy to find balancing points between conflicting goals. However, the perception of such balancing points is as yet varied among countries.

It is noteworthy that there is a lack of a systematic response manual that can be applied globally to deal with a world-wide disaster like the COVID-19 pandemic.[19] We could greatly benefit from a common response guide, and the fight against the global pandemic is only as successful as the weakest link.[20] Although application of AI could facilitate the global fight against the virus and help international policy alignment, the dearth of cooperation once again shows the difficulty of agreeing on a set of standards to apply the new technology. If the countries were to use AI to strengthen their authority, establishing legitimacy through an internationally agreed set of norms and regulations appears to be a big hurdle.

1.2 Displacement from Workplace and Employment

Another concern raised about human rights is that AI can be a disruptive force in the job market. According to a study by the United Nations (hereinafter UN) Department of Economic and Social Affairs (hereinafter UNDESA), the public's biggest fear regarding AI involves massive unemployment especially in the low-skilled sectors. The UNDEA notes that it could lead to an exacerbation of "the decline of middle-skilled jobs and rising wage inequality observed in the recent past, particularly in many developed countries."[21] The study also notes that even highly educated and skilled professionals may be at risk or replacement by the AI.[22]

[19] *See* Lee [46].
[20] Ibid.
[21] Ibid.
[22] Ibid.

Therefore, a revision of international labor standards, unemployment compensation and placement policy, together with training programs for workers in lieu of the impact of novel technology could be necessary in the adapting stage. The International Labour Organization (hereinafter ILO) runs several projects to mitigate the negative consequences of AI for workers.[23]

2 Infringement on Sovereignty of Other States

State sovereignty has been one of the most sacred concepts for modern international law, recognized not only by international customary law, but also the UN Charter. However, it faces dire challenges in the era of AI, as AI can be a serious threat to a state's sovereignty. Unbounded by territorial boundaries, AI facilitates transnational collection and processing of information which can then be used to create new value, or even have espionage purposes.

Not only do states have to worry about information leakage, but also their jurisdictions. While jurisdiction is a fundamental part of state sovereignty, its application may not be so clear in the digital economy and other AI related actions. This is the reason for which some countries require local data storage, thus making extraterritoriality an important topic.

2.1 Collection and Processing of Information of Other Countries

AI makes collection and processing of information of other countries easier. Because AI can produce and process large quantities of data at a higher speed with greater accuracy, there is simply more data available at a lower cost. Such a phenomenon is raising new challenges to be addressed by the international community. For instance, breaches of data collection and process standards among states without consent or notice via autonomic software and algorithms can be the subject of international disputes. A country could argue for a breach of its national sovereignty if another collects data within its borders without its consent or notice. The problem is that the advent of digital economy obscures the boundaries of data, encouraging freer flow of data between countries. Thus, data storage localization requirement and extraterritorial jurisdiction have become two important issues in the collection and processing of information of other countries.

To have better access to data and prevent data leakage, some governments require that the data be stored within their territory. For example, South Korea denied

[23] *See United Nations Activities on Artificial Intelligence (AI)*, ITU (2019), p. 9 available at https://www.itu.int/dms_pub/itu-s/opb/gen/S-GEN-UNACT-2019–1-PDF-E.pdf.

Google's request to use government mapping data outside the country.[24] As the importance of data grows, the geographical data restriction can be seen as the government's attempt to strengthen their control over information. One major anxiety has to do with the disadvantages for domestic digital corporations. While domestic providers are subject to a web of regulation, including taxation among others, their foreign competitors with foreign servers are basically not. So, regulatory restrictions for foreign companies are sometimes perceived (erroneously) as a counterweight to level the playing field.

Speculation has arisen whether these two practical considerations may have underpinned the debates over Google's request for permission to transfer the digital mapping data from Korea to its foreign servers. The data on point is the 1:5,000 scale map data compiled by the National Geographic Information Institute, as such information is in great demand for digital commercial services. The statutory deadline for a decision was in August 2016, and the South Korean government rejected the request.

It has been reported that national security concerns were the issue of contention. The digital information requested is the same information provided to Korean service providers (such as SK Telecom's T-Map, a major navigational map used in South Korea), but as exporting it to outside locations may let it fall into the wrong hands in terms of national security, the government cautions.

Information on locations of key facilities is already available on-line from various mapping service providers and satellite image companies. People marvel at the level of resolution and graphic details of structures and facilities available on the internet for free or that they can purchase. The question then is, what additional security risk does this provision bring? There may be some, but extra efforts are needed with all these digital images already stacked up and updated in the public domain.

So, a realization of this reality has fueled the said speculation. If the hidden agenda is the taxation issue for the global digital company, this may not be the best way to find a solution. The rule, for better or worse, is still based on the location of servers. Foreign corporations pay their corporate taxes when they have permanent establishments in Korea. The OECD has been contemplating introducing new rules here, but they are slow in coming. Some countries are trying out new rules, but there are still many things to be clarified.

If, on the other hand, the underlying reason is to somehow favor domestic service providers, it seems more problematic. Market competition through non-discriminatory treatment is the cardinal rule countries have agreed to. Dominant Korean service providers in this sector do not provide their mapping service in English, leaving a sizable segment of people—foreign residents and tourists—outside the service coverage. Competition may help change their service model.

Data storage localization requirement is also an important topic for the US at the moment. Unsurprisingly, the new US-Mexico–Canada Agreement (hereinafter USMCA) formalizes data localization requirements. *2020 National Trade Estimate*

[24] *See* Lee and Lee [55].

Report on Foreign Trade Barriers[25] mentions data localization requirements as potential trade barriers in more than 10 countries (Brazil, Canada, China, the EU, Ghana, India, Indonesia, Kenya, South Korea, Malaysia, Nigeria, Pakistan, Russia, Saudi Arabia, Turkey and Viet Nam). For example, the US Trade Representative (hereinafter USTR) criticizes China's implementation of *Cybersecurity Law*[26] and the EU's GDPR.[27] The US is also concerned about South Korea's licensing requirement for cross-border transfer of data and calls Korea "the only significant market in the world that maintains such restrictions on the export of location-based data".[28]

In addition to data location, location of headquarters also complicates judicial enforcements. For example, when complaints against Google were filed by two consumer watchdogs under the EU's GDPR, questions were raised about which

[25] *2020 National Trade Estimate Report on Foreign Trade Barriers,* USTR (2020), available at https://ustr.gov/sites/default/files/2020_National_Trade_Estimate_Report.pdf.

[26] Ibid., p. 122.

Various draft and final measures being developed by China's regulatory authorities to implement China's Cybersecurity Law, which took effect in June 2017, and China's National Security Law, which has been in effect since 2015, would prohibit or severely restrict cross-border transfers of information that are routine in the ordinary course of business and are fundamental to any business activity. These measures also would impose local data storage and processing requirements on companies in "critical information infrastructure sectors," a term that the Cybersecurity Law defines in broad and vague terms. Given the wide range of business activities that are dependent on cross-border transfers of information and flexible access to global computing facilities, these developments have generated serious concerns among governments as well as among stakeholders in the United States and other countries, particularly among services suppliers.

[27] Ibid., pp. 207–8.

The GDPR restricts the transfer of the personal data of EU data subjects (any natural or legal person whose personal data is being processed) outside of the EU, except to specific countries that the EU has determined provide adequate data protection under EU law or when other specific requirements are met, such as the use of standard contract clauses or binding corporate rules. Restrictions on the flow of data have a significant effect on the conditions for the cross-border supply of numerous services and for enabling the functionality embedded in intelligent goods (i.e., smart devices). The United States remains concerned that the implementation and administration of the GDPR create disproportionate barriers to trade, not only for the United States, but for all countries outside of the EU. The EU has so far found only a handful of countries to provide adequate data protection under EU law, which means that suppliers in the large majority of EU trading partners must rely on other arrangements or criteria to receive data from suppliers in the EU. Although the United States has received a determination of partial adequacy from the EU (for further information on the EU-US Privacy Shield Framework, see below), there are many other countries, including India and Korea, that have expressed interest in obtaining an adequacy determination to facilitate the exchange of data with the EU.

[28] Ibid., pp. 320–1.

Korea's restrictions on the export of location-based data have led to a competitive disadvantage for international suppliers seeking to incorporate such data into services offered from outside of Korea. For example, foreign-based suppliers of interactive services incorporating location-based functions, such as traffic updates and navigation directions, cannot fully compete against their Korean rivals, since their locally based competitors typically are not dependent on foreign data processing centers. Korea is the only significant market in the world that maintains such restrictions on the export of location-based data.

regulating body had the jurisdiction over the matter. Normally, the Data Protection Authority (hereinafter DPA) in the country where the 'main establishment' is located takes the leading role. However, the French DPA (the Commission Nationale de l'Informatique et des Libertés, hereinafter CNIL) reasoned that since the Google headquarters in Europe did not have 'decision-making power', its location (Ireland) was irrelevant and that CNIL had the authority to initiate the proceedings. It implies that non-EU companies could be at risk of enforcement by an authority in any EU country, not just where its headquarters are located in.[29] The incident also demonstrates that delineation of jurisdiction over digital firms is complicated as there are multiple layers at play.

Thus, as the meaning of international borders becomes less significant, extraterritorial application of domestic laws is expanding. For example, as states impose regulations on personal information and data, foreign corporations are subject to domestic regulations, which gives states indirect access to data in other countries. Since it is difficult to clarify whether extraterritorial application of domestic laws is in compliance with relevant international laws, at the moment there is room for each state to use the situation to its advantage.

The concept of 'extraterritoriality' stated in international laws basically means an area not in the jurisdiction of the sovereign state in question. Fundamentally, according to the principle of international laws, every country is eligible to exercise its jurisdiction over any person or corporation within its national territory regardless of nationalities and situations. For instance, the EU secures the jurisdiction over any situation or person within its territory, and the US is also eligible to exercise the jurisdiction over any situation or person within its territory. So is Korea.

Jurisdiction is the most important aspect of a state's sovereignty. In the event that another country blocks or restricts the exercise of juristdiction, it leads to the issue of infringement of the other party's sovereignty, raising the suspicion of violating relevant international laws. Specifically, such an act may be judged as violating the principle of sovereign equality and the principle of nonintervention specified in Article 2 of the UN Charter and the Customary International Law. It is highly likely, therefore, that a certain country's measure or law restricting another country's exercise of jurisdiction leads to an international dispute or is judged as a violation of relevant international laws.

According to present international laws, however, there are several exceptions to this principle of jurisdiction, which are often explained in a broad sense of 'extraterritoriality.' In such cases, it is viewed that a country does not (or cannot) exercise its jurisdiction over its territory. These cases are divided mainly into the following

[29] *See* Alvarez et al. [20].

three: (1) exceptions for warships and air fleets[30]; (2) diplomatic privileges and immunities[31]; and (3) sovereign immunity.[32]

In order to minimize contention regarding extraterritoriality jurisdiction, one criterion to be considered is whether there is a basic nexus between the nation that exercises the jurisdiction and the persons or situations that it is applied to. Recently, the entire world is connected as a single integrated system when it comes to the issue of personal information protection, and thus it is easier than before to confirm and verify the nexus, which is another important consideration in this respect. In the current situation, therefore, there is a relatively high probability that basic principles of international law are complied with as far as jurisdiction is concerned, at least with regard to domestic laws and measures for the protection of personal information. Thus, each country has a great deal of discretionary power on what type of approach

[30] Current international laws specify that a country's jurisdiction is not applicable to warships and air fleets of another country in its territory. This principle has been confirmed by the Customary International Law and confirmed again in 1982 by the United Nations Convention on the Law of the Sea. In other words, warships and air fleets of another country are not under the jurisdiction of the sovereign country of that territory. Accordingly, such warships and aircrafts are excluded from the application and forced execution of laws of the sovereign country. For example, in October 2018 when the Japan Maritime Self Defense Force that was a topic of discussion between Korea and Japan visited the Jeju Base, the Korean law was not applicable to warships of the Japan Maritime Self Defense Force in relation to its flag raising. Neither could the Korean judicial authorities enforce the domestic law. Even if a Japanese warship entered a domestic port as discussed at that time, it was unable for our authorities to prohibit the Maritime Self Defense Force from flag raising with its legal force or to forcibly have the flag lowered according to international laws. This is the case when our naval warships visit a foreign country as well. Under the current international laws, naval warships and military aircrafts of each country are regarded as all a part of its territory.

[31] Another example of exclusion from the exercise of jurisdiction in a national territory is related to "diplomatic privileges and immunities." According to the Customary International Law and "1961 Vienna Convention on the Diplomatic Relations," every diplomat is under the jurisdiction of the country that has sent him/her, not under that of a foreign country where they are stationed. This is an exceptional case of jurisdiction application that has been acknowledged in order to secure the position as a diplomat as well as to facilitate smooth diplomatic activity. Diplomats' privileges and exemptions have long been recognized as a principle in international laws, and this principle is still regarded as the most sensitive and important rule between countries. In the event that a North Korean defector enters a foreign embassy in Beijing, Chinese security police officers or soldiers are not allowed to chase and arrest them forcibly or execute laws in that embassy. This case as well is a type of 'extraterritoriality' in a broad sense that makes jurisdiction over the national territory invalid.

[32] Another case where jurisdiction over the national territory is excluded is sovereign immunity. Sovereign immunity means that one country is not subject to another country's jurisdiction according to international laws, and it is viewed as a result of applying the doctrine of sovereign equality. For example, Korea is not subject to Japan's jurisdiction, Japan not to the U.S. jurisdiction, and the U.S. not to Korea's jurisdiction. A country and an agency or person representing it are not under another country's jurisdiction. Thus, law application and forced execution on such entities become invalid in a certain country's domestic court. Such forced execution is possible only through an international court that reserves the valid jurisdiction. This arrangement has yet to be finalized with a separate treaty, but it is an international law that has long been recognized as a Customary International Law. Korea as well has complied with this norm and it has been confirmed by domestic courts so far. Therefore, sovereign immunity as well can be regarded as a form of 'extraterritoriality' in a broad sense that excludes the jurisdiction of a country in its national territory.

they will adopt. In order to address this problem, an argument that a certain action is asymmetric or non-proportional might be raised, but such assertions would be hardly viewed as persuasive since as yet there are no unified applicable international laws in this area.

In consideration of the current condition where a wide range of personal information has been collected, its availability, and the expected sensitive reactions of each nation, there seems to be a general consensus that a decisive response is necessary. In particular, since leakage and unjust utilization of personal information can cause serious and long-term problems, it is difficult to compare this matter simply with other general situations or incidents. If this is the case, simply applying the existing principle of jurisdiction in international laws that was introduced only with general or traditional situations and incidents in mind, to the matter of protection of digital contents or personal information as it is can result in problems. Therefore, a new concept of extraterritoriality needs to be introduced at least for the area of digital contents or personal information protection.

2.2 Creation of New Value from Collected Information

In the era of digital economy, data is power. The creation of new value—either commercial, business, political, scientific, or diplomatic—from collected data varies from artworks to political mobilization. Corporations can benefit from user data to improve their products or target their advertisements.

When new value is created from data collected by AI, intellectual property and legal responsibility issues can be complicated. AI may use various forms of collected information, including other individual's works and data. On the other hand, other individuals or AI entities may utilize creative works of AI in their new creative activities. In such cases, it is difficult to ascertain whether an infringement of intellectual property rights occurs and, if any, who should bear the legal responsibilities.

In the case of transnational information collection, the issue is even more complicated. For instance, the 2016 Cambridge Analytical scandal showed the power of data, and how its manipulation can be used in foreign interventions.[33] Exploiting and sharing new economic values from illegally collected data could constitute an infringement of sovereignty, and in this context, a regulatory framework should be established.

[33] *See supra* note 354.

2.3 Easier and Broader Monitoring of Other Countries

Just as AI makes it easier to collect data from individuals, AI implies easier and broader monitoring of other countries. Drones can go further, to more dangerous areas. Satellite images can be monitored constantly using AI at a faster rate and with higher accuracy. This is the same with internet user data. Moreover, as mentioned earlier, extraterritorial jurisdiction can be a means by which a country can exercise its power to monitor other countries. Against this backdrop, an international convention on limiting international monitoring can be beneficial for the international community. Autonomous artificial super-intelligence is difficult to track and control, and the issue can easily lead to an international dispute concerning infringement of national sovereignty. Therefore, international guidelines could discourage states to abuse the new technology.

Another issue that needs to be addressed is national security exceptions. Since most treaties being concluded recently include provisions of national security exception, now it is highly likely that new treaties to be concluded in the future also will adopt similar provisions concerning national security exception. Even if a treaty regarding information collection comes into effect, an agreement party has the right to be exempted from that duty under the agreement for reasons of a national security . In the event that certain information of a foreigner or a foreign company is judged as relevant to national security, it is highly likely that the agreement party in question will be able to retrieve and access that information regardless of the treaty provisions. This will be quite similar to what the US government claimed in the Snowden Case in terms of national security. Thus, even if a treaty is concluded with this condition, information might not be protected completely from the authority of a foreign government in critical moments.

3 State Responsibility

The issue of 'attribution reformulation' raises a fundamental question. To borrow the words of the European Parliament: it involves "creating a specific legal status for robots in the long run, so that at least the most sophisticated autonomous robots could be established as having the status of electronic persons responsible for making good any damage they may cause, and possibly applying electronic personality to cases where robots make autonomous decisions or otherwise interact with third parties independently."[34]

The discussion on state responsibility in terms of international laws is distinguished from the discussion on legal responsibilities in terms of domestic laws to

[34] *See* European Parliament Resolution of 16 February 2017 with Recommendations to the Commission on Civil Law Rules on Robotics (2015/2103(INL), doc. no. P8_TA(2017)0051), available at http://www.europarl.europa.eu/sides/getDoc.do?type=TA&reference=P8-TA-2017-0051&language=EN&ring=A8-2017-0005, para. 59;

some extent. The discussion on state responsibility in terms of international laws does not necessarily assume the state's act, intentional or negligent.[35] For an occurrence of performance or nonperformance, it is not discussed whether it involves the state's intentional act or negligence. Instead, only two elements—existence of a violation of international laws and whether the violation is attributable to the state—are investigated.[36] The discussion on state responsibility in terms of international laws is distinguished from that in terms of domestic laws since the latter focuses on whether it is intentional or negligent when it comes to who takes legal responsibilities concerning regulation on AI. Likewise, the fact that whether a loss occurs is not considered is another difference from the matter of legal responsibilities in terms of domestic laws.[37] This fact also indicates the difference from the discussion on the matter of state responsibility that involves the use of AI in terms of domestic laws.

Such characteristics of state responsibility norms present important implications in determining whether various activities that utilize AI will result in state responsibility. If state responsibility norms focusing on performance and nonperformance without assuming an intentional act or negligence on the part of a governmental official or agency are applied to the context of AI utilization, whether there has been an intentional act or negligence on the part of AI itself or its operator will not be taken into consideration. In other words, "the internal part (making judgment on behalf of a human)" which is the most complicated and sensitive issue with regard to AI is excluded in the context of state responsibility. Norms that focus on performance and nonperformance with no consideration of resulting loss as well exclude the question of who is responsible to compensate from the discussion of state responsibility, which is one of the main interests in relation to AI. Such aspects of state responsibility in the discussion on AI need to be noted.

The conventional method of attribution is not easily applicable to LAWS.[38] The HRW claims that "there is no fair and effective way to assign legal responsibility for unlawful acts committed by fully autonomous weapons".[39] There has been a number of suggestions on how to address the question of attribution when it comes to LAWS. Some argue that those people who are "sufficiently involved with the weapon (military commanders, designers, or manufacturers) can be held accountable,"[40] while others have pointed out flaws in such arguments.[41] Curiously, few have suggested invoking state responsibility.

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[35] *See* 2001 ILC Draft Articles, Art. 2.

[36] Ibid.

[37] Ibid.

[38] *See generally,* Alston [81], pp. 35, 51.

[39] Human Rights Watch, *Losing Humanity: The Case Against Killer Robots* (2012), p. 42, available at https://www.hrw.org/sites/default/files/reports/arms1112_ForUpload.pdf.

[40] *See,* for example, Marchant et al. [35], pp. 272, 280, Krishnan [8], pp. 103–05, *cited by* Hammond [22], p. 3.

[41] *See generally* Beard [45], p. 617, *cited by* Hammond [22], p. 3.

'Attribution' or accountability in terms of law of state responsibility is one of the most important pending questions in international law.[42] It has become a major issue in many disputes.[43] In particular, the matter of distinguishing state agencies from non-state agencies and on what occasions acts of a non-state agency are attributable to the state have long been subject to discussion and dispute. Such problems and issues still remain unsolved. As AI has emerged, it is necessary to examine regulations in view of international law concerning acts not of 'non-state' subjects but of 'impersonal' subjects. In the area of state responsibility, while major legal issues were to distinguish state agencies from non-state agencies and coming up with when and how acts of non-state agencies should be regarded as acts of the state, the future issues will be when and how acts of emerging impersonal subjects should be (or not be) regarded as acts of the state. Until now, even if an activity is of a non-state subject, the subject making that decision itself is a 'person' (natural or juridical person). Even if a juridical person makes a final decision, the decision-making subject is a manager or person authorized by the juridical person. Such subjects' decisions become decisions of the juridical person. The present issue, however, is distinguished from the past issues in that the decision-making of 'impersonal' subjects that have started to emerge and are predicted to increase are not a 'person' as has been specified so far.

The fact that a judgment, decision, or an act of AI can result in state responsibility has already been discussed. For instance, in the event a state agency commits a violation of international laws actively by means of AI (e.g. acts of torture by means

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[42] *Application of the Convention on the Prevention and Punishment of the Crime of Genocide (Bosnia and Herzegovina v. Serbia and Montenegro)* (hereinafter 'Application of the Genocide Convention'), Judgment, I.C.J. Reports 2007, p. 202, para. 385; *See* Bodansky and Crook [21], p. 791. ("For the invisible college of international lawyers, the ILC's completion of the articles is a considerable success.").

One Tribunal that handled a similar issue in an investment dispute stated as below in the final judgment:

> [w]hen assessing the merits of the dispute, the Tribunal will rule on the issue of attribution under international law, especially by reference to the Articles on State Responsibility as adopted on second reading in 2001 by the International Law Commission and as commended to the attention of Governments by the UN General Assembly in Resolution 56/83 of 12 December 2001 (the ILC Articles) as a codification of customary international law.

Jan de Nul NV and Dredging International NV v. Arab Republic of Egypt, ICSID Case No. ARB/04/13, Decision on Jurisdiction (Jun. 16, 2006), para. 89.

[43] *See Nicaragua* case, Merits, Judgment, I.C.J. Reports 1986, p. 65, para. 115 "effective control"; *Application of the Genocide Convention*, ibid., p. 43, (Confirms the jurisprudence of the *Nicaragua* case); WTO Appellate Body Report, *United States-Definitive Anti-Dumping and Countervailing Duties on Certain Products from China* (hereinafter 'US - AD/CVD'), WT/DS379/AB/R, Adopted Mar. 25, 2011, DSR 2011:V, p. 2869 ("Government ownership alone does not prove 'effective control'"); *Jan de Nul v. Egypt*, ibid., para. 173 ("International jurisprudence… requires both a "general control" of the State over [an] entity and a "specific control" of the State over the act"); *Gustav F W Hamester GmbH and Co KG v. the Republic of Ghana*, ICSID Case No. ARB/07124, Award, 18 June 2010, para. 179 ("The jurisprudence of the ICJ sets a very demanding threshold [by requiring] both general control of the State over the entity, and specific control of the State over the particular act in question. This is known as the "effective control" test").

of AI equipment) or a state agency fails to regulate a judgment, decision, or act by its AI system appropriately, and as a result, a violation of international law occurs (e.g. deployment of unmanned attacking devices despite the recognized danger to adjacent civilians), it may result in state responsibility. [44] Previous discussions mainly focus on a state's active violation of international law by means of AI. In other words, violation of individual international laws (violation of primary norms by means of AI) has been the main issue to be discussed while there has yet to be a review of how such performance or nonperformance by means of AI is 'attributable' or not to the state.

3.1 State Action Using AI

Governments are increasingly using AI in the exercise of their regulatory authority. There are many recent examples of utilization of AIby a government or government agency. Since January 2020, in the context of fighting COVID-19, countries are turning more and more to AI.

As a matter of fact, a group of scientists proposed an AI-based prognosis system to predict the deterioration of COVID-19 patients.[45] AI has been used to diagnose X-rays, and it is also contributing to the fight against the global pandemic. In addition to diagnosis, AI is a useful tool in many other aspects of the fight against pandemics, such as tracking, contact tracing, monitoring, and even developing vaccines.[46] For example, In the midst of such a global crisis, Nigeria has been using very simple, yet powerful AI-based tool to offer remote medical services to its people.[47] South Africa used an AI-based algorithm to predict how the virus will spread.[48] AI is also useful in finding asymptotic patients. Instead of randomly sampling asymptotic travelers, Greece used a machine learning system, called Eva, which estimated an individual's risk of infection based on travel history and demographic information.[49] Although Eva was developed in close consultation with lawyers so that it does not

[44] "The law of state responsibility is concerned with the determination of whether there is a wrongful act for which the state is to be held responsible, what the legal consequences are, and what reparations should be taken." Garcia [26], p. 104.

[45] Shamout et al. [33].

[46] Arora et al. [75].

[47] Abbany [98].

[48] Ibid.

[49] Nature, *Greece used AI to Curb COVID: What Other Nations Can Learn.* (Sep. 22, 2021), available at https://www.nature.com/articles/d41586-021-02554-y.

breach the EU's GDPR,[50] AI algorithms developed outside Europe may collect more information than the minimum amount required.[51]

A number of government agencies in the US, including the Internal Revenue Service (hereinafter IRS) and Department of the Treasury, are developing AI to detect fraud.[52] AI provides a useful tool to monitor COVID-19 relief funding; monitoring allocation of trillions of dollars is not easy with the antiquated traditional tools that have been used to track money flows.[53]

As the use of AI becomes more frequent in the public sector, there is also an increase in laws related to AI. For example, the state of Illinois passed the Artificial Intelligence Video Interview Act, which requires that when AI is used to determine whether an applicant is eligible for an in-person interview, the Department of Commerce and Economic Opportunity needs to analyze whether there is a racial bias in the use of AI.[54]

We can also find examples where a government delegates judgment to AI or something similar to it, instead of merely using AI for its convenience. Estonia pushed forward to delegate legal judgment to AI for small-claims cases.[55] Although other countries have incorporated AI to facilitate legal processes, Estonia is the first to give AI full decision-making authority.[56] American banks have been using AI to assess the credit-scores of individuals,[57] as well as the investment worthiness of small businesses.[58]

The Social Security Administration in the US utilizes AI to facilitate formal adjudication.[59] For example, a clustering algorithm based on machine learning is used to assign cases to adjudicators. The US Patent and Trademark Office also use AI in the examination process before patent examiners make a decision on whether to issue a patent or not.[60] However, these examples fall short of completely delegating the adjudicating authority to AI as it is only used to gather and analyze data prior to a decision.

Other examples in this regard are provided in the table below:

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[50] Regulation (EU) 2016/679, General Data Protection Regulation (GDPR), full text available at https://gdpr-info.eu/.

[51] *Supra* note 424.

[52] West [16].

[53] Ibid.

[54] IL H.B. 53. For a list of legislation related to AI in the US, *see* https://www.ncsl.org/research/telecommunications-and-information-technology/2020-legislation-related-to-artificial-intelligence.aspx.

[55] Niller [31].

[56] Ibid.

[57] Balogh and Johnson [86].

[58] Pazzanese [13].

[59] Engstrom et al. [24].

[60] Ibid., pp. 46–52.

	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Aroud Willems and Maryanne Kamau, *Of Binding Provisions and Trust Marks; Roadmap to a Global Legal Framework for the Digital Economy,* Legal Issues of Economic Integration, Vol. 46, Issue 3 (2019), pp. 225–246				Taxation on digital economy (240) Regulations concerning privacy and security (241–2) Intellectual property rights and related intermediary liability (244) Payment system (245)
Kaylynn Noethlich, *Artificially Intelligent and Free to Monopolize: A New Threat to Competitive Markets around the World,* American University International Law Review, Vol. 34, Issue 4 (2019), pp. 923-[ii]	Noethlich (2019: 955–6) points out that governments are not equipped to effectively regulate the use of AI, which actually leaves room for the government to use AI in regulating oligopolies			The author addresses a possibility of AI algorithmic collusion and a need to update the regulations against cartels
Barbora Jedličková, *Digital Polyopoly,* World Competition, Vol. 42, Issue 3 (2019), pp. 309–334	The authors point out that a digital economy is inevitably interdependent due to the inherent feature of algorithms (312). As a result, in the digital economy, there could be instances where there are many competitors, and yet parallel conduct, resembling that of oligopolistic firms, occurs—a digital polyopoly (313)		If the government were to regulate digital economy concerning algorithm-based collusion, the question of liability becomes an issue. (330–2). When the algorithm was not designed to create a collusion on purpose, the level of predictability could be an important issue to gauge the level of liability	

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Falco Kreis and Markus Kaulartz, *Smart Contracts and Dispute Resolution–A Chance to Raise Efficiency?* ASA BULLETIN 2/2019 (2019), p. 37	AI could be applied to many steps of a dispute settlement process, ranging from assessing the dispute, and identifying relevant cases to proposing potential settlements and collecting factual statements or parties' positions (349)	In theory, fully automated dispute resolution procedures are also possible (356)	Although most arbitration laws do not prohibit AI arbitrators, some explicitly require that only "natural persons" act as arbitrators (352)	Smart contracts allow for an automated initiation of dispute settlement mechanism. However, whether an outcome of such a process could be legally binding is questionable. For example, the Art. II(1) of the New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards 1958 requires an arbitration agreement to be "in writing" (343). It might be argued that an automated dispute settlement under the smart contracts does not qualify as a written agreement

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Maxi Scherer, *Artificial Intelligence and Legal Decision-Making: The Wide Open?-A Study Examining International Arbitration,* Journal of International Arbitration, Vol. 36, Issue 5 (2019), pp. 539–574			AI-based decision-making is also subject to biases, because data used to train the algorithm could be infected with biases. For example, there is a concern that if AI is used in the investment arbitration processes, it could reinforce the propensity to favor investors (559–560)	In 2016, a group of researchers tested whether the decisions by the European Court of Human Rights (ECtHR) regarding prohibition of torture, right to a fair trial, and right to respect for private and family life could be predicted by natural language processing and machine learning (547–8). The researchers found that they could predict the outcome of the Court's decision 79% of times (548). However, Scherer disagrees that their study supports AI's ability to accurately predict the judgments of the court ex ante (550) [The original study: Nikolaos Aletras et al., Predicting Judicial Decisions of the European Court of Human Rights: A Natural Language Processing Perspective, PeerJ Computer Science 2:e93 (2016).]

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Gauthier Vannieuwenhuyse, *Arbitration and New Technologies: Mutual Benefits,* Journal of International Arbitration, Vol. 35, Issue 1 (2018), pp. 119–130			Automation of arbitration could come at the cost of losing flexibility, one of the essential benefits of arbitration (126)	Smart contracts encompass self-executing procedures based on predetermined conditions (127). While smart contracts reduce the risk of litigation, the legal interpretation of the terms remains unclear (127–8) The current legal framework of arbitration, such as the United Nations Commission on International Trade Law (UNCITRAL) Model Law on International Commercial Arbitration, does not prohibit the use of AI in arbitration. (123)
Christine Sim, *Will Artificial Intelligence Take over Arbitration?,* Asian International Arbitration Journal, Vol. 14, Issue 1 (2018), pp. 1–14	AI could entirely replace human arbitrators, serve only as a part of the tribunal along with human arbitrators, or be used only for consultation		Sim questions whether AI could ensure due process and flexibility during an arbitration (10–11)	IBM's AI Ross is the first AI attorney, which can handle a law firm's bankruptcy practice (6)

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Lorna McGregor, Daragh Murray and Vivian Ng, *International Human Rights Law as a Framework for Algorithmic Accountability,* International and Comparative Law Quarterly, Vol. 68, Issue 2 (2019), pp. 309–343	The UK has introduced an automated algorithmic-based social security system (315–6). It may cause individuals with lower digital literacy level or internet connectivity to be subject to discrimination (316) London Councils use algorithms to determine the likelihood of neglect or abuse of children, which raises concern of privacy (316) Even if AI is only used to generate information and humans make the final decision, it may be difficult for human decision maker to rightly balance the recommendation from AI, overcoming its bias (317) Durham Constabulary in the UK uses 'Harm Assessment Risk Tool' ('HART') to assess individuals for out-of-court process eligibility (333) In a recent case of the *State of Wisconsin v Eric L. Loomis* in the US, the Wisconsin Supreme Court decided that a risk assessment tool based on AI may be consulted for a sentence decision, but not as a determinative source (339)		International human rights law requires states to prevent human rights violations, and both state actions and omissions could be held accountable to it (311–312). Because using algorithms to determine access to social security, to approve medical intervention, or to identify children at risk can affect human rights, states also have an obligation to regulate such use of AI in accordance with the international human rights law (310–312) There is also the issue of transparency of AI (318). Because the algorithms are difficult to understand, the transparency of decision-making process could be severely undermined. This implies that even with the presence of a human operator, it might not be enough to work as a safeguard against biased algorithmic decision-making process (338) Even if AI recommendations are not final or determinative, human decision-makers may be disinclined to go against the algorithmic decision in order to avoid greater accountability for such a decision (340)	

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Steven Livingston and Mathias Risse, *The Future Impact of Artificial Intelligence on Humans and Human Rights,* Ethics & International Affairs, Vol. 33, Issue 2 (2019), pp. 141–158			Super-intelligent AI could be put above the moral and legal standards of ordinary humans (152)	With further progress of AI, it might even be possible to redesign human DNA. While AI-enabled gene editing tool could lead to cures of many diseases, it could exacerbate the inequality between the rich and the poor (152)
Eileen Donahoe and Megan MacDuffee Metzger, *Artificial Intelligence and Human Rights,* Journal of Democracy, Vol. 30, Issue 2 (2019), pp. 115–126			"A range of practical problems that AI might cause for democratic accountability, due process, and the rule of law has been manifest in the recent adoption of AI by several U.S. jurisdictions that apply it in matters of criminal sentencing, parole, eligibility for social services, and employment decisions." (117)	

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Mariano-Florentino Cuéllar, *A Common Law for the Age of Artificial Intelligence: Incremental Adjudication, Institutions, and Relational non-Arbitrariness,* Columbia Law Review, Vol.119, Issue 7 (2019), pp. 1773–1792			Delegating decision making authority to AI in dispute settlement could make it difficult to defend and justify a judgment (1783). Again, the problem concerns the transparency of a decision (1784) "In *State v. Loomis*, the Wisconsin Supreme Court held that a trial court's use of an algorithm for risk assessment in sentencing didn't violate due process, even though the methodology used to produce the assessment remained undisclosed to either the defendant or the court. The *Loomis* court insisted on a mild procedural safeguard instead: a 'written advisement' accompanying presentencing reports. Irrespective of how one weighs the court's understandable concerns about practical constraints and avoiding excessive discovery, it's far from clear that its holding promotes the kind of meaningful deliberation a reasonable observer would naturally associate with relational non-arbitrariness—about the design of the algorithm, the data used, or even the user interface. That this ideal of practically informed reason-giving, shared deliberation, and manageable decision costs is difficult to achieve even without AI in the picture should be obvious." (1786)	

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Philipp Hacker, *Teaching fairness to Artificial Intelligence: Existing and Novel Strategies against Algorithmic Discrimination Under EU Law,* Common Market Law Review, Vol. 55 (2018), pp. 1143–1186	The US courts use COMPAS algorithm to predict the recidivism likelihood of criminal defendants, which affects sentencing decisions (1144). The problem is that there is a reason to suspect that COMPAS is biased against black offenders (1144)		Whether AI is used as a tool or as an autonomous decision maker, when it leads to direct or indirect discrimination, it could be subject to anti-discrimination law. For instance, although the EU anti-discrimination law attempts to minimize bias in AI, accidental discrimination in AI is not covered by it (1151–2). Moreover, the EU anti-discrimination laws are limited in both transactions and actors (1154) The GDPR could provide a framework to control algorithmic discrimination (1172–3)	

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Aaron D. Kirk, *Artificial Intelligence and the Fifth Domain,* Air Force Law Review, Vol. 80 (2019), pp. 183–236			If an AI is used as a weapon during an international fight, it could be subject to international humanitarian law. (211). However, computer codes, such as a malware, have distinct characteristics compared to physical weapons (212). The current state of international law does not unambiguously regulate cyberwarfare (214)	The US Joint Chiefs of Staff (JCS) defined cyberspace as "a global domain within the information environment consisting of the interdependent networks of information technology infrastructures and resident data, including the Internet, telecommunications networks, computer systems, and embedded processors and controllers." (197) The effects-based approach encompassed by the Tallinn Manual implies that existing international treaties could be extended to cover cyberspace. Moreover, commentary on Tallinn 2.0, Rule 4 states that cyberspace activities should also respect national sovereignty, but it is unclear whether it could serve as an independent rule; "Cyber operations that prevent or disregard another State's exercise of sovereign prerogatives constitute a violation of such sovereignty and are prohibited by international law. (Tallinn 2.9, at 19, Rule 4 commentary)" (209)

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Young-Yik Rhim and KyungBae Park, *The Applicability of Artificial Intelligencein International Law,* Journal of East Asia & International Law, Vol. 12, Issue 1 (2019), pp. 7–30	Daniel Katz developed a machine learning model that predicts the decision of US Supreme Court, with approximately 70% accuracy level (21) AI can also automatically translate texts from language to another. International institutions such as the UN or the EU committees have already begun to use Machine Translations to translate legal documents (23)		AI could one day act as an arbitrator in an international dispute settlement as well. The 1958 New York Convention does not prohibit AI from acting as an arbitrator, but there could be risks for algorithm bias (19)	Many AI expert systems have been developed, such as DENDRAL in chemistry, MYCIN in health care, and SHYSTER in intellectual property law (10–11). The legal field is experiencing an increase of AI use, and a fast-paced development of Legal Tech in general (13–14). IBM's Ross is a classic example (14)

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Sam Wrigley, *BOTS, Artificial Intelligence and the General Data Protection Regulation: Asking the Right Questions,* Trinity College Law Review, Vol. 22 (2019), pp. 199–211				Although the GDPR is not targeted to regulate AI, one article directly concerns automated data processing: "Under article 22, data subjects have a right not to be subject to 'a decision based solely on automated processing, including profiling, which produces legal effects concerning [the data subject] or similarly significantly affects him or her'. The article provides three exceptions: first, where the decision is necessary for entering into or performing a contract between the controller and the data subject; secondly where the decision is authorized by EU or Member State law; and thirdly where the decision is based on the data subject's 'explicit consent'." (204) However, the authors argue that the importance of article 22 is limited, and that we should expand the discussion on regulating AI to a broader set of topics (208)

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Matthijs M. Maas, *International Law Does Not Compute: Artificial Intelligence and the Development, Displacement or Destruction of the Global Legal Order,* Melbourne Journal of International Law, Vol. 20, Issue 1 (2019), pp. 29–57	Near ubiquitous surveillance allows close monitoring, facilitating law enforcement not only at national level, but also at international level (44) In 2018, it was reported that the Ministry of Foreign Affairs of the People's Republic of China started to use AI as a strategic decision support system (45) In domestic affairs, there is already some evidence of legal automation (46)			
Begoia Gonzalez Otero, *Evaluating the EC Private Data Sharing Principles: Setting a Mantra for Artificial Intelligence Nirvana?,* Journal of Intellectual Property, Information Technology and Electronic Commerce Law, Vol. 10, Issue 1 (2019), pp. 65–83				There is a need to foster a safe and healthy environment for data sharing as the importance of data grows with the development of AI technology
Reinmar Nindler, *The United Nation's Capability to Manage Existential Risks with a Focus on Artificial Intelligence,* International Community Law Review, Vol. 21 (2019), pp. 5–34			As the importance of AI grows, the UN may have to step in to provide a foundational framework to regulate the use of AI (34)	It may be possible to draw some parallels between biological weapons and AI, as AI could be abused to create serious harm to society which could be difficult to be detected ex ante (26). However, given the enormous benefits of AI, it would be more difficult to regulate AI (26–27)

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Peter Asaro, *What Is an Artificial Intelligence Arms Race Anyway, I/S*: A Journal of Law and Policy for the Information Society, Vol.15 (2019), pp. 45–64	Deputy Defense Secretary, Bob Work, tried to enhance US military digital technology, with a focus on "remote and autonomous platforms, big data and information processing, and information dominance" (61). AI used to support military planning and information warfare is such an example			One could argue that there is an "arms race" with respect to AI. First, companies from different countries are competing for international market advantage (53–4). Another view is that countries are fighting over technical dominance (55). Of course, AI can also be weaponized (59)
Ashley Deeks, Noam Lubell, and Daragh Murray, *Machine Learning, Artificial Intelligence, and the Use of Force by States,* Journal of National Security Law and Policy, Vol. 10, No. 1 (2019), pp. 1–26	AI is used in government for different types of risk analysis: "Risk analysis has long been recognized as a crucial tool in decision-making in other spheres of government, from health care provision to the criminal justice system, but it appears to be comparatively under-utilized in government decisions on the use of force." (6) For instance, machine learning algorithms can be developed to give early warnings of an attack (6). In fact, the US Central Intelligence Agency, and the US DoD, as well as the UN, have already begun to use public data to predict conflict initiations (7)	The US DoD is exploring automated cyber responses (8). It should be noted that an automated response can occur not only in cyberspace, but in the physical domain, such as air defense or underwater vehicles (9). In particular, when there is a cyber attack, delegating decision-making authority to a computer will be the most efficient response (18)	The concept of "meaningful human control" will be an important legal point when algorithms are used to make automated decisions (15–16). "In the context of resort to force, the role of algorithms in the underlying calculations could lead states to unwittingly make war-related decisions almost entirely based on machine calculations and recommendations. A key concern is that, even if the final war-making decision rests with humans (in this case, a state's executive branch), the degree to which that human involvement is considered 'meaningful' may be questioned. Two relevant factors are (a) the centrality of algorithms to the decision-making process, and (b) the deference given to algorithmically- produced decisions." (17)	

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Amandeep Singh Gill, *Artificial Intelligence and International Security: The Long View,* Ethics & International Affairs, Vol. 33, Issue 2 (2019), pp. 169–179	AI could be applied to enhance military capabilities, such as tracking down nuclear submarines, and delivering unmanned aerial or underwater vehicles		As AI weapons could be indiscriminate, there is a potential violation of international humanitarian law principles	
Antonio Missiroli, *The Dark Side of the Web: Cyber as a Threat,* European Foreign Affairs Review, Vol. 24, no. 2 (2019), pp. 135–152			There is a movement at the UN level to further regulate the cyberspace, such as the 2013 Report of the United Nations' Group of Governmental Experts (UN GGE and the Tallinn Manual (138). However, there is not yet a customary international law fully in place to regulate cyberspace, and the only international regime directly concerning the cyberspace is the Budapest Convention on Cyber Crime (140)	There have been a number of cases where public agencies, such as the German Parliament or the US Office of Personnel Management, were compromised by cyberattacks (*see* Annex)
Neil Robinson and Chelsey Slack, *Co-Operation: A Key to NATO's Cyberspace Endeavor,* European Foreign Affairs Review Vol. 24, no. 2 (2019), pp. 153–166			NATO is expanding its operation to cyberspace. However, since it is an emerging challenge, the specific responsibilities are not clear, even though there are some guidelines provided by previous research (158)	

(continued)

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	Government using AI as a tool	Government delegating decision making to AI	Legal responsibility of government/public official for using AI	Other issues
Lindsay Freeman, *Law in Conflict: The Technological Transformation of War and Its Consequences for the International Criminal Court,* New York University Journal of International Law and Politics, Vol. 51 (2019), pp. 807–869	The use of UAVs, or unmanned automated vehicles, is already common: By the end of 2005, the US had deployed over 2400 UAVs to Iraq and Afghanistan (837) The US military uses AI to make strategic decisions, such as Real-Time Adversarial Intelligence and Decision-Making (RAID), provides information on previous enemy actions within the area of operation (838) Governments are also interested in the information made available by big data, especially when it can affect the public image of a country, and thus soft power. For instance, Israel and Hamas are engaged in a media war to promote their own side of narratives on how the conflict is progressing (843). In the future, governments may also attempt to use AI to not only collect and analyze information, but also create and disseminate information of their preference. Russia is already suspected of using the new technology to further promote its propaganda (842). For instance, during the conflict with Ukraine, Russia has used mass dissemination of disinformation to justify its occupation of Crimea (845)		When automated weapons are used in combat, the attribution of such actions becomes one of the hot issues in terms of legal responsibility. (853) Jurisdiction over cyberspace is another problem, as digital activities are not bound by national borders. (860)	Governments can be a target of cyber attacks. For example, one hacker known as The Jester claimed that it has conducted over two hundred attacks on governments and other groups (836) The recent conflict in Syria provides an example where cyberwarfare was used in parallel to attacks in the physical domain (834–5). The Syrian Electronic Army (SEA) claimed to have hacked Twitter accounts of influential media such as the New York Times and Washington Post

3.2 Cross-Border Impact in a New Legal Dimension

Entering the twenty-first month as of September 2021, COVID-19 is still engulfing the entire global community.[61] Mindful of the fact that China was the originating place of the virus and that its failure to contain the virus in the early stage of the disease caused the pandemic, discussions to place state responsibility on China have been robust in various fora.[62] State responsibility in this respect can stem from two sources of international law: IHR 2005[63] and customary international law. Both sources of law, however, invite complex evaluations and pose difficult questions.

First of all, IHR 2005 is not detailed enough to impose specific obligations on states commensurate with the demands from the international community in the early stage of a pandemic. This phenomenon is not surprising because IHR 2005 was introduced without experiencing a pandemic with the scale and impact of the ones like COVID-19.[64]

On the other hand, customary international law imposes legal obligations on each state to manage its territory with due diligence and notify neighboring states of transboundary harm when one is suspected. These obligations from customary law, however, presuppose a physical flow of moving particles or something similar to it, which does not necessarily exist in the context of a pandemic which is in fact caused by human beings carrying a virus in their respiratory system arguably without any physical flow of particles.

It is questionable whether China fulfilled its duties to manage its territory with due diligence and notify neighboring states as the customary international law requires. As confirmed by the Trail Smelter Arbitration,[65] a state needs to manage its territory

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[61] United Nations [92].

[62] *Alters et al. v. People's Republic of China* et al. (Case 1:20-cv-21108-UU), United States District Court Southern District of Florida (Mar. 13, 2020), available at https://images.law.com/contrib/content/uploads/documents/392/85094/Coronavirus-China-class-action.pdf; Sanger [25],*Chinese State Media Accuses Pompeo of Spreading 'Political Virus'*, NBC News (Apr. 27, 2020), available at https://www.nbcnews.com/health/health-news/live-blog/2020-04-27-coronavirus-new sn1193086/ncrd1193331#blogHeader; *China Pushes Back Against US Claims that Coronavirus Originated from Wuhan Lab*, CNN (May 4, 2020), available at https://edition.cnn.com/2020/05/04/asia/china-us-coronavirus-spat-intl-hnk/index.html; *Pompeo Says 'Significant' Evidence That New Coronavirus Emerged from Chinese Lab*, Reuters (May 3, 2020), available at https://www.reuters.com/article/us-health-coronavirus-usa-pompeo/pompeo-says-significant-evidence-that-new-coronavirus-emerged-from-chinese-lab-idUSKBN22F0SC; Hass and Dong [85], Silver et al. [59].

[63] World Health Organization [95].

[64] The latest version of IHR was adopted in the aftermath of Severe Acute Respiratory Syndrome (SARS) in 2002–2003. World Health Organization [96]; World Health Organization, *International Health Regulations*, *available at* https://www.who.int/health-topics/international-health-regulations#tab=tab_1.

[65] *Trail Smelter Case (United States, Canada), 16 April 1938 and 11 March 1941*, United Nations, Reports of International Arbitral Awards, Vol. III, p. 1920.

with 'due diligence' [66] and 'use all the means at its disposal' [67] so that it does not create harm to neighboring states. It also has to notify the neighboring states once it is aware of the situation. The no harm principle is included in the "Draft Articles on Prevention of Transboundary Harm from Hazardous Activities".[68] Article 3 stipulates that "The State of origin shall take all appropriate measures to prevent significant transboundary harm or at any event to minimize the risk thereof."[69]

In a more recent case between Costa Rica and Nicaragua with respect to the development of San Juan, the ICJ confirmed that states are responsible to notify neighboring countries if there are risks for significant transboundary harm.[70] An important factor is that these cases deal with physical movement of activities across national territories and jurisdictions. This is also reflected in the *2001 Draft Articles on Prevention of Transboundary Harm from Hazardous Activities*. (hereinafter Draft Articles on Transboundary Harm)Article 1 of the Draft Articles on Transboundary

[66] See ibid., p. 1965.

[67] *Pulp Mills on the River Uruguay (Argentina* v. *Uruguay)*, Judgment, I.C.J. Reports 2010 (I), pp. 55–56, para. 101.

[68] International Law Commission [40].

[69] Article 2 of the Draft defines key terms as the following:

> Article 2. Use of terms
>
> For the purposes of the present articles:
>
> (a) "Risk of causing significant transboundary harm" includes risks taking the form of a high probability of causing significant transboundary harm and a low probability of causing disastrous transboundary harm;
>
> (b) "Harm" means harm caused to persons, property or the environment;
>
> (c) "Transboundary harm" means harm caused in the territory of or in other places under the jurisdiction or control of a State other than the State of origin, whether or not the States concerned share a common border;
>
> (d) "State of origin" means the State in the territory or otherwise under the jurisdiction or control of which the activities referred to in article 1 are planned or are carried out;
>
> (e) "State likely to be affected" means the State or States in the territory of which there is the risk of significant transboundary harm or which have jurisdiction or control over any other place where there is such a risk;
>
> (f) "States concerned" means the State of origin and the State likely to be affected.

[70] *Certain Activities Carried Out by Nicaragua in the Border Area (Costa Rica* v. *Nicaragua)* and *Construction of a Road in Costa Rica along the San Juan River (Nicaragua* v. *Costa Rica)*, *Judgment, I.C.J. Reports 2015* (henceforth "Certain Activities & Construction of a Road"), p. 665, para. 106.

Harm specifies that the harm comes "through their physical consequences".[71] In other words, one could argue that an activity without any physical consequences lies outside the scope of the Draft Articles on Transboundary Harm, as well as customary international law. This point is reiterated in the footnote of the Draft Articles on Transboundary Harm.[72]

With respect to COVID-19, it is questionable whether the spread of the virus is a physical consequence. The spread of a virus is not the same as the movement of pollutants: virus transmission occurrs naturally through the movement of people across borders. This is different from activities that take place within national borders that cause transboundary harm. Moreover, it is difficult to argue that there was a breach of due diligence. Tracing a virus, which is invisible and intangible, is different from tracing pollutants that are detectable by sight or smell. As a result, the level of control expected under the obligation of due diligence must be different for the spread of COVID-19. Thus, it seems inappropriate to apply a no harm principle straight onto the recent global pandemic.

As such, more thorough analyses should be conducted to confirm state responsibility for the spread of COVID-19 under international law, going beyond simply pinpointing a particular country as the origin and as the country with lax initial responses. Once the current pandemic is brought under control, debates on state responsibility based on international law jurisprudence will likely be conducted. Sound legal bases should guide these debates.

3.3 Application of Attribution Rules Under 2001 ILC Draft Articles

It is not difficult to think of instances of acts and omissions related to AI that constitute internationally wrongful acts. As shown by the 2001 ILC Draft Articles, the ILC

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[71] *See* International Law Commission, *2001 Draft Articles on Prevention of Transboundary Harm from Hazardous Activities.* Article 1.

> Article 1. Scope
>
> The present articles apply to activities not prohibited by international law which involve a risk of causing significant transboundary harm *through their physical consequences.* (emphasis added).

[72] *See* ibid., commentary Article 1. para. 16. ("The fourth criterion is that the significant transboundary harm must have been caused by the "physical consequences" of such activities. It was agreed by the Commission that in order to bring this topic within a manageable scope, it should exclude transboundary harm which may be caused by State policies in monetary, socioeconomic or similar fields. The Commission feels that the most effective way of limiting the scope of these articles is by requiring that these activities should have transboundary physical consequences which, in turn, result in significant harm.").

makes it clear that "[e]very internationally wrongful act of a State entails the international responsibility of that State."[73] According to the ILC, an "internationally wrongful action" can consist of either an act or omission, but it has to be "(a) attributable to the State under international law; and (b) constitutes a breach of an international obligation of the State."[74] It has been subject to debate what action is attributable to a state, but it is generally recognized that when an action is conducted by a body or agency of a state, it is regarded as attributable.[75] Following this logic, when an AWS operates under government authorities, its actions can be attributable to such institutions, and henceforth, the state itself.[76]

Breaches of international humanitarian law by indiscriminate use of automated weapons systems on civilians is an example. In such situations, the question is whether state responsibility can be invoked. It seems that under certain conditions, it is possible to fulfill the attributability requirement in the 2001 ILC Draft Articles. Using the 2001 ILC Draft Articles on the state responsibility, [22] a study explains:

> A state engages in an "internationally wrongful action" when an act or omission "(a) is attributable to the State under international law; and (b) constitutes a breach of an international obligation of the State." An action is attributable to a state when it is conducted by an "organ" of the state, which almost certainly includes the military as well as intelligence agencies. Therefore, assuming that [L]AWSs operate under the authority of these institutions, their actions would be attributable to the state.[77]

As a decision made by AI involves several actors—ranging from program developer to the policy-maker who decided to introduce the AI—it is hard to track down and pin-point one actor behind the decisions of an AI.[78] LAWS is especially tricky to attribute responsibility because of its so-called autonomy.[79] LAWS involves "the capacity of a machine to operate in the real-world environment without any form of external (human) control, once it is activated."[80]

The US DOD defines LAWS as "once activated, can select and engage targets without further intervention by a human operator."[81] By this definition, human-supervised systems that allow human intervention are also classified as a LAWS.[82]

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[73] Responsibility of States for Internationally Wrongful Acts, [2001] 2 Y.B. INT'L L. Comm'n 26, U.N. Doc. A/CNA/SER.A/2001/Add.1, *cited by*. Hammond [22], p. 11.

[74] Responsibility of States for Internationally Wrongful Acts, [2001]2 Y.B. INT'L L. Comm'n 26, U.N. Doc. A/CNA/SER.A/2001/Add.1 Art. II, *cited by* Hammond [22], p. 11.

[75] Responsibility of States for Internationally Wrongful Acts, [2001]2 Y.B. INT'L L. Comm'n 26, U.N. Doc. A/CNA/SER.A/2001/Add.1 Art. IV(1), *cited by* Hammond [22], p. 12.

[76] Hammond [22], p. 12.

[77] *See* Hammond [22], p. 668.

[78] *See* Alston [80], p. 51.

[79] *See supra* note 365, p. 652.

[80] Lin et al. [77].

[81] *Supra* note 160, pp. 619–620.

[82] US Department of Defense, *Autonomy in Weapon Systems,* Directive 3000.09 (Nov. 21, 2012), available at http://www.fas.org/irp/doddir/dod/d3000_09.pdf, *cited by* Schmitt and Thurnher Schmitt and Thurnher (2013), p. 3.

The Human Rights Watch (hereinafter HRW) refers to them as "human-out-of-the-loop weapons."[83] Thus, a machine's ability to 'select' and 'engage' seems to be a key defining feature of LAWS.[84]

There are other autonomous systems that are referred to as "automatic weapons defense systems."[85] They are only designed for defense. For instance, a 'close-in weapon system' (hereinafter CIWS, also known as "Sea Whiz") can automatically respond to inbound missiles, and is used to defend warships against missiles.[86]

Some question whether it is fair to attribute illegal actions to the commander of the LAWS when he or she does not have total control over its actions.[87] However, if we were to minimize the side effects of AI and encourage more cautious applications of the LAWS, it seems logical to attribute the faults of the AWS to a legal entity. Thus, for instance, the US DOD makes it clear that it considers the person behind the AI to be responsible for the AI's actions.

> [P]ersons who authorize the use of, direct the use of, or operate autonomous […] weapon systems must do so with appropriate care and in accordance with the law of war, applicable treaties, weapon system safety rules, and applicable rules of engagement.[88]

Key provisions of the 2001 ILC Draft Articles are briefly summarized below: First of all, Article 4, paragraph 1 of the 2001 ILC Draft Articles specifies that every act of a state agency is regarded as an act of the state. According to the relevant note in the 2001 ILC Draft Articles, the 'state agency' mentioned here includes "any person or entity" that forms the state organization and acts for the state.[89] Likewise, the "person or entity" stated in Article 4, paragraph 2 of the same is basically understood as including every natural person and every juridical person.[90] In summary, key provisions of the law of state responsibility are based on the assumption that a natural person or juridical person is involved in principle.

In actual situations concerning disputes, such provisions are applied according to the intent. Such disputes are examined on the assumption of whether there was a natural or juridical person in a certain position or status who was involved. In the case of Moses, for example, the arbitrator named Liber stated: "A public official or an authorized person represents his/her government or the body of all public officials and authorized persons in his/her scope of authority."[91] This statement assumes a natural person of a certain status. However, there is a possibility that similar issues may be handled differently among countries depending on their specific domestic

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[83] *Supra* note 422, Human Rights Watch, p. 30, *cited by* Hammond [22], p. 652.

[84] Anderson and Waxman [58], pp. 1100–1101.

[85] *Supra* note 422, p. 9, c*ited by* Schmitt and Thurnher [70], p. 4.

[86] *MK-15 Phalanx Close-In Weapons System (CIWS),* U.S. NAVY (Oct. 19, 2012) available at http://www.navy.mil/navydata/fact_display.asp?cid=2100&tid=487&ct=2.

[87] *See supra* note 365, p. 667.

[88] US Department of Defense, *Autonomy in Weapon Systems,* Directive 3000.09 (Nov. 21, 2012), available at http://www.fas.org/irp/doddir/dod/d3000_09.pdf; *See supra* note 365, p. 659.

[89] Commentaries, Art. 4; annotation, pp. 41–42.

[90] Commentaries, Art. 4; annotation, p. 46.

[91] Commentaries, Art. 4; annotation, p. 42.

laws since Article 4, paragraph 2 of the 2001 ILC Draft Articles states that "a person or entity in such a status according to domestic laws."[92] Thus, whether or not AI will be viewed as representing a state agency depends on the legal status that the state's domestic laws grant to it. Accordingly, whether or not Article 4 is applicable will also be determined.

According to Article 5 of the 2001 ILC Draft Articles, an act of 'a person or entity' entrusted by domestic law with the authority to exercise governmental power is regarded as an act of the state under international laws. According to the relevant note of the Draft, the term, "entity," in Article 5 means various entities, if not organizations, entrusted with the authority to exercise governmental power under domestic laws.[93] Such entities include public enterprises, public organizations, various types of state representatives, and even in special occasions, private enterprises.[94] In the same sense, acts of a natural person or entity entrusted by domestic law with the authority to exercise governmental power also may be attributable to the state.[95] Above-stated provisions as well basically assume that a natural person or juridical person is involved.

In addition, Article 7 of the 2001 ILC Draft Articles specifies that even if a 'person or entity' entrusted by a state agency or with governmental power engages in an act beyond its granted authority or in violation of a given command, such an act of that agency or entity may be attributable to the state.[96] Regarding the background of stipulating this provision, the annotation refers to 'agents' or 'public officials.' Thus, this provision also basically assumes the existence of a natural person.[97] Since Article 7 is applied to acts of an entity entrusted by a state agency or with an authority to exercise governmental power, this case is based as well on the assumption that there is a natural or juridical person concerned. Even if it is an act of an entity, whether or not that act was practiced by a natural person—"a person entrusted with governmental power" is important as mentioned by the US-Iran Claims Tribunal regarding the case of *Petrolane, Inc. v. Islamic Republic of Iran* (1991).[98]

In Article 8 of the 2001 ILC Draft Articles assuming a situation where a private subject is under the control of a governmental agency, the phrase, "a person or group of persons," is used.[99] This is the same in Article 9 of the 2001 ILC Draft Articles assuming a situation where the public power is not exercised properly for such purposes as revolution, armed conflict, and occupation by a foreign country.[100] In this context as well, the phrase, "a person or group of persons," is used regarding

[92] Commentaries, Art. 4; annotation, p. 45.
[93] Commentaries, Art. 5; annotation, p. 46.
[94] Ibid.
[95] Commentaries, Art. 5; annotation, p. 47.
[96] Commentaries, Art. 7; annotation, p. 51.
[97] Commentaries, Art. 7; annotation, pp. 51–52.
[98] Commentaries, Art. 7; annotation, p. 53.
[99] Commentaries, Art. 8.
[100] Commentaries, Art. 9; annotation, p. 57.

the subject. The above-stated provisions also basically assume that involvement of a natural person or juridical person is the fundamental basis.

Current provisions of state responsibility in the 2001 ILC Draft Articles are reliant on the basis of whether or not an act of a 'natural person' or 'juridical person' if any is attributable to the state.[101] Thus, it is necessary to examine this matter when it comes to the issue of state responsibility where AI is involved. Some argue that states are responsible for the actions of LAWSs by the operation of the traditional state responsibility principle as codified under the 2001 ILC Draft Articles assuming that LAWSs operate under the authorities of the state.[102] However, as mentioned earlier, this matter (of who should take legal responsibilities as a subject with a legal personality) is directly related to other discussions at home and abroad on AI that are developing currently in various legal forums. Fundamentally, therefore, the basic question is "What is AI?". This question is repeated with a similar emphasis in the area of international law.

3.4 In Case of State Actors

With internationally agreed verification and attribution procedures, unilateral or collective state measures could be subject to violations of international legal obligations stipulated in the 2001 ILC Draft Articles. When a state agency or governmental agency takes advantage of AI for its needs and purposes, it is an important matter in terms of domestic laws whether such a state agency or governmental agency bears legal responsibilities.

By way of example, in the financial sector, there is a possibility of damage caused by the malfunction of AI programs adopted by government regulators.[103] As long as the AI programs are merely used as an assisting tool for regulators and final decisions are made by the regulators, the measure at issue could still be attributed to the regulatory agency, thus the state at issue. The question, however, arises when the AI programs evaluate the circumstances and issue decisions automatically and autonomously. The answer to the question pretty much depends on the how much the government agency intervenes in the decision-making process, and thus how much independence the AI programs enjoy. Obviously, this is directly related to the technological development. In this instance, the issue of legal liability of the agency involved becomes murky and complex. So is the issue of state responsibility

[101] In relation to the *Nicaragua case,* the following statement of ICJ supports this idea: "[D]espite the heavy subsidies and other support provided to them by the United States, there is no clear evidence of the United States having actually exercised such a degree of control in all fields as to justify treating the contras as acting on its behalf.", *Nicaragua case, supra* note 138 para. 109.

The International Court of Justice confirmed the legal principle stated above with regard to the issue of *Armed Activities on the Territory of the Congo. Armed Activities on the Territory of the Congo (Democratic Republic of the Congo v. Uganda)*, I.C.J. Reports 2005, p. 168.

[102] Hammond [22], p. 2.

[103] *See* Dembosky et al. [61].

under international law. Indeed, even in the case of complete autonomy, attribution to state may still be found if the government agency at issue has delegated the authority to the AI program under domestic law,[104] has controlled the AI programs in specific circumstances,[105] or has accepted the decision of the program as if it were the government's own.[106]

An important issue is raised with regard to administrative law when it comes to the validity of the executive agency's measure and with regard to the civil law when it comes to compensation for a loss to the victims. In the same logical perspective, whether a state agency or governmental agency takes the legal responsibilities for the decision-making, the judgment and behavior of AI has important implications in terms of international law. This is because someone may gain profits or suffer a loss as a result of the decision or behavior and it matters whether the relevant country will bear responsibilities under international law.

Such issues are raised since the use of AI continues to increase in public sectors as well. The following are some of the main examples: The South Korean Ministry of Public Administration and Security announced that by 2019, it would adopt AI technologies such as 'AI administrative secretary' to the general e-government system. It also reported that it would handle governmental administrative services by means of AI. The Financial Services Commission as well announced that by the end of 2017, it would establish a review and analysis system by means of AI and big data analysis technology in order to prevent crimes in virtual currency and fintech-based money laundering. Such plans for financial supervision and regulation in common are based on the use of AI. The US Department of Homeland Security, Citizenship and Immigration Services has introduced AI for communication with humans (e.g. EMMA) and used the face recognition program in the process of visa issuance and airport security. In the US, AI is already utilized in policing to a large extent.[107] As such, important state services are performed via AI. Since the subject of examination is faced directly with the AI, it is suggested that the decision of AI is likely to be the governmental agency's decision. In addition, it is well known that in the administrative and security procedures of China, the AI based face-recognition technology is utilized widely. The Australian government as well entrusts computer programs with decision-making in some public sectors.[108] For example, the entry visa examination is implemented by means of AI, and controversy is rising within the country about entrusting AI with such important decisions.[109] Similar disputes are found in Canada as well. It is even predicted that in the future, AI will play such roles in interrogation, polygraph examination, and persuasion in investigation procedures of each country's judicial agencies.[110]

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[104] *See* ILC Draft Articles, Art. 5.

[105] *See* ibid., Art. 8.

[106] *See* ibid., Art. 9.

[107] *See supra* note 126, pp. 2539–40.

[108] *See supra* note 165, pp. 50–1.

[109] *See* Hendry [56], Wright [90].

[110] *See supra* note 126, pp. 2540–4.

Utilization of AI by governmental agencies are observed among local governments as well as central governments. Some states of the US have introduced AI in expressway safety management. In Korea, districts such as Seoul, Bucheon, and Paju utilize AI for various purposes such as water and sewage, subway, and fire safety management.

3.5 In Case of Non-state Actors

When the subject of an internationally wrongful act is a non-state actor, the attributability of state responsibility can be quite complicated, and when AI is involved in the process, the issue can be even more so.

Nevertheless, even when the requirements for attribution are not fulfilled, the concept of consequential responsibility may be another approach to invoke state responsibility. In consideration of AI characteristics, the government or state admits accountability for every situation resulting from deploying and utilizing an AI system right from the very beginning and takes state responsibility accordingly. This is a type of consequential responsibility. Even if a state or governmental agency fulfills its fiduciary duty, it takes the responsibility for consequences. If the focus is on the relief of victims, the validity of this policy is be admitted. Particularly, since it will become more difficult to verify the attribution of AI utilization to the state due to its unique nature,[111] it may be a reasonable alternative to view state responsibility according to the principle of consequential responsibility.

In comparison with situations where the principle of state responsibility is applied in a traditional line of command by persons, the operation of AI involves various complicated steps, directly or indirectly, such as programming, decision-making on introduction, judgment, extent of human participation, and so forth.[112] Particularly in the case that a series of acts and decisions are practiced in a way that is not programmed in advanced but on the basis of 'machine learning' or 'intelligent decision-making techniques,' it is unable to predict how to react at a certain point in time.[113] For better understanding and prediction in this regard, a high level of scientific expertise and deep understanding of relevant technology will be necessary. In this case, it may be more difficult to clarify the causality than in other normal occasions. Therefore, the approach of consequential responsibility could be one alternative. This perspective corresponds to the convention on compensation for damages in space activities that was concluded in 1972 in order to specify consequential responsibility for application of highly advanced technology.

However, the logical basis for introduction of this system would not be firm enough if it is determined that a certain subject makes decisions independently and separately from a governmental agency or official in charge and if it is possible to

[111] *See supra* note 364, p. 106.

[112] Schuller [3], p. 391.

[113] *See supra* note 160, p. 651.

clarify the causality between such decision-making and final results. In addition, AI may not fully accord with the concept of consequential responsibility since it comes into direct contact with the one subject to the final disposition or the other party and the process is traceable. In any case, such problems should be discussed in the international community.

3.6 New Challenges and Pending Questions

In addition to the above-mentioned complexities, there are other challenges and questions to be addressed. For instance, determining legal status of the autonomous artificial entity can have important implications in terms of international state responsibility. Even the existing rules, such as the attribution requirements, may need to be updated to fully and coherently deal with cases using AI programs. Similarly, developing an appropriate verification procedure to define the perpetrator is critical since it could be a hindrance. Answering these questions requires the accumulation of cases and thoughtful judgments.

In certain cases, it might be impossible to attribute the judgment and acts of AI to a state. For instance, there may be no relevance to the state in any of the perspectives stated above; or the AI system makes autonomous judgments and decisions based on big data collected independently from sources other than the state and government and a relevant performance or nonperformance is practiced based on such decisions. In other words, if a decision or act of AI is completely irrelevant to the state, the state is free from state responsibility under international law. One major example is the case where AI is used by a private subject in a private sector purely for personal or profit-making purposes. In this case as well, however, it is not difficult to separate it completely from the government if the governmental control is tightened over information or networking which is part of the AI neural network. The problem is that an action by LAWS is difficult to attribute to one particular person.[114]

Another case to be noted is that AI makes decisions directly and independently in technically more advanced environments. Additionally, the state or one of its governmental agencies may deny legal responsibilities. For example, the state may claim that it takes no state responsibility since it fulfills all of its duties under international law even though there has been an accident or occasion where AI was involved. In this case, the government may claim that it takes no legal responsibility since it deployed an AI system after verification based on reasonable effort and had completed its required obligation of management and supervision. It is highly likely that once the use of AI is expanded to general areas of governmental affairs, such arguments will often be presented. As a similar example, if a state purchased a vehicle after verification with a reasonable degree of effort and operated it for governmental affairs with the duty of care fulfilled in that process, the state would deny its state responsibility even if a traffic accident occurred in the process of transportation of a

[114] Hammond [22], p. 15.

diplomat stationed in that state by means of the vehicle and the diplomat died from that accident. In such cases, state responsibility is relevant to the obligation for a state to protect its diplomats according to 1961 Vienna Convention on Diplomatic Relations.[115] The result would be the same even after such vehicles are replaced with 'completely autonomous vehicles' in the near future. However, this logic is rational only when AI is utilized by a state agency or government agency as an instrument or assistant in such forms as automobile, electronic equipment, and computer. This basis is hardly applicable to non-instrumental and independent acts of AI that are expected to be practiced in the future. This is because in such cases, AI plays the role not of the vehicle transporting the diplomat itself but of the official in charge of the transportation at the relevant governmental agency. There may be situations where the relevant governmental agency has to take responsibility for certain choices such as transportation path and method.

As the use of AI is widely spread in the digital society, it becomes more and more difficult to classify and distinguish public areas from private because they share information and intelligence networks at home and abroad in carrying out their tasks. Joint development and use of technology between public and private sectors is recommended. Besides, what is 'state function' stated in Article 4 of the 2001 ILC Draft Articles is uncertain in newly formed cyber systems. Functions and roles of pure private subjects as well involve vague aspects. In order to prevent the state from being accountable for acts of a certain subject under the present law of state responsibility, basically, there should be no connection with the state or government. In the digital society that has already arrived and will continue to in the near future, it is more difficult to secure such 'absence of connection with the state.' Verification of such absence is also another challenge. This is because both public and private sectors are connected through one system or network, and both the government and private sectors participate in establishing and utilizing big data. The use of AI that is emerging now is based on the assumption that various persons and subjects in the society participate in one network or system and act based on such media. As digitalization advances, this situation will be more and more facilitated.

Some even argue that particularly in the new society led by big data, a government needs to be involved more deeply in securing and utilizing personal information in private sectors and to take necessary measures in that regard. In addition, new information and communication technology has resulted in new power dynamics between traditional actors (e.g. state, government, and person) and new actors (e.g. private social media enterprises). Although information and communication technology can contribute significantly to producing a large amount of big data and forming public policies based on it, the risk of social chaos also increases due to the spread of distorted information. Accordingly, securing transparency and accountability in the process of collecting and processing information based on big data becomes more and more important.[116] It is required, therefore, for a government to take necessary

[115] United Nations, Art. 36 of *Vienna Convention on Consular Relations*, 18 April 1961, 500 UNTS 95, (hereinafter "Vienna Convention on Consular Relations").

[116] *See supra* note 165, p. 42.

measures so that this area is not dominated by the pursuit of private interests.[117] Governmental intervention and regulation is required in the long run.

On the other hand, while a government relies heavily on information and data of private sectors as one unified system and controls new private sectors for the purpose of various policies at the same time, it is more difficult to separate and distinguish the former from the latter. Rather, in the society of the future, it is highly likely that government and private sectors will be more closely unified over time. Since the present law of state responsibility was stipulated with the focus on a distinction between the two, it is probable that the gap between this law and reality will be widened gradually. This is another clue suggesting that new norms are to be considered and introduced in the future.

In particular, attribution of acts of non-state subjects and private subjects to the state and determination on state responsibility have been major issues in terms of state responsibility so far, and the scope of such discussion will be widened further[118] because the issue of state responsibility is raised not only concerning acts of non-state subjects but also concerning acts of 'impersonal' subjects.

4 Dispute Settlement Proceedings

Digital technology is also changing the way disputes are resolved. The advent of AI and the digital economy also means changes in dispute settlement as well. On the one hand, it certainly means more efficiency in settling disputes through utilization of these new technologies. The changes, on the other hand, also signal further complexity in dispute settlements as well. As AI becomes more ubiquitous than ever, the technical difficulty of high-tech products and services subject to disputes may require one or more experts with technical expertise to make adequate decisions.[119] Sometimes technical knowledge may be required to access digital evidence.[120] The importance of a timely arbitration process for disputes in the high-tech fields demands innovative ways to empower arbitrators to adequately deal with such disputes.[121]

Trade disputes are also expected to be affected by the new technology. In order to respond actively to the era of digital trade, the trade dispute settlement system

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[117] Ibid., p. 60.

[118] *See* Bodansky, and Crook, *supra* note 396, p. 782.

The degree to which states should be held responsible for conduct involving private actors is an increasingly significant contemporary issue, as non-state actors such as Al Qaeda, Somali warlords, multinational corporations, and nongovernmental organizations play greater international roles, and as governments privatize some traditional functions and enter into a variety of public–private collaborations with international organizations and private actors.

[119] United Nations Commission on International Trade Law, *Possible Future Work in the Field of Dispute Resolution in International High-tech Related transactions: Proposal by the Governments of Israel and Japan* (May 29, 2019). A/Cn.9/997 para. 4.

[120] Ibid.

[121] Ibid.

also needs to be reviewed. First of all, with so much uncertainty regarding digital trade, countries may be reluctant to raise issues directly with the formal dispute settlement bodies. Given the long process and uncertainties of trade disputes, it is unlikely that a trading partner will raise a problem officially at a dispute settlement body unless there is a substantial business loss to the foreign exporter or service provider concerned. It is necessary, therefore, to conduct a verification process that demonstrates the damage that the new legal measure has caused to a foreign exporter or a service supplier in the domestic market.

Moreover, it is uncertain yet whether new norms will be introduced, and, even if they will, it takes a long period of time until they are firmly established. Thus, it is appropriate to come up with a dispute settlement system in advance that is operated based on a consensus among countries. For instance, the WTO Trade Policy Review Mechanism (hereinafter TPRM) or any similar systems may be examined. Mediation procedures also may be investigated in detail.

With regard to the TPRM system, WTO member countries have played an important role in facilitating the improvement and maintenance of domestic laws and systems for the past 23 years in accordance with the WTO Agreements.[122] In particular, the TPRM system is recognized as having played a vital role in complementing dispute settlement procedures which forcibly require a country to improve the system after losing a court case since the former promotes voluntary and proactive systematic improvement of each member country.[123]

Such resilience and flexibility of TPRM procedures suggest that TPRM can contribute to digital trade as well. In order to ultimately address issues in the era of digital trade and settle disputes, it is sometimes more effective to review the national system and promote a consensus among member countries in utilization of the TPRM system rather than (or in addition to) dispute settlement procedures. As mentioned earlier, when it is sought to settle various disputes regarding the fourth Industrial Revolution or digital trade through the WTO dispute settlement procedures, it will take a long period of time. Besides, even after the completion of such settlement procedures, party countries rarely come to a genuine settlement of disputes. Rather, the conflict between countries becomes deepened with the enlarging market. In order to respond to changing situations in this era of digital trade effectively, objective evaluation of each country's relevant policies and the formation of a consensus are essential. It is thought that in this process, the TPRM system can contribute significantly. In the era of digital trade, it is appropriate to strengthen the TPRM system further, specifically its scope of application, and utilize it as an instrument to respond to emerging issues. If the TPRM system can be operated in this manner, it will be able to provide important infrastructures for establishing new trading environments as well as a global consensus.

In order to settle disputes concerning measures for digital trade, it is necessary to avoid the mechanical approach in conformity to a trade agreement and to empirically

[122] *See* Ghosh [9], pp. 436–9.

[123] *See* Rodriguez Mendoza et al. [68], p. 25, Kuijiper [79], p. 100; Betz and Koremenos [91], pp. 597–8.

examine the intention of introducing a certain problematic measure and its material impact on the market. Relevant contents may be included in a TPRM report in detail. Such reports can have the effect of 'naming and shaming,' contributing to changing problematic measures of each country in a natural manner.

Likewise, mediation procedures can also be reviewed thoroughly in this regard. Such procedures are also a form of dispute settlement focusing on establishing a consensus between countries rather than seeking to apply binding procedures. Recently, the effectiveness of this system has become an issue again. In terms of digital trade as well, alternatives may be developed in full consideration of the aspects stated above.

Another aspect that can be considered in new digital trade chapters in relation to dispute settlement procedures is to limit standing requirements according to certain conditions. Introducing this method can contribute to reducing the possibility that a certain country exploits the right to sue another country or raises an issue under an agreement regardless of the market reality. For example, countries that specify a sort of standard for international official prices stated above and maintain the prices of goods as high as such standards in a certain year may be allowed to invoke regulations for other countries' trade of digital goods and services. In this way, it may be possible to minimize situations where a country that is involved in sales at excessively low prices or in an unjust manner sues a competing enterprise in the competing country by invoking such clauses. This suggestion is worth consideration in a new area of digital trade although it may be a somewhat experimental attempt.

Dispute settlement processes related to investment agreements also need to be reviewed. As mentioned earlier, since growth in digital trade will encourage countries investment across countries, investment agreements and relative dispute settlement procedures will increase in importance. The problem is that in many countries, including Korea, investment agreements and dispute settlement procedures have drawn keen attention and remain a politically sensitive issue.

The fundamental cause of such domestic conflicts is the direct connectivity between a government's exercise of regulatory authority and the application of investment agreements as well as the occurrence of investment disputes. As a matter of fact, most international treaties and agreements are inevitably related to a government's exercise of regulatory authority whatsoever. In the case of investment agreements, however, the scope and depth are incomparable to those of other agreements since the former are international treaties introduced for comprehensive governmental regulations of every type. Among other international treaties, there are quite a few that are applied directly to the governmental regulatory authority. Although trade agreements are one of the exceptional cases in this regard, the extent of infringement of regulatory authority is relatively small in comparison with investment agreements.

Every governmental measure (whether taken by the executive branch, the legislature or the judiciary) is potentially subject to regulations of the FTA and the Bilateral Investment Treaties (hereinafter BITs), which is a branch of IIAs. This situation often leads to an infringement of the legitimate sovereignty or judicial sovereignty. Despite this sensitivity, however, investment dispute settlement procedures are conducted respectively by each tribunal and finished in the first trial without an appeal process.

It is common that tribunals take different positions on major concepts and legal principles, and as a result, such differences act as significant barriers in implementing and applying agreements stably among party countries of the BIT and FTA. Particularly in the case of investment dispute settlement procedures, the lack of appeals procedures decreases the consistency and predictability of judgments further. Against such backgrounds, there are fundamental limitations in that the government or people of a host country that has failed in an arbitral award accept that judgment as persuasive.

Above all, it is sometimes observed that foreign investors and legal experts representing them take the lead of interpreting and applying BIT/FTA regulations in a creative manner. increasing the possibility of unpredicted lawsuits from the perspective of host countries attracting foreign investment. In the case of investment disputes being raised recently, for example, agreement provisions are interpreted and applied in a new way that has not been expected before. Major examples include the evasion of restrictions forum double lawsuits through a paper company, extended application of agreements through an umbrella provision, and invocation of advantageous BIT provisions in reference to a relevant Most Favored Nation (hereinafter MFN) provision.

One of the reasons for which such situations occur is that basically, investment agreements specified basic aspects briefly and in general, and that, as seen currently, such provisions were not introduced with various types of disputes in mind. Besides, it is difficult to revise an investment agreement once it is concluded, just like other agreements. Updating such agreements on a regular basis may seem to be possible, but it is actually impossible to some extent. In this respect, there is a fundamental difference from domestic laws that are frequently modified through yearly revisions, which need to be taken into consideration when an investment agreement is concluded. Of course, such aspects are observed in other international treaties as well but frequently in investment agreements, which also often involve international disputes. Thus, this issue is raised quite seriously, particularly with regard to investment agreements.

Finally, investigation and decision on such sensitive aspects are handled by a tribunal consisting of three arbitrators, rather than by a domestic court of the host country attracting investment. A fundamental concern is raised regarding investment agreements that the final decision on aspects of a host country that are sensitive in terms of politics and legal principles is made by a kind of tribunal consisting of foreigners. This concern has remained with regard to cases of so-called judicial sovereignty infringement.

Meanwhile, the new technology is changing the dispute settlement proceeding itself as well. The most visible change is that physical courts are being transferred online. In early April 2020, there was a proposal for a digital court.[124] This proposal is to use blockchain technology to settle disputes without resorting to the costly and lengthy legal proceedings. The proponents say that "the court algorithmically aggregates the parties' opinions and judges who violated their agreement. If the digital

[124] *See* Matsushima and Noda [38].

court judges that a party violated the agreement, the party is fined by withholding a deposit made during the initial agreement."[125] They went on to state:

> We designed a digital court which identifies and punishes parties who deviate from legal obligations such as commercial activities, but could potentially be any kind of agreement. On suspected violation of some agreement, those involved post their opinions to this digital court. The court algorithmically aggregates the parties' opinions and judges who violated their agreement. If the digital court judges that a party violated the agreement, the party is fined by withholding a deposit made during the initial agreement.[126]

This idea is basically in line with smart contracts arbitration in the case of an alleged non-performance of a party.[127] Using blockchain technologies for arbitration may benefit banking and financial industries.[128] As the proposed court and arbitration use technology that is already there and available, the idea can be implemented now.[129] The proponents set forth the detailed step-by-step process of applying the current technology to resolve disputes involving smart contracts.[130] These are contracts "embedded in digital code and stored in transparent, shared data bases, where they are protected from deletion, tampering and revision."[131] It is expected that smart contracts will take up the major portion of commercial agreements in the near future.[132] In short, the digital court applies distributed ledger technology to solve disputes between parties with the use of algorithms.[133] It represents an extension of the technology applied for smart contracts and is currently recommended for business contracts, sales and auctions.[134] The mechanism is meant to be self-enforceable: the parties make an initial deposit to the court, the digital court will analyze the claims based on an algorithm and decides the party at fault, and the losing party will be fined by withholding the deposit made previously.[135] While traditional litigation is costly and time-consuming, arbitration using blockchain technology can provide a useful alternative to certain sectors such as banking and financing industries.[136] Issues of consent formality and confidentiality need to be contemplated in this respect as well, at a level that is different from conventional arbitration.[137]

The same technologies may also be applied to settle state-to-state disputes. It may not cover all the disputes involving states, but some disputes, particularly those

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[125] The University of Tokyo [89].

[126] Ibid.

[127] Henriques [29], pp. 187–190.

[128] Ibid, p. 191.

[129] *See supra* note 406, p. 3; *supra* note 409, pp. 183–186.

[130] *See supra* note 406, pp. 8–26.

[131] Iansiti and Lakhani [64].

[132] *See supra* note 409, p. 187.

[133] *See* ibid.

[134] *See* ibid., pp. 26–30.

[135] *See supra* note 406, p. 2.

[136] *See supra* note 409, pp. 194–195.

[137] Ibid., pp. 195–197.

relating to commercial transactions or trade relationships, could be receptive to the idea of introducing a new dispute settlement mechanism.

4.1 Utilization of AI Adjudicators

Since AI excels in analysis and processing of data without bias, discussion about adopting the technology into international adjudication deserves more attention than ever before. In the domestic legal sector, AI is already playing a variety of roles. Examples include Settify,[138] an Australian dispute management system for family disputes, Split Up,[139] a decision support system for property distribution after a divorce, and the Persuader,[140] a case-based AI for labor related negotiations.[141] AI can contribute to the prevention of international disputes *ex ante.* In other words, AI can contribute to war prevention by providing information on possible outcomes of a conflict. Already in the domestic courts, the use of AI is increasing in the prevention of conflicts. For example, in Columbia, Siarelis,[142] an AI-based system for societal disputes resolution, has been employed.[143]

As of now, AI is not actively used in any final decision-making process.[144] In the international arena, application of AI is rarer, but not unthinkable.

Most of the discussion regarding AI in international courts is focused on applying AI in the international arbitration for the moment. For example, AI can assist in choosing arbitrators and analyzing evidence in arbitration.[145] However, at present, most of the world's leading arbitral institutions are based on premises that the arbitration will be completed by human resources.[146] Therefore, if any major changes were to be made, some modification or amendments may be necessary. Nevertheless, AI can contribute to technical analysis in trade disputes concerning anti-dumping duties or safeguard measures. AI could offer a more efficient way to analyze patterns of trade and reveal hidden trade barriers.

It is understandable that states are cautious of applying AI at more formal international dispute settlement mechanisms that involve state to state relations. The advocates of AI will argue that it will provide the most objective analysis, independent from the power dynamics of international relations. However, AI is far from a perfect, bias-free system. The limits of AI include the possibility of ignoring hidden

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[138] *See* the website of *Settify* available at https://www.settify.com.au/.

[139] *See* Zeleznikow [54].

[140] *See* Lin [82].

[141] *See* Lozada-Pimiento [74], pp. 348–66.

[142] *See* the website of *Siarelis,* available at https://www.supersociedades.gov.co/delegatura_mercantiles/Paginas/siarelis.aspx (in Spanish).

[143] *See supra* note 423, pp. 348–66.

[144] Ibid.

[145] *See* Westgarth [14], Marrow et al. [78].

[146] Ibid.

variables, which could lead to biased decisions, and a desire for human intervention in adjudication processes.[147] Furthermore, given the technology gap between the developed and the developing world, the global South will be reluctant to resort to AI-based arbitration. Meanwhile, the more powerful countries will be reluctant to relinquish their source of influence and rely on AI arbitration. In other words, neither the powerful nor the weak have much interest in introducing a fully AI-based international dispute settlement mechanism.

4.2 Analysis of Precedents and Jurisprudence

Seeking an alternative pathway to settle an international dispute with AI technology could affect the way past cases and judgments are applied in the ongoing disputes. In the domestic area, AI contributes to dispute resolution as well: Do Not Pay,[148] LISA,[149] ROSS[150] are only a few recent examples of online dispute resolution (ODR)[151] services that provide legal services based on AI.[152]

Another more complex example in which AI is changing the judicial procedures is technical arbitration.[153] According to Lozada-Pimiento [74], it can be defined as "a dispute resolution process dependent on the decision of a technical, not a judicial, expert, which requires an appreciation of the facts that relies on structured, scientific knowledge." In the US, a Data-mining computer program, Lex Machina, has been already created in 2006 to look for patterns in court cases.[154]

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[147] *See supra* note 423, pp. 348–366.

[148] *See* the website available at https://donotpay.com/; *also see* Jon Porter, *Robert Lawyer DoNotPay Now Lets You 'Sue Anyone' via an App,* The verge (Oct.10, 2018), *available at* https://www.theverge.com/2018/10/10/17959874/donotpay-do-not-pay-robot-lawyer-ios-app-joshua-browder.

[149] *See Robot Lawyer LISA* website, available at http://robotlawyerlisa.com/.

[150] *See ROSS* website, available at https://rossintelligence.com/.

[151] For instance, Jeremy Barnett and Philip Treleaven mention the following about ODR and automation:

> ODR uses technology to facilitate the resolution of disputes between parties; involving negotiation, mediation or arbitration, or a combination of all three; and can be fully automated or involve human intervention. In ODR, automation is used both in the process of bringing the parties together for a 'neutral' decision or to propose a settlement; in the resolution i.e. as the 'Fourth Party'; or can be mandatory.
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> Barnett and Treleaven [52].

[152] *See supra* note 423, pp. 348–366.

[153] Ibid.

[154] *See* website available at https://lexmachina.com/. *See also* [49].

4.3 Evidentiary Materials

AI deployed in the procedural and substantive parts of evidentiary rule could bring changes in the way the proceedings are held. The technology allows for online remote court proceedings. In fact, the advent of COVID-19 has also accelerated the introduction of new technology into the courts, and so far, it has facilitated the transparency of court proceedings. The relative ease of livestreaming is now prompting courts to consider adopting livestreaming technology to broadcast court proceedings. For instance, in the US, all 13 federal appeal courts and several state courts are livestreaming oral arguments as of August 2020.[155] This change may well solidify the concept of trial in public: it is indeed possible that anyone with access to the internet can observe important court proceedings. The chief justice of the Hawaii Supreme Court, Mark Recktenwald, noted that live streaming has enhanced public accessibility to the courts, as it enables people from remote areas to join in.[156]

Online hearings and proceedings are convenient. But at the same time, they also raise novel legal questions of fairness and due process. There are studies that show that people perceive televised and live witnesses differently.[157] Furthermore, there is the growing concern over the digital divide where a group of people are better equipped or positioned than others to utilize and take advantage of virtual hearings and proceedings. Not only is there a problem of internet accessibility, people have different digital literacy levels and technological skills.[158] For example, the 2013 US Census showed that only 51.4% of limited English-speaking households have internet subscription, compared to 75.5% of English-speaking households.[159] Moreover, a study of immigration removal hearings found that bad transmission resulting in image freezing, delays, and poor sound quality could negatively affect the image of immigrants.[160]

5 Law of War

Law of war is perhaps the area of international law which is most immediately affected by the advent of artificial intelligence. Military weapons have always been quick to adopt the state-of-the-art technologies of the time,[161] and the temptation equally applies to artificial intelligence. Autonomous weapons collecting data, processing information and making their own judgment are now changing the landscape of modern warfare and military strategists. The US forces' withdrawal from Afghanistan

[155] Alder and Reed [66].
[156] Ibid.
[157] Cahn [4], p. 11, Quinn [67], p. 199.
[158] Ibid., p. 9.
[159] Ibid., p. 9.
[160] Ibid., p. 10.
[161] *See for example,* https://www.rand.org/topics/military-technology.html.

in August 2021, and continuing reliance on drones to carry out military operation are quite telling with regard to the fast-changing nature of military strategies in the current world.

In this respect, the existing jurisprudence also provides important guidance. For instance, the current jurisprudence prohibits two types of weapons in military conflicts—weapons attacking indiscriminate targets[162] and weapons causing unnecessary suffering[163] The ICJ in *Nuclear Weapons* opined that "[s]tates must never make civilians the object of attack,"[164] and accordingly indiscriminate weapons are by nature in violation of this principle. The ICJ also deems weapons causing unnecessary suffering illegal as they are not justified in achieving military objectives. In *Nuclear Weapons*, the ICJ thus further stated that "it is prohibited to cause unnecessary suffering to combatants: it is accordingly prohibited to use weapons causing them such harm or uselessly aggravating their suffering […]. States do not have unlimited freedom of choice of means in the weapons they use."[165] These basic principles constituting the bulwark of the law of war are all the more meaningful and important in the age of AI. Whatever AI technologies adopted in military weapons or employed in military tactics could be found in violation of law of war if they are directed at indiscriminate targets or designed to cause unnecessary suffering. To the extent that AI is premised on the collection of data and information and that these data and information in turn stem from a broad range of individuals and corporations from diverse countries, it should be deeply intertwined with these private sources of data and information of various nationalities. Once weapons using them are trained at foreign targets, they might also bring into the scope of attack, intended or unintended, a wide range of private constituents. This may trigger an indiscriminate attack claim. Similarly, as the data and information have become a critical platform of social infrastructure such as the provision of electricity, health, medicine or water, disruption of the provision resulting from an AI using weapon may trigger an unnecessary suffering claim as well. Conventional jurisprudence in the area of the law of war remains current and pertinent even in the AI era.

That said, here also the devil lies in the details. The broad brush of big principles is not sufficient or sophisticated enough to address the current reality of war. There are areas in the law of war that require a reflection of fast changing realities and rapid technological development. The advent of AI has changed diverse aspects of war. By way of example, what constitutes a hostile activity has now become more complex than before. Application of AI in the military sector stands to shape how future wars are waged and conducted.[166]

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[162] International Committee of the Red Cross International Committee of the Red Cross (1977) (hereinafter 'Additional Protocol I'), *cited by* Weaver [53], p. 7.

[163] International Conferences (The Hague) [43], Art. 23, *cited by* [53], p. 7.

[164] *See* Legality of the Threat or Use of Nuclear Weapons, Advisory Opinion. 1996 I.C.J. 226, 257, para 78, c*ited by* Weaver [53], p. 8.

[165] *See* ibid.

[166] Garcia [27], p. 334–5.

Despite the fast pace of technological development and the wide application of new technologies in the field, the law of war is still lagging behind; it does not effectively regulate AI-based weapons, and the legal vacuum is leading to problematic escapes from otherwise existing state responsibility. Most notably, LAWS' operation without human intervention raises a fundamental question of who should bear legal responsibility under international law.[167] Debates to build consensus in the international community are currently under way, but further attention and efforts should be mobilized to complete the project. New norms in this area will then be able to provide legal guidance in this crucial area including the confirmation and attribution of legal responsibility for the utilization of LAWS.

On one hand, as much as AI-enabled weapons target civilians during war or military conflict, it is also subject to international humanitarian law. On the other hand, as the fundamental human rights of individuals (combatants and civilians alike) are at issue, international human rights law needs to be brought into the landscape. For both the international humanitarian law and international human rights law, the core legal question with respect to AI weapons concerns the accountability issue—whom to blame and when.[168] Both branches of international law aim to prevent indiscriminate targeting and unnecessary suffering, which are the two main components of the law of war.[169]

As of October 2021, a treaty that regulates the operation of LAWS specifically has yet to be drawn up.[170] While the law of war certainly applies to LAWS as a kind of weaponry,[171] a treaty addressing LAWS to reflect its unique aspects and characteristics does not currently. Nor is there customary international law either that focuses on LAWS. Accordingly, it is existing customary international law in the areas of armed conflict, international humanitarian law and international human rights law that applies to LAWS as one type of weapon.[172] Indeed, states and individuals may then be made liable for violation of these norms of customary international law if evidence supports the satisfaction of the attendant requirements of relevant customary international law.[173] In this respect, it is also possible that a particular LAWS is deemed prohibited even under existing customary international law due to its extremely harmful nature, in which case the utilization of such a weapon in and of itself may lead to violation of the customary international law.[174]

Some experts have already underscored the need to adopt a new legal instrument on the international plane to deal with AWS, having realized the imminent danger

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[167] Hammond [22], p. 15.

[168] Bills [37], pp. 176–209, Hammond [22], pp. 652–88, *cited by* Garcia [27], p. 336.

[169] Christof Heyns. 2013. *Report of the Special Rapporteur on Extrajudicial, Summary or Arbitrary Executions.* UN Document A, HRC/23/47, *cited by* Garcia [27], p. 336.

[170] Marchant et al. [35], pp. 272, 289, *cited by* Hammond [22], p. 13.

[171] International Committee of the Red Cross [42], *cited by* Schmitt and Thurnher [70], p. 9.

[172] *Supra* note 160, pp. 634–635.

[173] Ibid.

[174] Ibid.

stemming from the absence or lack of appropriate norms in this area.[175] It is certainly not accurate to assume that there is no norm for the regulation of LAWS; again, all weapons used in a military conflict are subject to the applicable legal review under existing treaties and customary international law.[176] The question is whether these current norms provide robust legal guidance beyond mere baseline issues. The answer to this question seems to be in the negative. The fast pace of technological development is arguably further widening the gap between the norm and reality, in which case the need for a set of robust legal norms for LAWS will become more acute. If anything, this is not surprising; already there are treaties that address a specific type of weapon. The Chemical Weapons Convention[177] or the Landmine Prohibition Convention[178] offer good examples. Given the increasing attention to and debates over LAWS, it would be difficult to find a persuasive rationale not to adopt a separate set of norms for LAWS. These separate norms (or individual treaties) may be called an exception.[179] Even so, this does not necessarily support the proposition that conventional norms provide sufficient legal guidance. If anything, their exceptional nature would lend weight to the argument for the adoption of new norms, possibly a new legal instrument, for LAWS.

Setting aside for a moment the need to introduce new LAWS-specific legal instruments, conventional legal norms should also be re-examined in the context of the LAWS. After World War Two, international society managed to successfully condemn weapons of mass destruction, and it is generally agreed that its use is against international law.[180] Should any LAWS fall under the category, it would have to be equally condemned. Likewise, states must abide by the International Humanitarian Law (hereinafter IHL) framework when they resort to autonomous weapons systems.[181] As a matter of fact, under Article 36 of the Additional Protocol I to the Geneva Conventions of 1949 any new weapon should be examined to establish "whether its employment would, in some or all circumstances, be prohibited by this Protocol or by any other rule of international law."[182] Needless to say, the application of the existing legal framework to LAWS would be anything but easy. LAWS is fast changing the tactics, techniques and procedures (hereinafter TTP) of warfare.[183]

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[175] Anderson and Waxman [58], p. 1104.

[176] Ibid., p.1103.

[177] *The Convention on the Prohibition of the Development, Production, Stockpiling and Use of Chemical Weapons and on their Destruction* (hereinafter 'CWC'), full text available at https://www.opcw.org/chemical-weapons-convention.

[178] *The 1997 Convention on the Prohibition of the Use, Stockpiling, Production and Transfer of Anti-Personnel Mines and on their Destruction,* full text available at https://www.un.org/disarmament/anti-personnel-landmines-convention/.

[179] *Supra* note 555, pp.1103–1104.

[180] E.g., Fidler [23], pp. 39–88.

[181] *Supra* note 160, p.652.

[182] *Protocol Additional to the Geneva Conventions of 12 August 1949, and Relating to the Protection of Victims of International Armed Conflicts,* art. 36, opened for signature Dec. 12, 1977, 1125U.N.T.S.3, reprinted n16I.L.M.1391(1977), *cited by* Beard [45], p. 635.

[183] For example, *see* Maymí et al. [34].

The increasing capacity of new automated weapons systems by processing a vast amount of information (such as facial recognition, monitoring the flow of people, tracking down of vehicular traffic, scanning the operation of ICT networks) indicates both their massive-scale military utilization capability and their precision targeting capability. Drones and killer robots could be as destructive and as inhumane as any bio-chemical weapons, but at the same time they could save the lives of soldiers of utilizing states and avoid unnecessary killing of enemy combatants. Depending on how these new automated systems are programmed and operated, their utilization may bring either aggravating or mitigating effects in terms of abiding by the existing norms of warfare and IHL.

When it comes to LAWS and its review under the existing norms, both individual weapons and weapons systems count. The US DOD directive also confirms the need to conduct reviews of both individual weapons and weapons systems by stating:

> The acquisition and procurement of DoD weapons and weapon systems shall be consistent with all applicable domestic law and treaties and international agreements […] customary international law, and the law of armed conflict [.]... An attorney authorized to conduct such legal reviews in the Department shall conduct the legal review of the intended acquisition of weapons or weapons systems.[184]

The US DOD further stated in a recent instrument:

> Persons who authorize the use of, direct the use of, or operate autonomous and semiautonomous weapon systems must do so with appropriate care and in accordance with the law of war, applicable treaties, weapon system safety rules, and applicable rules of engagement.[185]

The fact that not only individual weapons but also weapons systems are covered by the attendant legal review is quite telling in this regard. It indicates potential legal responsibility for how information gathering, information processing, and decision-making steps are structured. The question here goes beyond merely addressing an incident of firing or misfiring of an unmanned sentry rifle. Once a state has put in place a weapons system, it is obligated to ensure that it operates in a way that complies with existing legal norms. It could be more complicated than before, as modern warfare is conducted under a system integrating a wide range of information from various sources. Such being the case, a weapons system could be far broader and wider than a simple missile launch operation system. In the era of the information age and digital warfare, legal obligations for a 'weapons system' may turn out to be complex and controversial.

The precautionary principle may also provide an important guardrail to regulate LAWS under the present legal norms.[186] Considering the fast pace of technological

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[184] US Department of Defense, *Autonomy in Weapon Systems,* Directive 3000.09 (Nov. 21, 2012), "Autonomous Weapon Systems: Response-to-Query talking Points 1, *cited by* Schmitt and Thurnher [70], p. 26.

[185] US Department of Defense, *Autonomy in Weapon Systems,* Directive 3000.09 (Nov. 21, 2012), p. 4b, available at http://www.fas.org/irp/doddir/dod/d3000_09.pdf, *cited by* Schmitt and Thurnher [70], p. 30.

[186] The precautionary principle justifies decision making process in uncertain situations. *See* Saladin [12], pp. 270–280.

development in this area, the possibly strong desire of a state to deploy LAWS on the field to avoid its own human casualties, and unavoidable uncertainty because of the lack of any record of actual applications, the precautionary principle of international law may loom large in this respect. Under the principle, an LAWS-utilizing state should take preventive action before harm is done. Should what type of AI-assisted weapons are permitted and what are not also be examined in this process? When accumulated over time, a consensus will emerge as to benign AI-assisted weapons and malign ones.[187] The problem with this approach is that it will take time. The unprecedented speed of technological development may not allow states and governments to have patience so that they can wait for the ripeness of norms for the new AI weapons. Introduction of a new legal instrument through tight debates and exchange of views would be a preferable option.

The utilization of LAWS does not mean that a weapon or a weapon system is error-free. It could be a human error on the part of a person operating the weapon or system, or a mechanical error on the part of the weapon or the system. How the weapon or the system is programmed to respond in the case of uncertainty would be meaningful in calibrating the legal responsibility. For instance, compliance with the rule of proportionality and the requirement to take precautions to avoid harm to civilian targets can be evaluated based on how such uncertainties are handled.[188] In order to avoid a vacuum of legal responsibility, it would be preferable to place legal responsibility on states utilizing weapons regardless of the nature of error, be it human or mechanical. That way, states would be incentivized to take a more cautious approach to the deployment and utilization of LAWS.

In short, LAWS is still subject to the current norms of international law as a type of weapon. Exiting norms do provide important legal guidelines. That said, a critical portion of LAWS utilization is not sufficiently covered by the current norms. The legal vacuum may only be filled by the development and adoption of new legal instruments, possibly treaties on LAWS specifically. The delay in the adoption of these new instruments stands to widen the gap between reality on the field and applicable legal norms, which would only help intensify the controversy over this new weapon. Based on the above discussion, the following sections examine how AI is being used in hostile activities in general, as well as specific discussions on drones and killer robots.

5.1 Utilization of AI in Hostile Activities

A weapon becomes illegitimate under international law in either of two ways. The first route is the situation where a specific treaty prohibits a particular type of weapon.

[187] Garcia [27], p. 338.

[188] Schmitt and Thurnher [70], p. 20.

The second one is the result of the operation of the two cardinal rules of IHL.[189] The two rules are the prohibition of weapons with particular characteristics: i.e., they are weapons that cause indiscriminate targeting[190] and weapons that cause unnecessary suffering.[191] Criticism is also presented to the effect that certain weapons are immoral, inhumane or unconscientious.[192] Such morality arguments are important, but it is not and cannot be the same as the legal argument. It is one thing to say that a weapon is immoral but it is another to pronounce it as illegal.

At present, as there is no specific treaty that regulates AI-utilizing weapons per se, the question posed now concerns only the second route, which is whether and how the utilization of AI affects the two cardinal principles of prohibition of indiscriminate weapons and weapons inducing unnecessary suffering. One of the unique characteristics of AI weapons in this regard is the possible elimination of the human factor in targeting process. AI weapons sometimes allow humans to be taken 'out of the loop' in the lethal targeting process, which causes debates whether such elimination of the human factor is legitimate.[193] The absence of the human decision-making factor, in and of itself, may not necessarily implicate indiscriminate targeting or unnecessary suffering. In fact, theoretically speaking, if programmed well and performed as programed, those weapons may well operate within the bounds of the two cardinal rules of IHL. That said, targeting through machines and killing by a machine's own decision without human intervention or supervision may entail an ethical question that is not covered by IHL. In other words, even if pinpoint targeting and minimized suffering are somehow attained, the fact that machines administer the whole process alone may trigger an enduring ethical problem. This ethical problem may expose the legal norm to continuing criticism. This offers one critical reason why a separate legal norm is indeed in order to regulate AI weapons.

More critically, fully autonomous weapons independent of human intervention invite the question of who should be held liable in what circumstances.[194] When something goes wrong and civilian casualties take place, is it the human operator that deployed the AI weapon on the field or the AI weapon itself that made the final call, or the manufacturer of the AI weapon that failed to address these faulty judgment processes? Or do we simply have to shift all the blame to the state that utilizes the AI weapon regardless of the sources of the problem, something akin to consequential liability? The complex nature of tracking down the specific reason for faulty operation among various stakeholders, the highly scientific traits of the matter, and the policy dimension to place governments on the cautious side of the deployment decision may well tip the balance for a system of consequential liability

[189] Legality of the Threat or Use of Nuclear Weapons, Advisory Opinion, 1996 I.C.J. Rep. 226, para. 78(July 8) *cited by* Beard [45], pp. 636–637.

[190] International Committee of the Red Cross [41] (hereinafter Additional Protocol I), *cited by* Weaver [53], p. 7.

[191] International Conferences (The Hague) [43], Art. 23, *cited by* Weaver [53], p. 7.

[192] *Supra* note 160, p. 637.

[193] Schmitt and Thurnher [70], p. 1.

[194] Alston [80], pp. 35, 51.

for states; whoever decides to deploy the AI weapons should be held liable for the result of the operation of such weapons. A future treaty may be able to clarify and stipulate this principle.

LAWS, by definition, means a weapons system that possesses a degree of its own decisional capability.[195] Use of autonomous weapons with an AI system could spark discussion on conventional international humanitarian law and *jus in bello.* It is suspected that the introduction of AI in military operations could drastically change the warfare landscape.[196] Some thus call it "the third major transformation in warfare", [197] after the invention of gunpowder and nuclear weapons[198] However, despite the general perception, AI is far from foolproof. In other words, an LAWS may lack "the cognitive or sensor capability to positively identify a target in the particular circumstances, as might be the case with its employment in a cluttered battle space or an electronic warfare intense environment".[199] Therefore, in terms of humanitarian laws, it is critical that in the case of uncertainties, protecting civilians should be accounted for in the system.[200]

Meanwhile, there are semi-autonomous weapons systems, which do not select the targets on their own but follow human operator selections.[201] They are already widely used in contemporary warfare. For example, when a pilot identifies a target for a 'fire and forget' missile, the missile follows the target without any further human engagement[202] In fact, a number of countries such as the US[203] and Israel[204] have already introduced weapons with a lesser degree of autonomy, which operates under some kind of human supervision. Furthermore, there are also automatic weapons defense systems that are activated when they detect incoming threats, but they are limited for defense purposes.[205] Thus, what matters in terms of autonomy seems to be the machine's ability to identify a target without any human operator.[206]

One of the fundamental problems or limitations regarding LAWS is how to handle uncertainty in the battlefield. Any battlefield is full of uncertainties and military combat entails handling a series of uncertain situations. The question for LAWS then

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[195] Anderson and Waxman [58], pp. 1100–1101.

[196] Schmitt and Thurnher [71], p. 231.

[197] *See* Garcia [27], p. 335.

[198] Future of Life Institute, available at http://futureoflife.org/open-letter-autonomous-weapons/, *cited by* Garcia [27], pp. 334–5.

[199] *See supra* note 443, p. 262.

[200] *See supra* note 443, p. 262; *supra* note 395, p. 411.

[201] *See supra* note 443, p. 236.

[202] Schmitt and Thurnher [70], p. 236.

[203] *See* Lockheed Martin, *Aegis Combat System*, available at http://www.lockheedmartin.com/us/products/aegis.html; Federation of American Scientists (FAS), *Patriot TMD*, available at http://www.fas.org/spp/starwars/program/patriot.htm; Ibid.

[204] *See* Inbal Orpaz, *How Does Iron Dome Operate?, HAARETZ* (Nov. 19, 2012), available at http://www.haaretz.com/news/features/how-does-the-iron-dome-work.premium-1.478988; Iron Dome, *RAFAEL,* available at http://www.rafael.co.il/marketing/SIP_STORAGE/FILES/6/946.pdf.

[205] *See Supra* note 422, p. 9.

[206] Schmitt and Thurnher [70], p. 236.

is whether it is programmed in a way to handle these inherent uncertainties properly, so as to be in compliance with the present norms of IHL.[207] In the programming process, the LAWS can be equipped with either a higher level of self-judgment with few human interventions or a lower level of self-judgment with tight human control. In a spectrum of two extremes, it can be placed somewhere in-between. Regardless, the question is whether the LAWS is programmed in a way to comply with IHL.[208] It is the obligation of the developers and utilizers of LAWS to ensure this compliance, taking into consideration both the technological capability and limitation under the unique environment of warfare where uncertainty is brought to an extreme level.[209] Developers and utilizers would have to prove that they have done their homework in this process to be absolved from legal liability, which is separate from the possible liability stemming from the operators and soldiers on the field who use the weapon. If he or she makes a mistake in judgment, legal liability will attach to him or her, and by extension to his or her supervisors or commanders as well. Those supervisors and commanders may assume their own legal or vicarious liability.

In this regard, an argument is also being presented that the unique nature of LAWS makes it almost impossible or impractical for it to abide by all the tenets of IHL.[210] It follows then that such weapons are prohibited by international law.[211] The structural uniqueness and inherent limitations posed by LAWS will not be able to be solved even if professional efforts are being mobilized during the manufacturing, deploying and utilizing process. The issues presented by LAWS would never be solved by any one professional field operating in isolation. Legal experts and scientific experts need to talk to each other in the whole process, and these experts will then have to examine the factual situation on the battlefield which requires yet another set of expertise: carrying out military operations. This combination of diverse expertise seems impractical, if not entirely impossible. Any legal standard based on such a smooth flow of collective judgment would be vulnerable to challenges.[212] This argument thus supports the proposition that AWS are by nature illegal.

There is indeed an argument that some LAWS are inherently illegal.[213] The argument is that according to the IHL, indiscriminate weapons are unlawful,[214] and that AWS are such weapons. Weapons in violation of IHL are considered unlawful and unjust in the eyes of the international community.[215] Others base their arguments on the grounds of morality and see LAWS as in violation of 'principles of humanity'

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[207] Schuller [3], p. 411.

[208] Ibid., p. 412.

[209] Ibid., p. 417.

[210] *See supra* note 422, *see also* Human Rights Watch and International Human Rights Clinic, Harvard Law School, *Advancing the Debate on Killer Robots: 12 Key Arguments for a Preemptive Ban on Fully Autonomous Weapons* (2014), *cited by* Schuller [3], p. 382.

[211] *Supra* note 422, pp. 30–35, *cited by* Weaver [53] p. 2.

[212] Schuller [3], p. 379.

[213] *See supra* note 160, pp. 634–5.

[214] *See* Schmitt [72], pp. 1–37.

[215] Ibid., p. 652; *See supra* note 443, p. 243.

[216] Another morality-based argument against automated weapons sees the lack of human judgment as a problem.[217] According to this view, because acts of killing inevitably require a moral judgement, it is immoral to take humans out of the loop. The further the act of killing takes place from the initial decision, the easier it is to dilute the human role of the decision.[218] The argument goes that even when the weapons are unlikely to be cause unnecessary suffering and are not inherently indiscriminate, the principle of distinction and proportionality calls for "a proximate and significant level of human judgment" when they are applied to LAWS.[219]

The Special Rapporteur on Extrajudicial, Summary or Arbitrary Executions to the UN Council on Human Rights, Mr. Christof Heyns, suggested the unlawfulness of LAWS as based on the idea that robots should not have 'the power of life and death over human beings.' [220] Furthermore, Nia Griffith, Vice-Chair of the All Party Parliamentary Group on Weapons and Protection of Civilians, claimed that "robots may never be able to meet the requirements of international humanitarian law, as its rules of distinction and proportionality require the distinctively human ability to understand context and to make subjective estimates of value."[221] HRW asserts that robots by themselves are "incapable of abiding by key principles of international humanitarian law, […] [and] unable to follow the rules of distinction, proportionality, and military necessity and might contravene the Martens Clause."[222]

However, some point out that even when no human is directly involved at the application stage, there is always a commander or a civilian supervisor who "knew or should have known that the autonomous weapon system had been so programmed and did nothing to stop its use, or later became aware that the system had been employed in a manner constituting a war crime and did nothing to hold the individuals concerned accountable".[223] Opponents to such a view argue that it is not fair to hold a commander responsible for an action that he was neither in control of nor could foresee.[224] Some even argue that LAWS takes away the entire possibility of holding anyone accountable for violating the laws of war. In particular, those who emphasize individual criminal liability as the main enforcement mechanism see LAWS as a major obstacle. The reason is that while computers or algorithms

[216] *See supra* note 160, pp. 637–639.

[217] *See* Anderson and Waxman [57].

[218] Schuller [3], p. 389.

[219] *See supra* note 160, pp. 665–6.

[220] Special Rapporteur on Extrajudicial, Summary or Arbitrary Executions, U.N.G.A., Human Rights Council, 23d Sess., U.N. Doc. A/HRC/23/47, *cited by* Beard [45], p. 619.

[221] *See supra* note 160, p. 680.

[222] *See supra* note 422, p. 30; Also, *see supra* note 443, p. 243.

[223] *See supra* note 443, p. 277.

[224] *Protocol Additional to the Geneva Conventions of 12 August 1949, and Relating to the Protection of Victims of International Armed Conflicts, June 8, 1977,* 1125 U.N.T.S. 3, *cited by* Hammond [22], p. 9.

cannot be 'punished,' civil product liability is fundamentally different from wartime criminal penalties.[225]

A categorical prohibition approach, however, would be challenged, as it does not accurately reflect current jurisprudence.[226] Currently, weapons with a highly destructive capability are not prohibited per se.[227] They become illegal when they are used indiscriminately or in a way that causes unnecessary suffering.[228] In other words, the current jurisprudence does not recognize inherent unlawfulness when it comes to military weapons. The same logic should also apply to AWS; its mere deployment does not lead to violation but how it is used may. Thus, violation of international law (i.e., IHL) occurs when LAWS that does not or cannot sift civilian targets from military targets are deployed and used.[229] An argument to prohibit LAWS categorically, thus, would be, at least, difficult to sustain under the current jurisprudence. Instead, the proper inquiry should be whether an LAWS leads to unnecessary suffering or indiscriminate targeting.

The level of autonomy is unrelated to the probability of unnecessary suffering or superfluous injury caused by AI.[230] In other words, LAWS can be programmed to only direct military personnel and facilities, just as an individual system can be programmed to attack indiscriminately. Nevertheless, critics call for an absolute ban on lethal LAWS as a preemptive measure because they fear that governments will not respect the IHL.[231] However, as a matter of fact, it should be noted that even the HRW does not call for the prohibition of all types of LAWS. The HRW is clearly opposed to fully autonomous weapons, even though they refer to them as "robots".[232] The HRW is particularly critical of what they call "human-out-of-the-loop weapons", which means that they are "capable of making decisions without further human input after activation".[233] The HRW is not opposed to weapons with lesser degrees of autonomy, such as 'human-supervised' autonomous systems used by the US.[234] Israel has also used a type of human-supervised autonomous weapon system, Israel's Iron Dome, to defend itself from rocket attacks in Gaza.[235] These

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[225] Anderson and Waxman [58], p. 1110.

[226] Schmitt [72], available at http://harvardnsj.org/wpcontent/uploads/2013/02/Schmitt-Autonomous-Weapon-Systems-and-IHL-Final.pdf *cited by* Beard [45], p. 636/.

[227] *Legality of the Threat or Use of Nuclear Weapons*, Advisory Opinion. 1996 I.C.J.

[228] *Supra* note 160, p. 636.

[229] Hammond [22], p. 15.

[230] Schmitt and Thurnher [70], p. 31.

[231] Schmitt and Thurnher [70], p. 30.

[232] *Supra* note 422, p. 30, *cited by* Schmitt and Thurnher [70], pp. 1–2.

[233] *Supra* note 422, p. 5.

[234] Lockheed Martin, *Aegis Combat System*, available at http://www.lockheedmartin.com/us/products/aegis.html; Federation of American Scientists (FAS), *Patriot TMD,* available at http://www.fas.org/spp/starwars/program/patriot.htm, *cited by* Schmitt and Thurnher [70], p. 4.

[235] Inbal Orpaz, *How Does Iron Dome Operate?,* HAARETZ, (Nov. 19, 2012), available at http://www.haaretz.com/news/features/how-does-the-iron-dome-work.premium-1.478988; Iron Dome, *RAFAEL*, available at http://www.rafael.co.il/marketing/SIP_STORAGE/FILES/6/946.pdf, *cited by* Schmitt and Thurnher [70], p. 4.

systems operate with human monitoring agents who have overriding capability.[236] In other words, with enough safeguards, we do not need to completely ban weapons with some degree of autonomy. Nor do we have to abandon the existing laws of war. What should be pursued is to develop weapons so that the laws of armed conflict are not violated.[237]

It should be noted that at the end of the day, humans as decision makers will always play an important role. For instance, humans will decide the parameters of utilizing LAWS.[238] The lawfulness of LAWS will thus depend on human decisions. This also implies that even when more authority is delegated to the automated algorithms, the responsibility lies within the human operators and decision makers.[239] In other words, LAWS is not free from the existing laws. All in all, current legal norms can still regulate LAWS to a certain extent. Current legal norms thus can cover LAWS to the extent that it is a type of weapon.

It is true that even if the existing legal criteria are applied, LAWS deserves special consideration because of its unique characteristics, which are quite distinct from conventional weapons.[240] Nonetheless, to the extent that such special consideration is accorded within the existing framework of jurisprudence, a more persuasive argument would be that current jurisprudence, though insufficient in many respects, still applies to LAWS. The current legal norms relating to LAWS are not reflective of the reality in the field.

Again, the fact that current jurisprudence still applies to LAWS and provides general legal guidance does not mean that new norms are not necessary or current norms are sufficient. As noted above, the current norms provide only a limited set of legal guidance on the premise that LAWS is simply another type of military weapon. It turns a blind eye to the unique traits of LAWS; a machine equipped with an independent decision-making process. These traits bring LAWS outside the bounds of existing legal norms and raise a new question. Most notably, from the legal perspective, LAWS raises the issue of accountability. When AI weapons are inserted, the traditional pursuit of legal liability based on IHL and international human rights law may be disrupted.[241] Those responsible may try to hide behind newly developed AI weapons.

As a matter of fact, weapons that are installed to wait for the target so as to kill or maim it are not new, and have been a subject of domestic regulation. For instance, some states of the US enacted laws that seek criminal and civil sanctions for those who install traps targeting intruders or trespassers.[242] As these traps are operated by dint of pure mechanical response (such as touching a hook), they are quite different from the AI-enabled weapons that are operated by programs and data. Thus, autonomy in

[236] Schmitt and Thurnher [70], p. 4.

[237] International Committee of the Red Cross [42], *cited by* Schmitt and Thurnher [70], p. 31.

[238] Schmitt and Thurnher [70], p. 31.

[239] Schmitt and Thurnher [70], p. 31.

[240] *Supra* note 160, pp. 665–666.

[241] Bills [37], pp. 176–209 and Hammond [22], pp. 652–88, *cited by* Garcia [27], p. 336.

[242] *See* generally, Perovich [47], *cited by* Beard [45], p. 635.

the context of AWS relates to data collecting and processing rather than autonomy resulting from mechanical reaction. To the extent that an installer is held legally liable for the action of installment knowing the imminent bodily harm, the domestic booby-trap legislation offers some guidance for LAWS as well.

AWS introduces complex legal problems as regards individual responsibilities for programmers. While a programmer designs a program for LAWS, he or she does not control or determine how the LAWS behaves in a particular context based on specific circumstances. A probability based on data will be available but there would be no knowing how the LAWS responds at a particular point in time in a particular situation. It would be difficult to attribute legal liability to programmers in these circumstances unless the programmer's fault or error is somehow proven.[243]

What about the installer? An installer may also escape legal liability if he or she shows that all reasonable steps have been taken in testing LAWS based on the information provided by the manufacturer/programmer. Most likely the installer might want to say that he accepted the explanation and guarantee of the manufacturer/programmer on its face and relied upon it. As with a programmer, an installer would also be unable to determine how LAWS acts in a particular situation based on a set of information collected right before the strike. Reasonable calculation would be possible but precise prediction would not.

A commanding officer and a manufacturer/programmer may be held liable when certain conditions are met.[244] If they had been aware of the aberrational behavior of the LAWS and still sold and deployed the system, they would have to be found legally liable. In the view of the US DOD, "[p]ersons who authorize the use of, direct the use of, or operate autonomous [...] weapon systems must do so with appropriate care and in accordance with the law of war, applicable treaties, weapon system safety rules, and applicable rules of engagement."[245] In other words, if all these bases are covered before the deployment, then the manufacturer, programmer, and commanding officer are probably off the hook. So are the officers on the field using the system as long as they carry out these duties.

Suppose an officer knows the high likelihood of malfunction as regards distinguishing civilians and enemy combatants, he or she would have to be held liable under the current jurisprudence. But as these officers are not experts on science and machines, they would be unable to make an informed judgment as to a reasonably accurate probability of malfunction as a result of the designs and programs of the LAWS in question.[246] Even if they did, it could only happen after data and information through an operation is accumulated.[247] Nor could they try to re-program

[243] Stewart [17, 18], pp. 271, 290, *cited by* Beard [45], p. 651.

[244] *See* Sparrow [83], pp. 62, 69–71,*Supra* note 422, pp. 42–44; Schmitt and Thurnher [70], pp. 231, 277–78, *cited by* Hammond [22], p. 8.

[245] US Department of Defense, *Autonomy in Weapon Systems,* Directive 3000.09 (Nov. 21, 2012), p. 3, available at http://www.fas.org/irp/doddir/dod/d3000_09.pdf, *cited by* Hammond [22], p. 9.

[246] *Supra* note 422, p. 43, *cited by* Hammond [22], p. 9.

[247] *See* ibid.

the system for improvement.[248] The more sophisticated an LAWS becomes with technological development, the more difficult it becomes for officers to intervene and change. This lack or absence of control or foreseeability may also prevent the establishment of legal liability, be it individual liability or command liability, for the officers.[249] Finding legal liability for a soldier operating LAWS under the command of an officer would be equally difficult.

The HRW raised a concern that the difficulty of assigning legal responsibility would lead to a severe blow to IHL "[s]ince there is no fair and effective way to assign legal responsibility for unlawful acts committed by fully autonomous weapons, granting them complete control over targeting decisions would undermine yet another tool for promoting civilian protection."[250] However, such a conclusion is not inevitable, as the absence of a direct control does not signify the absence of legal responsibility.[251] As mentioned earlier, the process of deploying AWS involves multiple human decision makers, ranging from the programmer to the commander.[252] Just as with conventional weapons, had a commander requested AWS be used in a way that would be in violation of IHL, he or she can be held responsible.

On the other hand, the AI itself cannot be held responsible for war crimes or otherwise be punished, since it is not a human being; criminal liability attaches to a human being. Civil liability also accrues when human action or inaction is found to be at fault. AI, on the other hand, is a machine or s system, so that such legal liability would not attach to it. Of course, once AI is regarded as another type of person (in

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[248] *See* ibid.

[249] Protocol Additional to the Geneva Conventions of 12 August 1949, and Relating to the Protection of Victims of International Armed Conflicts, June 8, 1977, 1125 U.N.T.S. 3, *cited by* Hammond [22], p. 9.

[250] *Supra* note 422, p. 42.

[251] Fenrick [97], *cited by* Schmitt and Thurnher [70], p. 29.

[252] *See*, e.g., Convention (I) for the Amelioration of the Condition of the Wounded and Sick in Armed Forces in the Field, art. 49, Aug. 12, 1949, 75 U.N.T.S. 31; Convention (II) for the Amelioration of the Condition of Wounded, Sick and Shipwrecked Members of Armed Forces at Sea, art. 50, Aug. 12, 1949, 75 U.N.T.S. 85; Convention (III) Relative to the Treatment of Prisoners of War, art. 129, Aug. 12, 1949, 75 U.N.T.S. 135; Convention (IV) Relative to the Protection of Civilian Persons in Time of War, art. 146, Aug. 12, 1949, 75 U.N.T.S. 287; Additional Protocol I. Protocol Additional to the Geneva Conventions of 12 August 1949, and Relating to the Protection of Victims of International Armed Conflicts, art. 49, June 8, 1977, 1125 U.N.T.S. 3 Arts. 86–86; Hague Convention for the Protection of Cultural Property in the Event of Armed Conflict with Regulations for the Execution of the Convention, art. 28, May 14, 1954, 249 U.N.T.S. 240; Second Protocol to the Hague Convention of 1954 for the Protection of Cultural Property in the Event of Armed Conflict, art. 15(2), Mar. 26, 1999, 2253 U.N.T.S. 212; Additional Protocol I non-Party status. Statute of the International Criminal Court, July 17, 1998, 2187 U.N.T.S. 90 Arts. 25(3) (b) & 28; Statute of the International Criminal Tribunal for the Former Yugoslavia, art. 7(1), S.C. Res. 827 annex, U.N. Doc. S/RES/827 (May 25, 1993); Statute of the International Criminal Tribunal for Rwanda, art. 6(1), S.C. Res. 955 annex, U.N. Doc. S/RES/955, (Nov. 8, 1994); Prosecutor v. Blaški, Case. No. IT -95–14-T, Trial Chamber Judgment, PP 281–282 (Int'l Crim. Trib. for the Former Yugoslavia Mar. 3, 2000); Prosecutor v. Krstic, Case No. IT-98–33-T, Trial Chamber Judgment, P 605 (Int'l Crim. Trib. for the Former Yugoslavia Aug. 2, 2001); Prosecutor v. Kayishema & Ruzindana, Case No. ICTR 95–1-T, Trial Chamber Judgment, P 223 (Int'l Crim. Trib. for Rwanda May 21, 1999); COMMANDER'S HANDBOOK, P6.1.3, *cited by* Schmitt and Thurnher [70], p. 29.

addition to a natural person and a juridical person), then a new debate may begin, but at this point in time it is mainly regarded as a machine or a system. In fact, this possibility is already also being discussed.[253] Such being the case, LAWS may also expose a legal vacuum when it comes to legal liability for a particular person in the cycle. Addressing this problem may require a new treaty on how to handle individual legal liability when an LAWS is involved.[254]

A relevant question then becomes whether we could attribute the decision of LAWS to a particular person. It would be a challenging task, however.[255] The decision-making process can be a multi-layered one in which pinpointing one person may be difficult. According to one description:

> [T]he culpability of individual designers, engineers, programmers, or developers of specific components of autonomous weapon systems is problematically linked (for criminal law purposes) to a collaborative and advanced technological achievement. The complex and polymorphic nature of such systems is likely to defy attempts to assign clear responsibility for overall system failures. Decision-making by these machines will probably be distributed across several programs and processors, results will be derived from data input that originates from many different types of sensors, and no single agent may be able to be identified as the "decision-maker" in a complicated system of concurrently interacting human and machine components.[256]

As the above paragraph explains, the operation of LAWS may not be attributed to a single person.[257]

The designer or manufacturer can be held liable for violation of international law by their LAWS.[258] They then would have to pay for the damage and compensate victims.[259] When the negligence in the design is deemed to be very severe, the basic civil product liability may not be sufficient.[260] The problem is that as the LAWS decision-making algorithms go beyond simple 'if/then' flowcharts, then ensuring IHL compliance becomes less straightforward.[261]

> Regardless of the specifics of the design process, autonomy presents its own challenges for establishing individual responsibilities with respect to programmers. The essence of autonomy in machines means that inputted data and programming prior to a particular operation will not necessarily "result in a specific outcome in response to any given set of circumstances" and that, absent establishing the aberrant behavior of command programmers, "it would be almost impossible to attribute the autonomous system's behavior directly to a particular human."[262]

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253 Krishnan [8], pp. 105, *cited by*. Hammond [22], p. 8.

254 *See* Anderson and Waxman [58], p. 1110.

255 *See supra* note 365, p. 625.

256 Finn and Scheding [5], p. 183.

257 Hammond [22], p. 15.

258 *Supra* note 422, pp. 43–44, *cited by* Hammond [22], p. 10.

259 Krishnan [8], pp. 103–04, *cited* Hammond [22], p. 10.

260 *See supra* note 447, p. 1110.

261 *See supra* note 395, p. 417.

262 Stewart [18], pp. 271, 290.

The problem with the negligence approach to designer or manufacturer liability is that no designer could perfectly predict all the consequences of LAWS.[263] For instance, even though AI had not been programmed for war crimes, a commander could misuse it to commit illegal acts.[264] It seems unjust to hold the designers accountable for abuses of LAWS that they could not have controlled. Only those who had knowingly and willingly created and used LAWS for war crimes should be held accountable under the law of armed conflict.[265]

Most importantly, the legal liability debates regarding LAWS were focused on individual criminal liability. While this is important, it turns a blind eye to a more important issue of legal liability—state responsibility under customary international law as reflected in the 2001 ILC Draft Articles. The possibility of holding a state itself accountable for the war crimes of LAWS has not been scrutinized enough.[266] Individual criminal liability is only one way of addressing legal problems involving the LAWS.[267] Invoking state responsibility is not only another viable option, but also even normatively a better option than holding an individual accountable. It could contribute more effectively to enhancing compliance to international law.[268]

As such, a new treaty covering this issue is in order. In essence, a meaningful connection with the effective exercise of human judgment will be the key issue to link the state responsibility and individual culpability of related personnel.[269] For individuals such as designers, engineers, and developers, the complexity of LAWS makes it difficult to establish a meaningful connection.[270] In fact, by nature, the LAWS makes it difficult to argue that any one person can perfectly predict its actions.[271]

Unfortunately, each of the above proposed solution to addressing individual accountability over LAWS crimes has some problems. As a consequence, it is easy to imagine that a 'responsibility vacuum' would be likely to emerge.[272]

Thus, the emergence of AI-based weapons is causing large waves of disruption in the current IHL architecture.[273] The confusion is exacerbated due to the lack of specific convention-based regulations regarding AI-based weapons.[274] If the issue

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[263] *Supra* note 422, p. 43, *cited by* Hammond [22], p. 11.

[264] The law of armed conflict imposes a duty to investigate possible war crimes. Schmitt [69], pp. 31, 43, *cited by* Schmitt and Thurnher [70], p. 30.

[265] Schmitt and Thurnher [70], p. 30.

[266] Hammond [22], p. 3.

[267] Anderson and Waxman [58], p. 1110.

[268] Nissel [2], pp. 355, 369, *cited by* Hammond [22], p. 12.

[269] *Supra* note 160, p. 680.

[270] Finn and Scheding [6], *cited by* Beard [45], p. 651.

[271] Sparrow [83], pp. 62, 70, *cited by* Hammond [22], p. 11.

[272] Christof Heyns, *Report of the Special Rapporteur on Extrajudicial, Summary or Arbitrary Executions*, U.N. Doc. A/HRC/23/47 (Apr. 9, 2013), p. 80 available at http://www.ohchr.org/Documents/HRBodies/HRCouncil/RegularSession/Session23/AHRC-23-47_en.pdf, *cited by* Hammond [22], p. 11.

[273] *See* Bills [37], pp. 176–209, *supra* note 365, pp. 652–88; *supra* note 444, p. 336.

[274] *See supra* note 160, pp. 634–5.

of responsibility attribution is not addressed promptly, there may be a 'responsibility vacuum' leading to 'impunity for all [LAWS] use'.[275]

5.2 *Drones*

Weaponized drones and the requirement of human control needs further feedback to apply traditional rules of distinction, proportionality, military necessity and more. Drones are being used in different tasks, ranging from surveillance to weapons delivery.[276] In fact, drones offer a cheaper alternative to manned aircrafts with a lower risk of losing pilots. In particular, US drone strikes have received a lot of international attention, leading to fierce debates about its legality.[277] In particular, it was questioned whether the US drone strikes (e.g. in Pakistan, Yemen and Somalia) could be classified as 'self-defense' or the rightful use of weapons in an 'armed conflict'.[278]

From the IHL perspective, an important issue at hand is not whether drones are unmanned, but the types of weapons that they carry.[279] International humanitarian law condemns unnecessary use of lethal forces, such as missiles and bombs outside battlefields. While the use of drones in battlefields is lawful, allowing for higher accuracy and precision, the ease of killing with drones should be take into consideration.[280] The separation of killing and its decision may decrease the ethical burden of the decision-makers to use lethal weapons. This argument is closely related to the moral argument against killer robots and other automated weapons.

Another point that needs to be discussed is the use of drones before an armed conflict. Article 51 of the UN Charter permits self-defense under strict conditions.[281] Some argue that use of drones should be allowed to attack all ships of a unit, if one of them attacks, as "an asserted right for ships at sea to exercise 'unit self-defense'".[282] Another argument is the use of drones in exercising the right of hot pursuit.[283] Some even argue for the right of pre-emptive self-defense, but the current IHL does not recognize such a right.[284]

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[275] Christof Heyns, *Report of the Special Rapporteur on Extrajudicial, Summary or Arbitrary Executions,* U.N. Doc. A/HRC/23/47 (Apr. 9, 2013), p. 80, available at http://www.ohchr.org/Documents/HRBodies/HRCouncil/RegularSession/Session23/AHRC-23-47_en.pdf, *cited by* Hammond, *supra* note 365, p. 667.

[276] O'Connell [65].

[277] Brooks [84], p. 89.

[278] Ibid., 95.

[279] *Supra* note 471.

[280] Ibid.

[281] Ibid.

[282] Ibid.

[283] Ibid.

[284] Ibid.

Meanwhile, governments have incentives to regulate domestic use of drones as well. Drones can cause an infringement of privacy and be a safety hazard if they are flown too close to residential areas. As a consequence, South Korea has imposed strict regulations on drones. Under the current regulation, even small, hobby drones (including toys) need to get flight permission when they fly in the 'flight regulation zone' which takes up a significant portion of the Seoul Metropolitan Area. So, hobbyists flock to the banks of Han River to fly them. Reacting to the pleas for a change, the government issued proposed rules in early October. After the comments and hearings, the new rules will go into effect early next year, and they are more practical. Toy and leisure drones, more specifically, drones of 250 g or less, are now exempt from regulation. However, the core parts of the regulation remain, even if adjustments have been made according to the weight and capability of drones. For instance, flying them beyond eyesight is still prohibited. So is flying them higher than 150 m. Prior permission is required for these flights. Risks to pedestrians and security threats are cited as a main reason for the regulation.

5.3 Killer Robots

Killer Robots are programmed to engage in warfare and human control is another hindrance to applying rules of international Law of War. Some in the international community, such as the HRW has called for a ban on autonomous weapons. They argue that humans should be involved in the targeting decision.[285] In November 2012, HRW released a report containing sharp criticisms of the LAWS, entitled *Losing Humanity: The Case against Killer Robots*:

> With the rapid development and proliferation of robotic weapons, machines are starting to take the place of humans on the battlefield. Some military and robotics experts have predicted that "killer robots"—fully autonomous weapons that could select and engage targets without human intervention—could be developed within 20 to 30 years. At present, military officials generally say that humans will retain some level of supervision over decisions to use lethal force, but their statements often leave open the possibility that robots could one day have the ability to make such choices on their own power. Human Rights Watch and Harvard Law School's International Human Rights Clinic (IHRC) believe that such revolutionary weapons would not be consistent with international humanitarian law and would increase the risk of death or injury to civilians during armed conflict. A preemptive prohibition on their development and use is needed.[286]

The HRW views that fully autonomous weapons are inherently in violation of international humanitarian law which requires safeguards for civilians.[287] In particular, they believe that killer robots will not be able to respect the key principles of international humanitarian law, such as the rules of distinction, proportionality,

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[285] Weaver [53] p. 2.

[286] *Supra* note 422..

[287] *See* ibid., p. 30, *cited by* Schmitt and Thurnher [70], pp. 1–2.

and military necessity.[288] Thus, the HRW views that LAWS are in violation of the Martens Clause and demands that they be completely banned.

The IHL prohibits both indiscriminate weapons[289] and weapons that cause unnecessary suffering.[290] The *Additional Protocol I to the Geneva Conventions of 1949,* as well as customary international law asks for a state to consider whether the introduction of a new weapon is in compliance with the Protocol or any other relevant international law.[291] The US, which is not a party to the Protocol, has its own requirement of a legal review for new weapons "to determine whether the weapons or weapon systems or their intended use in combat are consistent with the obligations assumed by the US Government under all applicable treaties and with customary international law."[292] Some of the state laws in the US involve criminal and civil liability for the use of autonomous weapons.[293] Opponents of the LAWS argue that if the LAWS cannot adequately assess the lawfulness of a use of force, it lies outside the scope of what is allowed under the IHL.[294] In addition to the IHL, some argue that the precautionary principle of international law calls for prohibition of introducing AI based weapons.

> The precautionary principle of international law includes three domains: to prevent harm, to shift the burden of proof to supporters of a probably harmful activity, and the promotion of transparent decision-making that includes those who would be affected. The precautionary principle calls for action to be taken before harm is done. Artificial intelligence presents such a case: there is scientific certainty and consensus regarding the danger of its weaponization. There is already strong consensus in the international community that not all weapons are acceptable and that some have the potential to be so harmful that they should be preemptively banned. Such is the case with the prohibition on blinding laser weapons, a class of weapons that was banned while still in development. In the same way, the weaponization of AI can be halted before its full deployment in the battlefield, and international efforts can instead be focused on its peaceful uses.[295]

A major issue of LAWS is the uncertainties regarding who would bear the legal responsibility of illegal use of AWS or illegal actions by LAWS. When it is difficult to attribute the actions of a LAWS to one individual, invoking state responsibility is a possibility. While state responsibility and individual criminal and civil liability function differently, "a meaningful connection with the effective exercise of human judgment" is the key to between the two regimes.[296] One may even argue that it is

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[288] *See* ibid., p. 30, *cited by* Schmitt and Thurnher [70], p. 9.

[289] *See supra* note 681, p. 7.

[290] Ibid.

[291] *Protocol Additional to the Geneva Conventions of 12 Aug. 1949, and Relating to the Protection of Victims of International Armed Conflicts,* art. 36, opened for signature Dec. 12, 1977, 1125U.N.T.S.3, reprinted in 16I.L.M.1391 (1977), *cited by* Beard, *supra* note 160, p. 635.

[292] Army Regulation 27–53, *Review of Legality of Weapons Under International Law* (Jan. 1, 1979), para. 4.e.(1), *cited by* Beard, *supra* note 160, p. 635; *see also* Solis [36], p. 751.

[293] *See supra* note 160, p. 635.

[294] *See supra* note 395, p. 389.

[295] *Supra* note 444, p. 338.

[296] *See supra* note 160, p. 680.

normatively preferable to liability of individuals such as the commander, designer, or manufacturer.[297] For instance, while some suggest that the designers should be held accountable for war crimes committed by LAWS, it should be the responsible superiors who knowingly use the system who should be held accountable (unless the system was purposefully designed to commit war crimes).[298] Afterall, the concept of autonomy suggests that the LAWS can surpass the immediate predictions of its own designers and manufacturers.[299] Any commander or supervisor who ordered the actions resulting in war crimes could be held responsible, but so could those who failed to take action to prevent them.[300] In the latter circumstance, how much the commander or supervisor could intervene may be questionable.[301]

According to the conventional state responsibility principle, it is possible to attribute the responsibility of the LAWS to a state that has authority over it.[302] Furthermore, state responsibility contributes to better compliance with international law.[303] Currently, there is an absence of legal architecture to address autonomous systems.[304] Demand for creating and updating the legal architecture to adequately harness AI is growing.[305] Some states may be tempted to take advantage of the legal vacuum in the international community with respect to the AWS to bypass state responsibility for its illegal use of force. However, if there is a possibility of invoking state responsibility, the incentive to take advantage of the vacuum will decrease.

6 Deepening North–South Divide

Despite the positive opportunities that AI entails, there is a high risk of aggravated inequality between developed and developing countries due to the emergence of AI. As such, more attention should be directed to finding a way to support developing states and least developed states in the new ear of AI.

The developed countries cannot easily disregard the difficulties laid in front of the developing countries in the AI era. The developing countries have a lot of potential as a new, extensive market for businesses. Moreover, global inequality could lead to instability, hindering global economic growth. As AI brings everyone and every society closer, it is in everyone's interest to promote a global establishment of transparent and responsible use of AI.

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[297] *See supra* note 365, p. 656.

[298] *See supra* note 443, p. 279.

[299] *See supra* note 365, p. 666–7.

[300] The law of armed conflict imposes a duty to investigate possible war crimes. *See supra* note 443, p. 279.

[301] *See supra* note 365, pp. 664–5.

[302] Ibid., p. 668.

[303] Ibid., p. 669.

[304] Ibid., p. 671.

[305] *See* Wadhwa [93], Breland [1], Walsh [94], *cited by* Erdélyi and Goldsmith [76], p. 1.

6.1 Developing Countries in a Disadvantageous Position

Arguments on AI compounding socioeconomic division in the world is a lingering projection, entailing rapid changes and thus must be dealt with in multiple dimensions. While AI can contribute to the development of developing countries,[306] difficulties abound.

Lack of appropriate infrastructure is one. For example, the UN Food and Agriculture Organization (hereinafter FAO) states that some basic conditions for a society to benefit from the digital revolution include, but are not limited to: connectivity (e.g. mobile subscriptions, network coverage, internet access), affordability, and educational attainment (e.g. literacy, ICT education).[307] Without enough investment in infrastructure, developing countries may not be able to benefit from the new technologies.

There are many signs that AI and technological advancement can actually solidify and deepen the North–South divide and global inequality. As AI is predicted to replace labor-intensive, low-skilled jobs, the developing countries could suffer significantly due to job losses.[308] As the tech companies and high-skilled workers in the AI sector are concentrated in the developed world, it seems unlikely that developing countries will be in the front seats of the AI-era.[309] There are studies that suggest that AI and robots decrease social mobility, and thereby solidify social inequality structure even within a country.[310] Noting that the benefits of the digital revolution may be unequally distributed, with urban areas benefiting much more than rural areas, the FAO calls for special attention to connecting marginalized and remote communities.[311]

Another negative side-effect of the digital revolution is that the new technology can be easily abused to strengthen authoritarianism. For example, many developing countries do not have legal foundations to protect personal data. India finally passed its Personal Data Protection Bill in December 2019, after two years of intense debate.[312]

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[306] For example, the UNDP is applying digital technologies to assist development: it has deployed drones to map Mauritius, applied satellite and drone imagery to monitor Mali's post-crisis recovery, and launched mobile payment for rural villages in Fiji. For more, *visit* the UNDP website, available at https://www.undp.org/content/undp/en/home/news-centre/news/2019/Digital_technologies_are_revolutionizing_global_development_UNDP_chief_says.html.

Many other organizations such as the World Bank, the Food and Agricultural Organization, and the UNDP apply AI and machine learning technology to enhance productivity and to obtain better predictions of world resource situations. For a comprehensive list of activities of international organizations applying AI to achieve SDGs, *see supra* note 348.

[307] *See* Nikola M. Trndov, Samuel Varas, and Meng Zeng, *Digital Technologies in Agriculture and Rural Areas Briefing Paper,* FAO, pp. 3–6, available at http://www.fao.org/3/ca4887en/ca4887en.pdf.

[308] *See* Goldin [39].

[309] Ibid.

[310] Vincent [48].

[311] *See supra* note 703.

[312] *See* Basu and Sherman [7].

While it is applaudable that the bill attempts to provide a foundation for information governance, it reflects some 'protectionist and authoritarian-leading data policies'.[313] It suggests that the AI revolution could strengthen authoritarian regimes in developing countries.

6.2 *Capacity Building Projects*

Given a high possibility of future international tension arising from disparities accrued to diverging levels of technology, international collaboration on capacity building is an important agenda. In order to level out the difference between the developed and developing nations, the international community is calling for capacity building projects. For example, the UN High-Level Panel on the post-2015 development agenda called for a 'data revolution'.[314] The data revolution includes two objectives—to integrate statistics for better public and private decision making and to enhance transparency and accountability for more trust between society and state.[315] In other words, there is a recognition that AI can lead to better policies insofar as it could contribute to global development. In order to promote best practice cases and mitigate the risks associated with AI, McKinsey Global Institute has collected more than a 100 cases of AI applications in pursuit of SDGs.[316]

Other international organizations offer numerous capacity building programs as well. For example, recognizing the benefits of AI, the World Bank organized a capacity building training workshop in Bangkok:

> The World Bank has been supporting the use of AI and big data to achieve our goals of decreasing poverty and increasing shared prosperity, and recently launched the AI for Development initiative and an Artificial Intelligence Lab. Boosting capacity is key in achieving these goals. In this regard, the World Bank organized a 5-day skills-building program focusing on big data, AI, and decision science in health and nutrition in Bangkok, featuring the participation of World Bank country teams as well as government and academic partners from Bangladesh, India, Indonesia, Lao PDR, Myanmar, Nepal, Pakistan, Papua New Guinea, Rwanda, Samoa, Tajikistan, Thailand, Uzbekistan, Vietnam, and Zimbabwe. Similar trainings were also held in Bucharest, Romania, and Pretoria, South Africa, engaging decision makers in public health from Latin America, Europe, Central Asia, and Africa.[317]

Having quality data could greatly contribute development, but developing countries lack the infrastructure for the data that they need. The new technology can change the way data is collected (surveys, satellites, mobile phone data, social media,

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[313] Ibid.

[314] *High-Level Panel of Eminent Persons on the Post-2015 Development Agenda: A New Global Partnership: Eradicate Poverty and Transform Economies through Sustainable Development*, *UN*, New York: United Nations Publications (2013).

[315] *See High-Level Panel on the Post-2015 Development Agenda* (Aug. 27, 2013), available at https://www.post2020hlp.org/wp-content/uploads/docs/What-is-the-Data-Revolution.pdf.

[316] *See* Chui et al. [73].

[317] Osornprasop [88].

etc.), analyzed (hypothesis testing, machine learning) and used for policy making.[318] Big data allows for real-time information that could enhance monitoring of a policy implementation. For example, UN Children's Fund (UNICEF) office in Kyrgyzstan applied newly available data to map schools with internet access as a first step to address digital gaps.[319] Another example is that the Africa Regional Data Cube (ARDC) collects satellite images, which enable governments to detect illegal mining and monitor rapid urbanization.[320]

Capacity-building is expected to be an important subject in the trade agreement as well. Developing countries have difficulties in participating in new digital trade and thus it always involves the risk of isolation from it. As the importance of digital trade is increasingly emphasized, corresponding supportive interventions need to be provided so that developing countries can also participate. Accordingly, it is expected that capacity-building of developing countries will be a major issue in this area of discussion. On the assumption that the international community reaches a common understanding on the agenda of 'development' brought out by present trade agreements, it may be possible to come up with new norms in the area of digital trade that reflect realities.

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Chapter 7
Legal Vacuum to Regulate AI



In contrast to rapidly developing AI, the international community has been struggling to come to an agreement on how to regulate it. Consequentially, there is a legal vacuum with respect to AI. The criticism is getting louder as the legal vacuum with respect to AI is getting bigger.[1] However, a legal vacuum does not mean that the international society is completely devoid of discussions on AI as well. This chapter gives an overview of discussions related to AI in various international institutions and proposes what else should be included in prospective treaties on AI.

Recently, the UN has tried to assist in the formulation of an international consensus on the topic, starting from the international security area. Since then, the discussion has been expanded to cover other issues such as human rights, development, and sustainability. Moreover, many international institutions, including both UN specialized agencies and other issue-specific international organizations, have started discussions about AI. Such fragmented works can provide a good starting point for a broader legal system on AI to develop.

Additionally, there are a couple of points that need to be addressed in the future treaties on AI. First, as data becomes more important on the one hand, and transnational transactions increase on the other hand in the AI era, the international community needs to introduce the extraterritoriality zone of personal information. Second, the uniqueness of the digital economy needs to be sufficiently reflected. The current international trade law cannot be applied to digital trade, because many issues (*e.g.,* the global-value-chain or governmental measures for support) will look different in the context of AI-related transactions.

[1] *See* Wadhwa [1], Breland [2], Walsh [3], Erdélyi and Goldsmith [4].

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1 Gaps Between AI and Legal Norms

There is a general consensus at home and abroad that the introduction and utilization of AI in various areas will cause significant ripple effects throughout our society. AI in its initial stage having just emerged has already resulted in significant change in our society. Such change inevitably requires the general law system of our society to be adjusted accordingly since new subjects and new modes of decision-making inconsistent with the existing law system emerge. At the moment, there is legal lacuna in terms of hard law, soft law and customary law regarding AI. Internationally agreed norms and regulations regarding AI is under the process of development at best.

As the digital revolution accelerates around the world, and the topic of AI appears ubiquitous in international discussions, the UN has recognized the legal vacuum and sensed the need to address the issue, especially in relation to international security. At first, the discussion was focused on the application of AI in weaponry. The UN Convention on Certain Conventional Weapons has been used as a forum to discuss the development of LAWS.[2]

In a 2014 Report to the UN Council on Human Rights, *The Special Rapporteur on Extrajudicial, Summary or Arbitrary Executions*, Mr. Christof Heyns, argued that the deployment of LAWS or lethal autonomous robots (LARs) "may be unacceptable because no adequate system of legal accountability can be devised, and because robots should not have the power of life and death over human beings."[3] Fearing a future in which 'killer robots' threaten humanity, several non-governmental organizations have issued calls for autonomous weapon systems to be outlawed, with one group's leader stating that "[o]ur concern is that humans, not sensors, should make targeting decisions."[4] However, many existing weapon systems already have sensors that "make targeting decisions."[5]

There have been meetings of experts on LAWS since 2014, and an open-ended Group of Governmental Experts (hereinafter GGE) was established in 2016 to address the emerging technologies of LAWS. The first meeting of GGE took place in 2017.[6] Furthermore, in 2015, the United Nations Interregional Crime and Justice Research Institute (hereinafter UNICRI) established Centre for Artificial Intelligence and Robotics to support and foster AI and robot development.[7] The Centre opened in 2017 and aims to "enhance understanding of the risk–benefit duality of Artificial

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[2] *See* Library of Congress [5].

[3] *Special Rapporteur on Extrajudicial, Summary or Arbitrary Executions,* UNGA Human Rights Council, 23rd Sess., U.N. Doc. A/HRC/23/47; *Cited by* Beard, *supra* note 160, p. 619.

[4] Marks [6].

[5] *See supra* note 160, p. 619.

[6] For the report of the latest meeting of the GGE in 2019, *see Report of the 2019 session of the Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons Systems,* CCW/GGE.1/2019/3, available at https://undocs.org/en/CCW/GGE.1/2019/3.

[7] *Refer to* the website of *UNICRI Centre for Artificial Intelligence and Robotics,* UNICRI, available at http://www.unicri.it/in_focus/on/UNICRI_Centre_Artificial_Robotics.

Intelligence and Robotics through improved coordination, knowledge collection and dissemination, awareness-raising and outreach activities".[8]

In October 2015, the seventieth session of the UN General Assembly discussed "Rising to the Challenges of International Security and Emergence of Artificial Intelligence".[9] The event was organized by Georgia, in collaboration with the UNICRI, with a focus on chemical, biological, radiological, and nuclear (hereinafter CBRN) national action plans. The UNICRI launched its program on AI and Robotics in the same year. The UNICRI acknowledges that technology is a double-edged sword when it comes to dealing with crime and law enforcement.[10] For instance, AI and robotics can be applied to identify persons or objects of interest, predict criminal activities, and optimize the allocation of resources for law enforcement. However, such application could also entail violations of human rights, such as the right to privacy. Moreover, terrorists and criminals can abuse the new technology to disrupt our society especially with digital attacks.

On October 26, 2018, the 19th Meeting of the 73rd Session of the UN General Assembly (GA/DIS/3611) discussed the potential risks of new technologies, including *lethal autonomous weapons, and cyberattacks.*[11] In a Report of the Special Rapporteur to the UN General Assembly on AI and its impact on freedom of opinion and expression,[12] the Special Rapporteur recommends that states and companies follow human rights principles when they procure or deploy AI.

Discussion about AI in the UN has been extended in other areas beyond security. In October 2017, a joint meeting was organized and held by the UN Economic and Social Council (hereinafter UN ECOSOC), entitled "The Future of Everything—Sustainable Development in the Age of Rapid Technological Change".[13] UN Secretary-General António Guterres recognized the fast speed of AI's development and its potential to contribute to world development: "Artificial Intelligence has the potential to accelerate progress towards a dignified life, in peace and prosperity, for all people. The time has arrived for all of us—governments, industry and civil society—to consider how artificial intelligence will affect our future."[14]

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[8] *See* Smith [7]; Library of Congress [5].

[9] A recording of the meeting available at http://webtv.un.org/meetings-events/general-assembly/watch/chemical-biological-radiological-and-nuclear-cbrn-national-action-plans-rising-to-the-challenges-of-international-security-and-the-emergence-of-artificial-intelligence/4542739995001.

[10] *See Artificial Intelligence and Robotics,* UNICR, available at http://www.unicri.it/topics/ai_robotics/.

[11] *See First Committee Weights Potential Risks of New Technologies,* UN, (Oct.26, 2018), *available at* https://www.un.org/press/en/2018/gadis3611.doc.htm.

[12] For the full report, *see Report of the Special Rapporteur on the promotion and protection of the right to freedom of opinion and expression.* 2018. A/73/348, available at https://undocs.org/A/73/348.

[13] *Looking to Future, UN to Consider How Artificial Intelligence Could Help Achieve Economic Growth and Reduce Inequalities,* UN (Oct. 9, 2017), available at https://www.un.org/sustainabledevelopment/blog/2017/10/looking-to-future-un-to-consider-how-artificial-intelligence-could-help-achieve-economic-growth-and-reduce-inequalities/, archived at https://perma.cc/9WGH-TSVT.

[14] Ibid.

In 2018, the UN Secretary-General established internal principles to provide guidelines of UN engagement with new technologies in line with the values of the UN Charter and the UDHR.[15] The five principles include (1) protect and promote global values; (2) foster inclusion and transparency; (3) work in partnership; (4) build on existing capabilities and mandates; (5) be humble and continue to learn.[16] Based on these, the UN Secretary General announced four commitments: (1) deepening the UN's internal capacities and exposure to new technologies; (2) increasing understanding, advocacy and dialogue; (3) supporting dialogue on normative and cooperation frameworks; (4) enhancing UN system support to government capacity development.[17]

At the special agency level, the International Telecommunication Union (hereinafter ITU), the UN specialized agency for ICT matters, offers a forum of discussion on the impact of AI. The ITU was established in 1865 to facilitate international communications networks. Now its activities can be organized into three big categories: radiocommunications (managing the radio-frequency spectrum and satellite orbits), standardizations (setting standards for telephone calls or the internet), and development (providing and generating information on the recent development of the ICT sector). Its membership includes not only national states but also 900 private companies, international and regional organizations and universities, which make it suitable for a vibrant discussion encompassing both the public and private sectors.[18]

The ITU is the organizer of "AI for Good Global Summits".[19] The 2017 Summit was the first "inclusive global dialogue on beneficial AI". Since 2018, the Summit became more action-oriented and focused more on benefits of AI in pursuance of the Sustainable Development Goals. The 2020 Summit is scheduled to take place in Switzerland in September. The Summit is meant to provide a forum of discussion on how AI can contribute to the UN's SDGs. It is action-oriented, focused on real-world innovations, and aimed at a diverse global audience.[20]

The ITU also has a number of focus groups related to AI such as the Focus Group on AI for autonomous and assisted driving (hereinafter FG-AI4AD), Focus Group on "Environmental Efficiency for Artificial Intelligence and other Emerging Technologies" (hereinafter FG-AI4EE) Focus Group on "Artificial Intelligence for Health" (hereinafter FG-AI4H). For instance, the FG-AI4H was established in partnership with the WHO. It aims to establish international standards of AI in the health issue

[15] *See UN Secretary-General's Strategy on New Technologies* (Sep. 2018), available at https://www.un.org/en/newtechnologies/images/pdf/SGs-Strategy-on-New-Technologies.pdf.

[16] Ibid., p. 3.

[17] Ibid.

[18] *See ITU* website, available at https://www.itu.int/en/about/Pages/default.aspx.

[19] *See AI for Good Global Summit 2018, ITU,* available at https://www.itu.int/en/ITU-T/AI/2018/Pages/default.aspx.

[20] *See AI for Good, ITU,* available at https://aiforgood.itu.int/.

area, such as AI-based diagnosis and treatment decisions.[21] Because of the sensitivity of health data, the FG-AI4H aims to consider policy and regulatory aspects of AI-based solutions.

2 The Growing Gap Between Norm and Reality in the Digital Age

With all its stunning graphics and a myriad of functions, Apple's iPhone has indeed redefined the term 'cell phone.'[22] The first generation cell phone in the early 1990s, a brick-shaped, heavy, military equipment-type cell phone, has evolved into a multifunctional personal communication and entertainment kit in less than two decades.[23] Although people still tend to categorize the iPhone as a cell phone because the oral communication function between people still remains the core element of the machine, it would not be an exaggeration to say that we are basically talking about almost two different products.[24] What the term 'cell phone' represents has certainly changed over the years.

This trend is not just confined to the cell phone industry. We observe fast technological development in all ICT sectors.[25] Given the current pace of technological development in the ICT sectors, this is probably the area where one could expect the most dramatic changes over a very short period of time. Today's top product will become an old-fashioned one in just a year with its price slashed by half. New products with new technological applications flood the market, which are soon replaced with newer products with newer technological applications. Each step of technological development usually adds new functions and capabilities to the existing hardware. This addition then leads to the following two consequences: (i) the added function and capabilities expand the technological scope and coverage of an existing hardware; or (ii) the added function and capability produce a hybrid hardware which combines the core elements of the two or more existing hardware. Consequently, it

[21] *See* Salathé et al. [8].

[22] The iPhone is a new electronic device that integrates multiple digital functions in a single handheld terminal. These functions include a conventional cellular telephone, a media player, a digital camera and an internet platform. *See t*he website of Apple Inc. available at http://www.apple.com/iphone/iphone-3gs/. The Time Magazine selected the iPhone as the invention of the year 2007. *See* The Time's website available at http://www.time.com/time/specials/2007/article/0,28804,1677329_1678542,00.html.

[23] As for the detailed description of the various functions of the iPhone of Apple inc., *see* the company's website available at http://www.apple.com/iphone/iphone-3gs/.

[24] See ibid.

[25] Information technology, as defined by the Information Technology Association of America (hereinafter 'ITAA'), is "the study, design, development, implementation, support or management of computer-based information systems, particularly software applications and computer hardware." *See* the definition of the word appearing in an on-line business encyclopedia, available at http://www.answers.com/topic/information-technology.

becomes ever more difficult to name or describe these new ICT products. Descriptions of these products get longer and more complicated. More importantly, we do not know how the descriptions will change or evolve next year.

This reality in the ICT sectors raises a critical question, a question as to how we should interpret the terms relating to these products in international law. The general rule of interpretation in internationally law is apparently premised on a static situation,[26] where things are considered to remain the same for a significant amount of time, or at least not to change in a short period of time. Then, the remaining question is whether this general rule can apply to a fluid situation as we see in the ICT sectors as well. If the existing rules of interpretation are unable to deal with the new situation in the ICT sectors, maybe we should seek a new tool of interpretation. On the other hand, if the existing rules of interpretation do cover the new situation, we will have to offer an analysis to show why this is the case. It is not clear whether the current rules of interpretation could still deal with the new situation in the ICT sectors in an effective manner, with proper clarification and orientation. We need to determine whether the current rules of interpretation already contain elements which could reflect the fast technological development that we observe in the ICT sectors.

The current rule of treaty interpretation may well cover a certain degree of technological development through its tool of evolutionary treaty interpretation. This scheme, however, may not work at some point when the gap becomes bigger because of the ever-faster development of technology. Contracting parties may face a tension between the norm stipulated in treaties and the reality unfolding in the market. In that situation, one way to overcome the structural challenge would be to manage the

[26] Article 31 of the *Vienna Convention* stipulates general rule of treaty interpretation as follows:

Article 31 of the *Vienna Convention* stipulates general rule of treaty interpretation as follows:

Article 31.

General rule of interpretation.

1. A treaty shall be interpreted in good faith in accordance with the ordinary meaning to be given to the terms of the treaty in their context and in the light of its object and purpose.

2. The context for the purpose of the interpretation of a treaty shall comprise, in addition to the text, including its preamble and annexes:

(a) any agreement relating to the treaty which was made between all the parties in connection with the conclusion of the treaty;

(b) any instrument which was made by one or more parties in connection with the conclusion of the treaty and accepted by the other parties as an instrument related to the treaty.

3. There shall be taken into account, together with the context:

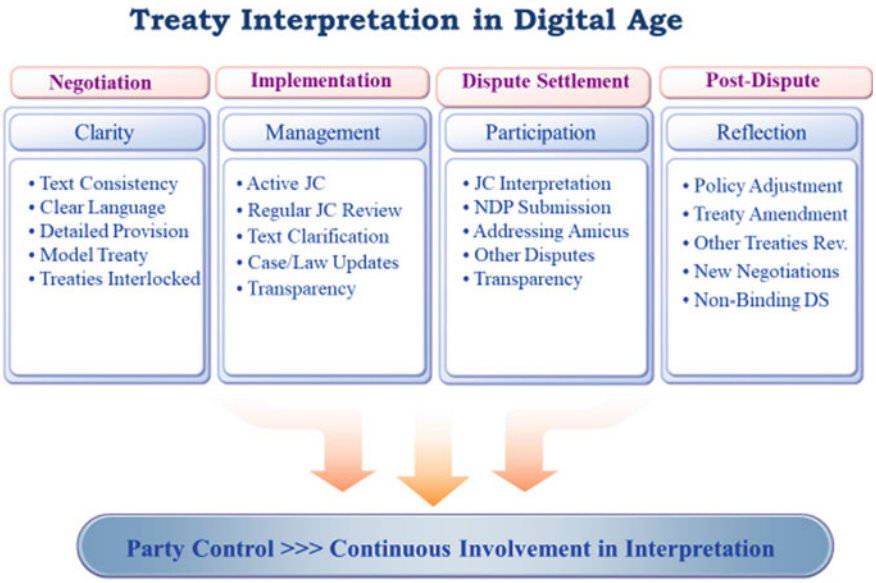
(a) any subsequent agreement between the parties regarding the interpretation of the treaty or the application of its provisions;

(b) any subsequent practice in the application of the treaty which establishes the agreement of the parties regarding its interpretation;

(c) any relevant rules of international law applicable in the relations between the parties.

4. A special meaning shall be given to a term if it is established that the parties so intended.

treaty interpretation on a rolling basis. To do so is to consider treaty interpretation in the digital as not static but fluid. Even within the confinement of the general rule of treaty interpretation, states may find routes to deal with treaty interpretation in the fast-changing digital age. Treaty interpretation in this sense does not have to stay static in a particular moment. Rather, it can be an on-going process, starting from negotiation all the way to the management after a dispute settlement proceeding (post-dispute phase). The diagram below illustrates this scheme of on-going interpretation scheme. JC in the diagram stands for Joint Committee of Joint Commission, an entity embedded in treaties and agreements which offers a forum for clarification of ambiguous textual meaning and provides sometimes binding interpretation as the case may be. NDP in turn stands for non-disputing party. A submission from an NDP is therefore a written instrument submitted by a third party in a dispute settlement proceeding between two other parties in the same treaty or agreement. The contents of an NDP submission may also shed helpful light on the meaning of the text. Accordingly, not only the dispute settlement proceeding *per se*, but also other legs of a continuing life cycle of a treaty or agreement plays an important role in treaty interpretation. This mechanism stands to operate effectively with respect to the digital society, digital economy and AI, where treaty texts are either vague or absent now and will remain so for the time being.



Governments and companies around the world are now facing the reality of having to navigate through a growing patchwork of national regulations. The patchwork situation could only be cured by the adoption of new global norms in this area. This effort, however, is still in the initial stage of development. The patchwork of domestic laws and absence of global norms loom larger each passing year because

of the rapid technological development in the ICT sectors. The current international legal regime, based on treaties and developed in the days of the analogue era, is not able to deal with the new digital landscape. It is almost like the norm is from Mars but the reality is from Venus. In the absence of the conclusion of new multilateral treaties, which is exceedingly difficult in the current climate of the global community with its dwindling trust in multilateralism, the awkward mismatch is bound to increase over the years.

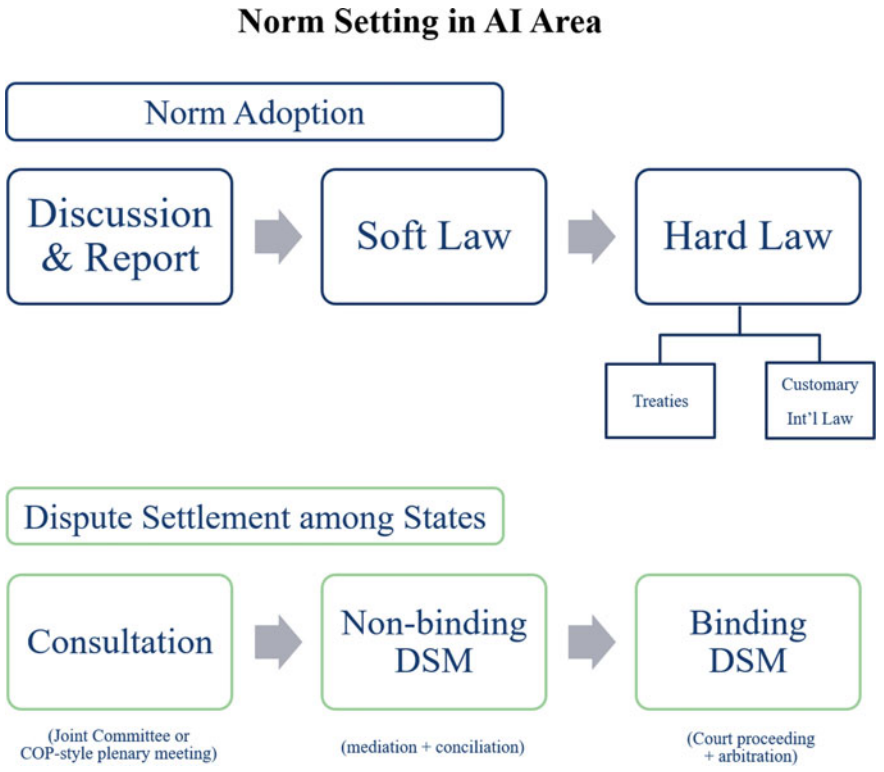
Should treaties be found to be tough challenges, something weaker could be pursued first. The global community could start with recommendations and guidelines for various issues arising from AI utilization by the global community, covering both private actors and government agencies. They could provide a set of soft laws to herd the private and governmental entities in a certain direction, which may also provide a groundwork for the discussion of hard laws in the form of treaties. Domestic legislation will then be informed and supplemented by the content and trajectory of these soft-law norms.

Indeed, even soft law is not an easy target to achieve. Introducing one also takes time and requires prior groundwork. A first step for the consideration of soft law may come as a candid discussion on a particular topic regarding artificial intelligence among countries. The discussion may then lead to a shared understanding of the topic, which will then serve as a basis for the consideration of a soft law. One recent example in this regard is a report issued by the Board of the International Organization of Securities Commissions (hereinafter IOSCO) on September 7, 2021. The IOSCO Board is the governing and standard-setting body of IOSCO with membership comprising of securities regulatory authorities around the world such as the United States Securities and Exchanges Commission. This is an effort to seek the development of more tailored supervisory approaches by securities regulators around the globe. The report of September 2021, entitled *The Use of Artificial Intelligence and Machine Learning by Market Intermediaries and Asset Managers*, aims to assist IOSCO members in supervising their regulated entities over the use of AI.[27] One might predict broad acceptance by regulatory agencies of the global community of the contents included in the report. The coordinated practice of the agencies of various countries will then help jump-start discussion of a soft law in the area. The soft law in turn will evolve into a treaty over time. It will indeed take time, but the process needs to start soon in a variety of sectors and areas.

New norms in the AI area will take root through an on-going process comprising multiple stages. Substantive norms will preferably go through an evolution process–proposal, testing, elaboration, amendment, and finalization. What started as a soft law will become a hard law. Disputes arising from the norms will also go through multi-layered stages comprising consultation, non-binding process and binding process. The outcome of these dispute settlement processes will then feed into the norm elaboration and amendment process above. Substantive norm development and dispute settlement proceedings will engage each other in a two-way communication scheme. The following diagram illustrates this. The term DSM in the diagram stands for

[27] Full text available at https://www.iosco.org/library/pubdocs/pdf/IOSCOPD658.pdf.

dispute settlement mechanism. COP in turn stands for conference of the parties, the supreme decision making entity for an international convention which meets regularly.



3 Recent Treaties Including AI Provisions

Recent treaties include provisions on Artificial Intelligence. FTAs now typically include e-commerce chapters and AI-related provisions started to appear in there. DEAs are becoming a new strand of treaties; they stipulate provisions on the digital economy and AI in a more detailed fashion than FTAs. DEAs purport to regulate the digital economy and AI separated and released from FTAs, so they are capable of covering a broader range of digital-specific, and AI-specific topics than FTAs.

Presumably mindful of the increasing importance, the USTR website posts a list of FTA chapters dealing with digital trade and e-commerce.[28] For instance, the

[28] Available at https://ustr.gov/issue-areas/services-investment/telecom-e-commerce/e-commerce-fta-chapters.

USMCA promotes open access to data, while ensuring protection for source codes.[29] Chapter 19 addresses digital trade, and in particular, Article 19.11 stipulates the following:

> Article 19.11: Cross-Border Transfer of Information by Electronic Means
>
> 1. No Party shall prohibit or restrict the cross-border transfer of information, including personal information, by electronic means if this activity is for the conduct of the business of a covered person.
>
> 2. This Article does not prevent a Party from adopting or maintaining a measure inconsistent with paragraph 1 that is necessary to achieve a legitimate public policy objective, provided that the measure:
>
> (a) is not applied in a manner which would constitute a means of arbitrary or unjustifiable discrimination or a disguised restriction on trade; and
>
> (b) does not impose restrictions on transfers of information greater than are necessary to achieve the objective.[5]
>
> Footnote 5: A measure does not meet the conditions of this paragraph if it accords different treatment to data transfers solely on the basis that they are cross-border in a manner that modifies the conditions of competition to the detriment of service suppliers of another Party.

On October 7, 2019, the US signed a digital trade agreement with Japan (US-Japan Digital Trade Agreement).[30] The US government saw the agreement as 'the gold standard' for digital trade agreements.[31] It calls for free flow of data, prohibits data localization requirements, and promotes consumer privacy, among many other provisions. However, it does not explicitly deal with AI. The agreement drew attention as a stand-alone agreement on digital trade (instead of e-commerce) between the two major countries in this field.

The Regional and Comprehensive Economic Partnership (hereinafter RCEP) concluded in November 15, 2020, also has an e-commerce chapter including provisions on free flow of data and prohibition of server localization requirements. But the mega-FTA spearheaded by China contains a broad exception clause for legitimate public policy objectives (hereinafter LPPO), which also includes a self-judging feature.[32]

Singapore has already concluded two DEAs: the Singapore-New Zealand-Chile DEPA and Singapore-Australia DEA. They are related to existing FTAs but were sought as independent agreements in effect. First, DEPA has an article on artificial intelligence as follows [33]:

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[29] *Agreement between the United States of America, the United Mexican States, and Canada 7/1/20 (USMCA)*, full text available at https://ustr.gov/trade-agreements/free-trade-agreements/united-states-mexico-canada-agreement/agreement-between. *See* Joshua New, *Here's What the USMCA Does for Data Innovation. Center for Data Innovation* (Oct 5, 2018). Available at https://datainnovation.org/2018/10/heres-what-the-usmca-does-for-data-innovation/.

[30] Full text available at https://ustr.gov/sites/default/files/files/agreements/japan/Agreement_between_the_United_States_and_Japan_concerning_Digital_Trade.pdf.

[31] CRS [9].

[32] Cutler and JP Meltzer [10].

[33] Full text available at https://www.mfat.govt.nz/assets/Trade-agreements/DEPA/DEPA-Signing-Text-11-June-2020-GMT-v3.pdf.

> Article 8.2: Artificial Intelligence
>
> 1. The Parties recognise that the use and adoption of Artificial Intelligence (AI) technologies have grown increasingly widespread in the digital economy.
>
> 2. The Parties recognise the economic and social importance of developing ethical and governance frameworks for the trusted, safe and responsible use of AI technologies. In view of the cross-border nature of the digital economy, the Parties further acknowledge the benefits of developing mutual understanding and ultimately ensuring that such frameworks are internationally aligned, in order to facilitate, as far as possible, the adoption and use of AI technologies across the Parties' respective jurisdictions.
>
> 3. To this end, the Parties shall endeavour to promote the adoption of ethical and governance frameworks that support the trusted, safe and responsible use of AI technologies (AI Governance Frameworks).
>
> 4. In adopting AI Governance Frameworks, the Parties shall endeavour to take into consideration internationally recognised principles or guidelines, including explainability, transparency, fairness and human-centred values.

DEPA was actually the first digital only trade agreement, aiming to facilitate cooperation in emerging areas, including AI.[34] DEPA calls for an ethical governance of AI so that AI can be used in a responsible manner.[35] In addition to the original member states, Canada has expressed its interest in joining DEPA and has been in exploratory discussion with the other parties since February 2021.[36] When the agreement was going through ratification process in Chile, International Economic Relations Undersecretary (SUBREI), Rodrigo Yáñez emphasized the importance of digital economy and that DEPA opens a new opportunity to provide better regulations.[37]

The ministers from the three member states of DEPA stated the following:

> We share the view that the DEPA is a comprehensive and forward-looking pathfinder agreement that addresses key issues in the digital economy, such as, use of electronic documentation in cross-border trade and business, personal information protection, cybersecurity, online consumer protection, digital identities, fintech, artificial intelligence, data flows and innovation, trade and investment opportunities for SMEs, and digital inclusivity. Through the DEPA, we can build on existing trade agreement commitments by leveraging on technology, facilitate business-to-business connectivity in the digital age, and collaborate on new issues intrinsic to the digital realm.[38]

In addition, Singapore and Australia concluded SADEA in August 2021, which is considered to be most comprehensive as of March 2022 in terms of coverage. Commensurate with the title, the treaty was concluded virtually, where the ministers of the two countries met online and signed the instrument online. For the first time, the

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[34] Falak [11].

[35] Ibid.

[36] Government of Canada [12].

[37] *Digital Economy Partnership Agreement (DEPA) is Approved by Congress,* Chile Reports (May 20, 2021), available at https://chilereports.cl/en/news/2021/05/20/digital-economy-partnership-agreement-(depa)-is-approved-by-congress.

[38] *Joint Ministerial Statement on the Substantial Conclusion of Digital Economy Partnership Agreement ("DEPA") Negotiations* (2020), available at https://www.imda.gov.sg/-/media/Imda/Files/About/Media-Releases/2020/Annex-B-Joint-Ministerial-Statement-on-the-Substantial-Conclusion-of-DEPA.pdf?la=en.

treaty describes in its final provision two cities as the place of signing. The SADEA, not surprisingly, also has a stand-alone clause on AI in Article 31.[39]

> Article 31 Artificial Intelligence
>
> 1. The Parties recognise that the use and adoption of Artificial Intelligence ("AI") technologies are becoming increasingly important within a digital economy offering significant social and economic benefits to natural persons and enterprises. The Parties shall cooperate, in accordance with their respective relevant policies, through:
>
> (a) sharing research and industry practices related to AI technologies and their governance;
>
> (b) promoting and sustaining the responsible use and adoption of AI technologies by businesses and across the community; and
>
> (c) encouraging commercialisation opportunities and collaboration between researchers, academics and industry.
>
> 2. The Parties also recognise the importance of developing ethical governance frameworks for the trusted, safe and responsible use of AI technologies that will help realise the benefits of AI. In view of the cross-border nature of the digital economy, the Parties further acknowledge the benefits of ensuring that such frameworks are internationally aligned as far as possible.
>
> 3. To this end, the Parties shall endeavour to:
>
> (a) collaborate on and promote the development and adoption of frameworks that support the trusted, safe, and responsible use of AI technologies ("AI Governance Frameworks"), through relevant regional and international fora; and
>
> (b) take into consideration internationally-recognised principles or guidelines when developing such AI Governance Frameworks.

In a joint statement, the government of Singapore and Australia expressed that SADEA would "support growth of the digital economy and enable firms to operate seamlessly using common and interoperable standards". As a matter of fact, SADEA is the most comprehensive and detailed DEA at the moment covering a wide range of digital sectors. Notably, SADEA has a dozen or so provisions stipulating bilateral cooperation and best endeavor in the field of AI.[40]

AI has also become an important topic in the military and national security sectors. The recent military conflict in Ukraine initiated by Russian invasion has brought to the fore the capability and potential of AI technology in modern warfare. The Russia-Ukraine military conflict is expected to further expedite discussions on AI, LAWS and cyber warfare for the future. Even before the recent development, some security alliances, such as the North Atlantic Treaty Organization (hereinafter NATO), is under discussion to create norms and policies to regulate the use of AI in military conflicts. It is interesting to watch how the NATO will react in response to the recent conflict in Ukraine and ensuing consequences.[41]

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[39] *Singapore Australia Digital Economy Agreement,* full Text available at https://www.dfat.gov.au/trade/services-and-digital-trade/australia-and-singapore-digital-economy-agreement.

[40] Joint Communique of the 12th Meeting of the Singapore-Australia Joint Ministerial Committee (Aug. 27, 2021), para. 8, available at https://www.foreignminister.gov.au/minister/marise-payne/media-release/joint-communique-12th-meeting-singapore-australia-joint-ministerial-committee.

[41] Allen and West [13].

4 Possible Consideration of Treaties and Institutions

Treaties and international institutions could be a useful vehicle in founding legal norms on AI, hence their formality and institutional arrangement are a top-notch issue in the field of international law. Already, many international organizations are discussing the impact of AI within their fields of activity. On one hand, they are actively applying the technology to achieve their goals, and on the other, they are contributing to the establishment of norms and standards regarding AI.

Although the ITU is the specialized agency for the ICT sector of the UN, given the wide applicability of AI, it is conceivable that other agencies will provide another forum of discussion. For instance, as AI is expected to have a big impact on the economic environment, a more general and political bodies such as the OECD or G-20 could be used to seek a consensus among the countries leading the development of the new technology. This chapter reviews how existing international institutions are attempting to take on new challenges and issues related to AI.

4.1 United Nations Educational, Scientific, and Cultural Organization (UNESCO)

The United Nations Educational, Scientific, and Cultural Organization (hereinafter UNESCO) organizes a series of debates on ethical issues of AI. The first was held in Marrakech, Morocco, on 12 December 2018, focused on AI and Africa, and the second in Paris, France, 2019. As a UN specialized agency with jurisdiction on "science", UNESCO is also well-versed with the various issues arising from AI and related technologies. It, however, hardly deals with technological aspects of AI. Instead, it places emphasis on educational issues and ethical matters. Audrey Azoulay, Director-General of the UNESCO, acknowledged the importance of AI in pursuit of the organization's mandate as well as the role of UNESCO to promote ethical development of technology as follows:

> The transformations arising from the technological revolution, and in particular from developments in AI, are relevant to every aspect of the mandate of the United Nations Educational, Scientific and Cultural Organization (UNESCO). Education is already being profoundly transformed by AI. Very soon, the tools of education—the way we learn, access knowledge and train teachers—will no longer be the same. From now on, the acquisition of digital skills stands at the centre of all our education programmes. Furthermore, we must "learn to learn" because the pace of innovation is rapidly transforming the labour market. Today, more than ever before, the humanities—history, philosophy, literature—are crucial to our ability to act in our rapidly changing world. In the field of culture, AI is already being employed extensively, for example, in the imagery used to reconstruct heritage. It is used in the sciences too, notably in our environmental programmes and underwater research. Communication and information are also directly dependent on progress in AI, particularly with regard to freedom of expression and access to information.
>
> Our organization [UNESCO] has many years of experience in the ethics of science and technology. Our advisory bodies have already produced numerous reports and declarations,

> including on robotics, such as the Report of the World Commission on the Ethics of Scientific Knowledge and Technology on Robotics Ethics in 2017. The advisory bodies also have experience in developing normative instruments, including the Universal Declaration on the Human Genome and Human Rights in 1997 and the Universal Declaration on Bioethics and Human Rights in 2005.
>
> UNESCO priorities must also guide our international action in this area. It is essential to ensure that Africa fully participates in transformations related to AI, not only as a beneficiary but also upstream, contributing directly to its development. In terms of gender equality, we must fight against the biases in our societies to guarantee that they are not reproduced in AI applications. Finally, we must empower young people by providing them with the skills they need for life in the twenty-first century for integration in a changing labour market […].[42]

4.2 United Nations Development Programme (UNDP)

In 2019, the United Nations Development Programme (hereinafter UNDP) Administrator, Achim Steiner, hosted a higher-level event on the side of the 74th UN General Assembly, entitled "The Digital Future of Development". During the event, leaders and experts from government, the private sector, and civil society joined to discuss how digital technologies can contribute in achieving SDGs.[43] Furthermore, the UNDP published its own Digital Strategy in 2019, describing how the UNDP will take on the digital transformation so that it is in line with the values captured in the UN Charter and the UDHR.[44] The two Digital Transformation Pathways cover how the UNDP can utilize digital technologies when it is working with outside partners, and how the UNDP can improve its internal work processes.[45]

In addition to developing its own Digital Strategy, the UNDP also participated in the consultation works for the Principles for Digital Development.[46] The Principles were developed to provide a unified set of principles to the international community working with digital development. While the specifics of the Digital Principles are subject to change, they include (1) design with the user; (2) understand the existing ecosystem; (3) design for scale; (4) build for sustainability; (5) be data driven; (6) use open standards, open data, open source, open innovation; (7) reuse and improve; (8) address privacy and security; (9) be collaborative. They are designed to assist

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[42] Azoulay [14].

[43] UNDP [15].

[44] *Future Forward—UNDP Digital Strategy* (2019), available at https://digitalstrategy.undp.org/assets/UNDP-digital-strategy-2019.pdf.

[45] *Visit* the UNDP's website, *Pathways to the SDGs,* available at https://digitalstrategy.undp.org/strategy.html.

[46] The Bill and Melinda Gates Foundation, the Swedish International Development Agency (SIDA), the UNICEF, the World Bank, the US Agency for International Development (USAID), and the World Health Organization (WHO) are the other organizations that participated in the consultation. *See* the website for the Principles for Digital Development, available at https://digitalprinciples.org/about/.

those working with the ICT tools in the development field so that the full potential of ICTs can be achieved, contributing to SDGs.[47]

4.3 Food and Agriculture Organization (FAO)

Recognizing the power of digital innovation in enhancing the efficiency of the agricultural sector including forestry, fishery and livestock production, the FAO strives to promote digital innovation to increase food security and to contribute to sustainable development.[48] In particular, the FAO emphasizes the need for basic conditions such as connectivity, affordability of digital technology for the digital transformation.[49] Moreover, a briefing paper by the FAO states:

> To unlock the full potential of digital agriculture transformation, governments need to create an enabling regulatory environment. Designing and managing digital government programmes requires a high level of administrative capacity which is beyond the capabilities of some countries, particularly LDCs [least developed countries] and developing countries. Addressing the digital divide must be made a policy priority and governments should make the socioeconomic case for digitalization of smallholder farming both to the farmers, and to potential private sector investors and start-up businesses. There will need to be significant capacity building among governments in developing countries and LDCs to facilitate this change in policy and regulation.[50]

In the context of UN's strategy for AI, too, the FAO emphasizes the importance of having adequate infrastructure. In a report assembled by the ITU on the activities of the UN and its special agencies, the FAO representative states:

> In terms of the four points mentioned: (a) infrastructure; (b) data; (c) human; and (d) policy/law/ human rights, we concur with all of them. However, we consider that the infrastructure area is a topic by itself, because it is important to bring other capabilities, technologies and solutions to eliminate the digital divide and promote innovation, jobs and fairness. We recognize that this is a good-to-have for AI, but it is not a necessary condition, because AI solutions could also be used offline. Therefore, we propose to have a different, separate note dedicated to the need to increase connectivity and reduce the digital dive.[51]

Against this backdrop, the FAO identifies a number of key factors that need more future work.[52] First, more systematic data on digital agricultural transformation, including information about government support and regulations, is needed. Second, the disparity between the developed and developing countries, as well as between small rural farmers and big firms, needs to be addressed. Third, given the importance of economies of scale, it is more difficult for small farmers to take advantage

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[47] Ibid.

[48] *Visit* the FAO website, available at http://www.fao.org/innovation/en/.

[49] *See supra* note 703, pp. 3–6.

[50] Ibid. p. 14.

[51] *See supra* note 348, p. 5.

[52] *See supra* note 539, p. 16.

of the new technologies. Based on these issues, the FAO seeks ways to assist in digitalization, while minimizing risks of a digital divide.[53]

The FAO, itself, is actively applying AI in its projects, such as mobile applications to detect any worm damage, satellite imagery to map palm tree and software (*iSharkFin*) to identify shark species.[54] These technologies are aimed at increasing production efficiency of farmers or fishers. In addition, there are more future projects that are under the process of development or planning, which include projects on more fish species identification, aquaculture mapping, or animal diseases identification from images.[55]

4.4 International Labour Organization (ILO)

The International Labour Organization (hereinafter ILO) is at the forefront at researching how AI will impact jobs and employment.[56] The introduction of AI and deep learning signifies that more complex tasks can be replaced by robots, insofar as robots can generate new economic values. As a consequence, it is expected to bring about many changes on the job market and employment situations. As ILO strives to ensure human-centered development of working life under the Decent Work Agenda, the impacts of the new technology are also bound to affect the goals of the organization. A study on the ILO's work on Decent Work notes:

> Due to robotization and artificial intelligence, the human role in production has become more one of planning, designing, quality control and services, transport and commercial, and marketing activities. The production of goods and materials will become assigned to automatic systems and robots. Some new branches of the economy will be generated, such as environmental and green economies, new energy economies, and knowledge economies. All these will have an impact on job contents, competences, and work ability profiles.[57]

Thus, the ILO's mission will also be affected by the changes brought by AI. Against this backdrop, the ILO analyzes how industrial robots in developing countries affect the manufacturing employment and patent data in robotics and AI.[58] Furthermore, The ILO also aims to assist policy responses to the distributional consequences of AI and prepare workers for future labor markets.[59] In partnership with the UNESCO, ILO aims to assist its member states to develop skills training strategies in the digitalization era. As the jobs required in the future are different, the ILO helps

[53] Ibid., p. 14.
[54] *See supra* note 348, pp. 3–4.
[55] Ibid., p. 4.
[56] Ibid., pp. 9–10.
[57] Rantanen et al. [16].
[58] *Supra* note 348, p. 9.
[59] Ibid., pp. 9–10.

members achieve a better skills anticipation strategy to minimize the gap between the skills needed and those offered in the job market.[60]

4.5 World Bank Group

Like other international institutions, the World Bank group is actively trying to take advantage of the newest technology in global development. The World Bank Group itself deploys AI to create better data on poverty and development. For instance, Development Analytics &Tools (DECAT) enables better data analysis to contribute to development.[61] It applies machine learning to better map poverty and minimize concerns of privacy through privacy-preserving methods. Furthermore, the World Bank strives to develop an AI-based policy-making guidelines for developing countries, by providing information on the pros and cons of utilizing AI.[62]

In addition to providing guidelines to the developing countries, the World Bank itself is actively applying AI to achieve sustainable development. For example, in 2018, the World Bank, with the UN and the Red Cross, as well as Amazon, Microsoft and Google, developed The Famine Early Action Mechanism, a new tool to predict famines around the globe.[63] Famines can be caused not only by natural disasters such as droughts, but by human conflicts, such as the civil war caused famine in South Sudan in 2017.[64] AI-based predictions can help the international community to move proactively to deal with global famines.

4.6 Organisation for Economic Co-Operation and Development (OECD)

As an international organization composed of 38 developed states, the OECD is also deeply entrenched in the discussion of the digital economy and AI. It lists "digital" as one of the core agenda of the organization, where various topics such as blockchains, online privacy, digital government and AI are robustly studied and discussed. The OECD adopted its *Principles on Artificial Intelligence* in May 2019.[65]

> The Recommendation identifies five complementary values-based principles for the responsible stewardship of trustworthy AI:

[60] For more information, visit ILO website, available at https://www.ilo.org/skills/areas/skills-training-for-poverty-reduction/lang--en/index.htm.

[61] *Supra* note 348, pp. 76–7.

[62] Ibid.

[63] Higgins [17].

[64] Ibid.

[65] Full text available at https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0449.

i. AI should benefit people and the planet by driving inclusive growth, sustainable development and well-being.
ii. AI systems should be designed in a way that respects the rule of law, human rights, democratic values and diversity, and they should include appropriate safeguards—for example, enabling human intervention where necessary—to ensure a fair and just society.
iii. There should be transparency and responsible disclosure around AI systems to ensure that people understand AI-based outcomes and can challenge them.
iv. AI systems must function in a robust, secure and safe way throughout their life cycles and potential risks should be continually assessed and managed.
v. Organisations and individuals developing, deploying or operating AI systems should be held accountable for their proper functioning in line with the above principles.[66]

The Principles were developed by a group of experts on AI coming from various sectors including business, labor, civil society and government representatives. The expert groups also announced public policy recommendations for responsible development and application of AI. Although they are not legally binding, the principles provide meaningful foundations for developments of future legal obligations for responsible use of AI.[67]

In February 2020, the OECD launched an online AI policy observatory, "OECD.AI Observatory", to assist countries in developing responsible governance of AI.[68] The Observatory gathers resources from a wide range of stakeholders across the OECD member countries.

4.7 World Trade Organization (WTO)

In its current form, the WTO Agreement does not explicitly address AI or the flow of data.[69] However, the fourth industrial revolution will not leave international trade

[66] *OECD Principles on AI,* OECD, available at https://www.oecd.org/going-digital/ai/principles/.

[67] Yeung [18].

Thus, despite its lack of legally binding force, the Recommendation represents an important political and moral commitment at the intergovernmental level in at least three respects. First, it reflects welcome recognition that, although AI has the potential to generate great benefits, these technologies have the potential for great harm—to individuals and to society more generally—including the potential to threaten foundational human and democratic values, thus making appropriate policies and governance mechanisms necessary. Second, the commitment to global cooperation is significant, both in potentially blunting the ferocity of an unbridled "AI arms race," and in paving the way for global cooperation to counter the cross-border risks of AI, which are inevitable given the capacity of AI systems to operate on a planetary scale. Third, the 20 largest industrial nations have expressed their support for the principles for responsible stewardship of trustworthy AI via the G20 Osaka Leaders' Declaration.

[68] *See OECD AI Policy Observatory, OECD,* available at https://oecd.ai/.

[69] *See* Aaronson [19]; Zheng [20].

unaffected,[70] and as mentioned earlier, countries are already entering fierce competition to take (or maintain) a leadership role in the new technology. Thus, it is not difficult to imagine that countries wish to pursue policies to nurture their digital economy as well as AI technology.[71]

The WTO recognizes the growing importance of the digital economy. The World Trade Report 2018 had a subtitle of "The Future of World Trade: How Digital Technologies are Transforming Global Commerce".[72] Recognizing potential benefits and risks related to AI and its impact on global commerce,[73] the report concludes with a concerning remark asking WTO members to consider adequate responses to the emerging challenges.[74]

The WTO Agreement as it stands is far from enough to address the new challenges arising from the growing digital economy and e-commerce.[75] For example, it is questionable whether the general exception clause in Article XX and (or) the national security exceptions of the GATT can justify a government's regulatory policies on data flow and information.[76] In particular, governance of AI can cause a lot of controversies. Ciuriak (2020) comments:

> The governance of AI is an area of extraordinary activity and ferment at the moment. Principles to govern the development of AI regulation have been articulated. As AI applications proliferate and are deployed in international commerce, the regulatory framework for these applications will become a major challenge for the rules-based system in terms of standards, interoperability, disguised barriers to trade, mutual recognition and so forth. In many ways, AI regulation promises to be as contentious in the digital realm as sanitary and phytosanitary standards have proven to be in agricultural trade.[77]

4.8 World Intellectual Property Organization (WIPO)

The WIPO provides a platform to discuss how AI could affect the current intellectual property rights (hereinafter IPRs) system. *The WIPO Technology Trends Report 2019* lays out how AI can affect our society and change the intellectual property system.[78]

> As seen in the previous chapters of this report, AI is expected to revolutionize processes across a wide range of fields. It is foreseen that AI will also affect intellectual property rights, in particular patent rights, and their management. This is likely to be a two-way process: on the one hand, AI developments will affect and be incorporated into IP rights

[70] *See* Meltzer [21].
[71] *See* Aaronson [22].
[72] WTO [23].
[73] Ibid., pp. 30–32; 140.
[74] Ibid., p. 204.
[75] *See* Ciuriak [24]; Zheng [25].
[76] Ibid.
[77] Ibid.
[78] WIPO [26].

management; on the other hand, IP policies and practices will interact with the strategy of managing innovation in AI.[79]

The first conversation on IP and AI was held in September, 2019, and the WIPO published a draft of its Issues Paper. Based on more than 250 submissions, a revised Issues Paper points out AI in IPR administration, IPR and AI Strategy Clearing House, and IPR policy as the three main topics that the WIPO will focus on to discuss the impacts of AI going forward.[80] A second session was held virtually in July 2020, after which the WIPO plans to publish written interventions submitted by participating parties.[81] The provisional agenda included issues such as IPR protection for AI-generated and AI-assisted works, patentability, disclosure and guidelines of AI inventions, copyright in training data, and further rights in data and trade secrets.[82] The WIPO plans to continue to offer a discussion platform on related matters for member states and other stakeholders.[83]

5 Topics to Be Covered by Prospective Treaties

The proposals for the treaty on AI could consider means of controlling information, legal status of AI related programs and activities, liability and responsibility framework, compliance method, dispute settlement mechanism, and more. Some of the issues can be addressed by updating the existing legal frameworks, but other issues call for creation of a new treaty. Having a comprehensive treaty dealing with the variety of issues is a possibility, but given the complexities of the issues at hand, having separate treaties per topic is another. This chapter will focus on two specific issues, the creation of extraterritoriality zone of personal information and digital trade.

5.1 Extraterritoriality Zones of Personal Information

One of the most challenging problems is how to protect personal information transferred to foreign countries. Information is transferred beyond the providing country's

[79] Ibid., p. 143.

[80] *See Revised Issues Paper on Intellectual Property Policy and Artificial Intelligence* WIPO/IP/AI/2GE/20/1 Rev. (May 21, 2020), para 2, available at https://www.wipo.int/edocs/mdocs/mdocs/en/wipo_ip_ai_2_ge_20/wipo_ip_ai_2_ge_20_1_rev.pdf.

[81] *See Artificial Intelligence and Intellectual Property Policy,* WIPO, available at https://www.wipo.int/about-ip/en/artificial_intelligence/policy.html.

[82] *See Provisional Agenda.* WIPO/IP.AI/2/GE/20/INF/1/PROV.2 (Jun. 19, 2020), available at https://www.wipo.int/edocs/mdocs/mdocs/en/wipo_ip_ai_2_ge_20/wipo_ip_ai_2_ge_20_inf_1_prov_2.pdf.

[83] *See Artificial Intelligence and Intellectual Property Policy,* WIPO, available at https://www.wipo.int/about-ip/en/artificial_intelligence/policy.html.

boundary of territorial jurisdiction and into the jurisdiction of a third-party country. While the providing country's jurisdiction is lost, the third party's jurisdiction comes into effect. This means that the third-party country has the basis for access to and use of such personal information anytime it wants. This situation results in worries among the providing country's people as well as its government.

To proactively address this issue, the international community needs to introduce a new concept of personal information extraterritoriality in the near future, if not now. Above all, now that companies and people in each country are worried that their information might be utilized or accessed overseas in reality, there is an increasing possibility that they may request their own government to take necessary measures to prevent a foreign governmental agency from obtaining an exclusive right of access to their personal information. This is a matter of time in a sense. There is a significant probability that in the near future various mechanisms will be adopted by major countries to prevent personal information of their citizens from coming under the control and jurisdiction of foreign governmental agencies. Thus, each country needs to take necessary measures to take part in the international discussion on such risks.

Particularly with regard to this issue, the international community has yet to seek various ways of coping in general recognition of all related problems. International agreements to address the issue of extraterritoriality may be concluded in either a bilateral or multilateral way. Although bilateral agreements may be relatively easy to be concluded, they involve fundamental problems such as limited effects because such agreements are valid only to the party countries. In this regard, concluding a multilateral treaty may seem to be more attractive. However, a multilateral treaty is likely to take a long period of time until its conclusion. Besides, it is highly likely that such treaties include only vague, compromise provisions due to the significant difference in position between parties. In such cases, it would be difficult to achieve the effects that should be intended.

It is up to every government to make efforts to examine this issue from the early stages of international discussions in order to play a leading role in the future ones. If the concept of digital or personal information extraterritoriality is introduced and stipulated in relevant treaties, it will certainly be recognized as a significant achievement. For example, in the case of Korea, if the government takes no appropriate measure while public concerns on protection of personal information is increasing, an argument might be raised that the government is negligent in leaving personal information under the jurisdiction of foreign governmental agencies. In view of domestic policies, therefore, it is necessary to actively come up with forward-looking alternatives.

In order to introduce the concept of digital or personal information extraterritoriality as discussed earlier, the following aspects need to be taken into consideration when an international agreement or bilateral treaty is to be concluded: (1) clarification of the concept of personal information and how it is to be applied in the agreement; (2) conditions for the application of the agreement; (3) confirmation of cases to be excluded from both the prescriptive jurisdiction and enforcement jurisdiction of the country; (4) specific statements of the exclusion (*e.g.*, whether the exclusion is

temporary or permanent); (5) how to handle cases of conflict of agreement; (6) clarification of irrelevance regarding treatment of foreigners or foreign companies; and (7) prevention of misuse of national security exception provisions.

When such agreements or treaties are concluded, most importantly, it is necessary to clarify the definition of personal information. As the term, 'personal information,' is used in various contexts, it is necessary to make clear what personal information means in an agreement. In addition to the general definition of personal information, it is also required to clarify the specific personal information that is to be protected under the agreement or treaty. In other words, it needs to be clarified whether every type of personal information or a certain type of personal information is to be protected in order to minimize the possibility of disputes.

Likewise, it is necessary to list specific conditions for a prospective agreement/treaty application. For instance, the following conditions may be considered in the debates and negotiations:

- Has that information been acquired by a company of one country directly concerned?
- Have domestic laws of both agreement parties been all complied with as that company acquired relevant personal information?
- Have individuals who provided the information agreed with all relevant protection procedures that are being taken?
- Is the acquired information located under the territorial jurisdiction of one agreement party?
- Has the information been already made public or provided to a third party in another way so that it is no longer worth protecting?
- Is the entity that requires access to the information a governmental agency on one side of the agreement party ?
- Is the governmental agency requesting access to the information one of the agencies specified in the agreement?
- Is the request for access to the information a just exercise of authority as a governmental agency?
- Does the access to information involve a risk of serious infringement of interests or privacy of individuals who provided that information?
- Has the governmental agency, which is a party of the agreement, obtained prior consent from individuals or companies regarding provision of their information?

It is required to consider and stipulate the various conditions listed above in treaties/agreements, confirm whether such conditions are satisfied, and determine whether such treaties/agreements will be applied.

Furthermore, there is a need to clarify that the country's jurisdiction is not applicable to such treaties/agreements and that both the prescriptive jurisdiction and enforcement jurisdiction of the country are excluded. As a matter of fact, the concept of jurisdiction basically includes both: The former judges whether the country's domestic laws are applied while the latter judges whether it is permitted to exercise legal force on the basis of such domestic laws including seizure, search, trial, and so forth. For complete exemption from the jurisdiction of the country concerning

personal information, it is required to specify in the agreement/treaty that both parties agree on the exemption. In general, merely indicating exemption from 'jurisdiction' with no such statement will be understood in a way that only the enforcement jurisdiction of the country is excluded while the prescriptive jurisdiction is still valid. In consideration of such aspects, it is required to specify the intent in the letter of an agreement as clearly as possible. For the same purpose, it is required to state specific occasions where the country's jurisdiction is excluded in detail. For example, it is required to make clear the scope: whether the country's jurisdiction is excluded in every step or any certain step where information provided by the government or any company and person in that country provided is accessed, used, and processed (more specifically, whether it is permitted to retrieve and access certain part of the information but not to use it in the process of legal execution).

If a phrase indicating that the country's jurisdiction is excluded is included in a treaty/agreement, the temporal scope of exclusion also needs to be clarified. For example, there must be a clear criteria that determines whether such exclusion is applied only temporarily (*e.g.,* within five years after provision) or permanently.

Another point that needs to be clarified in the treaty is the relationship with other existing and upcoming treaties. If it is the first agreement concluded in this specific area, there is a high possibility that will come into conflict with other existing agreements. In case of such conflict, therefore, it is appropriate to indicate the priority and the fact that this agreement precedes it. Provisions of this ilk are often found in recently concluded treaties. Such provisions indicate that in the case of conflict between the new agreement in question and other agreements, the former precedes. However, if the former violates provisions of other agreements, such violation will not be excused or justified (*e.g.,* upon violation of a trade agreement or an investment agreement).

It is important to include tools to prevent misuse of this treaty. For instance, it is of the greatest importance to specify that concluding and executing such a treaty/agreement does not guarantee a party's treatment advantageous or disadvantageous to foreigners or foreign companies and that the agreement parties do not intend such results. Including such statements will contribute to preventing discriminative treatment against foreign products, service suppliers, and investors under any trade agreement or investment agreement. Of course, the issue of 'treatment' under a governmental action applicable to trade agreements and investment agreements is handled not based on whether the government intended that result but based on its actual effects. Thus, including such phrases cannot be a perfect defense mechanism. Including such phrases, however, is still meaningful in that uncertain applications of a trade agreement or an investment agreement are minimized.

National security exception is another aspect that can be easily misused. Arguably the most critical aspect in concluding a treaty or agreement related to personal information extraterritoriality is that even if such a new documentation is prepared, it is easy to be exempted from its application by referring to provisions of exception for matters of national security. Particularly, the access to and utilization of personal information is often closely related to national security. If a relevant country relies on

such provisions of national security, it is highly likely that the other party faces difficulties in contradicting such assertions. It seems appropriate, therefore, to exclude provisions of exception for matters of national security, if possible, when a new treaty/agreement is adopted. Currently, however, most countries include provisions of exception for matters of national security in important agreements, and thus it is required to include such provisions of exception in new agreements regardless of our intention. Even when such provisions of exception are introduced, it is necessary to stipulate the scope of application as specifically as possible. In the case of provisions of exception for matters of national security as well, the scope may be varied depending on the agreements. For instance, the range of application of national security provisions in the Korea-US FTA is far wider than that in Article XXI of the GATT 1994. In consideration of such aspects, therefore, provisions of exception for national security need to be as finite and specific as possible.

5.2 Extended Domestic Jurisdiction for ICT Companies

The internet has given rise to a revolution in various areas of modern society, and one of the fundamental results of the revolution is borderless economy. Due to the openness and expandability of the internet, the era of ubiquitous business has been unlocked. Nowadays, some companies can operate their businesses wherever they want, even without physical offices.

Being the biggest beneficiary of the revolution, the platform business has currently taken center stage of the global economy. This new business focuses on 'connection', which is distinct from a traditional business model that puts 'production' at its center. This change has made it feasible for several successful platform companies, such as Facebook, YouTube and Netflix, to dominate the global community. All they need for their business is an internet connection without requiring any permanent facilities in foreign countries. These ICT giants are now providing their services in many countries by simply making contracts with domestic internet service providers (hereinafter ISPs).

The infinite expandability of the internet, however, has entailed a new concern: new crimes using internet-based connections bring incomparable damage both in the seriousness of crime and the number of victims. Cyber attacks and personal information leakages are other source of concern in the new age. Online platform companies lie at the center of the new phenomenon. They allow users to distribute almost any content they generate instantly, and the speed and scale of this distribution is beyond limits. They also collect user information to store and use it for their business development.

Such being the case, it is not surprising to see many countries scramble to introduce and apply new regulations for online platforms. Layers of new regulations invite legal controversies, one of which is the overreach of domestic statutes beyond national borders. Extraterritorial application of domestic statutes has been one of the most important subjects of international law, but the borderless nature of internet activity

and states' corresponding effort to regulate it freed from borders are now presenting a new controversy among states. Artificial intelligence is at the forefront of these internet-based business activities beyond territorial limits.

Purported extraterritorial application of domestic law is based on the concept of effect doctrine. The doctrine refers to the application of domestic law to a foreign company if its conduct has a significant effect on the country attempting to apply its domestic law. The effect doctrine was established in the process of extraterritorial application of US antitrust law, starting from *United States v. Aluminum Co. of America("Alcoa").*[84] The main issue of the case was whether applying US regulations to foreign defendants who were accused of violating the Sherman Act by setting up and executing an international aluminum cartel abroad is possible.[85] In this case, the court held that "any state may impose liabilities, even upon persons not within its allegiance, for conduct outside its borders that has consequences within its borders which the state reprehends".[86] The court then held that the Sherman Act could be applied to the defendants since their execution of cartels in foreign territory had affected the price and amount of aluminum imported to the US.

The effect doctrine has a unique place in the regulation of the platform giants. Due to the openness and extensibility, events occurring on a platform can easily have an effect in other countries, all connected by the global network of the internet. Indeed, the effect doctrine does not mean that any effect whatsoever satisfies the requirement. The legal threshold under the US law demands "a direct, substantial and reasonably foreseeable effect."[87] The business activities of the ICT giants will still be easily covered by this threshold: some of their activities carry direct, substantial and reasonably foreseeable effects in various countries even if they do not maintain physical facilities in those countries.

In this regard, consider the GDPR of the EU. The GDPR is a regulation concerning privacy and security in personal data which was drafted and passed by the EU in 2018. It imposes obligations onto organizations anywhere in the world that target or collect data related to people in the EU.[88] It has components of extraterritorial application. It does have a nexus with the EU in that it targets those foreign companies dealing with the information of EU citizens. And yet, the nexus with the EU is either diluted or thin, as personal data is collected by ICT companies on a global basis without reference to nationality, which then means any company using an internet database for a global business operation will be forced to put it under the reach of the GDPR. It would entail a classic phenomenon of extraterritorial application of domestic law.

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[84] United States v. Aluminum Co. of America and others, Court of Appeals of the United States, 2nd Cir, 148 F. 2d 416 (1945).

[85] Alford [27].

[86] Layton and Parry [28].

[87] *Foreign Trade Antitrust Improvements Act of 1982* (hereinafter FTAIA).

[88] *See* website, *GDPR. EU*, available at https://gdpr.eu/what-is-gdpr/.

5.3 *Digital Trade Treaties*

The proliferation of digital trade due to AI calls for updates of digital trade treaties. For example, regulations restricting the collection and processing of personal information by foreign companies may seem to contradict the national treatment principle embedded in the WTO and FTAs.

New norms are now being introduced, but they are mainly geared towards facilitation of cross-border trade via the internet, tariff elimination of ICT products, and the prevention of non-tariff barriers. However, it is important to consider the future of digital trade as a whole, and how new types of trade, combining goods and services, will arise. In particular, the core element of such trade will be the data flow. In consideration of a new international agreement on AI and international trade, the following details should be included to reflect the special characteristics of digital trade and the AI era.[89]

5.3.1 Adoption of Digital Trade Chapters and a List of Applicable Items

In consideration of the unique characteristics of digital trade, it is of great importance to introduce a separate relevant chapter. Some trade agreements specify 'electronic commerce' in a separate chapter, but it is quite different from a classification of digital trade. In the future, digital trade will advance not only in a way that sales and purchases are practiced by electronic means but also in a way that supply and consumption of goods and services are practiced at the same time. It is required, accordingly, to specify norms regulating such trade items in a separate chapter. In addition, such chapters need to list applicable digital trade items specifically. Such a list needs to be developed. Since various forms of goods and services are combined and the development rate is also quite high, it may not be easy to list all the specific items (*e.g.,* autonomous vehicle).

However, the way of listing major characteristics and suggesting representative goods and services of such characteristics should be considered. For example, the following major characteristics can be suggested: (i) To be embedded in goods; (ii) To be connected to the network via internet; (iii) Such goods are utilized as a medium for service supply and consumption at all times; (iv) Smooth provision of service is the core value of such goods; (v) The service itself relies on the existence of such goods; and (vi) The service corresponds to a certain item in the classification table of services. Additionally, major examples of trade items with such characteristics can be presented in an open-ended manner. For trade items that meet all these conditions, a newly introduced, separate chapter can be applied.

[89] *See* Weber [29]; Karachalios and McCabe [30].

5.3.2 A New Approach to GVC Matters

The matter of the the GSC or global value chain (hereinafter GVC) which is currently one major issue of international trade, also needs to be considered. The present GSC or GVC only assumes the case where 'division of work for goods production' is practiced in countries.[90] In this respect, digital trade embraces cases where entities in various countries participate in supply not only of goods but also of services. For example, a certain medical service may be provided through a smart-phone or a new electronic device. It is also possible to specialize tasks. For example, the database to accumulating personal information such as medical records managed by a Singaporean company; the initial consultation for detailed questions about patient conditions by a medical team in India; analysis of photographs such as x-ray pictures by a Japanese medical team; comprehensive judgment by a medical team in the US; and follow-up management and provision of medicines by a medical team in Korea. In this context, service suppliers in various countries participate in a certain supply of service. This is a type of GSC or GVC in service trade. Not only for goods trade but also for service trade, cooperation among multiple suppliers in different countries is possible and might even be required. In view of the importance of GSCs and GVCs in the digital era, such aspects also need to be considered.[91]

Furthermore, a fundamental issue is raised regarding how to calculate and regulate GSCc and GVCs in a condition where new trade items—goods and service—are integrated. If the two are inseparable due to their characteristics, how could each GSC or GVC be calculated? Since a GSC or GVC is an important criterion in the determination of origin which is a deciding factor whether to apply a trade agreement, the significance of this issue will be brought up continually. More details in this regard need to be specified in new trade agreements.

5.3.3 Review of Governmental Support Measures

Another critical factor is to secure appropriate governmental roles in the process. A number of businesses need certain forms of governmental intervention inevitably to some extent right from the initial stage in consideration of the basic nature (a tremendous amount of initial investment and the necessity of coordination among relevant industries) and additional effects (indirect ripple effects on many industries).

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[90] The concept of GVC is related directly to the structure of a global production network and any changes to that structure. *See* World Bank Group et al. [31].

[91] The OECD defines 'digital trade' as a type of trade based on transfer of information. Information or data is a means of production and an asset that can be transected. As a means of service provision, it also plays a key role in the global value chain structure. Information-based economy includes new types of service industry sectors based on customized service, IoT, AI, and so forth. *See* Aaronson and Leblond [32].

As new technology develops, significant changes are expected in production methods of goods and services as well. In use of AI and IoT, it is possible to control the entire process at once from selling products to serving customers at once. *See* Lyall et al. [33].

As shown in a series of cases related to Korea, however, governmental policies related to key industry sectors have long been a topic of trade disputes.[92] Such governmental coordination or interventions always involve the risk of being charged by competing countries for violation of the WTO Agreement (subsidy grant or domestic business preferential action). The present subsidy agreement, for example, includes no exceptions nor special provisions on R&D subsidies . Rather, R&D subsidies are regulated just like other subsidies.[93] If the core of the digital industry is connected with R&D businesses, it is inevitable that domestic policies related to the digital industry become a major issue in subsidy disputes. It is essential, therefore, to review such matters in a different perspective and introduce new norms accordingly. Governmental interventions with market-distorting effects need to be distinguished from fair R&D businesses. It is true even now, but in the digital age, the necessity should be given far more emphasis.

Even domestic business preferential measures that are in conflict with Article III of the GATT 1994 and the GATS Schedules of Specific Commitments may be justified according to Article XX of the GATT 1994 and Article XIV of the GATS, but the conditions specified in provisions of general exceptions are quite strict. After all, it is highly likely for digital industry promoting policies fail to be justified due to such provisions of general exceptions. The present trade agreement system always involves the probability for governmental leadership in conducting general projects regarding the fourth Industrial Revolution to face problems. The current system is inconsistent with the reality before it. It is accordingly necessary to introduce relevant provisions in a new chapter.

5.3.4 Comprehensive Review of Relevant Issues

The following are supplementary agreements that specify major items of goods trade: *WTO Agreement on Technical Barriers to Trade* (hereinafter TBT Agreement), *Agreement on Sanitary and Phytosanitary Measures* (hereinafter SPS Agreement), *Agreement on Agriculture*, *Agreement on Rules of Origin*, and *Agreement on Trade Facilitation.* Their provisions are included in major chapters of FTA's as well. New chapters regarding digital trade items also need to include modified or revised forms

[92] WTO Panel Report, *United States—Anti-Dumping Duty on Dynamic Random Access Memory Semiconductors (DRAMS) of One Megabit or Above from Korea,* WT/DS99/R (adopted 1999.03.19.); WTO Panel Report, *European Communities—Countervailing Measures on Dynamic Random Access Memory Chips from Korea,* WT/DS299/R (adopted Aug. 3, 2005); WTO Appellate Body Report, *Japan—Countervailing Duties on Dynamic Random Access Memories from Korea*, WT/DS336/AB/R (adopted Dec. 17, 2007); WTO Panel Report, *Korea—Measures Affecting Trade in Commercial Vessels*, WT/DS273/R (adopted Apr. 11, 2005).

[93] Originally, the subsidy agreement presented the provision of "non-actionable subsidy" in Part 4 and specified 3 types of allowable subsidies that are granted by each government: subsidy for R&D, subsidy for backward area development, and subsidy for environmental improvement. Since December 31, 1999 when this provision of allowable subsidies became ineffective, subsidies are treated all in the same way. It is noted that in various subsidy conflicts pending at present, different types of R&D subsidies are a main source of disputes.

of such material contents of these supplementary agreements. Such supplementary agreements or their certain chapters may be preserved and applied to general goods trade. As to digital trade items, however, it is necessary to modify them prior to their introduction or application.

For example, there should be a new type of a TBT Agreement provision that applies international standards for both goods and services fairly. In general, this agreement specifies technical regulations regarding packing, labeling, and standards for typical goods to be traded. In consideration of characteristics of digital trade items (goods and services), new regulations need to be introduced.[94] Special attention needs to be paid to how to select a market where acts of discrimination are to be investigated and how to manage standards. In order to prevent an importing country from excluding competing items from the market by applying its own technical standards, provisions that in principle allow technical standards for an actual global trade market need to be strengthened. Relevant provisions need to be adjusted so that if the importing country intends to apply other technical standards, it is required to clarify the reason and basis.

When it comes to the SPS Agreement as well, it is required to review the meaning of 'threats' to human life and health that result from a certain digital trade item and to revise relevant provisions accordingly. In case medical services provided for individuals through high-tech medical facilities advance to the point of threatening human life and health, it also needs to be considered if a type of SPS action can be taken. If home appliances, etc. are connected with the purchase of agricultural products and food materials, many provisions in an agricultural agreement also can be relevant. With regard to trade facilitation measures, conditions that are required for such new trade items in the customs procedures needs to be reviewed.

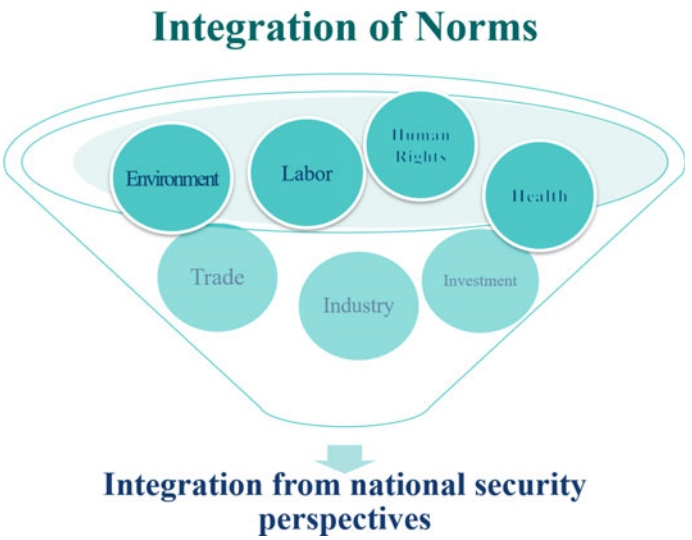
Such aspects that are fragmented in existing goods agreements are all to be integrated, rearranged, and reflected in appropriate new chapters of digital trade. In this process, the significance of intersecting points with services needs to be taken into careful consideration.

5.3.5 National Security Exception

Another area that needs clarification is the intersection between national security and free trade. National security is one of the core concerns of a state. As a result, under the principle of state sovereignty, national security is one of the most well-guarded areas. There has been a recent increase in attention to exceptions related to the national security concerns. As digital security becomes more important with the advent of AI, national security exceptions will also be invoked to limit digital trade. A national security exception is an example of how norms from different issue

[94] For example, there may be a situation where a medical diagnosis service and legal advice are provided through medical AI or legal AI mechanisms. In this case, it is uncertain whether such AI mechanisms are to be viewed as goods or service (or service supplier). Besides, it is uncertain who will be qualified to provide such medical diagnosis and legal advice services on what basis. Hence, the issue of technical regulations is also raised. *See supra 293*, p. 6.

areas—such as trade, investment, and industry—can be integrated mainly from the national security perspective.



The national security exceptions clause made its first appearance in the GATT 1947,[95] and since then, it has been included in numerous treaties (*e.g., Treaties on Friendship, Commerce and Navigation* (FCN Treaties),[96] the *Treaty on the Function of the European Union* (TFEU),[97] the *Comprehensive Nuclear Test Ban Treaty of 1996*,[98] the *Convention on the Physical Protection of Nuclear Material of 1980*,[99] and the *International Convention for the Suppression of Acts of Nuclear Terrorism of 2005*[100]), including a number of IIAs.[101] The CPTPP also contains a security exceptions provision, which stipulates as the following[102]:

Article 29.2 Security Exceptions

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[95] *See* UN Econ. & Soc. Council [34].

[96] As for the information on US FCN Treaties, *see Enforcement and Compliance*, Off. Trade Agreements Negot. & Compliance, available at https://tcc.export.gov/Trade_Agreements/All_Trade_Agree ments/index.asp.

[97] *See Consolidated Version of the Treaty on the Functioning of the European Union* (TFEU), Article 346 (Oct. 26, 2012), 2012 O.J. (C 326) 47.

[98] *See Comprehensive Nuclear-Test-Ban Treaty,* Article IV, para. 57 (Sep. 24, 1996), available at https://www.ctbto.org/fileadmin/user_upload/legal/CTBT_English_withCover.pdf.

[99] *See Convention on the Physical Protection of Nuclear Material,* Article 6 (Mar. 3, 1980), 1456 U.N.T.S. 101.

[100] *See International Convention for the Suppression of Acts of Nuclear Terrorism*, Article 7 (Sep. 14, 2005), 2445 U.N.T.S. 89.

[101] *See* Lee [35].

[102] Ibid., p. 191.

> Nothing in this Agreement shall be construed to:
>
> (a) require a Party *to furnish or allow access to any information* the disclosure of which it determines to be contrary to its essential security interests; or
>
> (b) preclude a Party from applying measures that it considers necessary for the fulfilment of its obligations with respect to the maintenance or restoration of international peace or security, or the protection of its own essential security interests (emphasis added).

The same provision is also included in the USMCA.

In particular, the "refusal to furnish information" provision, included in national security exceptions, is highly relevant in the AI era. With its extensive languages, the provision can be invoked by a state refusing to provide "any information to any entity in any proceeding", which could even severely derail international dispute settlements.[103] As the importance of data and information grows in the AI era, states will see a greater use of this provision.

The relationship and interaction between the national security exceptions and other exceptions are far from clear. For example, the exceptions for measures in pursuit of LPPO, included in the digital trade chapters of more recent trade agreements such as CPTPP and USMCA, may overlap with the national security exceptions. While the digital trade chapters address the issue of personal information, the scope and limits of the related LPPO provisions are unclear. The unsystematic development of digital trade norms may be at the root of such confusion. For a starting point, the overlaps between the LPPO and national security exceptions need to be clarified so that each could achieve its own initial objectives without harming the digital trade norms.[104]

5.4 The Increasing Importance of Data and Information

Data and information is becoming ever more important in the digital economy and in the age of AI. The commercial value and practical benefit of AI technology and machine stem from data and information, since they constitute the core ingredients and essential components of such technology and machine. In a sense, AI starts with data and ends with data. The advent of AI-era, therefore, is another way of saying the advent of the age of data and information. As AI becomes important in all corners of the society, so do data and information. No wonder major countries have initiated their respective national drive for the collection and utilization of data and information and, at the same time, have been wary of rival countries' drives as such. If anything, their importance will only continue to grow. At the same time, as data is capable of being applied and utilized in many different, creative, and unpredictable ways, once it is released from a source or provided by someone, its final destination or final

[103] Lee [36].

[104] Lee [37].

mode is beyond reasonable control. In fact, in a 2013 ICJ dispute between Timor-Leste and Australia involving information allegedly relating to national security of Australia, the ICJ pointed out that once data has been submitted to someone, it is difficult to protect its confidentiality completely, a statement which explains why states and governments sometimes attach significant importance to certain data and information relating to their national business.[105]

5.5 Different Views of States on Data and Information

While the importance attached to data and information is increasing, states and governments possess and portray different views on data and information in specific cases and instances. In other words, they all agree to the importance of data and information generally, but seldom agree to the level or degree of the importance of specific data and information in a specific instance. Such being the case, when a treaty provision stipulates the authority or discretion of a contracting party as regards a certain set data and information, it is very possible the two contracting parties may have differing views, even in good faith. It follows then that a contracting party's invocation of the provision based on its own assessment of the data and information at issue, the other party may oppose. Depending on which view is taken, violation or non-violation will be determined, which will in turn determine the disposition of significant economic interest in the digital society. It is a typical recipe for a dispute. By way of example, the nature of data and information is also provided for in national security exceptions of various treaties. If data and information concern national security matters, then their provision can be denied legitimately under these exceptions provisions. This provision may turn into a Pandora's Box because it can be invoked as a shield to any refusal of provision of data and information. Those exceptions clauses do not contain any qualifying words, so the exceptions can be invoked in an open manner as long as the invoking states believe in national security threats. This was not a critical issue in the previous era. In fact, with respect to Article XXI of the GATT 1947 and GATT 1994, there have been only a couple of instances in which countries have ever invoked Article XXI.: in 1949 by the US against Czechoslovakia, in 1961 by Ghana against Portugal, in 1982 by the EEC, Australia, and Canada against Argentina, and in 2016 by Russia against Ukraine. Even in these disputes, refusal to provide data and information was hardly mentioned. It is probably because data and information were not critical in those disputes in the analogue era. The situation is now changing and we are entering the new global landscape where data and information are always everything. If so, this particular provision in security exceptions in many treaties and agreements carries the potential for frequent invocation in the digital age, Different views of states on the nature of data and information, therefore, will stoke further disputes and confrontation in this regard.

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[105] *See Questions relating to the Seizure and Detention of Certain Documents and Data (Timor-Leste v. Australia), Provisional Measures,* Order of 3 March 2014, I.C.J. Reports 2014, para. 47.

5.6 Outdated Provisions

Furthermore, even those provisions on information in existing treaties and treaties are now outdated because the term "information" itself now has a new meaning and scope compared to its original understanding of the meaning and scope. The broad nature of "information" can be in fact confirmed in its dictionary meaning. The term "Information" means (i) knowledge obtained from investigation, study, or instruction, (ii) intelligence and news, and (iii) facts and data.[106] In addition, information includes signals and letters that represent the attributes or data that exist in order and combination, such as the nucleic acid sequence of DNA or the binary number system of computers.[107] In summary, any medium and means intended to convey facts or knowledge expressed in various ways such as letters, pictures, figures, numbers, or symbols correspond to this definition of "information." In fact, it covers a wide range of areas, issues and matters. Therefore, the term "information" or the phrase "any information" appearing in various provisions of existing treaties and agreements stand to cover various types of items, possibly much broader than our conventional thinking. The same also applies to the definition of "data" and its coverage. Existing provisions in the current treaties and agreements having relevance to data and information should now be re-visited and adjusted, with the new implications in mind.

5.7 Information's Close Relevance to National Security

As explained above, data and information possess close nexus to national security. Certain national security exceptions clauses contained in treaties and agreements allow states to "refuse to furnish any information" when the invoking state believe such information relates to national security matters: by way of example, 35.2% of the treaties with security exceptions clauses signed by South Korea between 2010 and 2020 include provisions for refusal to furnish information. Article XXI of GATT 1994 is the most representative in this respect. It permits governmental measures, even if inconsistent with other provisions of GATT 1994, when the measures are taken in relation to fissionable materials, the production and acquisition of weapons and military supplies, wartime and other emergencies, or fulfillment of the obligations under the UN Charter. In addition to these listed items, information concerning national security may also be withdrawn or may not

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[106] *See* Merriam-Webster Dictionary, *Definition of Information*, available at https://www.merriam-webster.com/dictionary/information. Oxford English Dictionary defines the term as follows: 1. The shaping of the mind or character; communication of instructive knowledge; education, training, advice, 2. Knowledge communicated concerning some particular fact, subject, or event that of which one is apprised or told; intelligence, news, 3. The action or fact of imparting the knowledge of a fact or occurrence; communication of news; notification and others. *See* Oxford English Dictionary, *Definition of Information*, available at https://www.oed.com/view/Entry/95568?redirectedFrom=information#eid.

[107] See ibid.

be provided. Considering the list of instances implicating national security in Article XXI of GATT 1994, at least information related to such instances can be understood as pertinent to national security. But the scope of the information falling under the refusal possibility is not confined to those instances. Again, the provision relating to refusal of provision of information states "any information" that the state considers to be contrary to national security of the state, which is arguably much broader and open-ended. Accordingly, whether the information at issue is confidential under domestic law is irrelevant here because it is still possible that the disclosure of information is nonetheless "contrary to essential security interests" under the provision. Noting the increasing frequency of states' invocation of national security exceptions these days, the textual linkage to the digital economy and AI (i.e., information) is indeed concerning due to its potential to undermine otherwise existing obligation to provide data and information and permit their movement to foreign countries. The security exceptions stand to be adjusted when opportunities arise.

5.8 AI's Relevance to National Security

Furthermore, not only data and information, AI itself is closely related to national security. First of all, AI is based on data and information which are closely related to national security, as understood by many states. Thus, AI is inseparable from national security due to its data and information component. If anything, the sensitivity of the same data and information would be amplified in the context of AI because of the new potential to process the source data to reach a next level or in a new dimension. Thus, a typical piece of ordinary information nay turn into information relating to national security when it is combined with other information or re-processed with other information. The critical question, therefore, is not just a single piece of information; it is rather about how, when and where the information can be utilized in the future. Arguably uncertainty is inevitable here. Uncertainty would make states more defensive and conservative in the new digital age when they evaluate specific information's national security implications. In addition, as AI is increasingly used in the core business of a government, national security concern arises naturally. Consider law enforcement agencies, intelligence agencies and armed forces in this regard. AI technologies and machines applied and utilized in these agencies are closely related to national security of the state at issue. All in all, an argument can be made that AI is inseparable from national security.

6 Shattered Multilateralism in the Age of AI

Finally, the advent of the digital society and AI also complicates the already staggering global trading regime. But they also offer an opportunity to revive the regime.

In all material respects, the global trading regime is now in a perilous situation. Multilateralism has been gradually losing its status and authority: arguably, the WTO is at its lowest point at the moment. And yet, there does not seem to be any breakthrough or grand settlement in the foreseeable future. In the absence of such a breakthrough, unfortunately the international community will be unable to break away from the downward spiral.

One of the major criticisms of the WTO regime is the increasing gap between reality and norms. The WTO Agreements are the outcome of the Uruguay Round negotiations between 1986 and 1994. The business and trade environment since then has undergone a sea change. At that time, there was no e-mail, no smartphone and no Google. AI and automated cars were simply unthinkable. So, legal texts concluded in those days apparently fail to reflect the business and trade reality of today. The digital society and AI dramatically showcases the wide gap between the analogue norms of late 1980s and the business realities of 2022. And yet, in principle the WTO dispute settlement system is forced to enforce the rules as they are written.[108] The frustration and dissatisfaction of states with the existing regime have been growing over the years. This widening fissure appears to underpin the US-China trade confrontation.

Unfortunately, the members of the WTO missed the opportunity to narrow the gap when they failed to conclude the Doha Development Agenda (hereinafter DDA). If this issue is not addressed effectively in the near future, frustration will intensify; norms will gradually lose their normative value; and disputes will continue to increase.

To address this problem, the WTO panels and the Appellate Body could and perhaps should have been given more discretion. If so, they could actively apply evolutionary interpretation to keep up with the changing environment, even within the confinement of the present texts. But they are legally restricted[109] and practically cornered.[110] Some proposals are being floated to further restrict the discretion of these WTO adjudicative bodies. In the meantime, the gap is growing ever wider and faster. The age of AI is pushing the envelope to the point of breaking.

It is hoped that even if DDA is going nowhere FTAs can help address this problem. True, the FTAs do contribute to updating and upgrading trade agreements, at least bilaterally or plurilaterally. But there is an inherent limitation. First of all, many of FTAs' key provisions focus on eliminating trade barriers and liberalizing trade among the bloc. Updating and upgrading trade norms have not been their top priority. The CPTPP is an exception in this regard, but its contribution can be said to be largely limited to the addition of selected new norms (such as SOEs and fisheries subsidies regulation) as opposed to updates across the board.

Besides, many critical provisions of FTAs also simply refer to the WTO Agreements for norms and guidance. To the extent that there are loopholes and gaps in the

[108] *See* DSU, Articles 3.2, 19.2.

[109] See ibid.

[110] The Appellate Body will cease to function as of December 11, 2019 if the current impasse continues at the WTO's DSB.

texts of the WTO Agreements themselves, these FTAs are also operating in a strait-jacket. In short, regardless of FTAs, the WTO Agreements or their future replacement for a new multilateral regime if that turns out to be the case are still important to restore multilateralism in the context of the digital society and AI.

Then what can be done? A new multilateral trade agreement will certainly offer an effective solution. That objective, however, seems to be too ambitious at the moment considering the deteriorating trade environment and spread of protectionism. If that is the case, Plan B options can be contemplated in the interim, in which digital society and AI are actively reflected within the basic framework of the current WTO regime. Specifically following actions may be contemplated:

A. Facilitating Monitoring and Advisory Roles of the WTO—Enhancing and Expanding the TPRM Mechanism: new data collection and processing systems can be employed to expedite TPRM process, and flag problematic measures of the members. More information can be retrieved from the target member and such information can be disseminated among members more promptly. Similarly, follow-up measures after TPRM process can also be tracked down.
B. Facilitating Non-Binding Dispute Settlement Proceeding—Fostering Mediation and Conciliation (in addition to the binding dispute settlement mechanism): new types of non-binding dispute settlement applying and utilizing AI technology can be contemplated for certain types of trade dispute. States will be reluctant to use AI mediators or conciliators, but they will be more open to the situation where mediators and conciliators are supported by AI technology and machines. As these non-binding proceedings are not necessarily confined to the text of the WTO Agreements, AI technology can search broad sets of data and information relating to the matter at issue, and provide a solution package to the mediator or conciliator.
C. Exploring Other Avenues of Dispute Settlement—Resorting to Arbitration under Article 25 of the DSU: in the arbitration proceeding AI technology can be widely employed. The parties may agree in their arbitration agreement to adopt AI technology broadly. It may reduce workload and cost. Hearings can be held online with the support of new technologies. Most importantly, the parties may also agree to look to other relevant international law instruments for issues spanning over multiple areas of international law, which can be facilitated by AI technology.
D. Formulation and Distribution of Best Practices in Key Industries and Trade Sectors (see the OECD steel discussions and GAMS semiconductor discussions): members of the WTO may apply and utilize AI to disseminate and share information on best practices in diverse areas of industry and trade. The secretariat of the WTO may also rely on AI technology for the collection and distribution of information in this regard. It can be a effective tool for the capacity building projects for developing members and LDCs.
E. Agreeing on a Global 'Standstill' Agreement–Refraining from Protectionism during a New Round of Negotiation (after MC 12 in Kazakhstan in June 2020):

through AI technology trade restrictive measures may be identified and examined more promptly. The basic information of the measures and summary evaluation of the legal implication may also be available from the application and utilization of AI. The evaluation may not be legally binding, nor does it carry any legal effect in the absence of official dispute settlement proceedings. Nonetheless, the dissemination of the basic information and overall assessment of the measure from the perspective of similar measures examined by the panels and Appellate Body may offer some helpful guidance for members' discussion of the measures at issue.

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Chapter 8
Conclusion



As AI is expected to change our society across a wide range of issues in diverse sectors, it is necessary to examine more fundamental aspects of this new phenomenon in terms of legal norms, in particular from the perspective of international law. Currently, there is a widening legal gap in dealing with new phenomena and problems associated with AI. The gap appears in various branches of international law including IHL, international criminal law, international human rights law, international trade law, international environmental law, international investment law, law of treaties, and state responsibility, etc.

New technologies call for new regulations. Since automated weapons and automated vehicles could effectively change the nature and characters of war, IHL needs to be adjusted accordingly. Similarly, the power of data and information in the AI era calls for better protection of personal information under the international human rights law, as well as a harmonized set of international standards on personal data collection, provision, and utilization. Existing treaties and agreements need to be revisited and examined for possible updates through amendment, and new treaties and agreements should be explored to fill the legal gap. If left unaddressed, disputes are bound to increase and intensify given the enormous value of data and information in terms of commercial, business, and national security perspectives.

With regard to the Fourth Industrial Revolution currently unfolding and the advent of the digital society and economy, governments and enterprises stand to suffer from the lack or absence of reliable legal norms. Conventional norms are not tuned to the new living and business environment of the digital age. Nor do they acknowledge the usefulness and importance of data and information. Under these circumstances, the continuing application and mechanical interpretation of the conventional analogue treaties and agreements may only delay the technological development and hinder social innovation. What are coming on our way instead are disputes and confrontations among states. It is time to update existing treaties and agreements and explore adoption of new ones. Customary international law may not offer that much of help given the time required for the formulation of the law. The pressing need for a new norm can be best illustrated by the situation in the trade, investment and taxation

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agreements where reform efforts are already under way. Those who may suffer from the lack or absence of appropriate norms are the entrepreneurial producers and businessmen that produce new goods, services and items with advanced technology and release them to the market, domestic and foreign. In fact, they only have one market in mind—the global market. The digital society and digital economy operate on a global scale at a global level at the same time. Focusing on domestic markets, as before, would be missing the point now. Particularly, one of the most significant characteristics of the new digital society and economy is that goods and services are produced, sold, provided, and consumed in an integrated manner, and sometimes in the global market at the same time. All these activities are also intertwined with the critical components of IPRs. While existing treaties and agreements in these areas were originally formulated in accordance with the dichotomous approach, separating goods and services, it is not valid any more. While the existing treaties and agreements have viewed goods, services, investment, IPRs, taxation, environmental protection, labor standard adherence, and competition as distinct and separate, all of them emerge and operate simultaneously in the digital age. The linkage is provided by AI. It is time to explore a new paradigm of treaties and agreements in which these changes and demands are reflected adequately, so that international law can support the emergence and evolution of the digital society, and foster development of AI technologies and machines in due course. It may be called 'digitization' of international law. As with any other legal norm, international law should keep up with the changing reality and shifting environment in the global community.

Another important issue in this regard is state responsibility. As AI is utilized by the global community actively and robustly, and particularly by state agencies and governmental entities for a variety of public purposes in various ways and in various fields, it is almost certain that new legal issues never seen before will emerge gradually with respect to government measures adopted by AI or with the support of AI. When present and future norms of international law concerning AI are discussed on the international stage in due course, it is necessary to take into consideration the relevance to and connection with the law of 'state responsibility' in the application and utilization of AI for regulatory purposes and public objectives. One of the critical analyses in this respect is to analyze aspects of state responsibility that are related to acts of governments and agencies in light of the introduction and arrival of AI in the governmental business and in the public sector. The key question to be answered here is whether, when and how judgment and decision making of AI are attributable to the state in question under international law. This matter is also closely related to the philosophical and legal-philosophical questions of how to view and understand the nature of AI (*vis-a-vis* humans). Finding state responsibility in the context of AI under international law will be dependent upon the final conclusion of this question of the nature of AI. This issue is currently being examined in other branches of law as well. Under the current law of state responsibility, subjects whose acts are attributable to a state are supposed to mean either a natural person or a juridical person. If a new subject not corresponding to this binary assumption emerges (i.e., AI), the fundamental nature of the subject in question should be discussed and established. The need for clarification and confirmation of the nature

of AI is most pressing in the context of state responsibility. Since the law of state responsibility, as secondary norms, applies to and implicates all norms of international law, clarification and settlement of the jurisprudence relating to AI technology and machines utilized or adopted by state organs or governmental entities would be crucial. Even before the advent of the AI age, the issue of state responsibility already became a controversial one because of the broadening roles and increasing activities of non-state actors (i.e., private entities) and government-affiliated entities (such as SOEs). It has also become one of the core issues in the US-China disputes recently. Most importantly, the issue of state responsibility is becoming all the more complex and controversial in the digital age, since the fundamental nature of AI is still unclear and different accounts are being presented by different stakeholders. It means that more disputes are coming on our way, and that the bulk of the disputes will concern state responsibility. Future discussion, therefore, should take up this issue and provide clarity to it as soon as possible. Overall, digitization of international law should not be delayed any further. AI technologies and machines are progressing by leaps and bounds while the legal norms applicable to them are either stuck in the analogue age or are moving forward at snail's pace. It should be changed before it is too late.